A First-Principles Dynamic Model of a Rubber V-Belt CVT for General Transient Simulation

With Mechanical Actuation and Traction-Limited Torque Transfer

Kai Arseneau

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Abstract

This work develops a first-principles dynamic model for a mechanically actuated dry rubber V-belt continuously variable transmission (CVT) and its coupling to the vehicle drivetrain. The model is motivated by a persistent gap in the available literature: while quasi-static Baja-oriented models are common and detailed dynamic models exist for hydraulically actuated metal-belt CVTs, publicly available formulations for mechanically actuated rubber belt CVTs do not generally combine sheave inertia, flyweight actuation, torque-reactive secondary clamping, belt geometric closure, traction-limited torque transmission, and vehicle longitudinal dynamics within a single unified framework.

Starting from pulley geometry and belt-length closure, the derivation develops the CVT kinematics, the rotational dynamics of the primary and secondary pulleys, the axial equation of motion governing shift, and the torque-transfer closure required to distinguish sticking from traction-limited slip. The primary clutch model includes centrifugal flyweight actuation and spring preload, while the secondary model includes helix-generated torque feedback together with spring forces. A centrifugal belt correction is also derived to account for the reduction in effective normal force at speed. These elements are assembled into a closed nonlinear state-space model suitable for direct numerical time integration.

The resulting formulation is intended both as a simulation model and as a derivation document in which each major modeling step remains physically interpretable. Simulation results are used to examine whether the completed model behaves in mechanically sensible ways under transient launch conditions. Comparisons of launch shift trajectories, pulley axial-force balance, and torque admissibility across multiple tuning cases show a consistent mechanism chain from tuning changes, to clamping-force evolution, to stick–slip behavior, to overall launch outcome. The default tuning produces the most balanced launch response, combining strong early admissible coupling torque, controlled low-ratio retention, and a smoother main shift. More aggressive cases reveal renewed slip and visibly underdamped oscillatory transients during shift onset, highlighting both the usefulness of the model and the limitations introduced by idealized loss assumptions.

Overall, the formulation provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behavior during launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behavior remains traceable back to clear mechanical causes.

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Nomenclature and Notation

All quantities are expressed in SI units unless otherwise stated. Angles are expressed in radians. Angular velocities are expressed in rad/s. Torques are expressed in N m, forces in N, lengths in m, and masses in kg.

Symbol Summary

Symbols used throughout the derivation are summarized in Table 1.

Symbol	Meaning
State and kinematics	
$\boldsymbol{x}(t)$	state vector
t	time
$\omega_p,\ \omega_s$	primary and secondary angular speed
$s,\ \dot{s},\ \ddot{s}$	shift position, velocity, and acceleration
$R,\ \dot{R}$	CVT ratio and ratio rate
v_b	belt transport speed
$v_b^*,\ T_b$	slip-branch target belt speed and belt-speed relaxation time constant
v_Δ	slip metric: primary-imposed minus secondary-imposed belt-line speed
Geometry	
$r_{p,\text{outer}}(s),\ r_{s,\text{outer}}(s)$	primary and secondary outer radius
$r_{p,\text{eff}}(s),\ r_{s,\text{eff}}(s)$	primary and secondary effective torque radius
$r_{p,\text{outer,min}}$	minimum primary outer radius
$r_{s,\text{outer,max}}$	maximum secondary outer radius
$s_{\text{dz}},\ s_{\text{max}}$	deadzone threshold and maximum shift
β	sheave face half-angle (radial reference)
C	pulley center distance
L_b	total belt length
ϕ	belt wrap angle used in distributed contact model
$r,\ \dot{r}$	local belt path radius and its time derivative
r_{cm}	belt element center-of-mass radius
Constraint and mapping functions	
$f(r_p, r_s, C)$	belt-length compatibility root function ($f = 0$)
$\text{root}_{r_s}(\cdot)$	numerical root solve for secondary radius
$r_f(s),\ \dfrac{dr_f}{ds}$	primary flyweight radius map and gradient
$\theta_s(s),\ \dfrac{d\theta_s}{ds}$	secondary helix angle map and gradient

Symbol	Meaning
Forces and torques	
$F_{p,\mathrm{ax}},\ F_{s,\mathrm{ax}}$	primary and secondary axial mechanism force
$F_{c,p},\ F_{c,s}$	primary and secondary belt centrifugal axial contribution
F_{ax}	generic axial clamp force at one pulley contact
$\tau_{\mathrm{eng}},\ \tau_{\mathrm{load}}$	engine input torque and reflected load torque
$\tau_p,\ \tau_s$	coupling torque seen at primary and secondary
$\tau,\ \tau_c$	generic interface torque and applied coupling/friction torque
τ_{ns}	no-slip requested coupling torque
$\tau_-^{\mathrm{stick}},\ \tau_+^{\mathrm{stick}}$	aggregate lower and upper coupling-torque bounds for stick admissibility
$\tau_-^{\mathrm{slip}},\ \tau_+^{\mathrm{slip}}$	aggregate lower and upper coupling-torque bounds on slip branches
$\tau_{c,p,\pm}^{\mathrm{stick}},\ \tau_{c,s,\pm}^{\mathrm{stick}}$	primary/secondary pulley-wise stick-branch lower/upper bounds
$\tau_{c,p,\pm}^{\mathrm{slip}},\ \tau_{c,s,\pm}^{\mathrm{slip}}$	primary/secondary pulley-wise slip-branch lower/upper bounds
Mass/inertia and material parameters	
$I_p,\ I_s$	primary and secondary rotational inertia
m_f	effective flyweight mass
$m_{p,m},\ m_{s,m}$	equivalent primary and secondary translating mechanism mass
ρ_b	belt material density
A_b	belt cross-sectional area
μ	belt–pulley traction coefficient
$k_p,\ x_{p,0}$	primary spring stiffness and preload displacement
$k_{s,\theta},\ \theta_{s,0}$	secondary torsional stiffness and preload angle
$k_{s,x},\ x_{s,0}$	secondary axial stiffness and preload displacement

Table 1: Core symbols used throughout the derivation.

Constants and Parameters

All parameters in Table 2 are treated as constant values that do not vary through the model. They are either physical constants, fixed by the physical design of the CVT and vehicle or else are tuning parameters that are set and held constant during a given simulation run.

Symbol	Name	Unit	Description
Geometry and belt			
L_b	Belt length	m	Fixed total belt length
β	Sheave half-angle	rad	Cone half-angle used for radial/axial conversion
h_b	Belt thickness	m	Radial belt thickness
b_{out}	Belt outer width	m	Belt width at outer face
b_{in}	Belt inner width	m	Belt width at inner face
$r_{p,\mathrm{outer,min}}$	Primary minimum outer radius	m	Radius at fully open primary
$r_{s,\mathrm{outer,max}}$	Secondary maximum outer radius	m	Radius at fully closed secondary
s_{dz}	Shift deadzone threshold	m	Onset of effective radial motion

Symbol	Name	Unit	Description
$s_{\max}$	Maximum shift	m	Physical upper travel bound
Primary/secondary actuation			
m_f	Flyweight mass	kg	Effective flyweight mass
$r_{f,0}$	Initial flyweight radius	m	Flyweight radius offset at $s = 0$
k_p	Primary spring stiffness	N/m	Primary compression spring rate
$x_{p,0}$	Primary spring preload	m	Initial compression of primary spring
$k_{s,\theta}$	Secondary torsional stiffness	N m/rad	Torsional spring rate in helix assembly
$\theta_{s,0}$	Secondary torsional preload	rad	Initial torsional deflection preload
r_h	Helix reference radius	m	Effective cylindrical radius in helix kinematics
$k_{s,x}$	Secondary axial stiffness	N/m	Secondary axial compression spring rate
$x_{s,0}$	Secondary axial preload	m	Initial compression of secondary axial spring
Inertia and drivetrain			
I_{eng}	Engine inertia	$\text{kg}\,\text{m}^2$	Engine rotational inertia
I_{prim}	Primary pulley inertia	$\text{kg}\,\text{m}^2$	Primary CVT pulley inertia
I_p	Effective primary inertia	$\text{kg}\,\text{m}^2$	Implemented primary-side inertia in current setup
I_{sec}	Secondary pulley inertia	$\text{kg}\,\text{m}^2$	Secondary CVT pulley inertia
I_{gb}	Gearbox inertia	$\text{kg}\,\text{m}^2$	Gearbox/intermediate shaft inertia
I_{wheel}	Wheel inertia	$\text{kg}\,\text{m}^2$	Wheel/hub inertia about wheel axis
I_s	Effective secondary inertia	$\text{kg}\,\text{m}^2$	Lumped secondary-side inertia about secondary axis
G	Final drive ratio	1	Fixed ratio defined by $\omega_s = G\,\omega_w$
m	Vehicle mass	kg	Total vehicle mass
r_w	Wheel rolling radius	m	Effective tire rolling radius
Contact and environment			
ρ_b	Belt density	kg/m^3	Belt material density
μ_s	Belt–pulley static friction coefficient	1	Static traction coefficient used at impending slip
μ_k	Belt–pulley kinetic friction coefficient	1	Kinetic traction coefficient used on slip branches
C_{rr}	Rolling resistance coefficient	1	Tire-road rolling resistance coefficient
C_d	Drag coefficient	1	Aerodynamic drag coefficient
A_f	Frontal area	m^2	Vehicle aerodynamic reference area
ρ	Air density	kg/m^3	Ambient air density used in drag model
g	Gravitational acceleration	m/s^2	Gravity constant

Table 2: Canonical non-varying parameters used by the CVT model equations.

Chapter 1

Introduction & Background

1.1 Problem and Motivation

Mechanically actuated rubber V-belt continuously variable transmissions (CVTs) are widely used in compact powersport drivetrains, including snowmobiles, all-terrain vehicles (ATVs), and utility terrain vehicles (UTVs) (Messick, 2018), because they can vary the speed ratio automatically while transmitting power through a relatively simple mechanical assembly. Their visible layout is straightforward: two variable-radius pulleys are connected by a belt. Their transient behaviour, however, is not. The operating state of the transmission emerges from the interaction of its internal mechanisms, the belt–pulley contacts, and the mechanical conditions under which it operates.

Several coupled effects determine that response. Engine speed influences the centrifugal actuation of the primary pulley, while transmitted torque influences the torque-reactive response of the secondary pulley. Together, these mechanisms generate the clamping forces that press the belt into the sheave grooves. Clamping force determines the traction available at the belt–pulley contacts, which in turn determines how torque is carried and whether the belt and pulleys can remain coupled without gross slip. At the same time, pulley motion changes the belt radii and therefore the transmission ratio, which changes the speed and torque relationships between the two pulleys. These effects do not occur independently; each changes the conditions under which the others act.

This coupling makes the behaviour of a mechanically actuated CVT difficult to interpret from external measurements alone. A shift curve, pulley-speed trace, or observed loss of traction may reflect the combined effect of primary actuation, secondary reaction, belt traction, and shift motion. The underlying states responsible for that behaviour are largely internal to the transmission and are not usually available directly during testing. This is especially consequential because mechanically actuated CVTs are tuned by changing physical components such as springs, flyweights, ramps, and helices; determining the effect of a change can therefore require repeated hardware changes and testing (Skinner, 2020).

Addressing this problem requires a physically interpretable dynamic framework that preserves the connection between the internal CVT mechanisms and the behaviour they produce. A useful model should do more than reproduce an output trace: it should make it possible to follow how actuation, clamping, belt traction, shift motion, and inertia combine as the transmission responds to changing operating conditions. If an undesirable response can be traced to its mechanical cause, changes to the hardware or operating conditions can be made deliberately rather than through trial and error. The motivation is therefore not only to predict transient CVT behaviour, but to make that behaviour understandable enough to diagnose and improve.

The remainder of this chapter develops the context for that framework. Section 1.2 introduces the physical layout and operating principles of the belt–sheave CVT, including ratio change, traction, and mechanical actuation. Section 1.3 contains a literature review that organizes prior work by modelling emphasis and identifies the foundations available for mechanical actuation, belt contact, and transient CVT dynamics. Given this context, Section 1.4 then states the aim, scope, assumptions, and contributions of the present formulation.

1.2 Conceptual Overview of a Continuously Variable Transmission

This section introduces the operating principles of the continuously variable transmission (CVT) considered in this work. The goal is not yet to derive the model equations, but to build the mechanical intuition needed to understand them. In particular, this section explains why a CVT is useful, how a rubber V-belt CVT changes ratio, why clamping and traction matter, how the primary and secondary pulleys generate their forces, and how these mechanisms interact to determine the transmission operating state.

1.2.1 Why Use a Continuously Variable Transmission?

A transmission is responsible for matching the operating speed of an engine to the speed required at the vehicle wheels. In a conventional stepped gearbox, this is achieved using a fixed set of discrete gear ratios. Each gear imposes a fixed relationship between engine speed and wheel speed. As the vehicle accelerates, the engine speed rises through one gear, drops when the transmission shifts, and then rises again in the next gear. This repeated sweep through the engine speed range is a direct consequence of having only a small number of available ratios.

This matters because an internal combustion engine (ICE) does not produce the same torque and power at every speed. Instead, the engine has a useful operating region where it can produce power more effectively, as shown conceptually in Figure 1.1. A stepped gearbox can pass through this region, but it cannot generally stay there while vehicle speed continues to increase. Figure 1.2 contrasts this stepped behavior with the smoother operating path made possible by a continuously variable transmission.

A continuously variable transmission (CVT) addresses this limitation by allowing the transmission ratio to vary smoothly within a finite range rather than changing through a fixed set of gear steps. Since the ratio can change continuously, the engine can remain closer to a desired operating region while the vehicle accelerates. This is the main benefit of a CVT: it gives the drivetrain more freedom to match engine behavior to vehicle load.

SAE Baja Restricted Kohler CH440							
Engine Speed (rpm)	3600	3400	3200	3000	2800	2600	2400
Torque Net Corrected (ft-lb)	13.5	14.5	15.4	16.6	17.4	18.1	18.5
Power Net Corrected (HP)	9.3	9.4	9.4	9.5	9.3	9.0	8.5

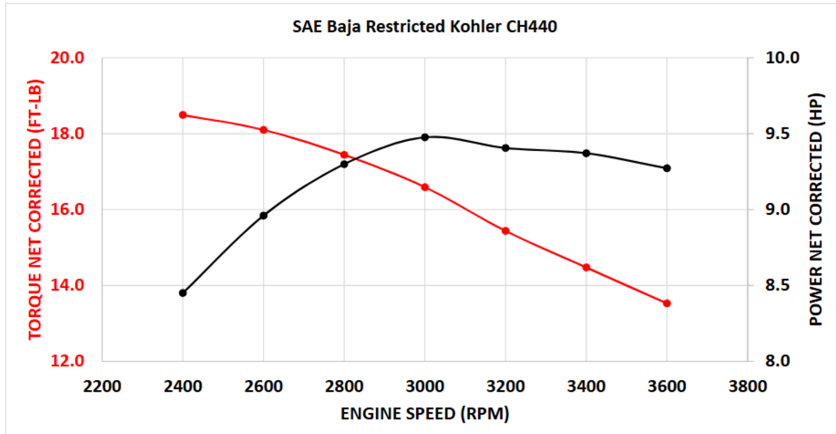

Figure 1.1: Torque and power of the Kohler CH440 engine as functions of engine speed. Both quantities vary substantially across the operating range, with peak torque occurring at a lower engine speed than peak power. (Baja SAE, 2022)

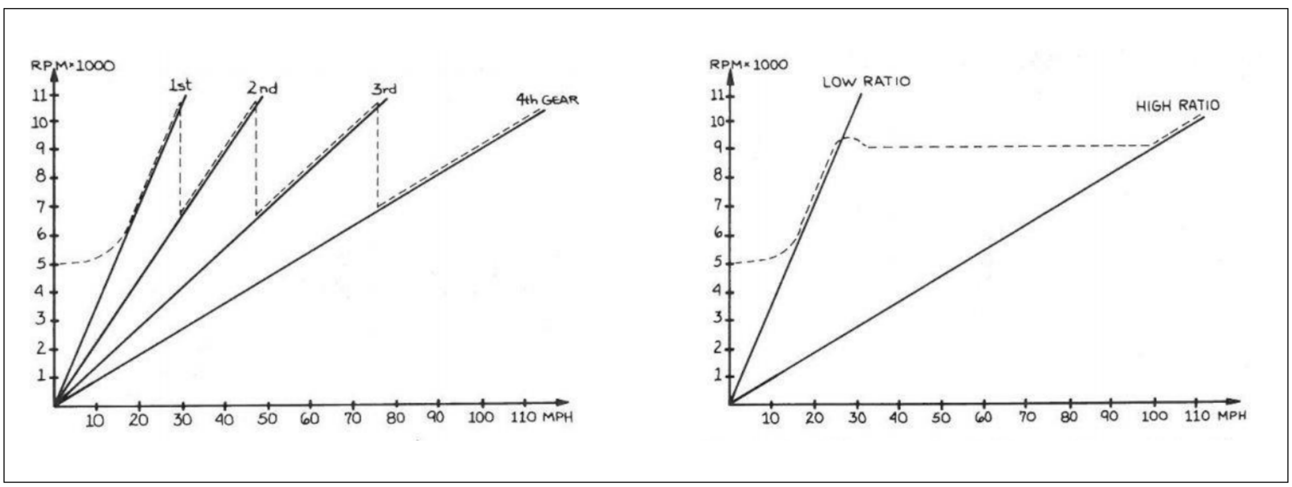

Figure 1.2: Conceptual comparison of engine-speed behavior during acceleration. A stepped gearbox requires repeated engine-speed sweeps and drops between gears, whereas a CVT can vary its ratio continuously to keep the engine closer to a desired operating speed as vehicle speed increases. (Aaen, 2007)

1.2.2 Physical Layout and Single-Pulley Geometry

There are several forms of continuously variable transmission, including different mechanical layouts and different ways of producing the variable speed relationship. This work focuses on a belt-sheave CVT, shown conceptually in Figure 1.3. In this arrangement, two variable-radius pulleys are connected by a rubber V-belt. The pulley connected to the engine side is called the primary (or driving) pulley. The pulley connected to the downstream drivetrain is called the secondary (or driven) pulley. The belt transmits motion and torque between these two pulleys.

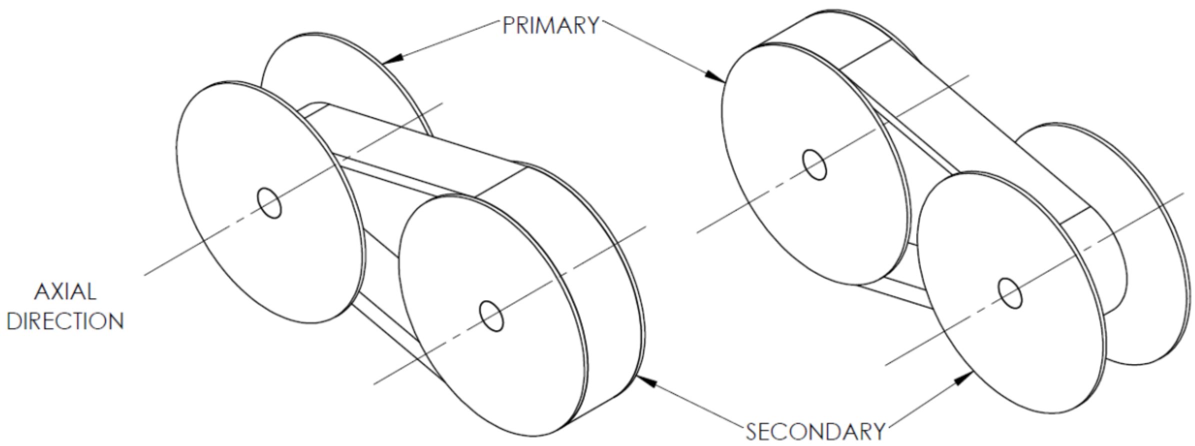

Figure 1.3: Two sample configurations of the belt-sheave CVT considered in this work. The primary pulley is connected to the engine side, the secondary pulley is connected to the downstream drivetrain, and the rubber V-belt transmits motion and torque between them. (Messick, 2018)

Before considering the full two-pulley system, it is helpful to look at the geometry of a single pulley. Both the primary and secondary pulleys share the same basic anatomy. Each pulley is made from two opposing conical sheaves that form a V-shaped groove. One sheave is fixed axially to the shaft, while the other can move along the shaft. The rubber V-belt sits between the sheave faces. This basic layout is shown in Figure 1.4.

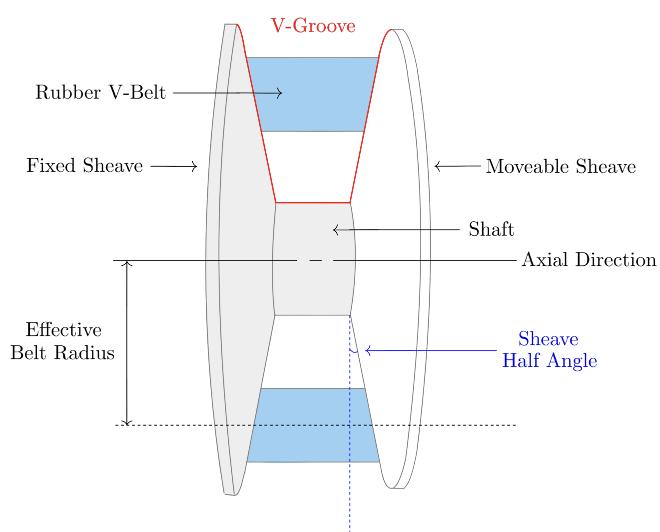

Figure 1.4: Generic anatomy of a belt-sheave CVT pulley. Each pulley consists of opposing conical sheaves that form a V-groove. Motion of the movable sheave changes the groove width and therefore changes the radius at which the belt rides.

The key geometric idea is simple: changing the axial separation between the sheaves changes the effective radius of the belt. If the movable sheave moves toward the fixed sheave, the groove becomes narrower. Since the belt has a finite width, it cannot occupy the same radial position in the narrower groove, so it is pushed outward along the conical sheave faces. The effective belt radius therefore increases.

If the movable sheave moves away from the fixed sheave, the groove becomes wider and the belt can fall deeper between the sheaves. This reduces the effective belt radius. At this level, the mechanism is just wedge geometry: axial motion of the sheave becomes radial motion of the belt. Closing the sheaves makes the belt radius larger, while opening the sheaves makes the belt radius smaller, as shown in Figure 1.5.

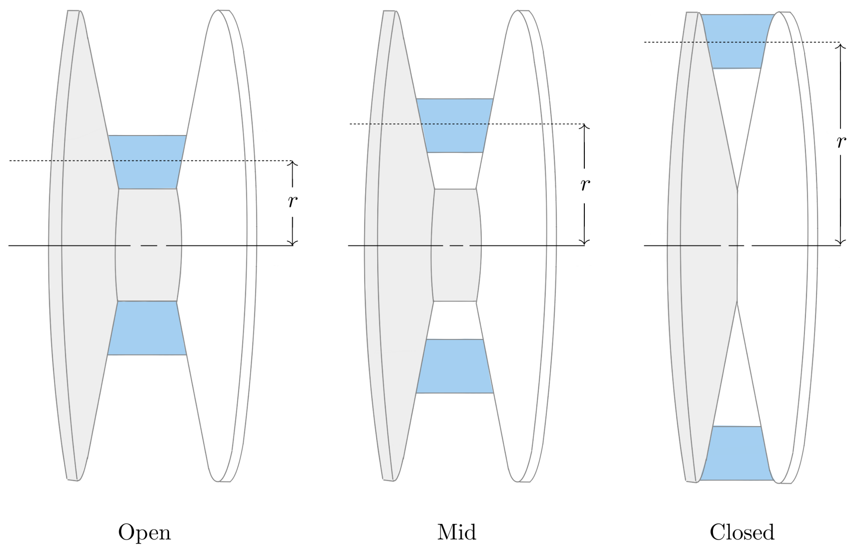

Figure 1.5: Single-pulley radius change. As the sheaves close, the V-groove narrows and the belt is pushed outward to a larger effective radius. As the sheaves open, the belt can fall deeper in the groove and the effective radius decreases.

1.2.3 Two-Pulley Coupling and Ratio Change

The full CVT contains two of the variable-radius pulleys described above. The important difference is that the two pulleys are connected by the same belt. Because the belt length is approximately fixed, the effective radii of the two pulleys cannot change independently.

If the belt rides outward on the primary pulley, it must ride inward on the secondary pulley. Likewise, if the primary radius decreases, the secondary radius must increase. This opposite motion is the core geometric behavior that allows the CVT ratio to change. In the physical mechanism, the actual motion of the sheaves occurs through contact forces, springs, actuator mechanisms, and the axial force balance of the pulleys.

The CVT begins launch in a low-speed, high-reduction configuration. In this state, the primary pulley has a small effective radius and the secondary pulley has a large effective radius. This behaves like a low gear in a conventional drivetrain: the engine-side pulley turns relatively quickly while the downstream side turns more slowly, amplifying torque through the drivetrain.

As the CVT upshifts, the primary sheaves close and the primary effective radius increases. At the same time, the secondary effective radius decreases. The transmission therefore moves from its launch configuration toward a higher-speed configuration. By the end of the shift, the primary radius is larger and the secondary radius is smaller. This coupled motion is shown in Figure 1.6.

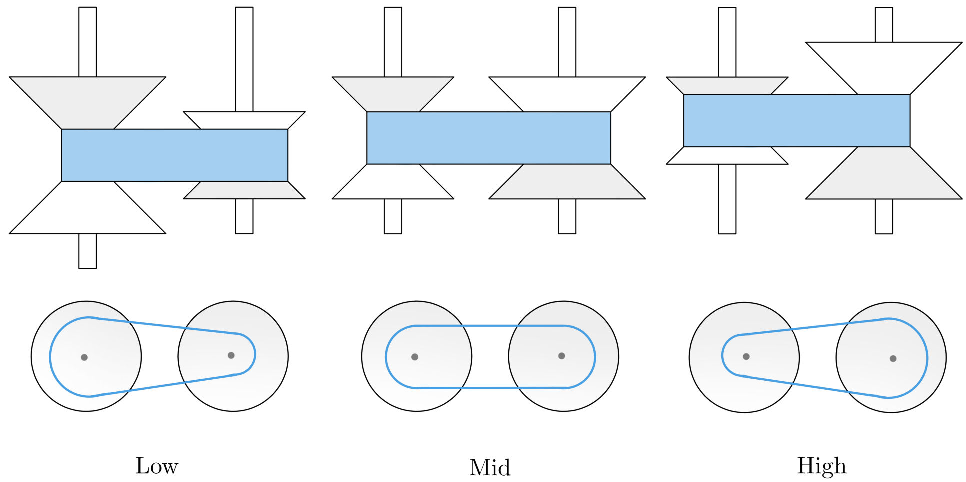

Figure 1.6: Low-, intermediate-, and high-ratio states of a belt–sheave CVT. As the belt moves outward on the primary pulley and inward on the secondary pulley, the effective radii vary in opposite directions while the belt length remains approximately constant. The lower views highlight the corresponding change in transmission ratio.

1.2.4 Torque Transfer, Clamping Force, and Slip

The geometry described so far determines the speed relationship between the primary and secondary pulleys. However, a speed relationship alone does not guarantee that torque can be transmitted. Unlike a gear pair, the belt and pulley are not mechanically locked together by teeth. Torque is transmitted through frictional traction between the belt and the sheave faces.

For traction to develop, the belt must be physically squeezed between the sheave faces. Figure 1.7 shows how this happens at a small portion of the belt–pulley contact. The circled region on the pulley wrap is enlarged to show the belt cross-section within the pulley V-groove. At that location, the movable sheave presses the belt toward the fixed sheave. This clamping action produces contact normal forces on the two sides of the belt, creating the contact pressure needed for frictional traction.

The same local interaction occurs continuously around the wrapped portion of the belt. At each position along the wrap, the normal forces between the belt and sheave faces allow a tangential friction force to act. Together, these small tangential traction forces transfer torque between the pulley and the belt. In practical terms, transferring more torque through the CVT requires enough clamping force to create enough available traction at the belt–pulley interface. More clamping force increases the normal contact force, which in turn increases the tangential force that friction can support.

If the available traction is not sufficient for the torque being transferred, the belt slips relative to the pulley. Slip reduces useful torque transfer and can produce heat, wear, and efficiency loss. A working CVT must therefore do two things at once: it must allow the pulley radii to change so the ratio can shift, and it must maintain enough clamping force for the belt to transmit torque without excessive slip.

This distinction is central to the rest of the model. The pulley geometry determines how the speed ratio changes. The contact mechanics determine how much torque can actually be carried through the belt–pulley contacts. A CVT is therefore not only a variable-geometry mechanism; it is also a clamping and traction problem.

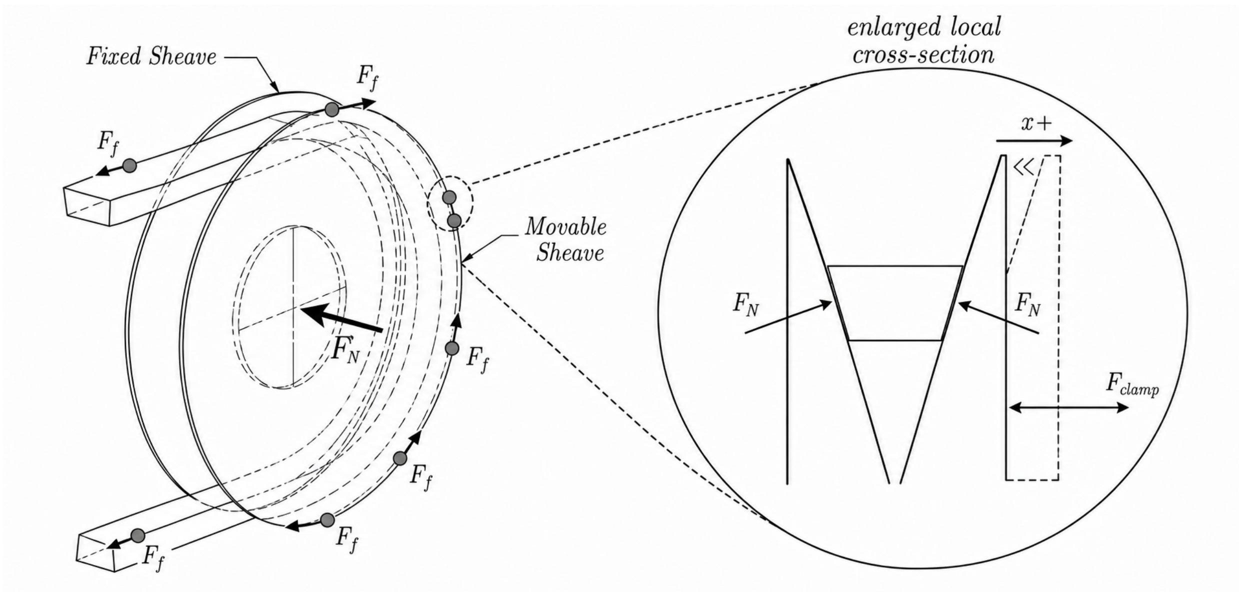

Figure 1.7: Clamping and traction in a rubber V-belt CVT. A representative location on the wrapped belt–pulley contact is circled and enlarged to show the local belt cross-section within the V-groove. Motion of the movable sheave toward the fixed sheave creates normal contact forces on the belt faces. These normal forces allow tangential frictional traction to act continuously along the wrap, collectively transmitting torque between the pulley and belt. The indicated tangential forces are representative local contributions to this distributed traction.

1.2.5 Mechanical Actuation in the Present CVT

Different CVTs generate radius change and clamping force in different ways. Some systems use electronic or hydraulic actuation to command pulley motion or clamping pressure directly. The system studied here is mechanically actuated: its pulley forces arise from mechanisms inside the primary and secondary pulley assemblies rather than from an externally commanded ratio.

These mechanisms are important because they determine both how the belt moves through the ratio range and how much clamping force is available for torque transfer. The primary pulley is mainly speed-sensitive, while the secondary pulley is mainly torque-reactive.

The primary pulley uses flyweights, ramps, and a spring. As engine speed increases, the flyweights move outward under centrifugal effects. The ramp surfaces convert this outward flyweight motion into a closing force on the primary sheaves. The spring resists this motion and helps set the speed at which the primary begins to close. As the primary closes, the belt is pushed outward, the primary effective radius increases, and the CVT shifts in the upshift direction introduced in Figure 1.6. The primary actuation mechanism is shown conceptually on the left side of Figure 1.8.

The secondary pulley uses a helix mechanism and spring loading. The helix can be understood like a screw-like interface: when torque is transmitted through the secondary, the angled helix surfaces convert part of that torque reaction into a clamping force between the sheaves. In this way, torque through the secondary helps create the clamping needed to carry torque through the belt. The spring provides preload and helps set the baseline clamping behavior.

The helix also means that secondary shift is not purely axial. As the secondary moves through its shift range, axial motion is coupled with relative rotation between secondary sheave components. This detail becomes important later when the secondary force model is developed, but the basic intuition is simply that the helix converts torque and relative rotation into axial clamping. The secondary actuation mechanism is shown conceptually on the right side of Figure 1.8.

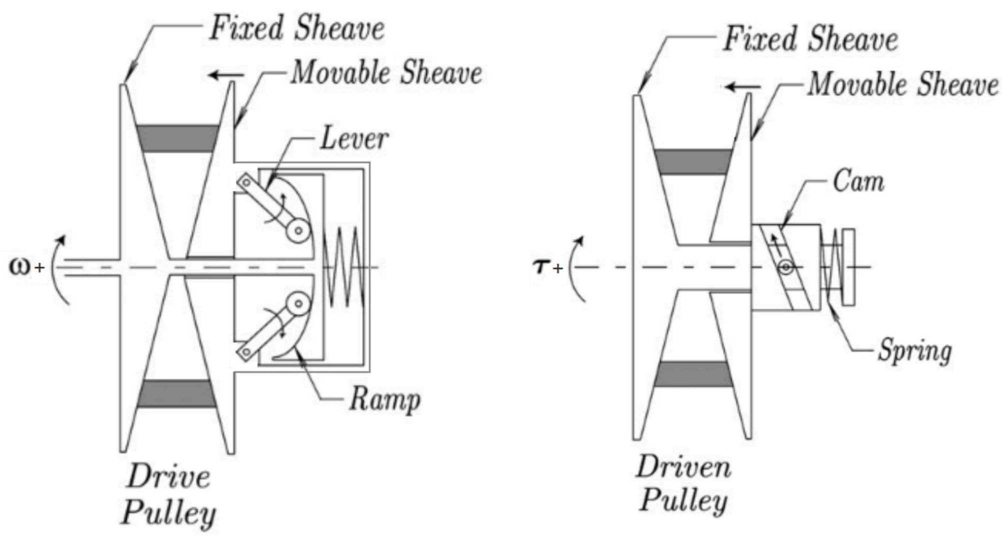

Figure 1.8: Conceptual illustration of the mechanical actuation mechanisms in the present CVT. Left: the primary (drive) pulley, where increasing rotational speed moves the flyweights outward and the ramp geometry converts that motion into a closing force on the sheaves, opposed by a spring. Right: the secondary (driven) pulley, where transmitted torque loads a cam or helix mechanism that converts part of the torque reaction into sheave clamping force, together with spring preload. In this way, the primary is mainly speed-sensitive and the secondary is mainly torque-reactive. (Julió and Plante, 2011)

Although the primary and secondary mechanisms respond most strongly to different physical quantities, they act on the same belt and the same shift process. Primary speed affects centrifugal actuation, transmitted torque affects the torque-reactive secondary, the resulting clamping forces determine the available belt traction, and the belt geometry couples motion of the two pulleys. As the operating conditions change, the balance among these effects can change as well, allowing the transmission to shift in either direction or to enter a different traction state. These coupled relationships form the physical foundation for the dynamic behaviour developed later in this work.

1.3 Motivation and Literature Review

CVT modelling is not a single, uniform problem. A model developed to size a variator, estimate belt efficiency, design a control strategy, or predict transient behaviour may involve the same physical components, but it will usually make different choices about which quantities are prescribed and which are solved. Some models are primarily design tools. Some focus on mechanical actuation and force balance. Some study belt forces, axial thrust, slip, or efficiency at fixed operating points. Others attempt to resolve transient shift behaviour or detailed belt–pulley contact mechanics.

The progression followed here is from design intuition, to component force balance, to mechanical actuation, to belt contact, to ratio change, and finally to general transient CVT behaviour. This ordering mirrors the way the modelling problem becomes difficult. A belt–sheave CVT can first be understood as a variable-radius transmission, then as a collection of mechanical actuators that generate axial force, then as a friction-limited belt contact problem, and finally as a coupled dynamic system in which pulley rotation, shift motion, belt behaviour, and contact state evolve together. The literature contains strong foundations for each of these pieces, but those foundations are distributed across several modelling traditions.

1.3.1 Framework for Classifying CVT Models

A useful way to compare CVT models is to ask what part of the system is being allowed to move, and what part is being held fixed. The same physical transmission can be treated as a geometry problem, a force-balance problem, a belt-contact problem, a shift-dynamics problem, or a coupled system-dynamics problem. Each viewpoint can be useful, but each one answers a different question. This is why the literature is best read by tracking the modelling choices that define each formulation, rather than by treating all CVT models as attempts at the same final system.

One of the first choices is how the ratio is represented. In the simplest case, the transmission is evaluated at a fixed ratio. A slightly broader approach computes the ratio from a quasi-static force balance at each operating point. Other models prescribe a ratio trajectory, or use an empirical relationship for the rate of ratio change. More detailed shift models treat ratio change as a mechanical process in its own right, involving radial belt motion, pulley motion, and contact forces (Kim et al., 2002; Sorge, 2008; Cammalleri and Sorge, 2009; Sorge and Cammalleri, 2011). These approaches are not interchangeable. A fixed-ratio model can explain forces at a known condition, while a shift model can explain how the system moves from one condition to another. Even then, a formulation that describes the mechanics of an imposed or already active shift does not necessarily determine when a shift begins, which direction it takes, or what operating condition follows it.

A second choice is how axial force is produced. Some models take pulley clamping forces as known inputs. This is natural when the goal is to study belt response under specified loading, or when the variator is externally controlled. Other models include the mechanism that generates those forces, such as centrifugal weights, springs, torque-sensitive cams, hydraulic pressure, or electronic actuation (Cammalleri, 2005; Kim et al., 2002; Skinner, 2020; Sorge and Cammalleri, 2011). The important distinction is not only the actuator type, but how completely its mechanical action is resolved. A clamping force may be prescribed directly, obtained from an instantaneous force balance or empirical relation, or produced by a mechanism whose own motion is treated dynamically. Each choice answers a different question about how the pulley force arises.

A third choice is how much of the belt–pulley contact is represented. At low fidelity, the belt may be treated through a capstan-style tension relation, often adjusted for the wedge action of the V-groove.

More detailed models include radial and tangential friction, sliding angle, belt penetration, bending stiffness, local normal pressure, tension variation, slip mechanisms, and efficiency losses (Gerbert, 1996; Kim and Kim, 1989; Sorge, 1996; Kong and Parker, 2005, 2006; Chen et al., 1998; Bertini et al., 2014; Messick, 2018). At still higher fidelity, the belt may be discretized into nodes or elements so that local belt motion and contact behaviour are solved directly (Julió and Plante, 2011; Ballew, 2015). Increasing contact fidelity can make the local mechanics more realistic, but it does not by itself determine how much of the complete transmission response is being solved. Local mechanical fidelity and global dynamic closure are therefore separate modelling choices.

A fourth choice is how the rotating system and its interaction with the surroundings are treated. Many models prescribe one or both pulley speeds, or impose the speed ratio algebraically. This is often appropriate for steady contact analysis, efficiency prediction, or controlled operating-point studies. Other models solve rotational equations of motion so that pulley speeds and transmitted torques evolve dynamically (Srivastava and Haque, 2007; Ballew, 2015; Duan et al., 2016). Any physical transmission must, of course, interact with whatever machinery is attached to it. The important distinction is whether those external systems supply only the mechanical loading associated with that interaction, or whether quantities belonging to the internal CVT mechanics—such as pulley-speed histories, clamping-force histories, axial pulley motion, or ratio trajectories—must also be prescribed. This distinction determines how much of the transmission operating state is actually predicted by the model.

A further choice is the operating region over which the same formulation remains valid. Some models are intended for steady or fully engaged operation. Others describe transient shifting once the variator is already engaged, focus specifically on clutching or engagement, or move through a predefined sequence of operating phases. A broader formulation must determine which mechanical regime is active as conditions change and remain valid as the transmission enters or leaves those regimes. A model may therefore reproduce behaviour accurately within one operating phase without necessarily providing the mechanics that determine how the transmission reaches, leaves, or reverses that phase.

These questions form a useful progression for understanding the modelling problem, but the associated modelling choices are partly independent. A formulation may resolve belt contact in great detail while prescribing global pulley motion. Another may solve pulley rotation dynamically while receiving clamping force as an external input. Another may model mechanical actuation carefully while treating ratio change through quasi-static or empirical relations. None of these choices is inherently wrong; they reflect the intended modelling objective. The important point is that model fidelity in one part of the CVT does not by itself imply closure of the complete transmission response.

The following review therefore considers both what each formulation was developed to predict and what quantities or operating conditions remain prescribed. It proceeds from practical and quasi-static design models, through mechanical actuation and belt-contact mechanics, to transient rubber-belt formulations and related dynamic models in metal-belt and chain CVTs. This provides a consistent basis for identifying which parts of the CVT dynamics are already well established and where the remaining modelling gaps lie.

1.3.2 Practical Design and Tuning Foundations

The practical understanding of mechanically actuated rubber V-belt CVTs developed largely through machine design and tuning practice. General machine design references such as Budynas and Nisbett

(2015) provide the background needed to treat the CVT as a real mechanical assembly involving belts, springs, clutches, shafts, friction interfaces, uncertainty, and design iteration. Not all of these effects are included in a compact model, but they provide useful context for deciding which behaviours are retained and which are idealized.

Practical tuning literature provides a more application-specific foundation. Aaen (2007) describes powersport CVTs using the language of engagement speed, shift speed, flyweights, primary springs, secondary springs, torque sensing, backshift, and speed variance. Its value is not that it provides a complete dynamic model, but that it builds component-level intuition from the perspective of a tuner or racer. It explains how changes to weights, springs, ramps, and helices are expected to affect behaviour, while leaving much of the coupled system response to experience and testing.

These sources are valuable because they describe the physical machine before it is reduced to equations. They also show why prediction is difficult: the same hardware change can affect engagement, shift rate, belt grip, backshift, engine speed, and vehicle acceleration at the same time. Qualitative tuning rules are therefore useful starting points, but they are not enough to determine the complete transient response of the CVT under changing operating conditions.

1.3.3 Design-Oriented and Quasi-Static CVT Models

Design-oriented CVT models commonly evaluate the transmission at a fixed operating condition or through a sequence of quasi-static operating points. Rather than asking how every transient state evolves in time, these models usually ask which geometry, actuator, belt, or tuning parameters should be selected to satisfy a desired operating requirement. In a fixed-operating-point analysis, quantities such as ratio and pulley speed may be specified while the corresponding forces, tensions, or capacities are calculated. In a quasi-static model, the operating condition may instead be obtained from an instantaneous force balance, while inertia and the time required to move between neighbouring equilibria are neglected. These approaches commonly relate vehicle requirements, pulley geometry, belt length, wrap angle, speed ratio, flyweight force, spring preload, helix force, belt tension, and torque capacity. Their main value is that they connect tunable hardware to expected operating behaviour.

A useful starting point is the numerical design method of Arango and Muñoz Alzate (2020), which begins from vehicle requirements, road specifications, and desired performance before working toward the CVT design. This is valuable because it places component selection in its proper context: the belt, pulley geometry, speed-ratio range, engine or motor selection, and mechanical centrifugal actuation are not independent choices, but responses to the requirements imposed by the vehicle and its operating route. In this sense, the work provides a design-level framework where system requirements drive CVT component choices.

Other design-oriented works focus more narrowly on particular parts of the variator design. Cammalleri (2005) develops a speed-torque-controlled rubber V-belt variator design using equilibrium relationships for a centrifugal driver and torque-sensitive driven pulley. La Battaglia et al. (2022) focuses on the kinematic design of motorcycle rubber V-belt CVTs, emphasizing the roller sliding profile of the front pulley as a critical design variable. Both works show how simplified mathematical models can be used to choose actuator or geometry parameters without requiring a complete transient simulation of the transmission.

Skinner (2020) provides a practical force-balance and tuning-oriented model. The work emphasizes quantifiable tuning objectives such as engagement speed, shift-out speed, static RPM shifting, and efficient power transfer, and relates those objectives to the primary and secondary subsystems. It is useful as an applied tuning reference because it connects the variables adjusted by teams to predicted operating behaviour, while remaining primarily an operating-point and force-balance model.

The common strength of these works is that they make CVT design less arbitrary. They identify the parameters that matter and show how geometry, springs, flyweights, helices, belt forces, and vehicle requirements can be related. They also highlight a recurring tension in CVT modelling: models that are simple enough to support early design may omit important coupled behaviour, while models that resolve more detailed mechanics can become difficult to apply directly as design tools. In this sense, design-oriented and quasi-static models occupy an important middle ground. They provide usable relationships for component selection and tuning, but they do not generally determine the complete time evolution of engagement, traction, shift motion, and pulley response.

1.3.4 Mechanical Actuation and Clamping Force Generation

After the basic design variables are identified, the next modelling question is how the CVT generates the axial forces that clamp the belt and drive ratio change. In externally controlled CVTs, these forces may be treated as prescribed hydraulic or electronic inputs. In mechanically actuated rubber V-belt CVTs, however, the forces are produced by the hardware itself. The primary force is typically speed-sensitive, while the secondary force is torque-sensitive.

Cammalleri (2005) provides an important design treatment of this problem through a speed-torque-controlled rubber V-belt variator. The centrifugal driver and torque-sensitive driven pulley are treated as part of the variator design, with equilibrium relationships connecting actuator geometry, spring loading, pulley torque, and the intended operating behaviour of the CVT. This establishes a quantitative link between the mechanical hardware and the clamping forces it is designed to produce.

Kim et al. (2002) also study a rubber belt CVT with mechanical actuators, including both steady-state and transient characteristics. Their driver and driven thrusts are obtained from the mechanical actuator geometry and instantaneous force balances, while the transient ratio response is represented separately by an experimentally identified first-order relationship. The work therefore connects mechanical primary and secondary actuation to transient CVT behaviour, but the motion of the actuator hardware itself is not what determines the shift acceleration.

Sorge and Cammalleri (2011) address the secondary side in substantially greater mechanical detail through the helical shift mechanics of rubber V-belt variators. Their formulation allows the fixed and movable driven faces to rotate differently and resolves how the belt-contact state determines the torque carried by the movable face. The resulting helix response therefore depends on spring loading, helix geometry, transmitted torque, and the internal relative motion of the driven pulley rather than on a simple assumed torque split. The instantaneous pulley radius and shift speed are nevertheless supplied to the contact calculation, so the resulting actuator relation determines the forces associated with a specified shift state rather than the acceleration of the movable secondary assembly itself.

Together, these works establish much of the mechanical foundation needed to model passive CVT actuation. They show how centrifugal primary mechanisms and torque-reactive secondaries generate configuration-dependent axial forces, and how detailed actuator geometry can materially change those forces. They also illustrate an important distinction introduced in Section 1.3.1:

including the actuator mechanism does not necessarily mean that the actuator is itself a dynamically propagated subsystem. In these formulations, the actuator force is generally closed configuration by configuration, while transient ratio motion is either outside the model or supplied through a separate shift description. This distinction becomes important when the actuator hardware has inertia and its motion is coupled directly to the shift and shaft dynamics.

1.3.5 Rubber V-Belt Contact, Slip, and Efficiency Models

Once axial force is generated by the actuation mechanisms, the belt-pulley interface determines what that force actually does. The sheaves may clamp the belt, but the resulting torque capacity, axial reaction, radial belt motion, slip, and loss all depend on how the rubber belt interacts with the pulley groove. A rubber V-belt therefore cannot be treated as an ideal kinematic constraint. It wedges into the sheaves, develops a tension distribution around the wrap, experiences radial and tangential friction, deforms under load, and dissipates energy through several mechanisms.

Classical belt theory gives the starting point: friction and wrap angle determine the maximum tension ratio that can be supported before gross sliding occurs. For V-belts, this picture is modified by the groove angle and the associated wedge effect. However, real belt slip is more complicated than a single capstan-style traction limit. Gerbert (1996) presents a unified treatment of belt slip in which creep, radial compliance, shear deflection, and seating/unseating effects all contribute to speed loss. The important point is not that every model must include all of these effects, but that "slip" is not one physical mechanism. A model should be clear about which slip and loss mechanisms it represents and which it leaves outside its scope.

The contact problem also determines the axial reaction felt by the sheaves. Kim and Kim (1989) developed a theoretical analysis of axial forces in a rubber V-belt CVT using simplified assumptions about the contact regions and belt force distribution. Sorge (1996) developed a compact model for axial thrust in V-belt drives, emphasizing radial penetration and the connection between belt forces and pulley thrust. These works are useful because they connect the belt contact state back to the sheave force balance: axial force is not only generated by the actuator, but also shaped by how the belt seats, slides, and transmits load in the groove.

A more detailed treatment is given by Kong and Parker (2005), who formulate the steady mechanics of belt-pulley systems using distributed belt equations. Kong and Parker (2006) extend this framework to pulley grooves and sliding friction. This line of work resolves local belt mechanics at much greater fidelity than simple force-balance models, including the distribution of forces and sliding behaviour around the contact arc.

Messick (2018) extends this contact-mechanics tradition for rubber V-belt CVTs by building on the Kong and Parker framework and validating axial-force and efficiency predictions experimentally. Messick's work is valuable for its contact-region formulation, belt property measurement, operating-range analysis, and dynamometer validation. It is also clear about its modelling scope: shifting is treated as a quasi-equilibrium process rather than as a transient response-time problem. In this sense, it is a strong reference for belt force and efficiency modelling, while leaving the time evolution of the complete CVT operating state to a different class of model.

Efficiency-focused studies provide another view of the same interface. Chen et al. (1998) experimentally investigated speed and torque losses in a rubber V-belt CVT, while Bertini et al. (2014) developed an analytical model for rubber V-belt CVT power losses including sliding, hysteresis, and entry/exit effects. These works show that transmission losses arise from several interacting mechanisms whose relative importance depends on the operating condition.

The contact and efficiency literature therefore provides the foundation for understanding how belt force, axial thrust, slip, and loss arise. It also reinforces the distinction between local fidelity and global dynamic closure. A high-fidelity contact model may be necessary for efficiency prediction, axial-force prediction, or detailed local belt mechanics. A lower-dimensional contact model may instead be appropriate when the modelling objective requires the belt to remain coupled transparently to actuator dynamics, pulley rotation, shift motion, and the broader transient operating state of the CVT.

1.3.6 Transient Rubber-Belt Models and Dynamic Closure

The contact and force-balance literature establishes how a rubber belt CVT behaves at a specified operating condition. Transient models extend this picture by asking how the transmission moves between operating conditions and which parts of that motion must be resolved in time. In the rubber-belt literature, this has been approached through models of shift mechanics, distributed belt motion, pulley rotational dynamics, and mechanically actuated vehicle response. A useful comparison among these works is therefore which parts of the CVT state are advanced dynamically and which quantities are supplied as operating conditions or inputs.

Sorge (2008) provides an important bridge from quasi-static analysis to transient shift mechanics. Rather than representing ratio change as a sequence of neighbouring equilibria, the formulation treats the radial migration of the belt through the pulley grooves as a time-dependent process coupled to circumferential belt motion, mass conservation, and sliding. Cammalleri and Sorge (2009) later develop approximate closed-form solutions for the same class of shift problem. These works establish a mechanical basis for transient ratio change, while taking the pulley loading and active shift condition as part of the problem definition.

A more detailed transient treatment is given by Julió and Plante (2011), who discretize the rubber belt and apply Newton's law to the individual belt elements. Their model predicts the time response of transmission ratio under specified drive-pulley axial force, drive-pulley speed, and load torque, while resolving belt elasticity, bending, damping, and frictional contact directly. The formulation is validated for steady and shifting conditions and provides substantially greater local belt detail than reduced force-balance models. The authors also report difficulty obtaining a stabilized prediction for the low-torque start-up condition in which belt clamping itself acts as the clutch, indicating that the operating range of the formulation is narrower than the full sequence of CVT engagement and shift behaviour.

Ballew (2015) extends this discretized-belt model family by removing the prescribed input-pulley speed used in earlier formulations. Applied torques, pulley inertias, and belt forces instead determine the pulley angular accelerations, allowing both shaft speeds to evolve dynamically. Engine and vehicle submodels are also included so that changes in external loading can feed back through the transmission rather than being represented by a prescribed pulley-speed history.

Axial actuation in Ballew (2015) remains a separate input to the belt–pulley model. The thesis notes that the axial-force inputs could be supplied either by a control system or by a model of a mechanically governed actuator. In the simulations presented, a PI controller with feed-forward gain generates the primary axial force, while the secondary axial force is held fixed. The formulation therefore extends the dynamic closure of the rotating system substantially, while leaving the source of the pulley clamping forces to a separate actuation model.

da Silva et al. (2023) present a reduced mechanically actuated rubber-belt CVT model coupled to a longitudinal vehicle simulation. Axial equations of motion are written for the movable sheaves, with flyweight loading on the primary and a torque-dependent secondary contribution. The operating sequence is divided into a clutching phase followed by a fully engaged phase, and the model is validated against a full-throttle acceleration. The work therefore provides an example of movable-sheave dynamics and mechanical actuation being retained in a transient application, while using a simplified, phase-specific representation of the CVT behaviour.

Taken together, these works show a clear progression in transient rubber-belt modelling. Ratio change has been treated as a mechanical process, belt motion and contact have been propagated directly, pulley rotational motion has been allowed to respond to applied torques, and movable-sheave inertia has been introduced in mechanically actuated applications. The formulations differ, however, in which parts of the transmission remain prescribed or are restricted to a particular operating phase. This makes the broader CVT literature useful for examining how other variator architectures have divided the boundary between prescribed actuation and dynamically resolved transmission behaviour.

1.3.7 Adjacent Metal-Belt and Chain CVT Dynamic Models

The broader CVT literature contains an extensive modelling tradition for metal push-belt and chain variators, particularly in automotive applications. These systems are not direct equivalents of dry rubber V-belt CVTs, since their load-carrying mechanisms, deformation, lubrication, and actuation systems differ. They nevertheless provide useful comparisons for transient ratio change, frictional contact, pulley dynamics, and the treatment of applied clamping forces.

Srivastava and Haque (2009) review the dynamics and control of belt and chain CVTs and summarize a wide range of steady and transient formulations, including slip and traction models, control-oriented shift laws, and detailed belt or chain mechanics. The review illustrates both the maturity of dynamic CVT modelling outside the rubber-belt literature and the importance of keeping the underlying transmission architecture in view when comparing models.

The Carbone–Mangialardi–Mantriota (CMM) family provides a well-developed example of mechanically derived shift dynamics for metal-belt and chain variators. Carbone et al. (2001, 2005) derive ratio-changing behaviour from belt or chain kinematics, mass conservation, friction, and pulley deformation, including creep-mode and slip-mode shifting. Experimental work by **?** validates the resulting CMM shift dynamics under load. In these tests, the predicted quantity is the speed-ratio response to a prescribed change in clamping-force ratio, while drive-pulley angular speed, driven-side clamping force, and load torque are specified operating quantities. The formulation is therefore directed at the variator shift response under known actuation and loading.

Other metal-belt studies similarly treat transient ratio change and traction behaviour directly. **?**Srivastava and Haque (2008) develop dynamic formulations that retain transient belt motion and examine the effects of friction characteristics and band-pack slip. **?** study variator slip and its relationship to transmitted torque and clamping. These works extend the treatment of shift and traction beyond steady operating-point analysis while retaining the controlled clamping architecture typical of metal-belt automotive CVTs.

Schindler et al. (2012) develop a spatial transient multibody model of a pushbelt variator with detailed representation of the pulley sheaves, ring packages, friction, and unilateral contact. The model is used to study the internal transient response of the pushbelt system, including deformation and three-dimensional contact behaviour. In the simulations presented, the output pulley receives prescribed clamping force and load torque, while the input pulley is assigned angular-velocity and clamping-velocity histories. The detailed belt and pulley response is therefore resolved under a specified global variator excitation.

Chain CVTs provide another example of strongly coupled transient mechanics. Duan et al. (2016) include pulley rotational dynamics, movable-pulley axial dynamics, chain radial motion, chain and pulley deformation, and a speed-dependent friction model spanning micro-slip and macro-slip. The axial and rotational pulley dynamics are coupled directly through the chain forces. The movable-pulley equation is driven by an applied clamping force, however, so the generation of that clamping force remains outside the variator model.

These adjacent models show how transient shift mechanics, detailed contact behaviour, pulley rotation, and movable-pulley motion can be represented at substantially different levels of detail. They also illustrate a recurring modelling boundary: in hydraulically or otherwise externally actuated variators, clamping force and pulley actuation are naturally supplied as inputs to the transmission model. For a mechanically actuated rubber V-belt CVT, those same functions are produced by hardware contained within the transmission, making the treatment of the actuation mechanism itself an important part of how the model boundary is defined.

1.3.8 Synthesis of Prior Work

The literature does not point to a single missing ingredient in CVT modelling. Instead, several mature lines of work resolve different parts of the same machine. Practical and design references establish the component relationships used to size and tune rubber V-belt CVTs. Mechanical-actuation studies explain how centrifugal and torque-reactive hardware generate pulley forces. Contact and efficiency models resolve the belt–pulley interface at progressively greater local detail. Transient formulations add radial belt motion, ratio change, pulley rotation, movable-sheave dynamics, or distributed belt and chain motion. The resulting body of work provides strong foundations for the individual mechanisms required in a dynamic transmission model.

A recurring distinction among these formulations is the location of the model boundary. Any CVT must receive mechanical interaction from the systems attached to it, but the quantities treated as external inputs differ substantially among models. Depending on the modelling objective, pulley speed, axial force, clamping pressure, ratio motion, or actuator commands may be prescribed while the remaining transmission response is solved. These choices are appropriate for many component, control, and contact studies, but they also determine how much of the CVT itself is contained within the dynamic system.

For a mechanically actuated rubber V-belt CVT, the mechanism that generates clamping is itself part of the transmission. Its motion therefore cannot be separated completely from the response that it produces: the actuation, pulley, belt, and rotational mechanics form one coupled physical system. No formulation identified in the reviewed literature was found to close that complete mechanically actuated system dynamically from the loading applied at the primary and secondary shafts. Treating the clamping mechanism dynamically is one part of this gap, but the broader requirement is that the internal CVT response be solved as a coupled whole rather than assembled from separately prescribed or separately closed pieces.

A separate gap concerns the range of operating behaviour covered by the same formulation. Much of the literature is developed for an already engaged variator, a specified shift, or a prescribed sequence of operating phases. Individual works address clutching, upshift and backshift, ratio limits, or traction behaviour, but no formulation identified in the reviewed literature was found to remain mechanically valid across the complete operating sequence of disengagement and engagement, seated ratio limits, free shifting, and traction loss and recovery without imposing a predefined operating phase or trajectory. This is a different requirement from accurately modelling any one of those operating conditions: it requires the same physical formulation to remain valid as the active mechanical constraints change.

The remaining modelling opportunity is therefore defined by two independent forms of closure. The first is system-level closure of the complete mechanically actuated CVT, so that its internal state evolves as one coupled mechanical system in response to the loading applied at the primary and secondary shafts. The second is operating-domain closure, so that the same model remains valid as the transmission moves between its major mechanical regimes rather than being constructed around one specified phase or trajectory. These needs do not require maximum local belt-contact detail; instead, they motivate a reduced formulation that retains the mechanics needed to represent the complete transmission as a general transient component.

Table 1.1 summarizes selected formulations that most directly establish this progression. The table is not a ranking of model quality; each formulation reflects a different modelling objective. It instead shows which parts of the transmission are resolved within each model and which quantities remain prescribed as operating conditions or inputs.

Formulation	Principal behaviour resolved	Actuation / clamping	Quantities supplied or separately closed	Primary operating scope
Cammalleri (2005)	Mechanical variator design and equilibrium force relationships	Centrifugal primary and torque-sensitive secondary represented mechanically	Operating speed, torque, and equilibrium condition	Steady / quasi-static design
Kim et al. (2002)	Steady mechanical response with an empirical first-order ratio transient	Mechanical primary and secondary thrust relations	Operating conditions and experimentally identified transient shift relation	Engaged upshift and downshift response
Sorge (2008)	Transient radial belt migration, sliding, and ratio-change mechanics	Pulley loading forms part of the specified shift problem	Pulley loading and active shift condition	Engaged transient shift
Sorge and Cammalleri (2011)	Torque-reactive secondary helix mechanics with differential sheave rotation	Helix and spring represented mechanically	Instantaneous pulley radius and shift motion supplied to the secondary calculation	Engaged secondary shift mechanics
Messick (2018)	Distributed rubber-belt contact, axial thrust, and efficiency	Axial loading prescribed	Pulley speed, torque, clamping, and operating point	Steady / quasi-equilibrium contact and efficiency
Julió and Plante (2011)	Transient discretized rubber-belt motion and contact	Drive-pulley axial force prescribed	Drive-pulley speed, axial force, and load torque	Engaged steady and shifting operation
Ballew (2015)	Discretized belt dynamics with pulley rotational response to applied torques	Primary axial force controlled; secondary axial force fixed in reported cases	Axial-force inputs; actuator mechanism remains separate	Engaged transient operation with dynamic shaft response
Carbone et al. (2001, 2005); **?**	Mechanically derived metal-variator ratio dynamics including creep and slip modes	Clamping-force ratio prescribed	Drive speed, clamping quantities, and load torque	Engaged underdrive / overdrive shifting
Schindler et al. (2012)	Detailed spatial pushbelt, pulley, friction, and unilateral-contact dynamics	Global pulley actuation prescribed	Input angular and clamping velocities; output clamping force and load torque	Detailed transient response of an engaged variator
Duan et al. (2016)	Coupled chain, pulley axial and rotational dynamics, deformation, and slip	Applied clamping force drives movable-pulley motion	Applied clamping and external loading	Engaged chain-CVT shift dynamics

Table 1.1: Selected CVT formulations illustrating the progression from operating-point and component models to increasingly coupled transient descriptions. The comparison emphasizes which behaviour is resolved within each formulation and which quantities remain prescribed according to its modelling objective.

1.4 Aim, Scope, and Contributions

The literature review in Section 1.3 motivates a specific modelling objective: to develop a physically interpretable dynamic formulation for a mechanically actuated dry rubber V-belt CVT coupled to a vehicle drivetrain. The formulation is intended to close the dominant system-level interactions within a single time-domain model, allowing the transmission and vehicle response to evolve together under changing operating conditions. It is not intended to provide the highest-fidelity representation of local rubber-belt contact or a complete efficiency-loss model; instead, it focuses on the coupled mechanisms needed to understand how actuation, shift behaviour, torque transfer, and vehicle loading produce transient CVT behaviour.

1.4.1 Aim

The aim of this work is to derive a coupled nonlinear dynamic model of a mechanically actuated rubber V-belt CVT and its longitudinal vehicle drivetrain. The model extends from the engine-side pulley, through the belt and downstream driveline, to the vehicle response. Pulley speeds, belt transport, torque transfer, axial shift position, and vehicle motion evolve from the governing equations rather than being prescribed independently.

The formulation is intended to describe how the drivetrain responds from specified admissible initial conditions under changing engine torque, road grade, external load, and other time-varying inputs. Primary actuation, secondary torque reaction, belt traction, ratio change, rotational inertia, and vehicle loading are treated as interacting parts of the same system: a change in one mechanism alters the conditions under which the others act. This allows launch, clutching, shifting, backshift, gross slip, engine braking, and vehicle acceleration to be studied as outcomes of a common coupled response.

A further aim is to preserve the physical interpretation of that response. The derivation is organized so that the role of pulley geometry, primary speed-sensitive actuation, secondary torque-reactive actuation, belt traction, rotational inertia, and vehicle loading remains visible throughout the model. Its assumptions are stated explicitly, together with the mechanisms they exclude, so that simulation results can be interpreted in terms of both the physics retained and the limits of the formulation. The goal is therefore not only to predict system behaviour, but also to clarify why a particular response occurs and how a mechanical or operating change propagates through the drivetrain.

1.4.2 Scope

The system considered in this work is a mechanically actuated dry rubber V-belt CVT coupled to a simplified longitudinal vehicle drivetrain. The CVT consists of a speed-sensitive primary pulley, a torque-sensitive secondary pulley, a rubber V-belt, and a downstream fixed-ratio driveline that transmits power to the vehicle wheels. The vehicle is represented through one-dimensional longitudinal motion.

The model includes the following effects:

- variable pulley geometry, belt-length closure, and ratio kinematics;
- a dynamic shift coordinate relating sheave motion to pulley radii, wrap angles, and transmission ratio;
- mechanically generated primary and secondary axial forces;

- flyweight, ramp, spring, helix, and torque-reactive effects within the pulley actuation mechanisms;
- effective inertia associated with axial shift motion;
- explicit rotational dynamics of the primary pulley, belt, and secondary pulley;
- belt–pulley torque transfer subject to traction limits and gross-slip conditions;
- engine torque and engine-side inertia;
- downstream fixed reduction, reflected driveline inertia, and longitudinal vehicle road load; and
- transient response to specified initial conditions and time-varying input histories.

The belt is represented through a reduced fixed-length geometry and wrap-level contact description rather than a fully distributed deformable-belt formulation. The principal consequences of this and the other model idealizations are collected in Section 2.2; they include omitted local belt deformation and migration, detailed actuator and contact tribology, flexible-hardware response, and thermal or wear evolution. Three-dimensional vehicle dynamics are also outside the present scope. Tire–ground contact is assumed to provide sufficient traction for the longitudinal force demanded by the drivetrain; wheel slip and tire-force limits are not represented.

The formulation retains these assumptions explicitly so that their consequences remain clear when interpreting a simulation result. Its structure also permits individual closures to be refined in later work. For example, the traction relation could be replaced by a distributed contact model, belt elasticity or pulley deformation could be introduced, or thermal, efficiency-loss, and tire-traction models could be incorporated while preserving the surrounding drivetrain framework.

1.4.3 Contributions

The primary contribution of this work is the closure of a coupled mechanism chain for a mechanically actuated dry rubber V-belt CVT operating within a vehicle drivetrain. Individual elements of the formulation build on established ideas in CVT and belt-drive modelling. The contribution lies in combining these elements into a physically interpretable time-domain framework in which the response of the transmission and vehicle emerges from their interaction.

The main contributions are as follows:

1. A vehicle-coupled dynamic formulation that closes the speed-sensitive primary mechanism, torque-reactive secondary mechanism, belt-mediated torque transfer, ratio change, pulley dynamics, and longitudinal vehicle response within a single system of governing equations.

2. A dynamic ratio-change and belt-transport treatment in which axial sheave motion, pulley radii, belt geometry, and belt speed evolve as physical states of the system rather than being imposed through a prescribed ratio map or instantaneous equilibrium condition.

3. A traction-limited torque-transfer formulation that compares the torque required for no-slip motion with the torque supportable by the belt–pulley contacts. This permits gross-slip behaviour to be represented when the sticking response is not admissible.

4. A transient drivetrain framework that evolves from specified admissible initial conditions under changing engine torque, road grade, external load, and other time-varying inputs. Launch, clutching, shifting, backshift, engine braking, traction loss, and vehicle acceleration can therefore be studied as responses of the same coupled system.

5. A derivation structure that preserves the physical role of the major mechanisms and states the modelling assumptions explicitly. This supports interpretation of the predicted response

and provides a modular basis for future extensions, including distributed belt contact, belt elasticity, pulley deformation, thermal effects, efficiency losses, and tire-traction limits.

Together, these contributions provide a framework for tracing how a change in component geometry, spring loading, actuator parameters, operating conditions, or vehicle load propagates through the CVT. The resulting effect on clamping, traction state, shift behaviour, torque transfer, engine speed, and vehicle acceleration can then be examined within the assumptions of the model.

Chapter 2

Model Definition and Simulation Problem

To predict the motion of a CVT from an arbitrary initial condition, the formulation must first turn the physical mechanism into a well-defined simulation problem. This requires a clear account of what the model contains, how the surrounding machinery acts on it, and what information describes the CVT at a given instant.

This chapter builds that description in the order required by the model. It first defines the CVT and its two shaft interfaces, then states the assumptions that determine the physical detail retained. Coordinates and sign conventions establish how the resulting motions are measured. From these definitions, the internal quantities that must be carried from one instant to the next are identified, together with the mechanical conditions that determine how those quantities may evolve.

The chapter concludes by separating the quantities used during a model evaluation according to their roles: fixed CVT data, the current internal condition, loading supplied at the shaft interfaces, and quantities solved only for the present instant. Together, these definitions establish the simulation problem used throughout the component-level derivation.

2.1 Modeled CVT and Shaft Interfaces

The CVT exchanges torque and rotational motion with the surrounding machinery through its primary and secondary shafts. An engine, motor, or test machine may be connected to the primary shaft, while a vehicle, dynamometer, or other load may be connected to the secondary. These two shaft connections therefore provide the natural interfaces between the CVT and the systems attached to it.

The modeled CVT includes the primary and secondary pulley assemblies, the complete belt loop, and both belt–pulley contacts. It also includes the primary flyweights, ramp, and spring and the secondary spring and helix. Any masses and inertias retained for these components are included with them. Their forces and inertias act together as parts of the same coupled mechanism.

The machinery attached to either shaft is not included as part of this mechanism. Instead, its effect is represented by the mechanical loading that it presents at the corresponding shaft interface. The two interfaces therefore mark the extent of the modeled CVT while allowing the same CVT description to be connected to different engines, vehicles, or test equipment.

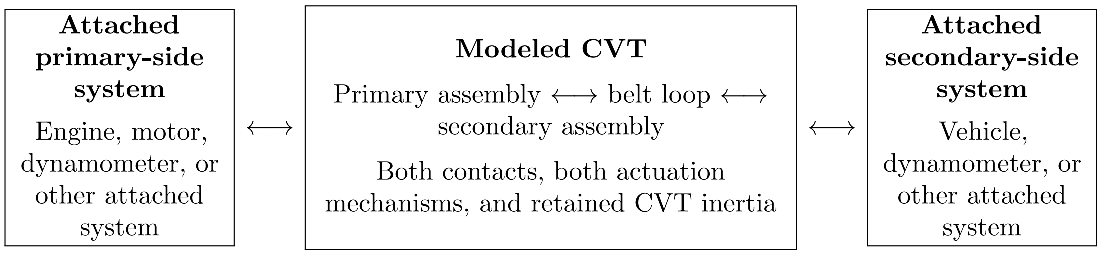

Figure 2.1: The modeled CVT and its two shaft interfaces. Everything in the centre box is resolved as part of the CVT. The attached system on either side supplies its mechanical loading through the corresponding shaft, while the shaft motion remains part of the modeled response.

The same interface description applies at either shaft. Let $j \in \{p, s\}$ identify the primary or secondary side. The attached system contributes two mechanical quantities:

$$\tau_{\mathrm{B},j}, \qquad I_{\mathrm{B},j}.$$

The applied torque $\tau_{\mathrm{B},j}$ is the signed torque exerted on the CVT through shaft j. The equivalent inertia $I_{\mathrm{B},j}$ is the inertia of the attached system as seen at that same shaft. A rotor connected directly to the shaft contributes its own inertia; a wheel, drivetrain, or other mechanism connected through gearing contributes the corresponding inertia reflected to the shaft.

These quantities describe how the attached system loads the CVT, not how the shaft must move. The shaft speed remains part of the modeled motion. An attached system may use the current shaft speed, time, and its own internal state to determine the torque and equivalent inertia that it presents at the interface. Those quantities then act together with the inertia and mechanics inside the CVT to determine the shaft acceleration. A prescribed engine-torque curve or vehicle-load law therefore supplies a mechanical load; it does not prescribe either shaft speed.

A positive applied torque acts in the positive rotational direction defined for the corresponding shaft, and the equivalent inertia is expressed about that same coordinate. The equivalent inertia contains only inertia belonging to the machinery attached beyond the shaft interface. Any pulley, sheave, flyweight, helix, or belt inertia retained inside the CVT is therefore excluded from $I_{\mathrm{B},j}$, so no component inertia is counted twice.

The terms *primary-side interface* and *secondary-side interface* identify the two shaft connections rather than a required direction of power flow. Torque may act through either interface in either rotational direction without changing which components belong to the CVT. Changing the attached driver or load therefore changes the loading at the interface, not the definition of the modeled CVT.

2.2 Modeling Assumptions

The assumptions below define the physical resolution of the formulation rather than a particular operating case. Each assumption is stated together with its principal consequence for the mechanics that can and cannot be represented. The consequence summaries are not intended to enumerate every secondary effect of an idealization; they identify the main physical resolution removed by that choice so that later results can be interpreted against a clear model boundary.

2.2.1 Material and Mechanism Idealizations

Assumption 1 (Time-Invariant Material and Surface State)

Model statement

The material properties and surface condition assigned to the CVT remain unchanged during a simulation. Intrinsic component dimensions are likewise not allowed to evolve through wear or thermal growth. The selected properties describe the hardware for the duration of the simulated event rather than evolving with its operating history.

Consequences for model fidelity

Temperature-dependent stiffness or friction, thermal state evolution, wear, glazing, conditioning, degradation, and other feedback from accumulated contact history are not represented. The formulation therefore describes short-term mechanical response for a specified material and surface state rather than the progressive evolution of that state.

Assumption 2 (Rigid Pulley and Actuation Hardware)

Model statement

The pulleys, shafts, actuation hardware, guides, and supporting CVT structure are represented as rigid bodies. Their prescribed geometry does not deform under the loads developed during operation.

Consequences for model fidelity

Pulley bending, shaft bending or torsion, guide and housing deflection, load-dependent groove geometry, and structural vibration generated by those compliances are outside the model. More detailed CVT formulations show that pulley deformation can alter local belt mechanics and axial loading; those effects would require compliant component models rather than changes to the rigid-body balances alone (Carbone et al., 2005; Duan et al., 2016).

Assumption 3 (Idealized Actuator Mechanics)

Model statement

The primary ramp and flyweight mechanism, secondary helix, guides, and springs follow their prescribed kinematic and force laws without additional contact friction, backlash, or local compliance unless such an effect is introduced explicitly by the selected component law.

Consequences for model fidelity

Ramp or roller friction, pin–helix friction, roller or pin spin, guide friction, backlash, helix-flank transitions, contact compliance, and unmodelled spring hysteresis or internal damping are omitted. Relaxing this assumption would add actuator-specific tribology and compliance to the same mechanism free bodies; for example, contact friction appears explicitly in more detailed centrifugal and helix treatments (Kim et al., 2002; Sorge and Cammalleri, 2011).

Assumption 4 (Fixed, Uniform Belt Cross-Section)

Model statement

The same prescribed cross-section is assigned everywhere along the belt and is carried unchanged as the belt runs and shifts. This description fixes the cross-sectional shape and dimensions and treats mass density as uniform through the section and along the belt. The belt therefore has a constant mass per unit length, and its cross-sectional centre of mass coincides with the geometric centroid. The cord layer is retained only as a prescribed tension-carrying line at the fixed depth d_{cord}; its distinct density and contribution to the through-section mass distribution are not resolved. As the section follows the curved modeled belt path, it does not compress, bulge, shear, warp, or grow under centrifugal or sheave loading.

Consequences for model fidelity

Accordingly, neither spatial variations caused by cogs, ribs, joints, local reinforcement, manufacturing variation, or nonuniform wear nor load-induced changes caused by lateral and radial compression, bulging, cross-sectional shear and warping, centrifugal growth, or local cog and rib deformation are represented. Concentrated cord mass and differences between the densities of the belt layers cannot shift the modeled centre of mass away from the geometric centroid. Changes in belt width, height, contact geometry, mass distribution, and internal strain energy caused by these omitted effects therefore lie outside the present formulation. Detailed belt models retain some of these spatial and compliance effects explicitly (Gerbert, 1996; Kong and Parker, 2005; Messick, 2018).

2.2.2 Belt Geometry and Kinematics

Assumption 5 (Aligned Pulley Axes and Fixed Center Distance)

Model statement

The primary and secondary pulley axes remain parallel at a fixed spacing. The pulley centers and shaft directions therefore do not change as the CVT shifts.

Consequences for model fidelity

Shaft misalignment, eccentricity, geometric runout, bearing-center motion, and the associated changes in belt path and face loading are not represented. Removing this assumption would require the belt geometry and contact loading to respond to the additional shaft and bearing motion.

Assumption 6 (Fixed Belt Length)

Model statement

The modeled belt has a fixed total length. Longitudinal extension is not treated as an independent deformation of the belt, so every admissible pulley configuration must be compatible with the same installed length.

Consequences for model fidelity

Tension-dependent elongation, longitudinal strain energy, extension waves, permanent elongation, and the longitudinal component of elastic creep are not represented. Detailed traction-belt models can retain these effects through a longitudinal compliance field; the present formulation instead assigns ratio change to the pulley and contact motions of a fixed-length belt (Gerbert, 1996; Kong and Parker, 2005; Messick, 2018).

Assumption 7 (Single Belt Transport Coordinate)

Model statement

The circumferential motion of the complete belt is represented by one global transport coordinate. Individual material points are not tracked around the loop, and the local tangential belt speed is not given an independent spatial field.

Consequences for model fidelity

The geometry-induced redistribution of local tangential material speed during shift, material-point migration through the wraps, and the additional momentum transport associated with that spatially varying velocity field are not resolved. The model still retains acceleration of the global belt transport and the motion of the prescribed wrap geometry. Shift-mechanics and discretized belt formulations resolve the omitted local transport in greater detail (Sorge, 2008; Cammalleri and Sorge, 2009; Julió and Plante, 2011).

Assumption 8 (Prescribed Planar Belt Path)

Model statement

At each instant, the belt is taken to wrap each pulley along a circular arc and to pass between the pulleys along two straight tangent spans in one common plane. The belt path is therefore set by the current pulley geometry rather than solved as a flexible dynamic curve.

While both pulleys are engaged, the belt is treated as taut and perfectly flexible. It carries load along its length through a single longitudinal tension, but it carries no compression, transverse shear force, or bending moment. The belt also remains in continuous contact with the sheave faces over each prescribed wrap. This engaged description is physically admissible only while the belt tension and distributed normal contact loading remain nonnegative.

Consequences for model fidelity

Slack or buckled spans within the engaged CVT, local belt lift-off, partial wrap, and contact reattachment are not represented. Bending stiffness, flexural strain energy, and bending hysteresis are also absent even though the prescribed belt path is curved. A calculated negative tension or normal loading therefore indicates that the assumed engaged path is no longer physically admissible; it does not represent a physical negative belt load.

Progressive or spiral radial migration within a wrap, seating and unseating trajectories, span sag, transverse belt vibration, and transverse inertia associated with straight-span reorientation are omitted. The same planar description also excludes the out-of-plane span skew, twist, and axial belt inertia that would arise when the two physical groove centerplanes move differently during shift. More detailed belt and shift-mechanics models resolve subsets of these path and contact effects explicitly (Sorge, 2008; Kong and Parker, 2005; Messick, 2018).

Assumption 9 (Reduced Belt Motion While the Primary Is Disengaged)

Model statement

While the primary sheaves are separated from the belt, no primary sheave normal resultant, tangential traction, or belt torque is retained. Any radial support provided at the minimum primary radius is not assigned a tangential contact law. Belt pretension and the changing shape of the slack spans are not retained as additional mechanical states.

The belt is instead taken to remain seated on the fully closed secondary and to be carried without slip at the secondary effective radius. The belt transport speed and secondary speed are therefore joined by one kinematic constraint throughout the disengaged-primary regime.

Consequences for model fidelity

The disengaged belt and secondary do not have independent transport coordinates. Slack-span motion, sag, belt whip, intermittent primary rubbing, secondary slip, and tension waves created as slack is taken up cannot be predicted. The model also cannot determine deadzone belt pretension or the transient contact loads produced by a detailed re-engagement history. Instead, the entire belt moves at the speed imposed by the secondary contact.

2.2.3 Belt–Pulley Contact Representation

Assumption 10 (Gross Tangential Coulomb Traction)

Model statement

During continuous motion, each belt–pulley interface is represented by a dry gross stick–slip law with fixed static and kinetic traction capacities for the selected surface state. The frictional contact action retained by the model acts in the circumferential belt-transport direction.

Consequences for model fidelity

The formulation can distinguish torque-carrying stick from gross kinetic slip, but it does not resolve radial friction, local creep or microslip, detailed presliding behaviour, lubrication or Stribeck effects, or friction variation with pressure, temperature, conditioning, or local sliding speed. Friction associated specifically with radial migration, seating, and unseating is also outside the retained contact direction. These mechanisms are treated separately in higher-fidelity rubber-belt contact models (Gerbert, 1996; Kong and Parker, 2006).

The static and kinetic traction limits are force-level laws for continuous motion. They do not supply a separate impulse-level friction law during an instantaneous change in mechanical constraint.

Assumption 11 (Wrap-Level Contact Resolution)

Model statement

The model may resolve how belt tension and normal loading vary around a pulley wrap, but each wrap is assigned one overall traction state at an instant. The contact is not resolved through the belt width or depth, and no separate local traction state is carried for different portions of the same wrap.

Consequences for model fidelity

Local adhesion and sliding subregions, spatial traction reversals, detailed pressure concentrations, edge loading, and cross-sectional contact-stress fields cannot be predicted. Resolving those effects requires a distributed or discretized contact description of the type used in higher-fidelity belt models (Kong and Parker, 2006; Messick, 2018; Julió and Plante, 2011).

Assumption 12 (Symmetric Sheave-Face Sharing)

Model statement

The two sheave faces at a pulley are assumed to share the total normal contact load equally. On the secondary pulley, the belt torque is also divided equally between the fixed and movable faces for the purpose of the movable-sheave balance. The primary sheave faces rotate together with their shaft and therefore have one common angular speed. On the secondary, the wrap-level tangential contact law does not resolve the separate surface speeds produced by helix motion of the movable face. The secondary shaft speed is used as the representative speed for the complete secondary wrap.

Consequences for model fidelity

Face-to-face differences in normal loading and transmitted torque are not resolved. On the secondary, the equal torque split and the use of one shaft-referenced contact speed also remove any difference between the tangential states of the two faces. The helix-induced rotation of the movable face relative to the shaft remains in the actuation and rotational balances, but it is not carried as a second surface speed in the stick–slip law. Secondary sticking therefore denotes shaft-referenced gross sticking of the reduced wrap, rather than simultaneous pointwise adhesion to both physical faces during helix motion. Face-resolved shift models show that the two secondary face torques and sliding conditions can depart substantially from this aggregate description during shifting (Sorge and Cammalleri, 2011).

2.2.4 Boundary and Transition Idealizations

Assumption 13 (Instantaneous, Perfectly Inelastic Constraint Activation)

Model statement

When a mechanical stop or a new kinematic constraint becomes active at finite relative velocity, its finite contact duration and local compliance are replaced by an instantaneous, perfectly inelastic event. This treatment applies to arrival at a mechanical travel stop, first belt engagement, and activation of the deadzone belt lock.

The generalized positions remain continuous. The generalized velocities may change, and their post-event values are determined by impulse–momentum balance over all retained inertial coordinates subject to the newly active kinematic constraints. Applied torques, spring forces, damping forces, and other bounded forces act over no resolved time during the event and therefore contribute no finite impulse. All masses and inertias retained by the continuous mechanics remain part of the momentum balance. A constraint release produces no impulse unless another constraint activates at the same instant.

Consequences for model fidelity

The event model does not resolve contact duration, local indentation, peak impact force, stress waves, rebound, chatter, or the high-frequency vibration created by compliant contact. Individual velocities are neither preserved nor set independently; incoming momentum is redistributed among the velocities that remain compatible with the new mechanical topology. Because the event is perfectly inelastic, kinetic energy need not be conserved. The decrease in kinetic energy is retained as the impact or topology-capture loss, but the model cannot divide that loss among local belt deformation, material damping, sound, heat, or structural vibration. Predicting those details would require a finite-duration compliant-contact model. An impulsive load applied through an external shaft boundary would likewise require an explicit boundary impulse rather than the continuous torque description.

Assumption 14 (Tangential Contact Through Boundary Events)

Model statement

A belt–pulley contact that is sticking immediately before an instantaneous boundary event remains constrained against tangential slip through the event. Contacts that are sliding, together with contacts created by the event, are not assigned a new impulsive no-slip constraint.

Their relative tangential motion after the event, together with the ordinary continuous traction limits, determines the stick–slip state that follows.

Consequences for model fidelity

An existing sticking contact may transmit whatever tangential constraint impulse is required to preserve no slip; that impulse is not checked against a separate impulse-level friction bound. The model therefore cannot predict impact-induced loss of an existing stick, brief impulsive slip followed by readhesion, or a transition caused by exhausting an impulse-level friction capacity.

Conversely, normal contact or kinematic capture does not automatically glue a new belt–pulley interface tangentially. In particular, a newly formed primary contact may retain tangential relative motion and begin its continuous motion in slip. Resolving the omitted impulsive friction behaviour would require an impulse-level traction law coupled to the normal constraint impulse.

2.2.5 Other Omitted Losses

Assumption 15 (Negligible Unmodeled Parasitic Losses)

Model statement

Bearing, seal, windage, and auxiliary-hardware losses are neglected unless they are introduced explicitly by a component or external boundary model. No global empirical efficiency factor is inserted to represent these omitted mechanisms.

Consequences for model fidelity

Energy loss and damping arise only from mechanisms represented explicitly by the formulation, including gross kinetic slip and perfectly inelastic boundary or topology-capture events. Absolute efficiency may therefore be optimistic, and real hardware may exhibit additional damping or resisting torque when the omitted parasitic losses are significant. Adding such losses does not require changing the kinematic structure of the CVT; it requires supplying the corresponding physical torque or force laws.

Resulting fidelity Taken together, these assumptions retain the dominant rigid-body actuation, pulley rotation, sheave-shift motion, global belt transport, wrap tension and contact resultants along an admissible taut belt path, clamping, traction limitation, gross stick–slip mechanics, and momentum redistribution when a new mechanical constraint activates. They also retain the kinetic-energy loss associated with gross slip and perfectly inelastic capture.

The formulation does not reproduce the complete deformable belt and local contact problem, spatial variation within the belt construction, slack or buckled engaged spans, partial-wrap lift-off and reattachment, belt bending mechanics, slack-belt dynamics while the primary is disengaged, actuator tribology, flexible-structure response, finite-duration impact and rebound, impulse-level friction limits, thermal or wear evolution, or miscellaneous parasitic losses. The assumptions are separated by physical mechanism so that a higher-fidelity model can replace one closure without obscuring which additional behaviour that change is intended to recover.

2.3 Coordinate and Sign Conventions

This section defines the coordinates and sign conventions used throughout the formulation.

2.3.1 Rotational and Belt-Transport Directions

Figure 2.2 shows the two pulley assemblies face-on, with their parallel shafts extending away from the viewer. Let ω_p and ω_s denote the angular velocities of the primary and secondary pulley assemblies, respectively. In this view, counterclockwise is positive for both pulleys, as shown by the arrows in the figure. Thus, $\omega_p > 0$ and $\omega_s > 0$ denote counterclockwise rotation, while negative values denote clockwise rotation.

Let v_b denote the signed transport speed of belt material around the loop. The positive direction is the direction shown in Figure 2.2: from the secondary pulley to the primary pulley along the upper span, and from the primary pulley to the secondary pulley along the lower span. Thus, $v_b > 0$ denotes transport in this direction, while $v_b < 0$ denotes transport in the opposite direction.

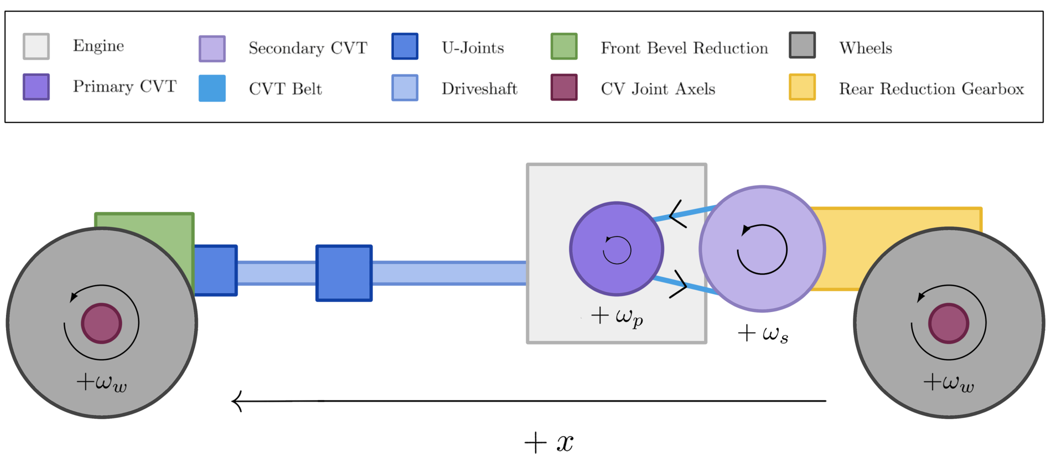

Figure 2.2: Rotational and belt-transport sign convention viewed from the outboard pulley faces, looking inward along the parallel shaft axes. Counterclockwise rotation is positive for both pulleys. With the primary pulley on the left and the secondary pulley on the right, positive belt transport runs from secondary to primary along the upper span and from primary to secondary along the lower span.

Torque follows the corresponding shaft convention. A torque acting on shaft $j \in \{p, s\}$ is positive when it acts in the rotational sense defined by $\omega_j > 0$. In particular, the interface torque $\tau_{\mathrm{B},j} > 0$ acts on the CVT in this positive direction. A torque that opposes positive shaft rotation is negative, but the torque coordinate itself does not change if the shaft reverses.

An attached engine, vehicle, dynamometer, or other system may use its own coordinates internally. The shaft speed and applied torque exchanged at its CVT interface are expressed using the convention defined here.

2.3.2 Local Axial Sheave Coordinates

Each movable sheave is assigned a local axial coordinate. Let x_p denote the position of the primary movable sheave and x_s the position of the secondary movable sheave. More generally, x_j denotes the corresponding coordinate for pulley $j \in \{p, s\}$.

For either pulley, $x_j = 0$ denotes the fully open position. Increasing x_j moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing x_j opens it. If $x_{j,\max}$ denotes the available travel, then

$$x_j = 0 \quad \text{is fully open,} \qquad x_j = x_{j,\max} \quad \text{is fully closed.}$$

The velocity follows the same convention: $\dot{x}_j > 0$ denotes closing motion and $\dot{x}_j < 0$ denotes opening motion. The fully open position is only the geometric origin of the coordinate; it does not prescribe the configuration from which a simulation must begin.

The two coordinates are local to their respective sheave pairs. Their positive directions may therefore point in opposite physical directions along a common global axis. What they share is their mechanical meaning: positive axial motion closes the corresponding pulley.

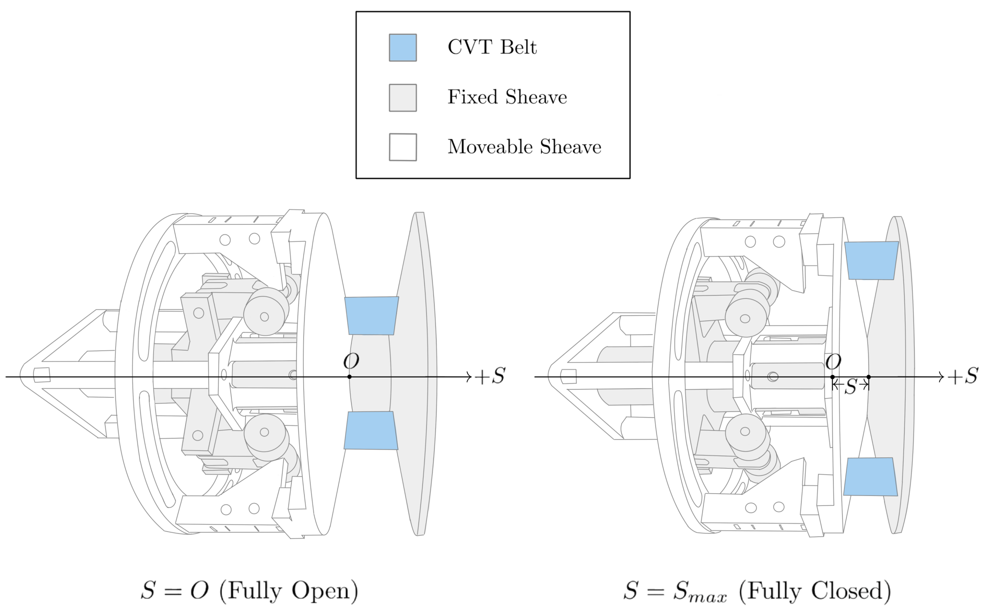

Figure 2.3: Local axial convention for the primary and secondary movable sheaves. For either pulley j, $x_j = 0$ denotes the fully open position and increasing x_j moves the movable sheave toward the fixed sheave. The fully closed position is $x_j = x_{j,\max}$. Because the coordinates are local, the positive primary and secondary directions need not point in the same global direction.

2.3.3 Local Radial, Tangential, and Axial Directions

At either pulley, the radial and tangential directions lie in the plane of pulley rotation. At any point around the pulley, the radial unit vector $\hat{\mathbf{e}}_r$ points outward from the pulley axis. The tangential unit vector $\hat{\mathbf{e}}_\theta$ is perpendicular to $\hat{\mathbf{e}}_r$ and points in the direction produced by the positive rotation defined in Section 2.3.1. A radial or tangential component is positive when it points along its corresponding unit vector.

These vectors form a local polar basis rather than a fixed Cartesian pair. Their directions therefore change with the point around the pulley, while $\hat{\mathbf{e}}_r$ remains outward and $\hat{\mathbf{e}}_\theta$ remains tangent in the positive rotational direction. The inset in Figure 2.4 shows this local basis.

The positive axial direction for pulley j is the direction of increasing x_j defined in Section 2.3.2. It lies along the pulley shaft and is perpendicular to the radial–tangential plane. In the face-on view of Figure 2.4, the shaft axis therefore passes through the page. That figure illustrates only the radial and tangential directions; the axial sign remains the local sheave-closing direction shown in Figure 2.3.

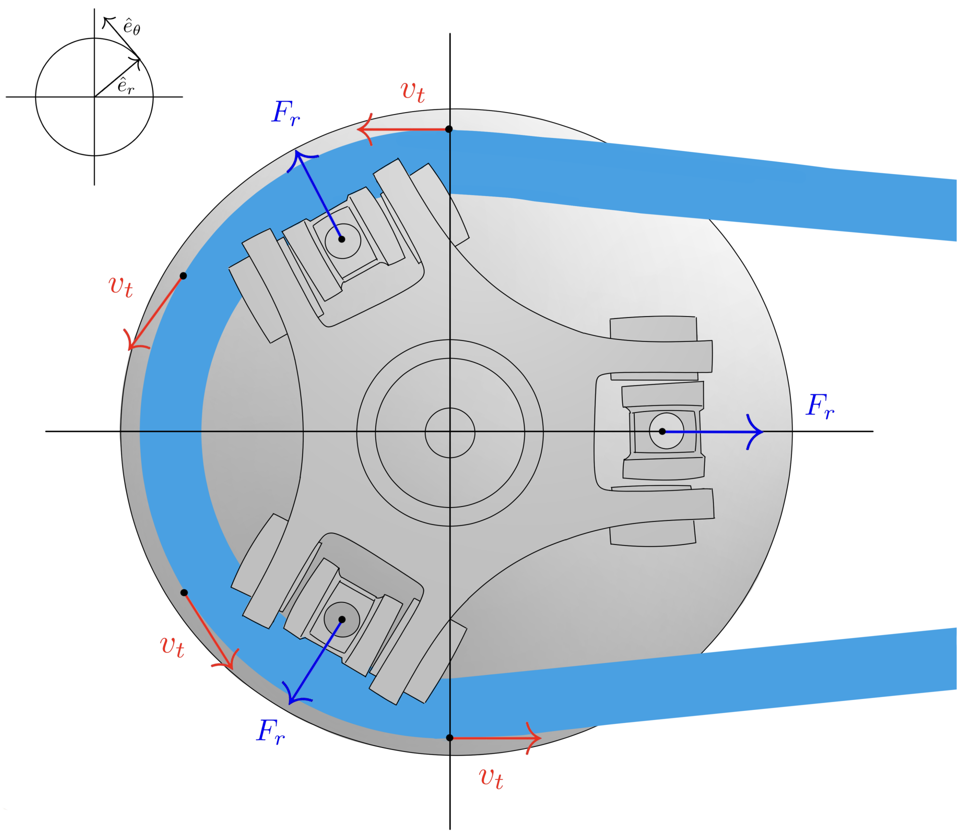

Figure 2.4: Radial and tangential directions in the plane of pulley rotation. The inset shows the local polar basis: $\hat{\mathbf{e}}_r$ points outward from the pulley axis, while $\hat{\mathbf{e}}_\theta$ is tangent in the positive rotational direction defined in Section 2.3.1. The basis changes orientation with the point around the pulley rather than remaining fixed like a Cartesian pair.

2.4 CVT State Variables

At any instant, the modeled CVT's motion and configuration are described by five internal numerical quantities. These quantities are collected in the state vector,

State Vector **(2.4.1)**

$$\boldsymbol{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ v_b \end{bmatrix}$$

Here, ω_p and ω_s are the primary and secondary shaft speeds. The shift coordinate s uniquely identifies the positions of both movable sheaves, while $\dot{s}$ is the corresponding shift velocity. The geometric relation between s and the local coordinates x_p and x_s is established in Section 3.1.3. The remaining entry, v_b, is the belt transport speed.

The primary and secondary pulley assemblies rotate about separate shafts. Although the belt couples their motion, neither shaft speed can generally be reconstructed from the other. Both ω_p and ω_s must therefore be retained. Absolute shaft angles are not required because the internal CVT mechanics depend on the current angular velocities, but not on the angular orientation of either pulley about its shaft.

The current pulley geometry and the configurations of the retained actuation mechanisms depend on the shift position, so s must also be retained. Shift motion has inertia and is governed by a second-order balance. Its present velocity $\dot{s}$ is therefore required alongside its position.

The belt transport speed is retained independently because belt motion need not equal the surface speed of either pulley when tangential slip is present. Under the uniform-belt and single-transport-coordinate assumptions, the mechanics do not depend on which material point occupies a particular location around the loop. No belt-phase or accumulated belt-displacement coordinate is therefore required.

Together, these five values contain all of the internal position and velocity information that must be carried from one instant to the next. For the selected CVT, these values, the current shaft-interface conditions, and the active mechanical constraints are sufficient for the formulation to determine

$$\dot{\boldsymbol{x}}(t) = \begin{bmatrix} \dot{\omega}_p \\ \dot{\omega}_s \\ \dot{s} \\ \ddot{s} \\ \dot{v}_b \end{bmatrix}. \tag{2.4.2}$$

This derivative gives the time rate of change of every state quantity. While the active constraints remain unchanged, advancing the state using these rates produces its value at the next instant, where the same evaluation is repeated. No other internal position or velocity variable must be retained.

The vector in Equation (2.4.1) describes only the modeled CVT. An engine, vehicle, dynamometer, or other attached system may carry additional states in its own model and use them when determining the torque and equivalent inertia presented at a shaft interface.

2.5 Mechanical Regimes and State Evolution

Consider the shift coordinate at one of its travel limits with $\dot{s} = 0$. The same position and velocity can describe two different mechanical conditions. If the surrounding forces press the movable sheave against the limit, a stop reaction is required and further motion into the stop is prevented. If the unconstrained tendency is away from the limit, the stop reaction vanishes and the sheave is free to accelerate away. The values stored in $\boldsymbol{x}$ are identical in both cases, but the motion that is allowed is not.

The missing information is which contacts and mechanical constraints are currently active. Their active set is called the *mechanical regime*. For the modeled CVT, a regime identifies which belt–pulley contacts exist, whether the shift coordinate is free or supported at a unilateral boundary, and whether each existing tangential contact is sticking or sliding. Let $q(t)$ denote the current regime. This label is not another position or velocity and does not introduce another moving part; it records the mechanical arrangement under which the state is evolving.

The complete internal condition needed to continue the model is therefore the pair $\big(\boldsymbol{x}(t), q(t)\big)$. The state vector records the current motion and configuration, while the regime identifies the constraints that determine which motion is admissible. For the selected CVT and a fixed regime, the current state and shaft-interface conditions determine the state derivative. At this level, that relation may be written as

$$\dot{\boldsymbol{x}} = \mathcal{F}_q(t, \boldsymbol{x}, \tau_{\mathrm{B},p}, I_{\mathrm{B},p}, \tau_{\mathrm{B},s}, I_{\mathrm{B},s})\,, \tag{2.5.1}$$

The function $\mathcal{F}_q$ represents the mechanics of the selected CVT under the current mechanical arrangement. Its subscript indicates that the active contacts and constraints select the applicable form of the mechanical balances. While that regime remains admissible, the state advances continuously according to Equation (2.5.1).

Every regime is defined by physical conditions that must remain satisfied. An event occurs when the evolving motion reaches the boundary of those conditions: a contact forms or separates, a travel limit is reached or released, or an existing tangential contact can no longer maintain its current stick–slip condition. At that instant, the old mechanical arrangement no longer provides the admissible continuation of the motion and the regime must change.

Not every regime change alters the state instantaneously. If the event changes the active force or constraint description without producing an impulse, the positions and velocities pass through the event unchanged,

$$\boldsymbol{x}^{+} = \boldsymbol{x}^{-}, \tag{2.5.2}$$

where the superscripts $(-)$ and $(+)$ denote the values immediately before and after the event. The new regime changes the derivative obtained from Equation (2.5.1), even though the state itself is continuous.

A different transition is required when a rigid kinematic constraint becomes active with a nonzero velocity into that constraint. The configuration remains continuous, but one or more velocity components may change over the effectively instantaneous event. In general, the compatible post-event state may be written as

$$\boldsymbol{x}^{+} = \Delta_{q^{-}\to q^{+}}\left(\boldsymbol{x}^{-}\right). \tag{2.5.3}$$

The transition law enforces the newly active kinematic constraint together with impulse–momentum balance for the retained inertia. It does not require each velocity or each coordinate momentum to remain unchanged. A constraint impulse can change the velocity components, and kinetic energy may decrease when the capture is inelastic. Once the transition is complete, continuous evolution resumes under the new regime.

The resulting motion has the repeated structure

$$\text{continuous evolution} \longrightarrow \text{event} \longrightarrow \text{constraint-consistent transition} \longrightarrow \text{new continuous evolution}. \tag{2.5.4}$$

Because the CVT combines continuous state evolution with discrete changes in its active mechanical arrangement, its motion forms a *hybrid initial-value problem.* A complete initial condition specifies both an initial state and an initial regime,

$$\boldsymbol{x}(t_0) = \boldsymbol{x}_0, \qquad q(t_0) = q_0, \tag{2.5.5}$$

with $\boldsymbol{x}_0$ satisfying the contacts and constraints represented by q_0. The regimes are therefore not separate CVT models. They are different admissible arrangements of the same pulley assemblies, belt, actuation mechanisms, and shaft interfaces. The state vector records how that mechanism is moving; the regime records which motion is presently allowed.

2.6 Model Quantities and Evaluation Flow

Advancing the CVT from one instant to the next requires more than the stored internal condition alone. The calculation also uses data that define the selected CVT and loading supplied through the two shaft interfaces, and it produces quantities that are needed only for the current evaluation. These quantities serve different roles in the model and should not all be treated as state.

Table 2.1 separates the four roles used throughout the formulation. The fixed data represented by its first row are collected in **??**, while the instantaneous quantities represented by its final row are listed by mechanical regime in Table 3.1. Fixed CVT data may be specified directly or calculated once from the selected hardware during initialization. In either case, they define the CVT being modeled and remain unchanged while that simulation is advanced. The time-varying internal condition is instead carried by the state and regime introduced in the preceding sections.

Table 2.1: Roles of the quantities used during a CVT model evaluation.

Quantity role	**Contents**	**Use during evaluation**
Fixed CVT data	Physical and geometric data for the selected hardware, including any values calculated once during initialization	Define the particular CVT and the mechanical relations used to represent it; they are not advanced through time.
Current CVT condition	State vector $\boldsymbol{x}(t)$ and mechanical regime $q(t)$	Record the time-varying internal information retained from the preceding instant and identify the currently active constraints.
Shaft-interface data	Applied torque $\tau_{\mathrm{B},j}$ and equivalent inertia $I_{\mathrm{B},j}$ at each shaft	Supply the current mechanical effect of the systems attached to the primary and secondary shafts.
Instantaneous CVT results	State derivative, current internal mechanical quantities, and regime admissibility measures	Determine how the state can advance from the current instant and whether the active regime can continue; they are recomputed rather than retained as independent state.

The pair $(\boldsymbol{x}, q)$ is therefore the only time-varying internal CVT condition passed from one model evaluation to the next. The shaft-interface quantities are supplied for the current evaluation. An attached system may use time, shaft speed, and states belonging to its own model to determine those quantities, but those external states do not become states of the CVT.

Using these roles, one evaluation proceeds as follows:

1. The current time, state $\boldsymbol{x}(t)$, and regime $q(t)$ identify the internal CVT condition at the start of the evaluation.

2. The systems attached to the primary and secondary shafts supply the torque and equivalent inertia presented at their respective interfaces under the current shaft conditions.

3. The CVT mechanics use the fixed CVT data, current internal condition, and shaft-interface data to determine the instantaneous mechanical solution. This solution provides $\dot{\boldsymbol{x}}(t)$ together with the conditions used to test whether the current regime remains admissible.

4. If the regime remains admissible, the state is advanced using $\dot{\boldsymbol{x}}$ and the regime is retained. If an event is reached, the event state is established, the applicable transition is applied when required, and the new regime is selected.

5. The resulting state and regime become the current internal condition for the next evaluation, and the cycle repeats.

Chapter 3

Derivation

3.1 Pulley Geometry

The conceptual overview introduced the CVT as two variable-radius pulleys connected by a single belt. Moving the pulley sheaves changes where the belt sits on each pulley and therefore changes the transmission ratio.

Both pulleys contain a movable sheave. Let x_p denote the axial position of the movable primary sheave and x_s the axial position of the movable secondary sheave. The geometric problem is therefore twofold: how does the axial position of each movable sheave set the belt radius on its own pulley, and how does the shared belt constrain the two pulley positions to move together?

The first question can be answered from the local geometry of each pulley. The second requires following the belt around the complete loop. Once these position relationships are known, they can be differentiated to obtain the corresponding radius rates and accelerations.

The natural place to begin is the geometry of a single pulley.

3.1.1 Local Pulley Geometry

The first part of the geometric question can be answered on a single pulley: how does the axial position of its movable sheave determine the pulley radius? The same conical geometry appears on both pulleys, so the relationship can be derived once and then applied to the primary and secondary.

The axial direction and primary-sheave convention were introduced in Section 2.3.2 and illustrated in Figure 2.3. The local coordinates x_p and x_s follow the same convention on their respective pulleys.

For either pulley $j \in \{p, s\}$, $x_j = 0$ denotes the fully open configuration. Increasing x_j moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing x_j opens it. The corresponding velocity follows the same sign convention: $\dot{x}_j > 0$ denotes closing motion and $\dot{x}_j < 0$ denotes opening motion.

The primary therefore begins at $x_p = 0$ and moves in the positive direction as it closes. The secondary begins in a closed position, with $x_s > 0$, and moves toward smaller values of x_s as it opens.

Consider a belt seated between two straight sheave faces with half-angle β, as shown in Figure 3.1. Let b_{in} denote the width of the belt at its inner surface. As assumed in Assumption 4 (Fixed, Uniform Belt Cross-Section), this width remains constant.

Suppose the movable sheave closes the pulley by a distance Δx_j. At the original belt radius, the available space between the sheaves is therefore reduced from b_{in} to

$$b_{\text{in}} - \Delta x_j.$$

The belt responds by moving outward through a radial distance Δr_j. At its new position, the angled sheave faces provide an additional axial distance d on each side.

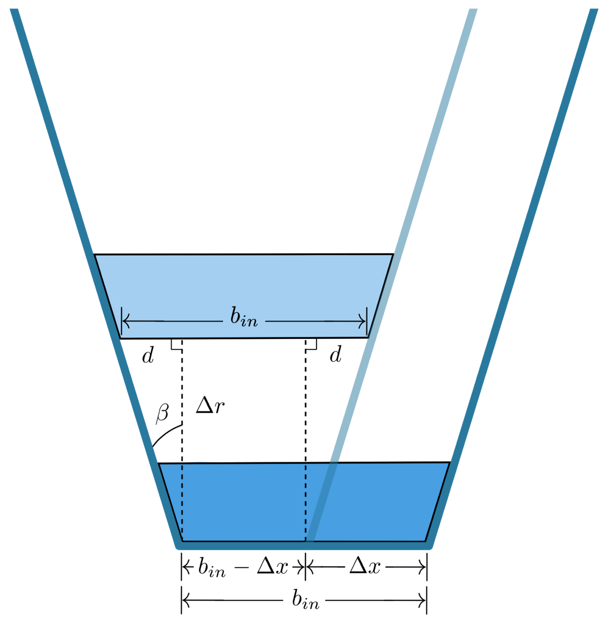

Figure 3.1: Local geometry of a belt moving outward between the sheave faces of pulley j. Closing the pulley by Δx reduces the available width at the original belt position to $b_{\text{in}} - \Delta x$. Moving outward by Δr provides an additional axial distance d at each sheave face.

At the new radius, the total space between the sheave faces must again equal the fixed belt width. Reading directly from Figure 3.1,

$$b_{\text{in}} = 2d + (b_{\text{in}} - \Delta x_j)\,. \tag{3.1.1}$$

The right triangle beside either sheave face relates d to the radial movement. For the sheave half-angle β,

$$\tan\beta = \frac{d}{\Delta r_j}, \tag{3.1.2}$$

and therefore

$$d = \Delta r_j \tan\beta. \tag{3.1.3}$$

Substituting Equation (3.1.3) into Equation (3.1.1) gives

$$b_{\text{in}} = 2\Delta r_j \tan\beta + (b_{\text{in}} - \Delta x_j)\,. \tag{3.1.4}$$

The constant belt-width terms cancel, leaving

$$\Delta r_j = \frac{\Delta x_j}{2\tan\beta}. \tag{3.1.5}$$

This is the basic axial-to-radial relationship of the pulley. Closing the pulley moves the belt outward, while opening it moves the belt inward. A smaller sheave angle produces more radial movement for the same axial motion, whereas a larger sheave angle produces less.

Taking the infinitesimal limit of Equation (3.1.5) gives

$$\frac{dr_j}{dx_j} = \frac{1}{2\tan\beta}. \tag{3.1.6}$$

Because the sheave faces are straight, this slope is constant.

Radius convention The derivation above follows the inner surface of the belt, where b_{in} is defined. The belt also has the finite radial height h_b, so its inner and outer surfaces occupy different radii. Let $r_{j,\text{inner}}$ and $r_{j,\text{outer}}$ denote those radii. They are related by

$$r_{j,\text{outer}} = r_{j,\text{inner}} + h_b. \tag{3.1.7}$$

The fixed-cross-section assumption in Assumption 4 (Fixed, Uniform Belt Cross-Section) makes h_b constant, so the axial-to-radial displacement derived above can be read from either surface. An absolute pulley radius must nevertheless identify which surface is being measured. The outer surface is used here because it provides a convenient and directly measurable reference. Throughout the geometric development,

$$r_j \equiv r_{j,\text{outer}}. \tag{3.1.8}$$

Using the outer surface gives a clear measure of where the belt is seated. It does not mean that every physical property of the belt acts at that radius.

The belt tension is carried along the cords embedded within the belt, so it acts through the centre of the cord layer. The distance from the pulley shaft to this line is therefore the lever arm through which belt tension transmits torque, making it the effective radius for gearing. This cord-line radius is commonly called the pitch radius (Gerbert, 1996).

Let d_{cord} denote the radial depth of the cord-layer centre from the outer surface, with $0 < d_{\text{cord}} < h_b$. Because the cord line lies this far inside the outer surface, its effective radius is

$$r_{j,\text{eff}} = r_{j,\text{outer}} - d_{\text{cord}}. \tag{3.1.9}$$

The belt mass is spread throughout the whole cross-section rather than concentrated at the outer surface or the cord line. To describe where this mass is with one radius, we use its centre of mass and denote the radius to that point by $r_{j,\text{cm}}$. Under the uniform-density approximation, the centre of mass is the geometric centroid of the trapezoid.

Let y measure radially inward from the outer surface, with $y = 0$ at the outer surface and $y = h_b$ at the inner surface, and let y_{cm} denote the depth of the centre of mass. These two depths and their corresponding radii are shown in Figure 3.2.

Draft figure placeholder: trapezoidal belt cross-section showing the outer and inner surfaces, b_{out}, b_{in}, and h_b. Measure y radially inward from the outer surface. Mark the cord-layer centre at d_{cord} and the mass centroid at y_{cm}, together with the radial lines $r_j = r_{j,\text{outer}}$, $r_{j,\text{eff}}$, and $r_{j,\text{cm}}$.

Figure 3.2: Radial reference lines within the belt cross-section. The outer-surface radius r_j locates the seated belt. The cord-line radius $r_{j,\text{eff}}$ is the lever arm through which belt tension transmits torque, while $r_{j,\text{cm}}$ locates the belt's centre of mass. The depths d_{cord} and y_{cm} are measured radially inward from the outer surface.

If b_{out} and b_{in} are the outer and inner widths, respectively, the straight sides of the trapezoid give

$$b(y) = b_{\text{out}} - \frac{b_{\text{out}} - b_{\text{in}}}{h_b}\, y. \tag{3.1.10}$$

Because the density is uniform, a thin strip at depth y contains mass in proportion to its area $b(y)\,dy$. The centroid depth is therefore the weighted average of y over the cross-section:

$$y_{\text{cm}} = \frac{\displaystyle\int_0^{h_b} y\, b(y)\, dy}{\displaystyle\int_0^{h_b} b(y)\, dy}. \tag{3.1.11}$$

The denominator is the belt cross-sectional area,

$$A_b = \int_0^{h_b} b(y)\,dy = \frac{h_b}{2}\left(b_{\text{out}} + b_{\text{in}}\right). \tag{3.1.12}$$

Using this area in the weighted average gives

$$\begin{aligned} y_{\text{cm}} &= \frac{1}{A_b}\int_0^{h_b} y\,b(y)\,dy \\ &= h_b \frac{b_{\text{out}} + 2b_{\text{in}}}{3\left(b_{\text{out}} + b_{\text{in}}\right)}. \end{aligned} \tag{3.1.13}$$

The result has the expected limiting behaviour. If $b_{\text{out}} = b_{\text{in}}$, the cross-section is rectangular and $y_{\text{cm}} = h_b/2$. For the usual trapezoid with $b_{\text{out}} > b_{\text{in}}$, the additional area near the outer surface places the centroid slightly closer to that surface, so $y_{\text{cm}} < h_b/2$. The belt mass therefore follows the radius

$$r_{j,\text{cm}} = r_{j,\text{outer}} - y_{\text{cm}}. \tag{3.1.14}$$

The outer-surface radius map still requires one known configuration to establish its absolute placement. Let $x_{j,\text{ref}}$ denote a known axial position at which the belt is seated, and let $r_{j,\text{ref}}$ denote the outer radius measured at that same position. The outer radius at any other seated position is then

$$r_j(x_j) = r_{j,\text{ref}} + \frac{x_j - x_{j,\text{ref}}}{2\tan\beta}. \tag{3.1.15}$$

The pair $(x_{j,\text{ref}}, r_{j,\text{ref}})$ simply anchors the linear relationship to a known physical configuration.

For the secondary pulley, a convenient reference is its known closed starting position. The secondary begins with the belt seated at a large radius and opens as the CVT shifts. Since opening corresponds to decreasing x_s, Equation (3.1.15) produces the corresponding decrease in r_s.

The primary introduces an important geometric asymmetry: it can open farther than the width of the belt. At $x_p = 0$, the distance between the primary sheaves can therefore exceed b_{in}. Initial positive motion of the movable primary sheave only removes this clearance, leaving the belt at the same radius. The primary radius begins to increase only once the movable sheave reaches the belt.

This interval is the *primary deadzone*. Let $x_{p,\text{dz}}$ denote the primary position at which the movable sheave first reaches the belt. Within the deadzone,

$$0 \le x_p < x_{p,\text{dz}},$$

the primary coordinate changes, but the belt remains at its minimum primary radius. At $x_p = x_{p,\text{dz}}$, the clearance has been removed and the belt becomes seated between the sheave faces. Further closure then moves the belt outward according to the local geometry derived above.

The first-contact position provides the natural reference for the seated primary branch:

$$x_{p,\mathrm{ref}} = x_{p,\mathrm{dz}}, \qquad r_{p,\mathrm{ref}} = r_{p,\min}.$$

The complete primary radius map is therefore

$$r_p(x_p) = \begin{cases} r_{p,\min}, & 0 \le x_p < x_{p,\mathrm{dz}}, \\ r_{p,\min} + \dfrac{x_p - x_{p,\mathrm{dz}}}{2\tan\beta}, & x_{p,\mathrm{dz}} \le x_p \le x_{p,\max}. \end{cases} \tag{3.1.16}$$

The linear axial-to-radial relationship in Equation (3.1.15) is standard in reduced CVT geometry models (Duan et al., 2016; Ballew, 2015). It applies to both pulleys whenever the belt is seated, while Equation (3.1.16) extends the primary description through the deadzone before contact.

The local geometry now tells us the outer-surface radius associated with either movable-sheave position. The two local descriptions belong to the same closed belt, so only certain primary and secondary positions are geometrically compatible.

3.1.2 Complete Belt-Loop Geometry

The local pulley geometry describes how axial motion changes the belt's outer-surface radius on each pulley. The belt, however, passes around both pulleys. To connect those radii, the belt must now be followed around the complete loop.

Under Assumption 8 (Prescribed Planar Belt Path), the modeled belt path consists of a circular wrap on each pulley joined by two straight tangent spans. The aligned, fixed-center geometry of Assumption 5 (Aligned Pulley Axes and Fixed Center Distance) makes the two spans equal. If each straight span has length l, and the primary and secondary wrap angles are ϕ_p and ϕ_s, the total geometric length of the loop is

$$L_{\mathrm{geo}} = r_p\phi_p + r_s\phi_s + 2l. \tag{3.1.17}$$

The quantities l, ϕ_p, and ϕ_s follow directly from the geometry shown in Figure 3.3.

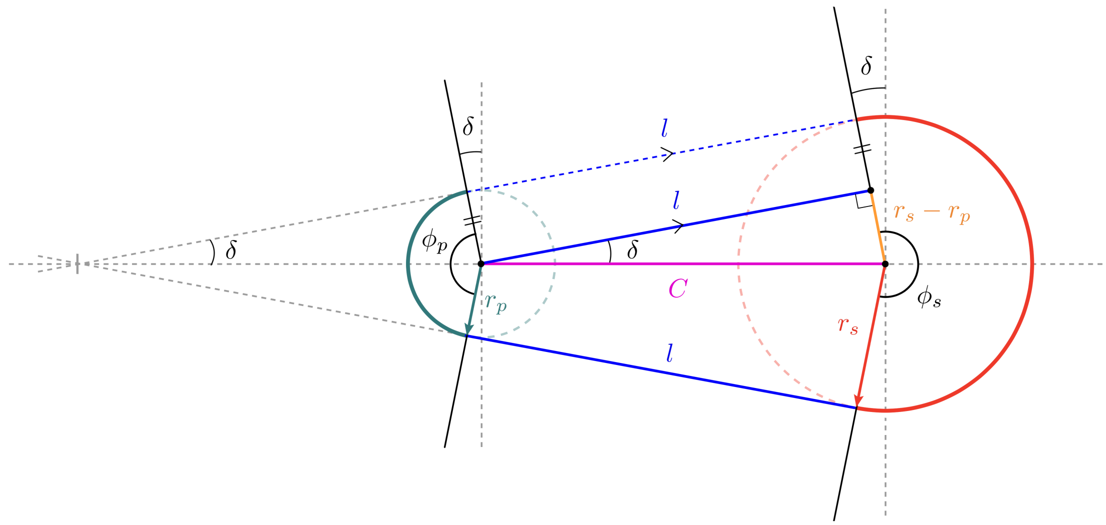

Figure 3.3: Geometry of the complete belt loop. The belt path consists of two wrapped arcs and two equal straight spans of length l. Unequal pulley radii incline the straight spans by the angle δ, producing the primary and secondary wrap angles ϕ_p and ϕ_s.

Consider the right triangle in the upper span in Figure 3.3. The line joining the pulley centres forms its hypotenuse and has length C. One side lies along the straight belt span and therefore has length l.

The other side is perpendicular to the span. The primary and secondary radii drawn to their respective contact points are also perpendicular to this span and are therefore parallel. As shown by the orange construction in Figure 3.3, the difference between their lengths gives the remaining side of the triangle, $r_s - r_p$. The triangle therefore satisfies

$$l^2 + (r_s - r_p)^2 = C^2, \tag{3.1.18}$$

and the length of either straight span is

$$l = \sqrt{C^2 - (r_s - r_p)^2}. \tag{3.1.19}$$

This expression requires

$$|r_s - r_p| < C. \tag{3.1.20}$$

This condition is already enforced by the physical separation of the pulleys. The pulley circles must remain apart, requiring $C > r_s + r_p$, and $r_s + r_p$ is necessarily greater than $|r_s - r_p|$.

The straight-span length accounts for only two parts of the belt path. The two wrapped portions still depend on how the straight spans meet the pulleys. If the pulley radii are equal, the spans are parallel to the line joining the pulley centres and the belt wraps halfway around each pulley. Both wrap angles are then equal to π.

When the radii differ, the straight spans rotate through the angle δ shown in Figure 3.3. The same right triangle gives

$$\sin\delta = \frac{r_s - r_p}{C}, \tag{3.1.21}$$

and therefore

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{3.1.22}$$

The angle δ is signed. For the configuration shown, $r_s > r_p$, so $\delta > 0$. The primary wrap loses an angle δ at each end of its contact arc, while the secondary wrap gains the same amount at each end. Their wrap angles are therefore

$$\phi_p = \pi - 2\delta, \qquad \phi_s = \pi + 2\delta. \tag{3.1.23}$$

If $r_p > r_s$, then $\delta < 0$, and the same equations automatically reverse which pulley has the larger wrap angle.

The complete belt length follows by adding the two wrapped arcs and the two straight spans. Substituting Equations (3.1.19), (3.1.22) and (3.1.23) into Equation (3.1.17) gives

Exact geometric belt length **(3.1.24)**

$$\begin{aligned} L_{\text{geo}}(r_p, r_s, C) = \pi\,(r_p + r_s) \\ + 2\,(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \\ + 2\sqrt{C^2 - (r_s - r_p)^2} \end{aligned}$$

Each term still corresponds directly to part of the belt path. The first is the combined half-circle contribution from the two pulley wraps. The second accounts for the unequal wrap angles, and the third is the combined length of the two straight spans.

For any chosen values of r_p, r_s, and C, Equation (3.1.24) tells us how much belt is required to follow that path. The installed belt cannot change its total length as the pulleys shift. Under Assumption 6 (Fixed Belt Length), every possible pulley position must therefore satisfy

$$L_{\text{geo}}(r_p, r_s, C) = L_b, \tag{3.1.25}$$

where L_b is the fixed length of the same outer-surface path described by r_p and r_s, consistent with the convention in Equation (3.1.8). This is the geometric compatibility condition for the belt loop. Once any two of r_p, r_s, and C are known, the condition determines the third.

For a particular CVT, the first quantity to determine is the centre distance C. The pulley radii change as the sheaves move, but the pulley shafts do not move with them. Once the CVT is assembled, C remains fixed throughout operation.

The fixed centre distance can be found from any installed pulley position that can be measured. At that position, the belt length and the two pulley radii are known. Denoting the measured radii by $r_{p,\text{ref}}$ and $r_{s,\text{ref}}$, the only unknown in Equation (3.1.25) is C.

In the general unequal-radius case, C appears both inside the square root and inside the inverse sine. The exact relation therefore cannot be rearranged into a useful explicit expression for C. Instead, the centre distance is found with a one-dimensional root solve.

For a trial centre distance, the difference between the belt length required by the geometry and the belt length actually available is

$$\mathcal{R}_L(r_p, r_s, C) = L_{\text{geo}}(r_p, r_s, C) - L_b. \tag{3.1.26}$$

A positive residual means that the trial geometry requires more belt than is available. A negative residual means that it requires less. The correct centre distance is the value for which the required and available lengths match:

$$\mathcal{R}_L\left(r_{p,\text{ref}}, r_{s,\text{ref}}, C\right) = 0. \tag{3.1.27}$$

The root solver varies C until this condition is satisfied.

Remark (Simplified Belt-Length Relation)

The exact root solve is required because changing pulley radii alter both the straight-span lengths and the wrap angles. When the radius difference is small relative to the centre distance,

$$\frac{|r_s - r_p|}{C} \ll 1,$$

the straight-span angle is also small:

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

A common simplification is to neglect the resulting wrap-angle correction while retaining the exact straight-span length. The compatibility relation then becomes

$$L_b \approx \pi\left(r_p + r_s\right) + 2\sqrt{C^2 - \left(r_s - r_p\right)^2}.$$

Unlike the exact relation, this approximation can be rearranged directly. For example, the centre distance is

$$C \approx \sqrt{\left(r_s - r_p\right)^2 + \frac{1}{4}\left[L_b - \pi\left(r_p + r_s\right)\right]^2}.$$

The present model retains the exact relation. The effect of using the simplified form is examined in Chapter A.

Once the initial root solve has determined C, both the centre distance and the belt length are known constants. The compatibility condition can then be used to connect the two pulley radii at every subsequent position.

Suppose the primary radius r_p is known. Substituting r_p, C, and L_b into Equation (3.1.26) leaves the secondary radius as the only unknown. Its compatible value is found from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \qquad \text{solved for } r_s. \tag{3.1.28}$$

The resulting r_s is the secondary radius required for the same fixed belt to pass around both pulleys. The calculation also works in the opposite direction. If r_s is known instead, the compatible primary radius follows from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \qquad \text{solved for } r_p. \tag{3.1.29}$$

Thus, once C has been established, either pulley radius can be used to determine the other. The two radii remain local geometric quantities, but they cannot vary independently while belonging to the same fixed-length belt loop.

The arc-and-tangent construction used above is standard in belt-drive geometry. Budynas and Nisbett (2015) presents the same underlying construction for relating pulley radii, straight spans, wrap angles, and total belt length. The exact geometry is retained here because it provides the compatibility relation between the two shifting pulleys.

The local relations $r_p(x_p)$ and $r_s(x_s)$ now describe how each pulley produces its radius, while Equation (3.1.25) describes which pairs of radii can belong to the same belt loop. Combining these results gives the relationship between the two local axial coordinates.

3.1.3 From Local Pulley Coordinates to the Shift Coordinate

The preceding geometry gives a local relation for each pulley: x_p determines $r_p(x_p)$, and x_s determines $r_s(x_s)$. A complete CVT configuration, however, cannot be formed by choosing these positions independently. Both radii must belong to the same belt loop of length L_b. The remaining problem is therefore to identify the pairs (x_p, x_s) that satisfy this shared geometry.

For any proposed pair, the local pulley relations first give the two radii. Substituting those radii into the belt-length residual gives

$$\mathcal{R}_x(x_p, x_s) \equiv \mathcal{R}_L(r_p(x_p), r_s(x_s), C) = 0. \tag{3.1.30}$$

Every pair that makes Equation (3.1.30) equal to zero is a possible position of the two pulleys. Together, these pairs form the compatibility curve.

To reveal the shape of this curve, either local coordinate could be chosen as a starting point. Take x_p simply as an arbitrary choice. For each value of x_p, the primary relation gives $r_p(x_p)$. The shaft spacing fixes C, and L_b remains fixed by Assumption 6 (Fixed Belt Length), so the belt-length residual can then be solved for the required secondary radius r_s. The local secondary relation finally gives the corresponding position x_s:

$$x_p \longrightarrow r_p(x_p) \longrightarrow r_s \longrightarrow x_s.$$

Repeating these steps across the primary travel produces the representative compatibility curve in Figure 3.4.

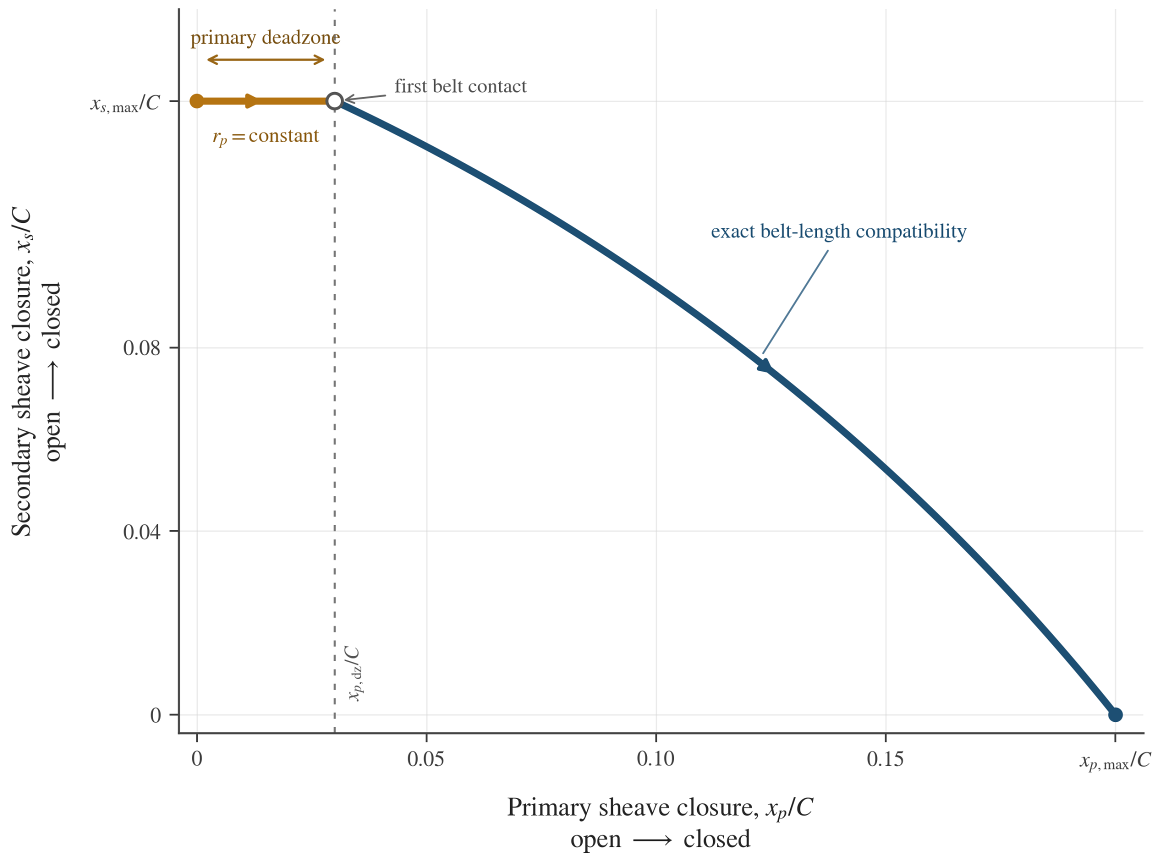

Figure 3.4: Representative compatibility curve between the local primary and secondary axial coordinates for an admissible belt–pulley geometry. Both coordinates are normalized by the same centre distance C, so their relative travel and the changing slope of the engaged branch are shown on a common scale. The dimensions are illustrative and do not represent a particular CVT.

The horizontal portion is created by the primary deadzone. While $0 \leq x_p < x_{p,\mathrm{dz}}$, the primary sheave closes only the clearance to the belt. The primary radius remains fixed, so the compatible secondary radius and position remain fixed as well.

Once the primary reaches the belt, further closure increases r_p. The fixed belt loop then requires r_s to decrease, which means that the secondary opens and x_s decreases. The two sheaves do not move by equal amounts, and the resulting branch is not linear. Changing the radii alters both wrap lengths and both straight spans, so the compatible secondary displacement changes throughout the shift.

The curve also reveals which local coordinate can represent the complete geometry. A vertical line at any x_p meets the curve once, giving one compatible x_s. At the deadzone height, however, a horizontal line meets every x_p between 0 and $x_{p,\text{dz}}$. Thus x_s is a unique function of x_p over the full travel, whereas x_p is not a unique function of x_s.

The primary coordinate is therefore adopted as the system shift coordinate. This is the same s included in the state vector of **??**:

System shift coordinate **(3.1.31)**

$$s \equiv x_p, \qquad x_p(s) = s, \qquad x_s = x_s(s)$$

The secondary mapping is then

$$x_s(s) = \begin{cases} x_{s,\max}, & 0 \le s < x_{p,\text{dz}}, \\ \underset{0 \le x_s \le x_{s,\max}}{\text{root}} \left[\mathcal{R}_x(s, x_s) = 0\right], & x_{p,\text{dz}} \le s \le x_{p,\max}. \end{cases} \tag{3.1.32}$$

The first branch holds the secondary fixed while the primary closes its clearance; the second returns the unique compatible position after belt contact. Both give $x_{s,\max}$ at $s = x_{p,\text{dz}}$, so the mapping is continuous. The shift coordinate therefore determines both movable-sheave positions and, through the local pulley relations, both pulley radii. The position geometry is now complete. What remains is to determine how these positions and radii move when s changes with time.

3.1.4 Pulley-Radius Rates

The shift coordinate now determines the positions of both movable sheaves and the corresponding pulley radii. Its rate form asks the same geometric question over a small motion: if the primary closes by an amount ds, how far must the secondary open, and how quickly do the two pulley radii change?

We first find how each local coordinate and pulley radius changes per unit s. Throughout this subsection, a prime denotes differentiation with respect to s. For either pulley $j \in \{p, s\}$, the corresponding time rates are then

$$\dot{x}_j = x_j'(s)\dot{s}, \qquad \dot{r}_j = r_j'(s)\dot{s}. \tag{3.1.33}$$

Primary radius rate The primary side follows immediately from the definition $x_p(s) = s$:

$$x_p'(s) = 1, \qquad \dot{x}_p = \dot{s}. \tag{3.1.34}$$

The corresponding radius map is constant through the primary deadzone and linear after the movable sheave reaches the belt. Differentiating Equation (3.1.16) therefore gives

$$r_p'(s) = \begin{cases} 0, & 0 \le s < x_{p,\text{dz}}, \\ \dfrac{1}{2\tan\beta}, & x_{p,\text{dz}} < s \le x_{p,\max}. \end{cases} \tag{3.1.35}$$

Multiplying this slope by $\dot{s}$ gives

Primary Radius Rate **(3.1.36)**

$$\dot{r}_p = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ \dfrac{\dot{s}}{2\tan\beta}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

Thus, the primary sheave can move while crossing the deadzone even though the primary belt radius does not. After contact, the sheave angle fixes the radial change produced by each unit of primary closure.

Secondary coordinate and radius rates The secondary requires one additional step. Its local pulley relation tells us how r_s changes when x_s changes, but it does not tell us how far the secondary must move. That motion is set by the fixed belt loop.

At every compatible position,

$$L_{\mathrm{geo}}(r_p(s), r_s(s), C) = L_b. \tag{3.1.37}$$

The centre distance C is fixed by the pulley shafts, while L_b is constant under Assumption 6 (Fixed Belt Length). A small change in s therefore cannot change the geometric length of the belt path:

$$\frac{dL_{\mathrm{geo}}}{ds} = 0. \tag{3.1.38}$$

Both radii change during this motion. The change caused by r_p is the sensitivity of the belt length to r_p, multiplied by the actual rate r_p'. The secondary contributes in the same way. Applying the chain rule to Equation (3.1.37) gives

$$\frac{\partial L_{\mathrm{geo}}}{\partial r_p} r_p' + \frac{\partial L_{\mathrm{geo}}}{\partial r_s} r_s' = 0. \tag{3.1.39}$$

The two partial derivatives have a direct purpose: each tells us how much additional belt path would be required if one pulley radius increased while the other radius remained fixed. The actual pulley motion must make these two changes cancel. We therefore calculate the two sensitivities next.

The span angle and straight-span length entering the belt-length relation were derived in Equations (3.1.19) and (3.1.22). Writing them together in the form needed for differentiation,

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right), \qquad l = \sqrt{C^2 - (r_s - r_p)^2}, \tag{3.1.40}$$

Here, δ sets the wrap angles through Equation (3.1.23), while l is the length of either straight span. Using these quantities, the exact path length from Equation (3.1.24) is

$$L_{\mathrm{geo}} = \pi(r_p + r_s) + 2(r_s - r_p)\delta + 2l. \tag{3.1.41}$$

First, hold r_s and C fixed and increase r_p. The radius difference $r_s - r_p$ then decreases at the same rate:

$$\frac{\partial(r_s - r_p)}{\partial r_p} = -1. \tag{3.1.42}$$

Using this result in the definitions of δ and l gives

$$\begin{aligned}
\frac{\partial \delta}{\partial r_p} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(-\frac{1}{C}\right) \\
&= -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}} = -\frac{1}{l}, && (3.1.43) \\
\frac{\partial l}{\partial r_p} &= \frac{1}{2\sqrt{C^2 - (r_s - r_p)^2}} \left[2(r_s - r_p)\right] \\
&= \frac{r_s - r_p}{l}. && (3.1.44)
\end{aligned}$$

We can now differentiate the complete path length in Equation (3.1.41). The first term differentiates directly, the wrap correction requires the product rule, and the final term contains the changing straight spans:

$$\begin{aligned}
\frac{\partial L_{\text{geo}}}{\partial r_p} &= \pi + 2\left[(-1)\delta + (r_s - r_p)\left(-\frac{1}{l}\right)\right] + 2\left(\frac{r_s - r_p}{l}\right) \\
&= \pi - 2\delta \\
&= \phi_p. && (3.1.45)
\end{aligned}$$

The two terms containing $(r_s - r_p)/l$ cancel. The remaining sensitivity is the primary wrap angle.

The secondary calculation follows the same path, but increasing r_s increases the radius difference:

$$\frac{\partial(r_s - r_p)}{\partial r_s} = 1. \tag{3.1.46}$$

The corresponding derivatives of δ and l are

$$
\begin{aligned}
\frac{\partial \delta}{\partial r_s} &= \frac{1}{\sqrt{1-\left(\dfrac{r_s-r_p}{C}\right)^2}}\left(\frac{1}{C}\right) \\
&= \frac{1}{l},
\end{aligned}
\tag{3.1.47}
$$

$$
\begin{aligned}
\frac{\partial l}{\partial r_s} &= \frac{1}{2\sqrt{C^2-(r_s-r_p)^2}}\left[-2(r_s-r_p)\right] \\
&= -\frac{r_s-r_p}{l}.
\end{aligned}
\tag{3.1.48}
$$

Differentiating the full path length with these signs gives

$$
\begin{aligned}
\frac{\partial L_{\text{geo}}}{\partial r_s} &= \pi + 2\left[(1)\delta + (r_s-r_p)\left(\frac{1}{l}\right)\right] + 2\left(-\frac{r_s-r_p}{l}\right) \\
&= \pi + 2\delta \\
&= \phi_s.
\end{aligned}
\tag{3.1.49}
$$

Again, the terms involving the changing straight-span length cancel. The sensitivity to the secondary radius is the secondary wrap angle. Together,

$$
\frac{\partial L_{\text{geo}}}{\partial r_p} = \phi_p, \qquad \frac{\partial L_{\text{geo}}}{\partial r_s} = \phi_s.
\tag{3.1.50}
$$

This result also has a simple geometric interpretation. If one pulley radius increases by a small amount while the other is held fixed, the required belt path increases by that radial change multiplied by the angle over which the belt wraps the pulley. The remaining changes in the tangent geometry cancel one another.

Returning to the fixed-length condition Equation (3.1.39), the two radial changes must therefore satisfy

$$
\phi_p r_p' + \phi_s r_s' = 0.
\tag{3.1.51}
$$

After primary contact, the local pulley relation gives the same axial-to-radial slope on both pulleys. Since $x_p' = 1$,

$$
r_p' = \frac{1}{2\tan\beta}, \qquad r_s' = \frac{1}{2\tan\beta}x_s'.
\tag{3.1.52}
$$

Substituting these two slopes into Equation (3.1.51) gives

$$
\phi_p\left(\frac{1}{2\tan\beta}\right) + \phi_s\left(\frac{1}{2\tan\beta}x_s'\right) = 0.
\tag{3.1.53}
$$

The common local pulley slope cancels, leaving

$$\phi_p + \phi_s x_s' = 0. \tag{3.1.54}$$

Solving for the slope of the secondary mapping gives

Secondary Coordinate Slope **(3.1.55)**

$$x_s'(s) = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

Within the deadzone, the primary radius does not change, so neither the compatible secondary radius nor x_s changes. After contact, $x_s' < 0$: primary closure requires the secondary to open. Its magnitude is not generally one and changes with the two wrap angles, which is why the engaged branch in Figure 3.4 is curved rather than straight.

The time rate of the secondary coordinate is now obtained by multiplying this slope by $\dot{s}$:

$$\dot{x}_s = x_s'(s)\dot{s} = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}\dot{s}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases} \tag{3.1.56}$$

Applying the local axial-to-radial slope once more gives the secondary radius rate:

Secondary Radius Rate **(3.1.57)**

$$\dot{r}_s = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}\dfrac{\dot{s}}{2\tan\beta}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

The rates above were written for r_p and r_s, which denote the outer-surface radii. The inner surface, cord line, and mass centre remain at fixed depths within the belt cross-section. When the belt moves radially on either pulley, all four reference radii therefore change by exactly the same amount. For either pulley $j \in \{p, s\}$,

$$r_j' = r_{j,\mathrm{inner}}' = r_{j,\mathrm{eff}}' = r_{j,\mathrm{cm}}', \qquad \dot{r}_j = \dot{r}_{j,\mathrm{inner}} = \dot{r}_{j,\mathrm{eff}} = \dot{r}_{j,\mathrm{cm}}.$$

The choice of reference changes the absolute radius, but not its rate. The two pulley-radius rates are now determined by s and $\dot{s}$. However, the secondary multiplier $-\phi_p/\phi_s$ changes as the radii and wrap angles change. Even if $\dot{s}$ is constant, the secondary rate is not. Its acceleration must therefore include the changing slope of the compatibility curve.

3.1.5 Pulley-Radius Accelerations

The pulley-radius rates in Equations (3.1.36) and (3.1.57) were obtained by multiplying the slope of each geometric mapping by the shift velocity $\dot{s}$. Their accelerations follow by asking how those rates change. On the primary, the mapping slope is constant within each branch. On the secondary, the slope in Equation (3.1.55) changes as the pulley radii and wrap angles change. The secondary acceleration must therefore account for both the imposed shift acceleration and the changing geometric slope.

Primary radius acceleration The primary coordinate requires no additional geometry. The system-coordinate definition in Equation (3.1.31) gives $x_p(s) = s$, so

$$x_p'(s) = 1, \qquad x_p''(s) = 0, \tag{3.1.58}$$

and therefore

$$\dot{x}_p = \dot{s}, \qquad \ddot{x}_p = \ddot{s}. \tag{3.1.59}$$

The general shift-to-time relation in Equation (3.1.33) gives the primary radius rate in the form

$$\dot{r}_p = r_p'(s)\dot{s}. \tag{3.1.60}$$

Differentiating this product with respect to time gives

$$\begin{aligned} \ddot{r}_p &= \frac{d}{dt}\left[r_p'(s)\dot{s}\right] \\ &= \frac{dr_p'}{dt}\dot{s} + r_p'(s)\ddot{s}. \end{aligned} \tag{3.1.61}$$

Because r_p' is itself a function of s,

$$\frac{dr_p'}{dt} = r_p''(s)\dot{s}. \tag{3.1.62}$$

Substituting Equation (3.1.62) into Equation (3.1.61) gives

$$\ddot{r}_p = r_p''(s)\dot{s}^2 + r_p'(s)\ddot{s}. \tag{3.1.63}$$

Within either smooth branch of the primary radius map, $r_p'(s)$ is constant. Hence

$$r_p''(s) = 0. \tag{3.1.64}$$

Using the primary slopes from Equation (3.1.35), the primary radius acceleration is

Primary Radius Acceleration **(3.1.65)**

$$\ddot{r}_p = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ \dfrac{\ddot{s}}{2\tan\beta}, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

The result is direct: within the deadzone the primary radius remains fixed, while after contact the shift acceleration is converted into radial acceleration by the same constant sheave slope found in Equation (3.1.35).

Secondary coordinate curvature The secondary mapping is not linear after contact. Its first derivative was found from the fixed-length rate balance in Equation (3.1.54):

$$\phi_p + \phi_s x_s' = 0.$$

The balance in Equation (3.1.54) contains the information needed for the second derivative. We will differentiate it once more with respect to s, which first requires the rates at which the two wrap angles change with s.

The required geometric quantities were defined in Equations (3.1.22) and (3.1.23). Written together, they are

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right), \qquad \phi_p = \pi - 2\delta, \qquad \phi_s = \pi + 2\delta. \tag{3.1.66}$$

As s changes, both r_p and r_s change, so the radius difference inside δ changes at the rate

$$\frac{d}{ds}(r_s - r_p) = r_s' - r_p'. \tag{3.1.67}$$

Applying the chain rule to δ, and using the straight-span length from Equation (3.1.19) in the final line, gives

$$\begin{aligned} \delta' &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(\frac{r_s' - r_p'}{C}\right) \\ &= \frac{r_s' - r_p'}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \frac{r_s' - r_p'}{l}. \end{aligned} \tag{3.1.68}$$

After primary contact, the local pulley slopes are given by Equation (3.1.52):

$$r_p' = \frac{1}{2\tan\beta}, \qquad r_s' = \frac{x_s'}{2\tan\beta}.$$

Substituting these slopes into Equation (3.1.68) gives

$$\begin{aligned} \delta' &= \frac{\dfrac{x_s'}{2\tan\beta} - \dfrac{1}{2\tan\beta}}{l} \\ &= \frac{x_s' - 1}{2l\tan\beta}. \end{aligned} \tag{3.1.69}$$

The wrap-angle derivatives now follow directly from Equation (3.1.66):

$$\phi_p' = -2\delta', \qquad \phi_s' = 2\delta'. \tag{3.1.70}$$

We can now differentiate the secondary slope balance in Equation (3.1.54). The primary term contributes ϕ_p', while the secondary term requires the product rule because both ϕ_s and x_s' change with s:

$$\begin{aligned} 0 &= \frac{d}{ds}\left(\phi_p + \phi_s x_s'\right) \\ &= \phi_p' + \phi_s' x_s' + \phi_s x_s''. \end{aligned} \tag{3.1.71}$$

Substituting the wrap-angle derivatives from Equation (3.1.70) gives

$$-2\delta' + 2\delta' x_s' + \phi_s x_s'' = 0. \tag{3.1.72}$$

Solving first for x_s'',

$$\begin{aligned} \phi_s x_s'' &= 2\delta'(1 - x_s'), \\ x_s'' &= \frac{2\delta'(1 - x_s')}{\phi_s}. \end{aligned} \tag{3.1.73}$$

Using the expression for δ' from Equation (3.1.69),

$$\begin{aligned} x_s'' &= \frac{2}{\phi_s}\left(\frac{x_s' - 1}{2l\tan\beta}\right)(1 - x_s') \\ &= -\frac{(1 - x_s')^2}{\phi_s l\tan\beta}. \end{aligned} \tag{3.1.74}$$

On the engaged branch, the first-derivative result in Equation (3.1.55) gives $x_s' = -\phi_p/\phi_s$. The remaining factor can therefore be simplified:

$$\begin{aligned} 1 - x_s' &= 1 + \frac{\phi_p}{\phi_s} \\ &= \frac{\phi_s + \phi_p}{\phi_s} \\ &= \frac{2\pi}{\phi_s}, \end{aligned} \tag{3.1.75}$$

where $\phi_p + \phi_s = 2\pi$ follows from Equation (3.1.23). Substituting this result into Equation (3.1.74) gives the curvature of the secondary coordinate mapping:

Secondary Coordinate Curvature **(3.1.76)**

$$x_s''(s) = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{4\pi^2}{\phi_s^3 l \tan\beta}, & x_{p,\mathrm{dz}} < s \le x_{p,\mathrm{max}}. \end{cases}$$

The deadzone branch in Equation (3.1.32) is horizontal, so both x_s' and x_s'' are zero there. On the engaged branch, $x_s'' < 0$: as the primary closes, the secondary mapping becomes progressively steeper in the opening direction.

Secondary axial and radial accelerations The curvature in Equation (3.1.76) describes how the slope of the compatibility curve changes with position. To convert it into acceleration, begin with the secondary axial velocity from Equation (3.1.56):

$$\dot{x}_s = x_s'(s)\dot{s}.$$

Differentiating with respect to time gives

$$\begin{aligned} \ddot{x}_s &= \frac{d}{dt}\left[x_s'(s)\dot{s}\right] \\ &= \frac{dx_s'}{dt}\dot{s} + x_s'(s)\ddot{s}. \end{aligned} \tag{3.1.77}$$

Because x_s' changes only through s,

$$\frac{dx_s'}{dt} = x_s''(s)\dot{s}. \tag{3.1.78}$$

The secondary axial acceleration is therefore

$$\ddot{x}_s = x_s'(s)\ddot{s} + x_s''(s)\dot{s}^2. \tag{3.1.79}$$

Substituting the slope from Equation (3.1.55) and the curvature from Equation (3.1.76) gives

Secondary Axial Acceleration **(3.1.80)**

$$\ddot{x}_s = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}\ddot{s} - \dfrac{4\pi^2}{\phi_s^3 l \tan\beta}\dot{s}^2, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

The local radius derivative in Equation (3.1.6) is the constant $1/(2\tan\beta)$. Applying it to the secondary gives

$$\ddot{r}_s = \frac{1}{2\tan\beta}\ddot{x}_s. \tag{3.1.81}$$

Substituting Equation (3.1.80) into Equation (3.1.81) gives

Secondary Radius Acceleration **(3.1.82)**

$$\ddot{r}_s = \begin{cases} 0, & 0 \le s < x_{p,\mathrm{dz}}, \\ -\dfrac{\phi_p}{\phi_s}\dfrac{\ddot{s}}{2\tan\beta} - \dfrac{2\pi^2}{\phi_s^3 l \tan^2\beta}\dot{s}^2, & x_{p,\mathrm{dz}} < s \le x_{p,\max}. \end{cases}$$

The two terms have different physical origins. The first comes from the imposed shift acceleration, converted through the coordinate slope in Equation (3.1.55) and the local pulley slope in Equation (3.1.6). The second comes from the curvature in Equation (3.1.76): even when $\ddot{s} = 0$, moving through the engaged branch changes $-\phi_p/\phi_s$, so the secondary velocity continues to change.

As with the radius rates, the fixed offsets between the four radial references give, for either pulley $j \in \{p, s\}$,

$$\ddot{r}_j = \ddot{r}_{j,\mathrm{inner}} = \ddot{r}_{j,\mathrm{eff}} = \ddot{r}_{j,\mathrm{cm}}. \tag{3.1.83}$$

Remark (Deadzone Boundary)

The idealized maps in Equations (3.1.16) and (3.1.32) change slope when the primary first reaches the belt at $s = x_{p,\mathrm{dz}}$. Their first derivatives therefore change discontinuously, and their ordinary second derivatives are not defined at that single point. The acceleration relations in Equations (3.1.65) and (3.1.82) apply within the deadzone and engaged branches, while first contact is treated as the transition between them.

Remark (Practical Consequences of the Pulley Geometry)

The radius, rate, and acceleration relations derived in this section describe how the pulley geometry evolves as the CVT shifts. Chapter C explores the practical consequences of these results in more detail. In particular, it examines the resulting geometric CVT ratio and ratio-change rate across the shift travel, together with how the current pulley geometry and sheave angle shape the available ratio range and ratio-change sensitivity.

3.1.6 Geometric Closure and Model Scope

Specifying the shift state now determines the complete modeled pulley geometry. The definition in Equation (3.1.31) sets the primary position directly, while Equation (3.1.32) gives the unique compatible secondary position. The local pulley maps in Equations (3.1.15) and (3.1.16) convert those two positions into the outer-surface radii used to describe the belt path. Finally, the rate and acceleration relations in Equations (3.1.36), (3.1.57), (3.1.65) and (3.1.82) determine how those radii move. Thus, for any admissible shift state,

$$(s, \dot{s}, \ddot{s}) \quad \longmapsto \quad (r_p, \dot{r}_p, \ddot{r}_p, r_s, \dot{r}_s, \ddot{r}_s)\,.$$

Here r_p and r_s retain the outer-surface convention used throughout the geometric development. The corresponding inner-surface, effective, and centre-of-mass radii follow directly from their fixed positions within the belt cross-section. Nothing else has to be solved for.

These results follow from the geometry derived above under the idealizations stated in Section 2.2. The belt cross-section and total length remain fixed under Assumption 4 (Fixed, Uniform Belt Cross-Section) and Assumption 6 (Fixed Belt Length), while Assumption 8 (Prescribed Planar Belt Path) constrains the belt to circular wraps and straight tangent spans. Together with the aligned pulley axes and fixed centre distance of Assumption 5 (Aligned Pulley Axes and Fixed Center Distance), these conditions leave one modeled belt path for each admissible value of s.

Relation to prior transient models These choices also define the level of detail represented by the present geometry. It determines the complete belt path that is globally compatible with the imposed shift position, but it does not track the trajectory of each belt element through the pulley wraps.

Duan et al. (2016) begin from closely related open-belt arc-and-tangent geometry and the same wedge relationship between movable-sheave motion and nominal radius motion. Their chain-CVT formulation retains additional internal deformation. Chain extension and its rate contribute to the span tension, and pulley deformation allows the local radius followed by a chain element to differ from the nominal pulley radius. The present formulation instead fixes the belt length through Assumption 6 (Fixed Belt Length) and prescribes the arc-and-tangent belt path through Assumption 8 (Prescribed Planar Belt Path). Consequently, elastic path-length change and deformation-driven departure from that prescribed path are not represented in this geometric closure.

Rubber-belt transient models resolve the same shifting process at several levels of detail. Analytical shift-mechanics models, such as those of Sorge (2008) and Cammalleri and Sorge (2009), describe local radial belt migration through a pulley wrap and couple it to circumferential transport, sliding, and the mechanics of the contact region. Discretized rubber-belt models, such as those of Julió and Plante (2011) and Ballew (2015), instead obtain the evolving belt geometry from the motion and contact state of individual belt elements. These approaches can resolve local deformation, tension redistribution, and contact migration in greater detail, but require additional spatial or nodal states and the contact laws that govern them.

The distinction is therefore one of model resolution. The present formulation uses the global compatibility condition in Equation (3.1.25) to find the representative pulley-radius pair and its derivatives directly from s, $\dot{s}$, and $\ddot{s}$. The cited transient models additionally ask how the belt or chain deforms and migrates within that path. Those internal motions lie outside the present geometric layer; within the assumptions stated above, the global pulley geometry and its motion are fully determined.

This closure determines which sheave and radius motions are compatible with the modeled belt path, but it does not determine which compatible acceleration occurs. In particular, every acceleration relation above still contains the shift acceleration $\ddot{s}$. Its value must be consistent with the forces and inertias acting on the two movable pulley assemblies.

3.2 Axial Actuation and Sheave Balances

The preceding section established how the two movable sheaves must move together. It did not determine the motion itself. That requires the forces acting on each movable assembly and the inertia that those forces must accelerate.

The primary and secondary remain separate mechanisms even though the belt geometry couples their positions. Each has its own actuator, spring, moving hardware, and belt reaction, so each must satisfy its own axial equation of motion. We therefore complete the primary balance in its local coordinate x_p, then do the same for the secondary in x_s. Once each mechanism has been interpreted and every local force has been derived, the geometric maps developed above in Section 3.1 express both balances through the shared shift coordinate s. This final step shows how the two local free bodies act on one shift motion.

3.2.1 Primary Movable-Sheave Balance

The primary is easiest to understand by separating rotation about the shaft from motion along it. The entire primary rotates with the shaft at ω_p, but its parts do not all move axially. The fixed sheave remains at one axial position. Opposite it, the movable sheave, ramp carrier, and the hardware attached rigidly to them slide together along the shaft. Motion in the closing direction narrows the pulley groove and clamps the belt; motion in the opening direction widens the groove.

The flyweights provide the closing drive. They are attached to the fixed sheave and carried around the shaft with it, but each can also pivot about its attachment to the fixed sheave. As the primary rotates, centrifugal loading tends to swing each flyweight radially outward until it presses against a ramp carried by the movable sheave. Because the ramp surface is inclined, its contact reaction has an axial component that drives the movable sheave in the closing direction. The primary spring pushes the movable sheave in the opening direction, and the belt pushes back through its normal reaction on the movable face.

This leaves two distinct moving bodies in the model. The movable sheave, ramp carrier, and attached hardware form one translating assembly with axial coordinate x_p. The flyweights rotate with the primary and pivot separately, but do not translate rigidly with x_p; they deliver their closing action to the movable sheave through the ramp contacts. During a shift, the net axial force must accelerate the translating assembly, while the flyweights require their own rotational balance. We therefore begin by isolating the translating assembly, then determine the force transmitted to it by the flyweights. This mechanism and the resulting axial free body are shown together in Figure 3.5.

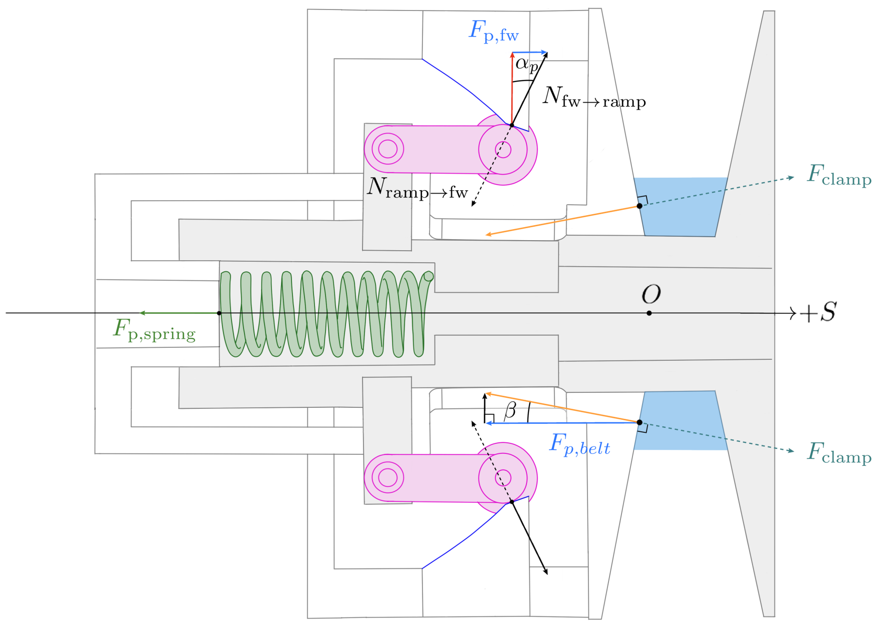

Figure 3.5: Axial free-body diagram of the primary movable assembly (unshaded). The symmetric sides of the cross-section are used to show the flyweight–ramp and belt–sheave force resolutions separately. Dotted arrows show the equal-and-opposite forces on the flyweight and belt, with F_{clamp} indicating the movable sheave's squeeze on the belt.

As established in Section 3.1, positive x_p closes the primary. Let m_p be the total mass of the movable sheave, ramp carrier, and every other component that translates rigidly with them. The three axial forces shown in Figure 3.5 are:

- $F_{p,\text{fly}}$, the closing force delivered to the ramp carrier by the flyweight contacts;
- $F_{p,\text{spring}}$, the opening force exerted by the primary spring; and
- $F_{p,\text{belt}}$, the opening force exerted by the belt on the movable face.

Newton's second law along the shaft is therefore

$$m_p \ddot{x}_p = F_{p,\text{fly}} - F_{p,\text{spring}} - F_{p,\text{belt}}. \tag{3.2.1}$$

This free body tells us exactly what remains to be found. The spring force will follow from its compression, and the belt force from the normal reaction on the conical face. The flyweight force is harder. It is the axial result of a contact on a body that pivots rather than translating rigidly with the sheave. We therefore determine $F_{p,\text{fly}}$ first.

Force Transmitted by the Flyweights

Flyweight free body Figure 3.6 isolates one flyweight in the radial–axial plane fixed to the primary. The flyweight is pinned at P and pivots about the circumferential axis through that point. Its pose is described by q_f, measured from $+x_p$ to the body-fixed reference line shown in the figure. At $q_f = 0$, this line is aligned with $+x_p$; at $q_f = \pi/2$, it points radially outward. Increasing q_f therefore describes the outward-pivoting direction.

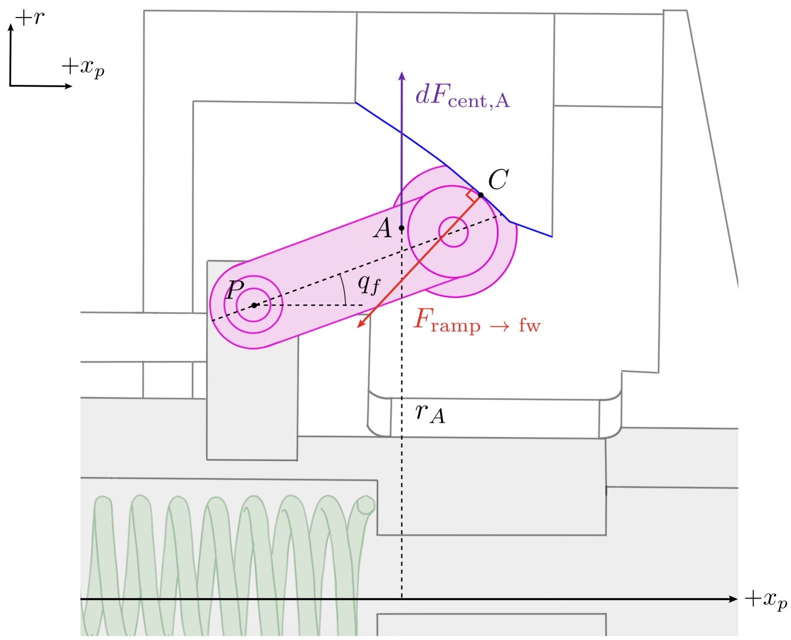

Figure 3.6: Free body of one primary flyweight in the frame rotating with the primary. Centrifugal loading acts throughout the flyweight mass; the marked element A identifies one infinitesimal mass dm at shaft radius r_A and its outward load. The ramp-contact normal at C and the pivot reaction at P complete the force system.

As the primary rotates at ω_p, the flyweight experiences centrifugal loading throughout its mass. Each infinitesimal element is loaded radially outward according to its distance from the primary shaft. The marked element A is one example: for mass dm at shaft radius r_A, its outward load is

$$dF_{\text{cent},A} = \omega_p^2 r_A \, dm. \tag{3.2.2}$$

Every other element is loaded in the same way at its own shaft radius. Together, these forces form the distributed centrifugal loading of the flyweight.

Because the flyweight's relative motion is confined to the radial–axial plane, the Euler and Coriolis inertial forces are circumferential and have no moment about the circumferential pivot axis. The q_f balance therefore depends on ω_p, but not on $\dot{\omega}_p$.

Taking moments about P separates the roles of these loads. The pivot reaction passes through P and therefore contributes no moment. The radial lines of action of the centrifugal loads generally do not pass through P, so together they produce a moment in the direction of increasing q_f. The ramp normal produces the opposing moment.

The figure shows one flyweight, whereas the primary contains a symmetric set whose members follow the same q_f. Let $\tau_{f,\text{cent}}$ and $\tau_{f,\text{contact}}$ denote the respective magnitudes of the centrifugal and contact moments summed over this complete set. Since their shared coordinate q_f is a rotation about the flyweight pivots, let I_f denote the sum of each flyweight's mass moment of inertia about its own pivot axis. With positive moment defined in the direction of increasing q_f, rotational Newton's second law gives

$$\tau_{f,\text{cent}} - \tau_{f,\text{contact}} = I_f \ddot{q}_f. \tag{3.2.3}$$

Mechanism geometry The moment balance is written in the flyweight coordinate q_f, whereas the force required by the movable-sheave balance must be expressed through the axial coordinate x_p. The ramp geometry connects the two: each sheave position fixes the corresponding flyweight pose.

The centrifugal moment in Equation (3.2.3) comes from the distributed loading shown in the figure, so its value depends on how the complete flyweight mass is distributed about the primary shaft. The marked element A contributes $r_A^2\,dm$ to the mass moment of inertia about that shaft. Adding the corresponding contributions from every element of every flyweight gives the complete shaft-axis inertia $J_{f,p}$. Since x_p fixes the flyweight pose, it fixes every element's shaft radius and therefore fixes $J_{f,p}$. This shaft-axis inertia should not be confused with the pivot-axis inertia I_f in Equation (3.2.3), which is constant for rigid flyweights.

The mechanism therefore supplies two configuration maps:

$$q_f = q_f(x_p), \qquad J_{f,p} = J_{f,p}(x_p). \tag{3.2.4}$$

Together with the constant I_f, these maps abstract the physical geometry of the flyweight mechanism into the quantities required by the general force balance.

Remark (Constructing the flyweight maps)

The mechanism-specific construction is not carried out here. **??** works through two common flyweight arrangements, showing how their geometry and mass distribution produce $q_f(x_p)$, $J_{f,p}(x_p)$, and I_f.

With these mechanism properties specified, the remaining task is to express the two moments and $\ddot{q}_f$ in Equation (3.2.3) through the primary motion. We begin with the moments.

Both could be found by resolving the forces and constructing their moment arms for the specific flyweight geometry. A more convenient approach is equivalent virtual work. Hold the mechanism at its current position and imagine a small compatible motion. During this motion, a moment τ_f acting through dq_f performs the work

$$\tau_f\,dq_f = \delta W_f. \tag{3.2.5}$$

We can therefore find each moment by calculating the work of its underlying load during the same motion.

For the centrifugal loading, at fixed instantaneous angular speed, the work associated with a change dJ in shaft-axis inertia is $\frac{1}{2}\omega^2\, dJ$. Applying this relation to the flyweights gives

$$\tau_{f,\text{cent}}\, dq_f = \delta W_{f,\text{cent}} = \frac{1}{2}\omega_p^2\, dJ_{f,p}. \tag{3.2.6}$$

For the contact moment, the corresponding work is the force–displacement work $F\, d\ell$. The movable sheave moves axially through dx_p, so only the axial component of the ramp-contact force performs work. This component is $F_{p,\text{fly}}$, the closing force introduced by the movable-sheave free body, and its work is therefore $F_{p,\text{fly}}\, dx_p$. Under Assumption 3 (Idealized Actuator Mechanics), the ramp contact is treated as an ideal frictionless constraint, so no work is dissipated at the contact. The work of the contact moment therefore appears entirely as axial work on the movable assembly. Using the positive magnitudes defined above,

$$\tau_{f,\text{contact}}\, dq_f = \delta W_{f,\text{contact}} = F_{p,\text{fly}}\, dx_p. \tag{3.2.7}$$

Flyweight acceleration The two moment terms are now written in forms that can be returned to the rotational balance. Its remaining term is the flyweight angular acceleration. Ramp contact makes this acceleration a consequence of the sheave motion rather than an independent state. Since the contact geometry already gives q_f as a function of x_p, its angular velocity is

$$\dot{q}_f = \frac{dq_f}{dx_p}\dot{x}_p, \tag{3.2.8}$$

and one more derivative gives

$$\ddot{q}_f = \frac{dq_f}{dx_p}\ddot{x}_p + \frac{d^2 q_f}{dx_p^2}\dot{x}_p^2. \tag{3.2.9}$$

The second term appears when the amount of flyweight rotation required per unit sheave travel changes along the ramp. Even at constant $\dot{x}_p$, the flyweight then changes angular speed as it moves.

Axial closing force There is now an expression for every term in the rotational balance. Multiplying Equation (3.2.3) by dq_f and substituting Equations (3.2.6), (3.2.7) and (3.2.9) brings them together,

$$\frac{1}{2}\omega_p^2 dJ_{f,p} - F_{p,\text{fly}} dx_p = I_f\left(\frac{dq_f}{dx_p}\ddot{x}_p + \frac{d^2 q_f}{dx_p^2}\dot{x}_p^2\right) dq_f. \tag{3.2.10}$$

Rearranging for the force transmitted to the movable sheave gives

$$\boxed{\begin{aligned} F_{p,\text{fly}} = \frac{1}{2}\omega_p^2 \frac{dJ_{f,p}}{dx_p} - I_f \left(\frac{dq_f}{dx_p}\right)^2 \ddot{x}_p \\ - I_f \frac{dq_f}{dx_p}\frac{d^2 q_f}{dx_p^2}\dot{x}_p^2. \end{aligned}} \tag{3.2.11}$$

The result now has the form required by the movable-sheave free body. The first term is the centrifugal drive and therefore uses the shaft-axis property $J_{f,p}$. The final two terms arise from acceleration about the flyweight pivots and therefore use the pivot-axis property I_f: the first reflects $\ddot{x}_p$ through the motion ratio, while the second appears when that motion ratio changes with position.

Remark (Admissible flyweight maps)

The completed force law uses $q_f''(x_p)$ and $J_{f,p}'(x_p)$, while the selected coordinate directions require primary closure to correspond to outward flyweight rotation. Over the admissible shift range, the supplied mechanism maps must therefore satisfy

$$q_f \in C^2, \qquad J_{f,p} \in C^1, \qquad 0 < \frac{dq_f}{dx_p} < \infty.$$

These are requirements on the fitted or constructed maps, not additional equations of motion.

The one-sided flyweight contact can transmit compression but not tension, so the force predicted by Equation (3.2.11) must satisfy $F_{p,\text{fly}} \geq 0$.

Spring and Belt Forces

Returning to the axial force balance, the two remaining forces in Equation (3.2.1) act directly on the translating assembly.

Primary spring Let k_p be the spring stiffness and $x_{p,\text{pre}}$ its installed compression when the primary is open at $x_p = 0$. Closing the sheave by x_p adds the same amount of compression, so the spring opening force is

$$F_{p,\text{spring}} = k_p \left(x_{p,\text{pre}} + x_p\right). \tag{3.2.12}$$

Belt reaction Let N_p be the total integrated normal load over both primary sheave faces. This load is coupled to the belt and the secondary balance, so it remains an unknown at this stage. Under Assumption 12 (Symmetric Sheave-Face Sharing), the movable face carries one half of this total load. Its normal is inclined by the sheave half-angle β from the axial direction, so the opening force on the movable face is

$$F_{p,\text{belt}} = \frac{N_p}{2}\cos\beta. \tag{3.2.13}$$

Completed Primary Balance

Each force in the original free body now has a mechanism-level expression. Substitute Equations (3.2.11) to (3.2.13) into Equation (3.2.1) and collect the terms multiplying $\ddot{x}_p$:

$$\boxed{\begin{aligned}\left[m_p + I_f\left(\frac{dq_f}{dx_p}\right)^2\right]\ddot{x}_p = {} & \frac{1}{2}\omega_p^2\frac{dJ_{f,p}}{dx_p} \\ & - I_f\frac{dq_f}{dx_p}\frac{d^2q_f}{dx_p^2}\dot{x}_p^2 \\ & - k_p\left(x_{p,\text{pre}} + x_p\right) - \frac{N_p}{2}\cos\beta.\end{aligned}} \tag{3.2.14}$$

The flyweights now appear through the same two axes introduced by their free body. Rotation about the primary shaft, described by $J_{f,p}$, provides the speed-dependent closing drive. Rotation about the flyweight pivots, described by I_f, adds to the inertia that the axial motion must accelerate. If the motion ratio dq_f/dx_p changes along the ramp, the flyweights also accelerate even when $\dot{x}_p$ is constant, producing the term proportional to $\dot{x}_p^2$. The spring and belt forces act directly on the movable assembly and oppose closure exactly as shown in the first free body.

This is the primary axial equation of motion. It states how the flyweight action is divided among spring compression, belt loading, and acceleration of the translating and pivoting hardware.

Point-Mass Limit and Relation to Existing Models

The general force law becomes the familiar classical actuator model when the distributed flyweight mass is collapsed to its centre of mass and the flyweight's pivot dynamics are neglected. Let m_f be the total mass of the symmetric flyweight set and let $r_f(x_p)$ be the common instantaneous centre-of-mass radius. Under the point-mass assumption, the shaft-axis inertia reduces to

$$J_{f,p} = m_f r_f^2. \tag{3.2.15}$$

Substituting this point-mass inertia into the centrifugal term of Equation (3.2.11), while omitting the pivot-acceleration terms, gives

$$F_{p,\text{fly}}^{\text{pm}} = m_f r_f \omega_p^2 \frac{dr_f}{dx_p}. \tag{3.2.16}$$

This is the familiar point-mass result. The factor $m_f r_f \omega_p^2$ is the total centrifugal loading, while dr_f/dx_p is the local radial-to-axial motion ratio that converts that loading into sheave force. The path need not be straight, and its motion ratio need not be constant. A sliding mass on a straight ramp is one simple case; for a pivoted flyweight, $r_f(x_p)$ instead follows from the flyweight pose and its mass geometry.

If the movable sheave is also in axial equilibrium, the point-mass actuator force balances the opening forces from the spring and belt:

$$m_f r_f \omega_p^2 \frac{dr_f}{dx_p} = k_p\left(x_{p,\text{pre}} + x_p\right) + \frac{N_p}{2}\cos\beta. \tag{3.2.17}$$

This reduction appears directly in the quasi-static model of Messick (2018). Their three wedge-shaped flyweights slide along linear ramps without changing orientation. Measured CAD geometry supplies the centre-of-mass radius and the radial-to-axial displacement ratio, giving a force of the same form as Equation (3.2.16). Friction is neglected, and the resulting primary force is supplied to the belt model as a boundary condition; shifting is treated as a sequence of equilibrium configurations rather than a time-dependent response. Arango and Muñoz Alzate (2020) use the same instantaneous force conversion to construct both sliding-weight ramps and pivoted flyweight-cam profiles. Their contact geometry is designed so that the available centrifugal force supplies the required clamping force through the local mechanical advantage.

Other treatments retain more of the contact geometry while keeping the same instantaneous-equilibrium closure. Kim et al. (2002) resolve the two contacts of a centrifugal roller and include contact friction when converting roller centrifugal force into driver thrust. The current roller radius and contact angles vary with ratio, but no roller acceleration enters the thrust equation. Skinner (2020) resolve a flyweight, its supporting link, and the ramp reactions in a complete free body, including the centrifugal loading of both the flyweight and the link. In the present notation, those separate mass contributions could be collected in $J_{f,p}(x_p)$. Skinner's derivation therefore retains a substantially richer equilibrium force path than a single fixed ramp factor, but still closes the flyweight system by force equilibrium at each configuration.

Despite their different levels of geometric detail, none of these actuator relations includes the acceleration of the flyweight mechanism. Messick does not model shift time, Skinner solves algebraically for the speed required at each configuration, and Kim places the time dependence in a separate experimentally identified first-order ratio law, using different fitted coefficients for upshift and downshift. Conversely, the transient belt model of Ballew (2015) imposes primary axial force through a controller rather than resolving the mechanical actuator. Across these examples, transient shift motion is therefore obtained without returning the flyweights' own acceleration to the primary axial balance.

The distinction of the present formulation is therefore not that it can describe a curved ramp or a detailed flyweight shape; the equilibrium models above can already retain substantial geometric detail. Here, the complete mass distribution enters through $J_{f,p}(x_p)$, so the centrifugal drive is not restricted to a centre-of-mass point model. More importantly, the flyweight moment balance is not closed at instantaneous equilibrium. Mapping $I_f \ddot{q}_f$ into x_p produces both the reflected pivot inertia and the changing-motion-ratio term in Equation (3.2.14). Consequently, the flyweight drive, spring force, belt reaction, translating mass, and pivoting mass all determine the same shift acceleration rather than being divided between an equilibrium actuator law and a separate shift-response model. No equivalent explicit reflection of the flyweights' pivoting inertia was identified in the reviewed mechanically actuated rubber-belt CVT formulations.

This extension does not retain every effect represented in the earlier models. Structural deformation is excluded by Assumption 2 (Rigid Pulley and Actuation Hardware), while Assumption 3 (Idealized Actuator Mechanics) treats the ramp contact as an ideal constraint rather than resolving its tribology or local compliance. The roller-contact friction retained by Kim et al. (2002), for example, therefore lies outside the present force law; roller spin and contact compliance are likewise not resolved. The added fidelity is in the coupled mechanism dynamics, not in frictional or compliant contact detail.

3.2.2 Secondary Movable-Sheave Balance

While the primary closes mainly in response to rotational speed, the secondary is torque reactive: torque transmitted through the pulley can also increase the axial force clamping the belt.

To see how, begin with the two secondary sheaves rotationally disconnected. The fixed sheave rotates with the secondary shaft. The movable sheave can slide along that shaft and rotate relative to the fixed sheave. Because the belt contacts the two faces separately, it applies torque to each body. With no connection between them, the two bodies can accelerate differently and rotate relative to one another.

Now place a pin on the movable sheave in a straight slot running along the shaft. The movable sheave remains free to slide, but the pin prevents relative rotation. Whenever the two sheaves tend to rotate differently, the pin presses against the side of the slot and a normal reaction develops. Because the slot is axial, this normal acts entirely in the circumferential direction. It transfers torque between the sheaves but produces no axial force.

A helix is the same slot tilted around the shaft. The contact normal remains perpendicular to the slot surface, but now has both circumferential and axial components. The circumferential component continues to couple the rotational motion of the sheaves, while the axial component acts on the movable sheave. Relative rotation is therefore allowed only when accompanied by axial travel along the helix.

These three cases and their contact reactions are compared in Figure 3.7.

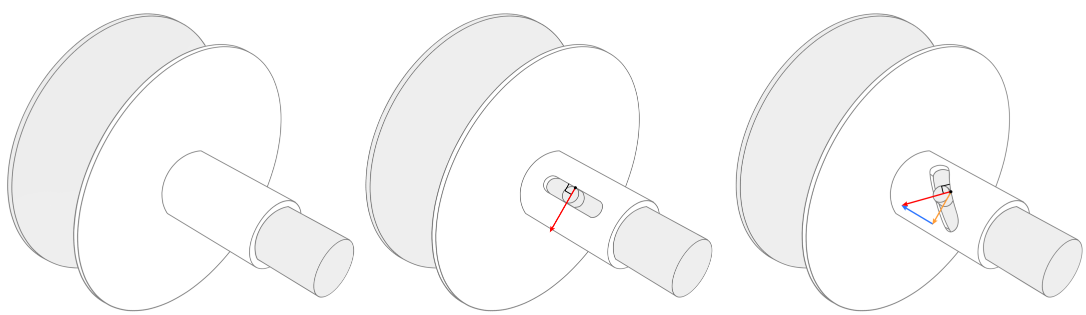

Figure 3.7: How the helix creates torque-reactive axial loading. With no connection, the two sheaves may rotate independently. A straight axial slot couples their rotation through a purely circumferential reaction; inclining the slot adds an axial component to that same reaction, coupling relative rotation to axial travel.

If the axial component of the helix reaction is balanced by the other axial loads, the movable sheave remains at its current position. Its relative angle then remains fixed, and the two sheaves rotate together with the shaft. If the axial loads do not balance, their resultant accelerates the movable sheave along the helix, and it rotates by the amount required by the slot geometry.

In the actual secondary, the belt and spring load this mechanism. The belt applies torque to the two sheave faces separately. The secondary spring is installed both compressed and twisted: its torsional preload tends to rotate the movable sheave relative to the fixed sheave, while its axial compression pushes the movable sheave toward closure. The torsional action loads the helix; the axial action enters the movable-sheave balance directly.

The axial free body gives the target for the derivation.

Axial Free Body

Figure 3.8 isolates the movable assembly in the axial direction.

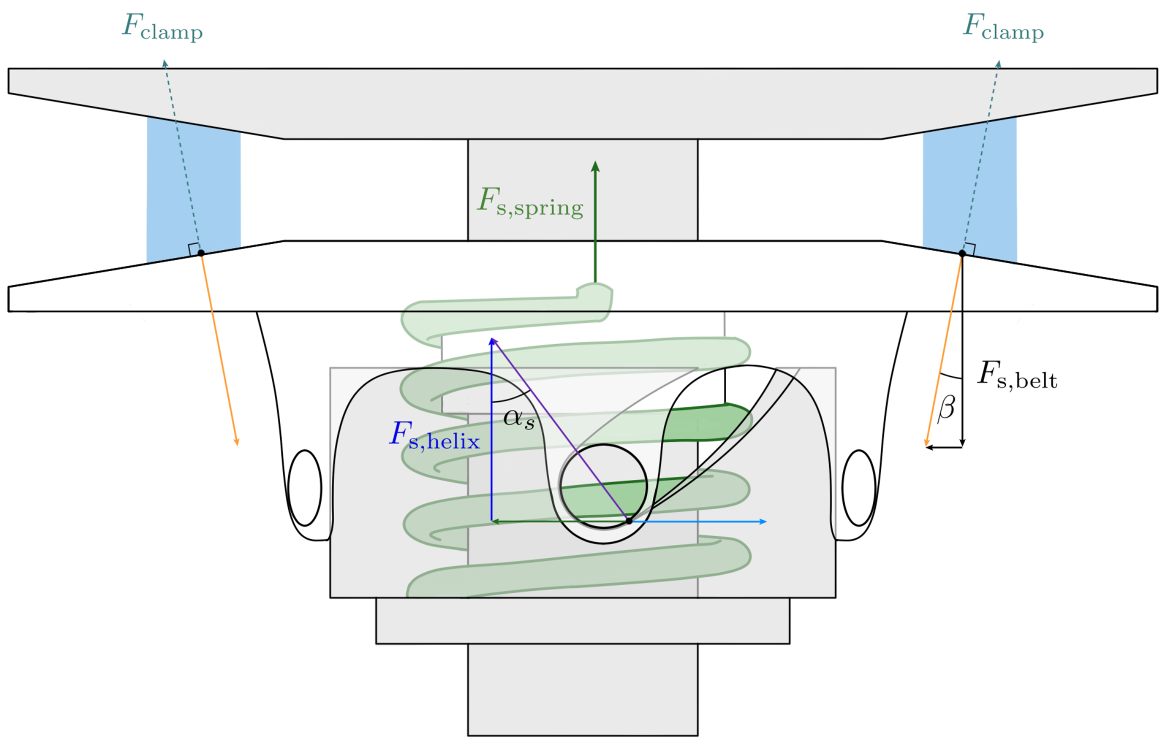

Figure 3.8: Axial free body of the complete movable secondary assembly. The axial spring force and belt reaction act directly on the translating assembly, while the secondary's torque-reactive behaviour enters through the pin–helix force $F_{s,\text{helix}}$.

Let m_s be the mass of the movable sheave and every component that translates rigidly with it. Three axial loads act on this assembly. The pin–helix contact contributes the signed force $F_{s,\text{helix}}$. The compressed spring pushes the movable sheave toward the fixed sheave in $+x_s$. The belt pushes the two sheave faces apart, so its reaction on the movable face acts in $-x_s$, just as on the primary. Newton's second law therefore gives

$$m_s \ddot{x}_s = F_{s,\text{helix}} + F_{s,\text{spring}} - F_{s,\text{belt}}. \tag{3.2.18}$$

This free body makes the remaining task clear. The direct spring and belt forces follow from compression and contact geometry. The helix force does not: the tilted slot has established that an axial reaction exists, but not its magnitude or sign. Because the same contact reaction must also carry the torque that constrains relative rotation, its rotational effect is resolved first. This sets the order of the derivation:

1. use the rotational balance to find the torque carried by the helix;
2. use the helix geometry to convert that torque into axial force; and

3. add this force to the direct spring and belt loads in the axial balance.

Rotational Loading of the Movable Sheave

As established in Section 3.1, x_s is measured from the open secondary toward the fixed sheave, so $+x_s$ closes the pulley. Let θ_s measure the rotation of the movable sheave relative to the fixed sheave in the same positive rotational sense already assigned to ω_s (see Section 2.3.1). Only the conventional torque-reactive helix orientation is considered: as the secondary closes in $+x_s$, the movable sheave turns relative to the fixed sheave in $+\theta_s$.

With these directions set, the first question is what loads the helix circumferentially. The belt applies torque to the movable face, and the twisted spring applies torque between the two sheaves. The pin–helix contact supplies the torque that constrains the resulting relative motion to follow the slot.

Our goal is to find the torque reacted at the pin–helix contact. To do so, isolate the movable assembly and apply Newton's second law for rotation. Use the following quantities:

- $I_{s,M}$: rotational inertia of the movable sheave and every component that rotates rigidly with it;
- $\omega_{s,M}$: absolute angular velocity of that assembly;
- $\tau_{s,M}$: signed belt torque applied to the movable face;
- $\tau_{s,\text{spring}}$: signed spring torque applied to the movable assembly; and
- $\tau_{s,\text{helix}}$: the contact torque at the pin–helix connection, produced by the circumferential component of the contact normal. A positive $\tau_{s,\text{helix}}$ acts on the movable assembly in $-\theta_s$.

The figures show one representative pin and track. The quantities listed above and later in this section are the totals over the complete set of loaded pin–track contacts, so no additional track multiplier is required.

The rotational balance is therefore

$$\tau_{s,M} + \tau_{s,\text{spring}} - \tau_{s,\text{helix}} = I_{s,M}\dot{\omega}_{s,M}. \tag{3.2.19}$$

Solving for the helix torque gives

$$\tau_{s,\text{helix}} = \tau_{s,M} + \tau_{s,\text{spring}} - I_{s,M}\dot{\omega}_{s,M}. \tag{3.2.20}$$

Equation (3.2.20) gives the contact torque we want, but its right-hand side still needs three relations:

- the share of belt torque applied to the movable face, $\tau_{s,M}$;
- the torque supplied by the wound spring, $\tau_{s,\text{spring}}$; and
- the absolute angular acceleration of the movable assembly, $\dot{\omega}_{s,M}$.

The belt term follows immediately from the face-torque assumption.

Movable-face belt torque Let τ_s be the total signed torque exerted by the belt on the secondary, so $\tau_s = \tau_{s,F} + \tau_{s,M}$. Under Assumption 12 (Symmetric Sheave-Face Sharing), the two faces receive equal torque shares. Only the movable-face share appears in its free body:

$$\tau_{s,M} = \frac{\tau_s}{2}. \tag{3.2.21}$$

This equal split is a reduced-contact assumption. During shift, the movable face rotates slightly faster or slower than the fixed face. The two belt–face contacts can therefore have different sliding conditions, so their torques need not divide equally.

Remark (Why the face-torque split can matter)

Sorge and Cammalleri (2011) resolve the two contacts separately and show that the face torques can depart substantially from an equal split; distributed tangential forces on the two faces can even act in opposite directions over parts of the wrap. Since $\tau_{s,M}$ enters Equation (3.2.20) directly, this is also a potential error in the torque reacted by the helix.

The remaining terms both depend on the helix kinematics: $\dot{\omega}_{s,M}$ depends on how the movable sheave rotates as it shifts, and the spring torque depends on how far that motion unwinds the spring. We therefore derive the relation between x_s and θ_s next.

Motion Imposed by the Helix

The helix geometry tells us how axial shift changes the relative angle of the movable sheave. Unwrap one track from its effective cylindrical radius r_h, as seen in Figure 3.9. A small relative rotation $d\theta_s$ moves the pin by $r_h\,d\theta_s$ circumferentially while the sheave moves by dx_s axially. Let α_s be the local angle of the unwrapped track, measured from the positive circumferential direction toward the positive axial direction. For the conventional helix orientation considered here,

$$0 < \alpha_s < \frac{\pi}{2}.$$

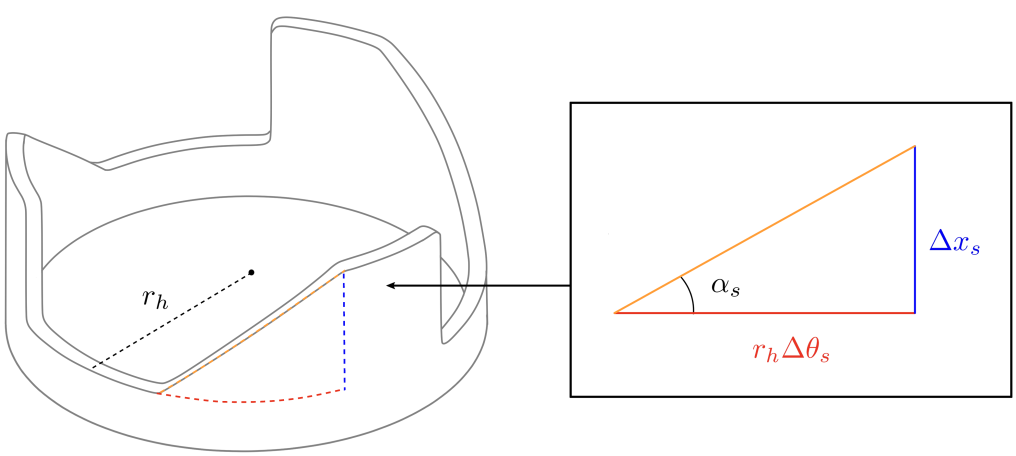

Figure 3.9: Unwrapped secondary-helix geometry. The same track angle relates relative rotation to axial travel and reaction torque to axial force.

Taking the infinitesimal limit of the triangle in Figure 3.9,

$$\tan\alpha_s = \frac{dx_s}{r_h\,d\theta_s}, \qquad \frac{d\theta_s}{dx_s} = \frac{1}{r_h\tan\alpha_s}. \tag{3.2.22}$$

Thus, $d\theta_s/dx_s$ is the relative rotation required per unit axial travel. It is positive for the conventional helix orientation assumed here. Returning to the straight axial slot from the opening example, $\alpha_s \to \pi/2$ and $d\theta_s/dx_s \to 0$, as expected.

Taking $\theta_s(0) = 0$ at the open secondary, integrate the local motion ratio to obtain

$$\theta_s(x_s) = \int_0^{x_s} \frac{1}{r_h\tan\alpha_s(\xi)}\,d\xi. \tag{3.2.23}$$

Equation (3.2.23) defines the helix as a function $\theta_s(x_s)$: for each axial position, it gives the relative angle enforced by the track. The function need not come from a constant helix angle. Any supplied function may be used provided it is single-valued and strictly increasing over the admissible shift range.

Rotation of the movable sheave We can now use this function to determine the two remaining quantities in the rotational balance. First consider the absolute motion of the movable sheave. The fixed sheave rotates with the secondary shaft at ω_s, while the movable sheave also has the relative rotation θ_s:

$$\omega_{s,M} = \omega_s + \dot{\theta}_s = \omega_s + \frac{d\theta_s}{dx_s}\dot{x}_s. \tag{3.2.24}$$

Because θ_s was defined in the same positive rotational sense as ω_s, the relative speed enters with the plus sign shown here. Differentiating gives the angular acceleration required by the rotational balance:

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2. \tag{3.2.25}$$

The shaft acceleration $\dot{\omega}_s$ remains unknown here because ω_s is a state of the coupled model. It is solved together with the axial motion rather than by the helix relation alone. The term proportional to $\ddot{x}_s$ is the angular acceleration caused by axial acceleration. The term proportional to $\dot{x}_s^2$ appears only when the motion ratio varies with position.

Torsional Spring Loading The same helix function tells us how much the spring has unwound. Let $k_{s,\theta}$ be the spring's torsional stiffness and $\theta_{s,\text{pre}}$ its signed twist at $x_s = 0$. Increasing θ_s reduces this twist, leaving $\theta_{s,\text{pre}} - \theta_s(x_s)$. The resulting torque on the movable assembly is

$$\tau_{s,\text{spring}} = k_{s,\theta}\left[\theta_{s,\text{pre}} - \theta_s(x_s)\right]. \tag{3.2.26}$$

For the installed winding assumed here, this quantity is positive while the spring remains preloaded. The opposite winding direction can be represented by a signed $\theta_{s,\text{pre}}$.

We now have relations for all three terms on the right-hand side of Equation (3.2.20). Substituting them gives the pin–helix contact torque:

$$\begin{aligned}\tau_{s,\text{helix}} = \frac{\tau_s}{2} &+ k_{s,\theta}\left[\theta_{s,\text{pre}} - \theta_s(x_s)\right] \\ &- I_{s,M}\left[\dot{\omega}_s + \frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2\right].\end{aligned} \tag{3.2.27}$$

This completes the rotational part of the derivation. The next step is to use the pin–helix contact geometry to find the axial force associated with this torque.

Axial Component of the Helix Reaction

The torque above is the moment produced by the circumferential component of the contact normal. Let $F_{s,\text{helix}}$ denote the signed axial force produced by the helix reaction, positive in $+x_s$.

Under Assumption 3 (Idealized Actuator Mechanics), the pin–helix contact is treated as an ideal frictionless constraint. The same compatible-work argument used for the primary therefore converts the rotational contact load into its axial effect without contact dissipation. Consider the motion of the movable sheave relative to the helix carrier. The contact normal does no net work in this relative motion. Its axial component contributes $F_{s,\text{helix}}\,dx_s$, while its circumferential component contributes $-\tau_{s,\text{helix}}\,d\theta_s$. Their sum is zero, so

$$F_{s,\text{helix}}\,dx_s = \tau_{s,\text{helix}}\,d\theta_s. \tag{3.2.28}$$

Using the motion ratio gives

$$F_{s,\text{helix}} = \tau_{s,\text{helix}}\frac{d\theta_s}{dx_s}. \tag{3.2.29}$$

The motion ratio $d\theta_s/dx_s$ is therefore the mathematical origin of the axial effect: it converts the contact torque into axial force. In the straight-slot limit used in the opening example, this ratio is zero, so the slot can react torque while the axial conversion vanishes.

Substituting Equation (3.2.27) gives the helix force in terms of the model variables:

$$\boxed{\begin{aligned}F_{s,\text{helix}} = \frac{d\theta_s}{dx_s}\Bigg\{&\frac{\tau_s}{2} + k_{s,\theta}\left[\theta_{s,\text{pre}} - \theta_s(x_s)\right] \\ &- I_{s,M}\left[\dot{\omega}_s + \frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2\right]\Bigg\}.\end{aligned}} \tag{3.2.30}$$

This gives $F_{s,\text{helix}}$. The belt and wound spring provide the driving torques inside the braces; part of that torque may be required to accelerate the movable assembly, and only the remainder is converted into axial thrust.

Remark (Admissible helix maps)

The completed force uses the first two derivatives of $\theta_s(x_s)$. Over the admissible shift range, the supplied helix map must therefore be single-valued and twice continuously differentiable, with a finite, nonzero motion ratio:

$$\theta_s \in C^2, \qquad 0 < \frac{d\theta_s}{dx_s} < \infty.$$

A manufactured multi-angle helix may still be represented, but nominal segments must be joined smoothly rather than by a mathematical corner.

The difficult force in the preliminary axial balance is now known. The two remaining forces act directly on the translating assembly.

Direct Spring and Belt Forces

Direct axial spring force The spring force follows directly from its compression. Let $k_{s,x}$ be the axial stiffness and $x_{s,\text{pre}}$ the compression at $x_s = 0$. Closing the secondary by x_s lets the spring extend by the same amount, so its remaining compression is $x_{s,\text{pre}} - x_s$. Hence,

$$F_{s,\text{spring}} = k_{s,x}\left(x_{s,\text{pre}} - x_s\right). \tag{3.2.31}$$

This force and the torque in Equation (3.2.26) come from the same installed spring. Following the reduced decomposition of Kim et al. (2002), the model represents its axial compression and torsional twist with separate linear laws. This use of prescribed, uncoupled spring laws is part of Assumption 3 (Idealized Actuator Mechanics); compression–twist coupling, hysteresis, and end-contact friction are not retained.

Belt opening reaction The belt opening force follows the same projection already used on the primary. Let N_s be the integrated normal load over both secondary faces. Under Assumption 12 (Symmetric Sheave-Face Sharing), the movable face carries one half of this total load, giving

$$F_{s,\text{belt}} = \frac{N_s}{2}\cos\beta. \tag{3.2.32}$$

As on the primary, N_s is not determined by this local free body. It remains a coupled unknown supplied by the belt-contact and remaining system equations.

Completed Local Secondary Balance

We now have a relation for every term in the axial free body. Substitute them into Equation (3.2.18) and collect the terms multiplying $\ddot{x}_s$:

$$
\boxed{\begin{aligned}\left[m_s + I_{s,M}\left(\frac{d\theta_s}{dx_s}\right)^2\right]\ddot{x}_s = &\left[\frac{\tau_s}{2} + k_{s,\theta}\left(\theta_{s,\text{pre}} - \theta_s\right) - I_{s,M}\dot{\omega}_s\right]\frac{d\theta_s}{dx_s} \\ &- I_{s,M}\frac{d\theta_s}{dx_s}\frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2 \\ &+ k_{s,x}\left(x_{s,\text{pre}} - x_s\right) - \frac{N_s}{2}\cos\beta.\end{aligned}} \tag{3.2.33}
$$

The completed equation separates into four physical pieces:

- $m_s + I_{s,M}(d\theta_s/dx_s)^2$ is the local axial inertia. Just as the flyweight pivot inertia appeared in the primary balance, the helix makes the secondary's relative-rotational inertia part of the load that axial motion must accelerate;
- the first bracket contains the belt and torsional-spring torques, less the portion required to accelerate the movable assembly with the secondary shaft;
- the term proportional to $\dot{x}_s^2$ accounts for a helix motion ratio that varies with position; and
- the last two terms are the direct spring closing force and belt opening force.

This is the axial equation of motion we set out to obtain. If its right-hand side remains zero and the sheave is initially stationary, x_s and θ_s remain fixed while the two sheaves rotate together. A nonzero right-hand side accelerates the sheave axially, and $\theta_s(x_s)$ fixes the accompanying relative rotation.

Classical Limit and Relation to Existing Models

As on the primary, the full dynamic balance should recover the established actuator equilibrium when the shift mechanism is stationary. Consider a fixed secondary configuration at constant shaft speed, $\dot{x}_s = \ddot{x}_s = \dot{\omega}_s = 0$. Under the equal face-torque split adopted here, the helix force reduces to

$$
F_{s,\text{helix}} = \left[\frac{\tau_s}{2} + k_{s,\theta}\left(\theta_{s,\text{pre}} - \theta_s\right)\right]\frac{d\theta_s}{dx_s}. \tag{3.2.34}
$$

Adding the direct spring and belt loads gives the corresponding equilibrium condition for the complete movable assembly:

$$
\left[\frac{\tau_s}{2} + k_{s,\theta}\left(\theta_{s,\text{pre}} - \theta_s\right)\right]\frac{d\theta_s}{dx_s} + k_{s,x}\left(x_{s,\text{pre}} - x_s\right) = \frac{N_s}{2}\cos\beta. \tag{3.2.35}
$$

These equations expose the classical mechanism directly. The axial spring supplies one closing force, while the belt torque and torsional-spring torque are converted into a second closing force by the helix motion ratio. The factor $1/2$ is not a property of the helix itself; it follows from the equal face-torque closure used here. A face-resolved contact model would instead use the torque $\tau_{s,M}$ actually applied to the movable face.

The important comparison is how this equilibrium relation is used. In the present formulation, Equations (3.2.34) and (3.2.35) are stationary limits of a dynamic balance. In much of the mechanically actuated CVT literature, an equivalent algebraic relation remains the actuator closure while the transmission is shifting.

The formulation of Kim et al. (2002) is a direct example. Their driven thrust is the sum of a spring force and a torque-cam force obtained from instantaneous equilibrium. The cam term retains the applied torque, torsional-spring load, cam geometry, and cam friction. Suppressing that friction and adopting the equal face-torque split used here gives the same spring-plus-torque-cam structure as Equation (3.2.35). Their shift transient is not obtained from the movable secondary's equation of motion, but from a separately identified first-order ratio law. Kim et al. therefore retain contact friction that is absent here, but do not use the secondary assembly's inertia to determine its shift acceleration.

The design-oriented sources examined here apply the same configuration-by-configuration logic at different levels of detail. Messick (2018) calculates the secondary thrust required by a steady-state belt solution and uses it as a target for helix design, explicitly excluding the shift response time. Skinner (2020) calculates the compression-spring, torsional-spring, and torque-feedback contributions to secondary clamping algebraically. Arango and Muñoz Alzate (2020) construct the torque-sensing cam profile from the transmitted torque, spring force, required axial force, and axial displacement over the operating range. These treatments are useful for sizing and tuning the mechanism, but they do not determine its motion from a movable-sheave equation of motion.

Among the reviewed sources, Sorge and Cammalleri (2011) provide the most detailed treatment of the belt–face–helix interaction during shifting. Their distributed contact model allows the fixed and movable faces to rotate at different speeds and determines the torque $M_{N,s}$ carried by the movable face. The resulting driven-side actuator relation combines the spring load with a helix contribution having the same torque-times-motion-ratio structure as Equation (3.2.34). It therefore replaces the assumed $\tau_s/2$ with a face-resolved torque and can capture substantial departures from an equal split.

That contact calculation is shift-aware, but the secondary actuator itself is still closed quasi-statically. Sorge and Cammalleri prescribe the instantaneous radius and the dimensionless shift speed $\rho = \dot{r}/(\omega_f r)$, then solve for the corresponding belt-contact state. A nonzero ρ changes the relative face speeds, sliding directions, contact tractions, and face-torque split, but it is not produced by the forces on the movable secondary. Because no axial or relative-rotational equation of motion is written for the movable half-pulley, its translational inertia and helix-reflected rotational inertia are absent by construction.

The models therefore retain complementary parts of the physics. Kim et al. retain cam friction, while Sorge and Cammalleri resolve the two belt–face contacts and their unequal torques. The present reduced model retains neither capability. It instead carries the movable assembly's rotational balance through the general constraint $\theta_s(x_s)$ and solves its free axial motion. This introduces the shaft-acceleration coupling, the reflected inertia $I_{s,M}(d\theta_s/dx_s)^2$, and the velocity-squared term generated when the helix motion ratio varies with position. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent explicit projection of the movable side's rotational inertia through a mechanical secondary helix was identified.

This distinction is deliberately narrow. The rigid-hardware, ideal-actuator, and symmetric-face idealizations are stated explicitly in Assumption 2 (Rigid Pulley and Actuation Hardware), Assumption 3 (Idealized Actuator Mechanics), and Assumption 12 (Symmetric Sheave-Face Sharing). They leave the compliant and frictional actuator details, and the face-resolved load redistribution of the models above, outside the present balance. The comparison is therefore not a claim of greater local contact fidelity; it isolates the added treatment of the movable assembly's inertia and helix-constrained dynamics.

3.2.3 Coupling Through the Shared Shift Coordinate

The primary and secondary balances have been derived in their natural local coordinates, x_p and x_s. The fixed-length belt geometry makes those motions compatible. Choosing the primary displacement as the shared shift coordinate leaves the primary balance unchanged and supplies the secondary motion through the map $x_s = x_s(s)$.

Local-to-Shared Kinematics

Because the shared coordinate is the primary displacement,

$$x_p = s, \qquad \dot{x}_p = \dot{s}, \qquad \ddot{x}_p = \ddot{s}. \tag{3.2.36}$$

The secondary displacement follows from the fixed-length belt map. Its first two time derivatives are

$$x_s = x_s(s), \qquad \dot{x}_s = \frac{dx_s}{ds}\dot{s}, \qquad \ddot{x}_s = \frac{dx_s}{ds}\ddot{s} + \frac{d^2x_s}{ds^2}\dot{s}^2. \tag{3.2.37}$$

Positive x_p and x_s both denote sheave closing. Increasing s closes the primary but opens the secondary, so $dx_s/ds < 0$. If the belt geometry is nonlinear, this motion ratio changes through the shift and $d^2x_s/ds^2 \neq 0$. Both derivatives are determined by the belt-geometry relation established earlier.

Two Conditions on One Shift Motion

Substituting Equation (3.2.36) into Equation (3.2.14) gives the primary condition directly:

Primary Condition on the Shared Shift Motion **(3.2.38)**

$$\begin{aligned}\left[m_p + I_f\left(\frac{dq_f}{ds}\right)^2\right]\ddot{s} = {} & \frac{1}{2}\omega_p^2\frac{dJ_{f,p}}{ds} \\ & - I_f\frac{dq_f}{ds}\frac{d^2q_f}{ds^2}\dot{s}^2 \\ & - k_p\left(x_{p,\mathrm{pre}} + s\right) - \frac{N_p}{2}\cos\beta.\end{aligned}$$

Because $x_p = s$, this is simply the local primary balance written using the shared coordinate.

The secondary condition follows just as directly. Substituting Equation (3.2.37) into Equation (3.2.33) and collecting terms gives

Secondary Condition on the Shared Shift Motion **(3.2.39)**

$$\left[m_s + I_{s,M}\left(\frac{d\theta_s}{dx_s}\right)^2\right]\frac{dx_s}{ds}\,\ddot{s}$$
$$= \frac{d\theta_s}{dx_s}\left[\frac{\tau_s}{2} + k_{s,\theta}\left(\theta_{s,\text{pre}} - \theta_s\right) - I_{s,M}\dot{\omega}_s\right]$$
$$+ k_{s,x}\left(x_{s,\text{pre}} - x_s\right) - \frac{N_s}{2}\cos\beta$$
$$- \Bigg\{\left[m_s + I_{s,M}\left(\frac{d\theta_s}{dx_s}\right)^2\right]\frac{d^2x_s}{ds^2}$$
$$+ I_{s,M}\frac{d\theta_s}{dx_s}\frac{d^2\theta_s}{dx_s^2}\left(\frac{dx_s}{ds}\right)^2\Bigg\}\dot{s}^2.$$

Here x_s, θ_s, and their local derivatives are evaluated at $x_s = x_s(s)$. Since $dx_s/ds < 0$, a force that closes the secondary drives the shift toward decreasing s, thereby opening the primary. Conversely, the secondary belt force acts in the opening direction and drives the shift toward increasing s. The kinematic map therefore carries the inverse coupling between the two pulleys directly into the acceleration balance.

The two terms proportional to $\dot{s}^2$ have different geometric origins. The term containing d^2x_s/ds^2 appears because the belt geometry can require a changing amount of secondary travel per unit primary travel. The term containing $d^2\theta_s/dx_s^2$ is inherited from the local secondary balance and appears when the helix motion ratio varies with position; substituting $\dot{x}_s = (dx_s/ds)\dot{s}$ supplies its squared local-to-shared motion ratio.

The two balances are not added into one axial free body. They remain separate conditions on the same shift acceleration: the same $\ddot{s}$ must satisfy both movable assemblies. The belt reactions N_p and N_s, together with the coupled shaft and contact quantities, allow both conditions to hold. Those unknowns are supplied by the rotational, belt, and contact relations developed later.

Relation to Existing Dynamic Mappings

Transient belt and pulley models provide broad precedent for dynamic ratio prediction. Julió and Plante (2011) discretize the rubber belt and integrate its motion, while Ballew (2015) allow both pulley speeds to evolve from the applied torques and inertias. In that model family, however, pulley axial force is prescribed and the radius follows from belt compression; a movable-sheave equation $m_j\ddot{x}_j = \sum F_{j,x}$ is not integrated.

Movable-sheave dynamics and shared-coordinate reductions also predate the present model. Duan et al. (2016) retain movable-pulley mass in a chain CVT and couple the pulley motions through chain deformation and tension. Most directly, da Silva et al. (2023) write axial equations for both movable sheaves of a mechanically actuated rubber-belt CVT, impose a linear relation between their displacements, and reduce the two translations to one shift equation. Thus transient shifting, axial sheave inertia, and a shared shift coordinate are not individually new.

The distinction here lies in how these ingredients are combined. The fixed-length belt geometry supplies the nonlinear map $x_s(s)$, so both dx_s/ds and d^2x_s/ds^2 remain in the secondary acceleration.

At the same time, the flyweight pivot inertia and the movable secondary's relative-rotational inertia remain tied to their own mechanism maps, $q_f(x_p)$ and $\theta_s(x_s)$. The result is a pair of local free-body balances expressed as simultaneous conditions on one shift acceleration.

This reduction also defines the tradeoff. It does not retain the spatial belt deformation fields of Julió and Plante (2011) and Ballew (2015), or the prescribed-shift, face-resolved torque distribution of Sorge and Cammalleri (2011). It instead keeps the actuator and movable-sheave inertias explicit while closing the nominal belt geometry without spatial discretization. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent combination of these features was identified.

3.2.4 What the Axial Balances Establish

The primary and secondary now have complete axial equations of motion in their natural coordinates. On the primary, shaft speed produces a closing action because the flyweight centres of mass move outward as the movable sheave closes. Part of that action accelerates the flyweights about their pivots before the remainder is transmitted to the movable assembly. On the secondary, the belt and torsional-spring torques act through the helix. Part of their action changes the rotation of the movable sheave relative to the shaft-side assembly before the remainder appears as axial force.

Although the two actuators are physically different, their constrained motions produce the same underlying structure. The map $q_f(x_p)$ reflects the flyweights' pivot inertia into the primary axial coordinate, while the helix map $\theta_s(x_s)$ reflects the movable-sheave rotational inertia into the secondary coordinate. In each case, the local motion ratio contributes an effective axial inertia, and variation of that ratio with position produces a velocity-squared term. The primary closing action is additionally set by dr_f/dx_p, while the secondary balance couples to the shaft acceleration because its movable sheave rotates relative to an assembly that is itself rotating.

The shared belt geometry adds a second layer of constrained motion. The primary map $x_p(s) = s$ carries its local balance directly into the shift coordinate, whereas the secondary map $x_s(s)$ reflects its local motion into the same $\ddot{s}$ and contributes its own velocity-squared curvature term. Both free-body relations must therefore be satisfied by one shift motion.

They do not, however, determine that motion in isolation. At this stage, the two axial equations contain the five unknown quantities

$$\ddot{s}, \qquad N_p, \qquad N_s, \qquad \tau_s, \qquad \dot{\omega}_s.$$

The belt normal loads, transmitted secondary torque, and secondary shaft acceleration follow only when the rotational and belt-contact mechanics are added. The two balances derived here therefore form the axial part of one simultaneous dynamic solution for the engaged CVT.

This completes the axial force description. Attention now turns from translation along the pulley shafts to rotation about them.

3.3 Rotational Dynamics of the Pulley Assemblies

The preceding section established how forces along the pulley shafts move the sheaves, but its equations still contain quantities belonging to rotation about those shafts. The primary flyweight force depends on the primary speed ω_p, while the secondary balance contains the transmitted torque τ_s and the secondary acceleration $\dot{\omega}_s$. The axial equations can use these quantities, but cannot supply them.

The primary pulley is carried around the primary shaft at ω_p, while the secondary rotates about its own shaft at ω_s. These are separate rotations, so their speeds need not change together. The mechanisms inside each pulley also move relative to these underlying shaft rotations. The primary flyweights pivot as the primary rotates, while the secondary movable sheave translates and rotates relative to its shaft through the helix. These internal motions are already tied to the shift coordinate, but they also change how angular momentum is distributed within each pulley as the CVT shifts.

The remaining question is therefore what governs the evolution of ω_p and ω_s, and consequently how shifting affects that evolution.

3.3.1 Following the Torque

To determine how ω_p and ω_s evolve, each pulley side requires a rotational equation of motion. That equation must account for the angular momentum carried about the corresponding shaft and the torques capable of changing it. The first step is therefore to identify the rotating systems represented by the two shaft speeds.

The primary-side system contains the primary shaft and the complete primary pulley assembly. The movable assembly and flyweights remain part of this system even while they move relative to the fixed sheave, since they continue to carry angular momentum about the primary axis. The secondary-side system is defined in the same way: it contains the secondary shaft and the complete secondary pulley assembly, including the helix-guided motion of the movable sheave. The belt passes between these two systems and is not included in either one.

The resulting division is shown in Figure 3.10.

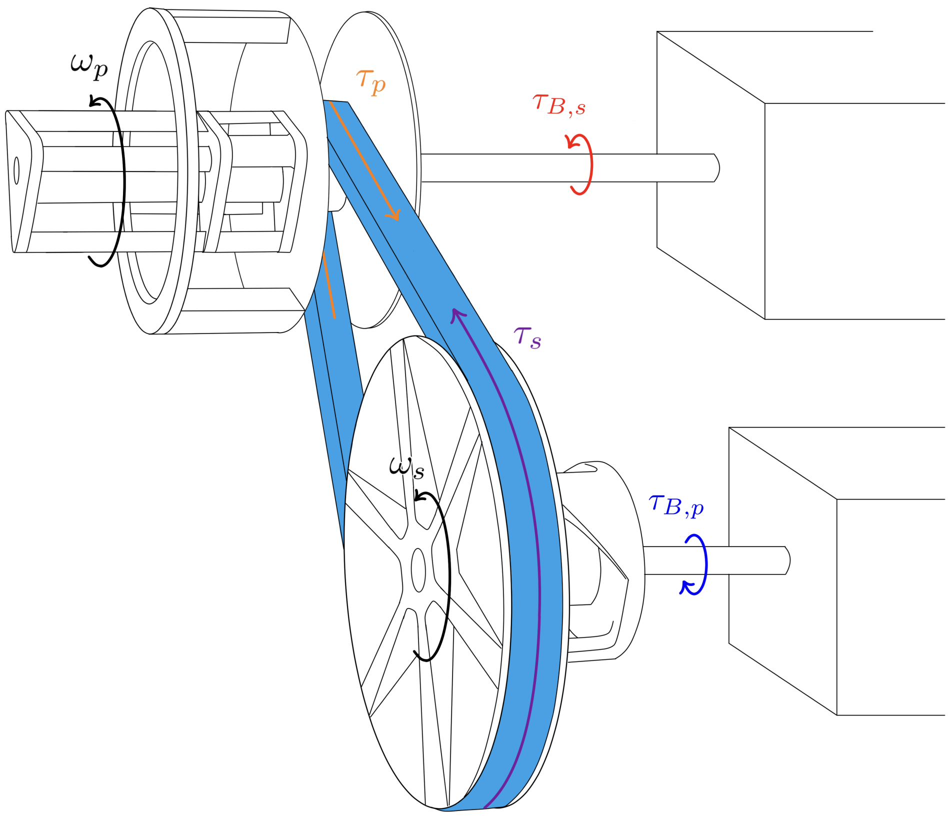

Figure 3.10: Rotating systems and torque path of the engaged CVT. The shaft-side boundaries represent the torque and equivalent inertia presented by the attached systems. The belt remains outside the two pulley-side systems and interacts with them through the contact torques τ_p and τ_s. The arrows show the signed directions associated with forward power transfer.

Begin with the primary. When its speed changes, the primary shaft must accelerate not only the pulley itself, but also the rotating hardware attached beyond the CVT boundary. As established in Section 2.1, that external hardware is represented at the shaft by the equivalent inertia $I_{\mathrm{B},p}$, while $\tau_{\mathrm{B},p}$ is the torque it applies to the primary side. The remaining external interaction is the belt, which exerts the torque τ_p on the primary pulley.

The secondary follows the same construction. Its attached hardware contributes the equivalent inertia $I_{\mathrm{B},s}$ and applies the boundary torque $\tau_{\mathrm{B},s}$, while the belt exerts τ_s on the secondary pulley.

The signs follow directly from the convention established in Section 2.3.1. During forward power transfer, the primary boundary drives the primary pulley and the belt resists it. The belt then drives the secondary pulley against the resisting secondary boundary. Thus,

$$\tau_{\mathrm{B},p} > 0, \qquad \tau_p < 0, \qquad \tau_s > 0, \qquad \tau_{\mathrm{B},s} < 0.$$

Although the primary and secondary contain different mechanisms, their rotational balances now have the same outer structure. Let $j \in \{p, s\}$ denote the pulley side under consideration.

The angular momentum on side j has two contributions. The fixed equivalent boundary inertia rotates at ω_j, so it carries angular momentum $I_{B,j}\omega_j$. Let $H_{CVT,j}$ denote the angular momentum carried by the corresponding shaft and pulley assembly, including the internal motions of its components. The total is therefore

$$H_j = I_{B,j}\omega_j + H_{CVT,j}. \tag{3.3.1}$$

Bearing, seal, windage, and other unmodeled resisting torques are omitted under Assumption 15 (Negligible Unmodeled Parasitic Losses). Within the resulting pulley-side free body, the applied boundary torque and belt torque are therefore the only external torques retained. Newton's second law for rotation gives

$$\tau_{B,j} + \tau_j = \frac{dH_j}{dt}.$$

Substituting Equation (3.3.1) writes this as

$$\tau_{B,j} + \tau_j = \frac{d}{dt}\left(I_{B,j}\omega_j + H_{CVT,j}\right).$$

The equivalent boundary inertia is fixed, so differentiating yields the common pulley-side equation of motion:

Common Pulley-Side Rotational Balance **(3.3.2)**

$$\tau_{B,j} + \tau_j = I_{B,j}\dot{\omega}_j + \frac{dH_{CVT,j}}{dt}, \qquad j \in \{p, s\}.$$

The way shifting changes the angular momentum carried within a pulley is contained in $H_{CVT,j}$. On the primary, the flyweights redistribute mass relative to the shaft as they pivot. On the secondary, the movable sheave rotates relative to the shaft-side assembly as it follows the helix. The two pulley sides therefore share the balance in Equation (3.3.2), but require separate expressions for their internal angular momentum. Beginning with the primary, each assembly can now be examined in turn.

3.3.2 Primary Pulley Assembly

The primary is the simpler of the two pulley assemblies because nearly all of its hardware shares the shaft speed ω_p. The shaft, sheaves, ramp carrier, and other coaxial components may translate during a shift, but that translation is along the shaft and therefore does not change their distance from its axis. Their combined shaft-axis inertia can consequently be collected into one constant quantity, $I_{p,\text{const}}$.

The flyweights are different. They are carried around the primary shaft at ω_p, but their mass moves radially as they pivot. The same motion was already used in Section 3.2.1 to determine the centrifugal action that closes the primary. There, the flyweight geometry was summarized by the shaft-axis inertia map $J_{f,p}(x_p)$. Since $x_p = s$, that map is now $J_{f,p}(s)$.

It is useful to recall what this inertia represents. The quantity $J_{f,p}$ measures the flyweight mass distribution about the *primary shaft*. It is therefore distinct from the constant pivot-axis inertia I_f that appeared in the flyweight moment balance of Equation (3.2.3). The two quantities describe the same flyweights about different axes and enter different equations of motion.

At a given shift position, the angular momentum carried inside the primary CVT boundary is therefore

$$H_{\mathrm{CVT},p} = [I_{p,\mathrm{const}} + J_{f,p}(s)]\,\omega_p. \tag{3.3.3}$$

If the shift position were fixed, differentiating this expression would simply multiply the current inertia by $\dot{\omega}_p$. During a shift, however, $J_{f,p}$ changes as well. Because its only configuration variable is s, differentiating Equation (3.3.3) gives

$$\frac{dH_{\mathrm{CVT},p}}{dt} = [I_{p,\mathrm{const}} + J_{f,p}(s)]\,\dot{\omega}_p + \frac{dJ_{f,p}}{ds}\dot{s}\,\omega_p. \tag{3.3.4}$$

The two terms describe different ways of changing the same angular momentum. The first accelerates the primary with the flyweights held at their current configuration. The second changes the angular momentum because the flyweight mass itself moves toward or away from the shaft while already rotating.

Substituting this result into the common pulley-side balance from Equation (3.3.2) gives the primary equation of motion:

Primary Pulley Rotational Balance **(3.3.5)**

$$\begin{aligned}\tau_{\mathrm{B},p} + \tau_p = {} & [I_{\mathrm{B},p} + I_{p,\mathrm{const}} + J_{f,p}(s)]\,\dot{\omega}_p \\ & + \frac{dJ_{f,p}}{ds}\dot{s}\,\omega_p.\end{aligned}$$

The connection to the axial flyweight balance is now explicit. In Equation (3.2.38), the derivative $dJ_{f,p}/ds$ determined how shaft rotation produces a centrifugal closing action. Here, the same derivative determines how that closing motion changes the shaft-axis angular momentum. The two appearances are not separate flyweight corrections: they are the axial and rotational consequences of the same radial redistribution of flyweight mass.

For the conventional primary mechanism, outward flyweight motion increases $J_{f,p}$. During an upshift, $\dot{s} > 0$, so the rotating primary must supply the angular momentum required by that outward motion in addition to accelerating the shaft. During a backshift the redistribution reverses, and the flyweights return angular momentum to the shaft-side motion.

If the shift is held fixed, $\dot{s} = 0$, and the flyweight mass distribution no longer changes. The additional term vanishes and the familiar fixed-inertia form is recovered:

$$\tau_{\mathrm{B},p} + \tau_p = [I_{\mathrm{B},p} + I_{p,\mathrm{const}} + J_{f,p}(s)]\,\dot{\omega}_p, \qquad \dot{s} = 0. \tag{3.3.6}$$

3.3.3 Secondary Pulley Assembly

The secondary differs from the primary in one important way. During a shift, the complete secondary remains one mechanical assembly, but not every part of that assembly has the same angular speed.

The secondary shaft, fixed sheave, and hardware locked to them rotate at ω_s. Let their combined shaft-axis inertia inside the CVT boundary be $I_{s,F}$. The movable sheave and the components rigidly attached to it have inertia $I_{s,M}$. As established while deriving the secondary axial balance, the helix makes this movable assembly rotate relative to the shaft-side assembly as it translates. Its absolute angular speed is therefore

$$\omega_{s,M} = \omega_s + \frac{d\theta_s}{dx_s}\dot{x}_s, \tag{3.3.7}$$

which is the relation already obtained in Equation (3.2.24). The angular momentum carried by the secondary must therefore retain the two angular speeds separately:

$$H_{\text{CVT},s} = I_{s,F}\omega_s + I_{s,M}\omega_{s,M}. \tag{3.3.8}$$

The free body here is larger than the movable-sheave free body used in Section 3.2.2. There, only the movable assembly was isolated, so the spring and helix torques crossed the boundary and had to appear explicitly. Here, the shaft-side and movable-side hardware are enclosed together. The torque exerted by the helix on the movable side is accompanied by an equal-and-opposite torque on the shaft-side assembly, and both actions lie inside the same free body. The torsional spring produces the same kind of internal pair. These torques therefore cancel from the moment balance of the complete secondary. Their mechanical effect has not disappeared: the helix still allows the movable side to rotate at $\omega_{s,M}$ rather than ω_s, and that difference remains in Equation (3.3.8).

The common pulley-side balance in Equation (3.3.2) requires the time rate of change of the angular momentum inside this boundary. Because Equation (3.3.8) contains two angular velocities, differentiating it requires the corresponding two angular accelerations:

$$\frac{dH_{\text{CVT},s}}{dt} = I_{s,F}\dot{\omega}_s + I_{s,M}\dot{\omega}_{s,M}. \tag{3.3.9}$$

The first acceleration, $\dot{\omega}_s$, is the shaft acceleration governed by the present rotational balance. The second is not another independent degree of freedom. The helix already tied the movable-side rotation to the axial sheave motion in Section 3.2.2. In particular, differentiating the helix kinematics there gave Equation (3.2.25):

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2. \tag{3.3.10}$$

The rotational balance has therefore led back to a kinematic relation already established by the axial mechanism. Substituting Equation (3.3.10) into Equation (3.3.9) gives

$$
\begin{aligned}
\frac{dH_{\mathrm{CVT},s}}{dt} = {} & (I_{s,F} + I_{s,M})\,\dot{\omega}_s \\
& + I_{s,M}\left[\frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2\right].
\end{aligned}
\tag{3.3.11}
$$

Substituting this rate into the common pulley-side rotational balance in Equation (3.3.2) produces the local secondary equation of motion:

Secondary Pulley Rotational Balance **(3.3.12)**

$$
\begin{aligned}
\tau_{\mathrm{B},s} + \tau_s = {} & (I_{\mathrm{B},s} + I_{s,F} + I_{s,M})\,\dot{\omega}_s \\
& + I_{s,M}\left[\frac{d\theta_s}{dx_s}\ddot{x}_s + \frac{d^2\theta_s}{dx_s^2}\dot{x}_s^2\right].
\end{aligned}
$$

The first term accelerates the rotation shared by the complete secondary. The bracketed terms change only the movable side's rotation relative to the shaft. They are the rotational counterpart of the helix-reflected inertia already encountered in the secondary axial balance.

The complete CVT uses the shared shift coordinate rather than x_s as an independent coordinate. The required mapping was already established in Equation (3.2.37). Substituting

$$
\dot{x}_s = \frac{dx_s}{ds}\dot{s}, \qquad \ddot{x}_s = \frac{dx_s}{ds}\ddot{s} + \frac{d^2x_s}{ds^2}\dot{s}^2
$$

into Equation (3.3.12) gives the same rotational balance in the coordinate used by the complete model:

$$
\begin{aligned}
\tau_{\mathrm{B},s} + \tau_s = {} & (I_{\mathrm{B},s} + I_{s,F} + I_{s,M})\,\dot{\omega}_s \\
& + I_{s,M}\frac{d\theta_s}{dx_s}\frac{dx_s}{ds}\ddot{s} \\
& + I_{s,M}\left[\frac{d\theta_s}{dx_s}\frac{d^2x_s}{ds^2} + \frac{d^2\theta_s}{dx_s^2}\left(\frac{dx_s}{ds}\right)^2\right]\dot{s}^2.
\end{aligned}
\tag{3.3.13}
$$

All local helix quantities in this expression are evaluated at $x_s = x_s(s)$. Its signs follow directly from the kinematics already chosen. For the conventional helix, $d\theta_s/dx_s > 0$, while the belt geometry gives $dx_s/ds < 0$ over the engaged shift range. Hence

$$
\frac{d\theta_s}{ds} = \frac{d\theta_s}{dx_s}\frac{dx_s}{ds} < 0.
$$

During an upshift, $\dot{s} > 0$, so $\dot{\theta}_s < 0$: the secondary opens and the movable side rotates more slowly than the shaft-side assembly. The sign of the corresponding angular-momentum term is therefore carried by the mechanism maps rather than assigned separately in the rotational balance.

If the shift is held fixed, $\dot{s} = \ddot{s} = 0$. The helix then produces no relative motion, and the fixed and movable sides rotate together. The balance reduces to

$$\tau_{\mathrm{B},s} + \tau_s = (I_{\mathrm{B},s} + I_{s,F} + I_{s,M})\,\dot{\omega}_s. \tag{3.3.14}$$

3.3.4 What the Rotational Balances Establish

The primary and secondary now have complete shaft-axis equations of motion. Both remain direct statements of $\sum \tau = dH/dt$. The additional terms appear because the hardware that actuates the pulleys continues to move while the pulley assemblies rotate, so shifting changes the angular momentum stored inside the assemblies themselves.

On the primary, this connection is carried by the same flyweight inertia map $J_{f,p}(s)$ that appeared in the axial derivation. There, $dJ_{f,p}/ds$ contributes to the centrifugal closing action because the flyweights move outward as the primary closes. Here, $J_{f,p}(s)$ and its derivative determine how much shaft-axis angular momentum those same flyweights carry as their radial mass distribution changes. The axial and rotational balances are therefore two views of the same flyweight motion: one asks how rotation drives the movable sheave, while the other asks how that moving flyweight mass changes the rotational response of the pulley.

The secondary shows the same reciprocity through $I_{s,M}$ and the helix map $\theta_s(x_s)$. In the axial balance, the helix reflects the movable assembly's rotational inertia into the sheave motion, so part of the available torque is required to change the movable side's rotation before the remainder appears as axial force. In the rotational balance, that same relative motion means the movable side carries angular momentum at a speed different from the shaft-side assembly, so the axial shift acceleration enters the shaft-axis balance. Again, the two equations are describing one constrained piece of hardware from different free bodies.

Mechanical actuation therefore creates a direct coupling between the axial and rotational dynamics inside each pulley assembly. This is distinct from the coupling carried by the belt reaction. Belt contact can connect the two kinds of balance because its normal action loads the sheaves axially while its tangential action transmits torque about the shafts. The present terms add a second path: the flyweight and helix mechanisms that generate the clamping response also carry inertia, so their own motion appears directly in the rotational equations. The actuator is therefore part of both balances, rather than acting only as a source of clamping force whose influence reaches shaft rotation through the belt.

This distinction gives a precise comparison with earlier transient models. In the rubber-belt formulation of Ballew (2015), belt-contact torque changes pulley speed through a fixed lumped pulley inertia, while pulley axial force is supplied to the belt model. Axial and rotational behaviour can still interact through the belt solution, but motion of the clamping hardware does not alter the pulley angular momentum itself. Duan et al. (2016) provide a stronger adjacent precedent in a chain CVT: rotational dynamics, movable-pulley axial dynamics, chain motion, and contact forces are solved as a coupled system. Their shaft balance nevertheless retains a fixed pulley inertia, while the movable-pulley mass appears in a separate axial equation. The coupling is therefore carried through the chain, contact forces, and changing radius rather than through actuator motion entering both balances.

A mechanically actuated rubber-belt comparison is provided by da Silva et al. (2023), who retain axial equations for both movable sheaves and reduce their translations to a shared shift coordinate. Their rotational treatment uses a fixed primary-pulley inertia. This establishes direct precedent for mechanical clamping, movable-sheave inertia, and coupled shift motion, but not for the additional shaft-axis angular-momentum terms produced by the moving actuator hardware.

The helical model of Sorge and Cammalleri (2011) provides a complementary secondary comparison. Their formulation resolves the relative motion of the driven sheaves and the generally unequal belt torques acting on the two faces in substantially greater local detail. That work therefore captures secondary helix and face-contact mechanics that are reduced here. The distinction in the present balance is different: the helix-constrained movable assembly is retained as part of the complete secondary angular momentum, so the same relative motion that appears in its axial dynamics also appears in the shaft-axis equation.

Taken together, these comparisons show that axial–rotational coupling itself is not the new ingredient. Existing formulations can produce that coupling through belt or chain reactions, changing radii, and shared shift motion. Among the reviewed reduced-order formulations for mechanically actuated rubber-belt CVTs, however, no equivalent pair of pulley balances was identified in which the moving clamping hardware itself provides this additional inertial coupling on both sides: the primary flyweights change their shaft-axis mass distribution as they generate closing action, while the secondary movable assembly changes its shaft-axis angular momentum through the same helix motion that produces its axial response.

The result is a single mechanical picture of each pulley rather than separate axial and rotational descriptions joined only through transmitted load. The same mechanism maps that determine how the pulley clamps also determine how its internal motion changes the shaft dynamics. When those actuator motions cease, the additional terms vanish and the familiar fixed-configuration inertia balances are recovered.

3.4 Belt Transport and Pulley Contact Mechanics

The preceding sections established the axial and rotational equations of motion of the two pulley assemblies, but in both cases the belt still appears through actions that those equations cannot determine. The axial balances contain the belt normal loads N_p and N_s, while the rotational balances contain the belt torques τ_p and τ_s. The belt also carries the transport state v_b, whose acceleration has not yet been established.

These quantities cannot be assigned independently. The contact that loads the sheave faces is also the contact through which torque is exchanged with the belt, and the same forces must account for the changing momentum of the belt itself. The remaining question is therefore how the motion and tension of one closed belt determine its own transport dynamics while producing the normal reactions and torques acting on both pulley assemblies.

3.4.1 Following Tension Around the Closed Belt

The belt transfers power between the pulleys through forces exchanged at the two belt–sheave interfaces. For power to cross either interface, the pulley and belt must exert tangential forces on one another. Those forces transmit torque, and the tension entering a pulley wrap will therefore generally differ from the tension leaving it.

The pulley contacts are not the only places where tension can change. The straight spans also contain belt mass. If the belt transport speed changes, a net tangential force is required to accelerate the belt contained within each span, so its two endpoint tensions can differ as well. The closed belt thus contains two repeated types of mechanical region: two pulley wraps, where load is exchanged with the sheaves through contact, and two free spans, where belt mass is carried between those contacts. The tangent points between the regions provide convenient boundaries at which to record the tension, as shown in Figure 3.11.

Draft figure placeholder: complete CVT belt loop showing the primary and secondary wraps, upper and lower straight spans, positive belt-travel direction, wrap angles, and the four boundary tensions $T_{p,u}$, $T_{s,u}$, $T_{s,l}$, and $T_{p,l}$.

Figure 3.11: Complete belt loop and the four tension values at the boundaries between the pulley wraps and straight spans. The arrows indicate the positive belt-transport direction.

The four tensions in Figure 3.11 are not additional dynamic states. They are the values taken by one continuous belt-tension field at the four boundaries between the spans and wraps. With the positive belt path introduced in Section 2.3.1, the upper span is traversed from the secondary to the primary, from $T_{s,u}$ to $T_{p,u}$, while the lower span is traversed from the primary back to the secondary, from $T_{p,l}$ to $T_{s,l}$.

Begin with a free span, where the belt is no longer in contact with either pulley and is acted on tangentially only through the tensions at its ends. Consider either span $i \in \{u, l\}$, with instantaneous length l_i, and orient its local tangential direction with positive belt travel. Under Assumption 7 (Single Belt Transport Coordinate), the retained motion along the span is the common belt transport speed v_b. Under Assumption 8 (Prescribed Planar Belt Path), the span remains the instantaneous straight tangent joining the two pulley wraps. With the belt cross-sectional area A_b defined in Equation (3.1.12), its mass per unit length is $\rho_b A_b$.

Isolate the belt occupying this span. Its tangential equation of motion follows from Newton's second law in momentum form,

$$\sum F_{i,t} = \frac{dP_{i,t}^{\text{sys}}}{dt}. \tag{3.4.1}$$

The force side is immediate from the free body: the two endpoint tensions pull in opposite directions, so $\sum F_{i,t} = T_{i,\text{out}} - T_{i,\text{in}}$. The momentum side requires more care. The tangent points that bound the span move as the pulley radii change, so the spatial region between them does not contain a fixed set of belt material. Its length changes while belt enters and leaves through the moving endpoint stations.

Let $v_{\text{st},i}^{\text{in}}$ and $v_{\text{st},i}^{\text{out}}$ denote the velocities of those stations projected along the instantaneous span direction. Evaluating the system momentum rate in Equation (3.4.1) with that moving control volume gives

$$T_{i,\text{out}} - T_{i,\text{in}} = \frac{d}{dt}\left(\rho_b A_b l_i v_b\right) + \rho_b A_b v_b \left[\left(v_b - v_{\text{st},i}^{\text{out}}\right) - \left(v_b - v_{\text{st},i}^{\text{in}}\right)\right]. \tag{3.4.2}$$

The first term is the change of tangential momentum stored between the two stations; the second is the net tangential momentum carried through those moving boundaries. The span-length rate is set by the same endpoint motion,

$$\dot{l}_i = v_{\text{st},i}^{\text{out}} - v_{\text{st},i}^{\text{in}}. \tag{3.4.3}$$

Substituting Equation (3.4.3) into Equation (3.4.2) cancels the term produced by the changing amount of belt stored in the span with the corresponding momentum flux through its moving endpoints. What remains is

$$\boxed{T_{i,\text{out}} - T_{i,\text{in}} = \rho_b A_b l_i \dot{v}_b.} \tag{3.4.4}$$

The compact result therefore does not require fixed tangent points. Their motion was retained in Equation (3.4.2); it simply makes no additional contribution to the final tangential balance once the associated momentum flux is included. The balance is only along the instantaneous span. As the pulley radii change, the span direction also changes in the pulley plane; the transverse dynamics associated with that prescribed reorientation are not retained under Assumption 8 (Prescribed Planar Belt Path).

Applying Equation (3.4.4) to the two spans in Figure 3.11 gives

$$\boxed{\begin{aligned} T_{p,u} - T_{s,u} &= \rho_b A_b l_u \dot{v}_b, \\ T_{s,l} - T_{p,l} &= \rho_b A_b l_l \dot{v}_b. \end{aligned}} \tag{3.4.5}$$

The straight spans can therefore be related from their endpoint tensions and the common belt transport acceleration. A pulley wrap is different. Between its two tangent points the belt is continuously redirected while the sheave faces apply distributed contact forces, so the entry and exit tensions cannot be related from the endpoints alone. Determining how the tension changes through the wrap requires a free body taken from within the contact itself.

3.4.2 Local Equations of Motion on a Pulley Wrap

The straight-span balances in Equation (3.4.5) describe the parts of the belt whose direction is constant between their two tangent points. Inside a pulley wrap, the belt direction changes continuously and the sheave faces act on it throughout the contact arc. Consider therefore the infinitesimal belt element shown in Figure 3.12 on a generic pulley $j \in \{p, s\}$. Let θ increase through the wrap in the direction of belt travel, and let the element subtend the angle $d\theta$.

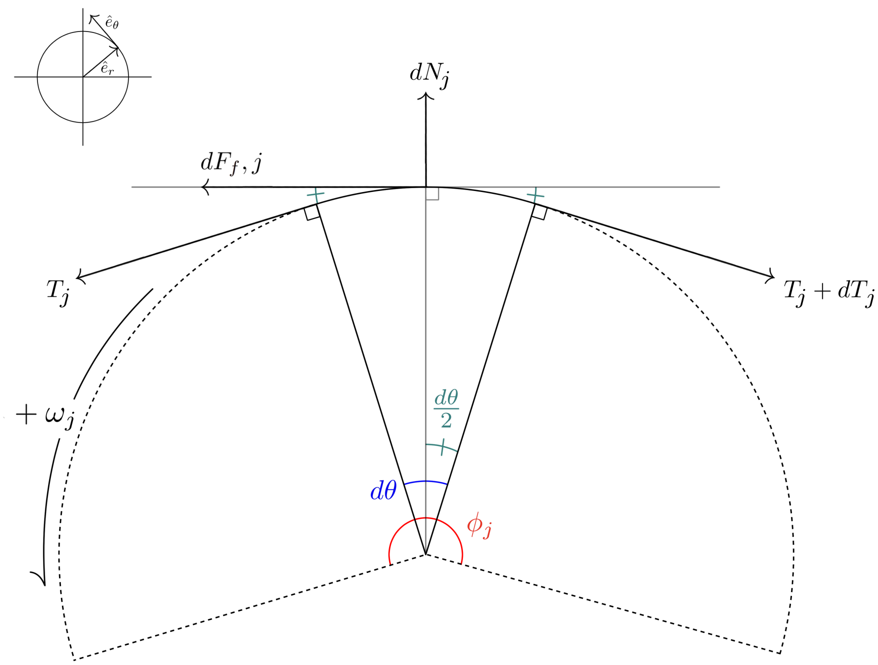

Figure 3.12: Free-body diagram of an infinitesimal belt element within the wrap of a generic pulley j. The local radial direction is positive outward from the pulley axis and the tangential direction is positive with belt travel. The belt tensions act at the two cut faces, while the sheave faces supply the normal contact and tangential traction.

The local axes are taken through the midpoint of the element. The belt entering it pulls with tension T_j, and the belt leaving it pulls with $T_j + dT_j$. Because the two tension directions are tangent to the wrap at opposite ends of the element, each is inclined by $d\theta/2$ from the midpoint tangent. The sheave faces apply a combined physical normal load dN_j. Under Assumption 12 (Symmetric Sheave-Face Sharing), each face carries $dN_j/2$; their axial components oppose one another on the belt, while their radial components add to $dN_j \sin\beta$. The retained tangential contact force is denoted by $dF_{t,j}$, taken as positive in the direction of belt travel.

The belt element can therefore accelerate in two in-plane directions. Applying Newton's second law in the midpoint radial and tangential directions gives, directly from the free body,

$$dN_j \sin\beta - T_j \sin\left(\frac{d\theta}{2}\right) - (T_j + dT_j)\sin\left(\frac{d\theta}{2}\right) = dm_j a_{r,j}, \tag{3.4.6}$$

$$(T_j + dT_j)\cos\left(\frac{d\theta}{2}\right) - T_j \cos\left(\frac{d\theta}{2}\right) + dF_{t,j} = dm_j a_{\theta,j}. \tag{3.4.7}$$

These balances expose what the free body alone does not determine. Both right-hand sides require the mass and acceleration of the belt element, and the tangential balance contains the additional contact force $dF_{t,j}$. The inertial quantities are common to both directions, so they can be established first.

Mass and motion of the belt element The radial references needed for this free body were established in Section 3.1.1. The geometric radius r_j locates the outer surface of the belt, while $r_{j,\text{cm}}$ locates its mass centroid. Because the present free body follows the belt mass, its arc length and inertia must be formed at $r_{j,\text{cm}}$. The required area and centroid radius are given by Equations (3.1.12) and (3.1.14).

Under Assumption 4 (Fixed, Uniform Belt Cross-Section), the centroid remains at a fixed depth within the belt. It therefore shares the radial motion of the outer surface:

$$\dot{r}_{j,\text{cm}} = \dot{r}_j, \qquad \ddot{r}_{j,\text{cm}} = \ddot{r}_j. \tag{3.4.8}$$

The rate relation follows from Section 3.1.4, and the corresponding acceleration relation is stated in Equation (3.1.83).

This is the first place where the radius derivatives developed in Section 3.1 enter the belt dynamics directly: the same changing pulley geometry that moves the belt radially also accelerates its mass.

Under Assumption 8 (Prescribed Planar Belt Path), the centroid follows a circular arc through the wrap. The element length is therefore $r_{j,\text{cm}} d\theta$, and its mass is

$$dm_j = \rho_b A_b r_{j,\text{cm}}\, d\theta. \tag{3.4.9}$$

The same element moves radially with the changing wrap and tangentially with the belt transport. Under Assumption 7 (Single Belt Transport Coordinate), its velocity is

$$\mathbf{v}_{b,j} = \dot{r}_j\, \hat{\mathbf{e}}_r + v_b\, \hat{\mathbf{e}}_\theta. \tag{3.4.10}$$

As the element travels around the pulley, the local basis rotates at

$$\dot{\theta}_j = \frac{v_b}{r_{j,\text{cm}}}, \qquad \dot{\hat{\mathbf{e}}}_r = \frac{v_b}{r_{j,\text{cm}}}\hat{\mathbf{e}}_\theta, \qquad \dot{\hat{\mathbf{e}}}_\theta = -\frac{v_b}{r_{j,\text{cm}}}\hat{\mathbf{e}}_r. \tag{3.4.11}$$

Differentiating the radial part of Equation (3.4.10) gives

$$\frac{d}{dt}\left(\dot{r}_j\hat{\mathbf{e}}_r\right) = \ddot{r}_j\hat{\mathbf{e}}_r + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}}\hat{\mathbf{e}}_\theta, \tag{3.4.12}$$

while differentiating its tangential part gives

$$\frac{d}{dt}\left(v_b\hat{\mathbf{e}}_\theta\right) = \dot{v}_b\hat{\mathbf{e}}_\theta - \frac{v_b^2}{r_{j,\text{cm}}}\hat{\mathbf{e}}_r. \tag{3.4.13}$$

Adding the two contributions separates the acceleration into

$$a_{r,j} = \ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}}, \tag{3.4.14}$$

$$a_{\theta,j} = \dot{v}_b + \frac{\dot{r}_j}{r_{j,\text{cm}}} v_b. \tag{3.4.15}$$

The radial acceleration therefore contains both the motion of the shifting wrap and the centripetal acceleration required to turn the belt. The tangential acceleration contains the direct change in belt transport together with the contribution created as the belt moves radially through the rotating local basis.

Radial balance With the mass and radial acceleration known, return to Equation (3.4.6). Combining the two inward tension projections gives

$$\begin{aligned} dN_j \sin\beta - (2T_j + dT_j)\sin\left(\frac{d\theta}{2}\right) \\ = \rho_b A_b r_{j,\text{cm}} d\theta \left(\ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}}\right). \end{aligned} \tag{3.4.16}$$

Dividing through by $d\theta$ gives

$$\sin\beta \frac{dN_j}{d\theta} - (2T_j + dT_j)\frac{\sin(d\theta/2)}{d\theta} = \rho_b A_b \left(r_{j,\text{cm}}\ddot{r}_j - v_b^2\right). \tag{3.4.17}$$

As the belt element shrinks,

$$\frac{\sin(d\theta/2)}{d\theta} = \frac{1}{2}\frac{\sin(d\theta/2)}{d\theta/2} \longrightarrow \frac{1}{2}. \tag{3.4.18}$$

Over the same vanishing interval, $dT_j = (dT_j/d\theta)d\theta \to 0$. The dT_j contribution to the radial projection therefore disappears, while the two leading tension terms leave one T_j. The radial equation of motion becomes

$$\boxed{\sin\beta \frac{dN_j}{d\theta} = T_j - \rho_b A_b \left(v_b^2 - r_{j,\text{cm}}\ddot{r}_j\right).} \tag{3.4.19}$$

The tension provides part of the inward force needed to curve the belt. The normal contact supplies the remainder required by the actual radial acceleration of the moving belt path.

Tangential balance and traction The tangential balance in Equation (3.4.7) begins more simply. The common T_j terms cancel, leaving only the tension change across the element,

$$dT_j \cos\left(\frac{d\theta}{2}\right) + dF_{t,j} = dm_j a_{\theta,j}. \tag{3.4.20}$$

The remaining contact term is frictional. Under Assumption 10 (Gross Tangential Coulomb Traction), the usual Coulomb description bounds its magnitude by the available static friction while sticking and fixes it at the kinetic-friction magnitude while sliding,

$$\begin{cases} |dF_{t,j}| \leq \mu_s dN_j, & \text{sticking}, \\ |dF_{t,j}| = \mu_k dN_j, & \text{sliding}. \end{cases} \tag{3.4.21}$$

That statement is naturally written as a magnitude inequality, but the force balance requires a signed force. A single signed traction utilization λ_j carries both pieces of information without introducing separate positive- and negative-friction balances:

Signed Traction Utilization **(3.4.22)**

$$dF_{t,j} = \lambda_j dN_j, \qquad \lambda_j = \begin{cases} \lambda_{j,\mathrm{st}}, & |\lambda_{j,\mathrm{st}}| \leq \mu_s, & \text{sticking}, \\ \sigma_j \mu_k, & \sigma_j \in \{-1, +1\}, & \text{sliding}. \end{cases}$$

For sliding contact, σ_j takes the sign that makes the traction oppose the relative sliding motion. Under Assumption 11 (Wrap-Level Contact Resolution), the same λ_j represents the complete wrap at a given instant.

Substituting Equations (3.4.9) and (3.4.15) and **??** into Equation (3.4.20) and dividing by $d\theta$ gives

$$\frac{dT_j}{d\theta} \cos\left(\frac{d\theta}{2}\right) + \lambda_j \frac{dN_j}{d\theta} = \rho_b A_b \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b\right). \tag{3.4.23}$$

As $d\theta \to 0$, $\cos(d\theta/2) \to 1$. The tangential equation of motion therefore reduces to

$$\boxed{\frac{dT_j}{d\theta} + \lambda_j \frac{dN_j}{d\theta} = \rho_b A_b \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b\right).} \tag{3.4.24}$$

The two equations describe different parts of the same local motion. The radial balance relates the belt tension, normal contact, and radial acceleration, while the tangential balance relates the tension change and tangential traction to the acceleration of belt transport. Within the adopted reduced wrap description, these are the radial and tangential equations of motion for an infinitesimal belt element on either pulley.

3.4.3 Tension Evolution Through a Pulley Wrap

The local balances now describe the mechanics at every point within a pulley wrap. The four tensions introduced in Section 3.4.1, however, are boundary values at the entrance and exit of those wraps. To connect the local mechanics back to the complete belt, the two local balances must therefore be reduced to a relation that describes how the tension changes with position through the contact.

Both local equations contain the same normal-load gradient. Solving the radial balance, Equation (3.4.19), for that gradient gives

$$\frac{dN_j}{d\theta} = \frac{1}{\sin\beta}\left[T_j - \rho_b A_b\left(v_b^2 - r_{j,\text{cm}}\ddot{r}_j\right)\right]. \tag{3.4.25}$$

Substituting this expression into the tangential balance, Equation (3.4.24), removes the local normal load and leaves a single differential equation for the belt tension,

$$\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin\beta}\left[T_j - \rho_b A_b\left(v_b^2 - r_{j,\text{cm}}\ddot{r}_j\right)\right] = \rho_b A_b\left(r_{j,\text{cm}}\dot{v}_b + \dot{r}_j v_b\right). \tag{3.4.26}$$

Collecting the tension term gives

Generic Pulley-Wrap Tension Equation **(3.4.27)**

$$\begin{aligned}\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin\beta}T_j = \rho_b A_b\Big[r_{j,\text{cm}}\dot{v}_b &+ \dot{r}_j v_b \\ &+ \frac{\lambda_j}{\sin\beta}\left(v_b^2 - r_{j,\text{cm}}\ddot{r}_j\right)\Big], \qquad j \in \{p, s\}.\end{aligned}$$

At a fixed instant, the pulley geometry and its radius rates are fixed with respect to the spatial coordinate θ. The belt transport quantities v_b and $\dot{v}_b$ are likewise common to the reduced belt, and Assumption 11 (Wrap-Level Contact Resolution) assigns one traction utilization λ_j to the complete wrap. The coefficients in Equation (3.4.27) are therefore constant as the equation is followed from the entrance of one wrap to its exit. The remaining problem is a first-order linear ordinary differential equation in θ.

It is useful to solve that mathematical form once before restoring the CVT quantities. Consider

$$\frac{dT}{d\theta} + aT = b, \tag{3.4.28}$$

where a and b are constants with respect to θ. Multiplying by $e^{a\theta}$ gives

$$e^{a\theta}\frac{dT}{d\theta} + ae^{a\theta}T = be^{a\theta}. \tag{3.4.29}$$

The left-hand side is now the derivative of a single product. From the usual product rule, together with the chain rule for the exponential,

$$\frac{d}{d\theta}\left(e^{a\theta}T\right) = e^{a\theta}\frac{dT}{d\theta} + ae^{a\theta}T. \tag{3.4.30}$$

Equation (3.4.29) can therefore be written as

$$\frac{d}{d\theta}\left(e^{a\theta}T\right) = be^{a\theta}. \tag{3.4.31}$$

Let the belt enter the wrap at $\theta = 0$ with tension T_{in}. Integrating from the entrance to an arbitrary position θ gives

$$\int_0^\theta \frac{d}{d\xi}\left(e^{a\xi}T(\xi)\right)d\xi = \int_0^\theta be^{a\xi}\,d\xi,$$
$$e^{a\theta}T(\theta) - T_{\text{in}} = \frac{b}{a}\left(e^{a\theta}-1\right), \qquad a \neq 0. \tag{3.4.32}$$

Solving for the tension gives the familiar constant-coefficient result

$$T(\theta) = T_{\text{in}}e^{-a\theta} + \frac{b}{a}\left(1-e^{-a\theta}\right). \tag{3.4.33}$$

For the pulley wrap,

$$a = \frac{\lambda_j}{\sin\beta}, \tag{3.4.34}$$

and the ratio b/a obtained from Equation (3.4.27) is

$$\frac{b}{a} = \rho_b A_b\left[v_b^2 - r_{j,\text{cm}}\ddot{r}_j + \frac{\sin\beta}{\lambda_j}\left(r_{j,\text{cm}}\dot{v}_b + \dot{r}_j v_b\right)\right]. \tag{3.4.35}$$

Substitution into Equation (3.4.33) gives the tension at any position through the wrap,

Pulley-Wrap Tension Distribution **(3.4.36)**

$$\begin{aligned} T_j(\theta) = T_{j,\text{in}}&\exp\left(-\frac{\lambda_j}{\sin\beta}\theta\right) \\ &+ \rho_b A_b\left[v_b^2 - r_{j,\text{cm}}\ddot{r}_j + \frac{\sin\beta}{\lambda_j}\left(r_{j,\text{cm}}\dot{v}_b + \dot{r}_j v_b\right)\right] \\ &\times\left[1-\exp\left(-\frac{\lambda_j}{\sin\beta}\theta\right)\right]. \end{aligned}$$

The division by λ_j in this closed form does not make zero traction singular. It entered only because Equation (3.4.33) was written for $a \neq 0$. If $\lambda_j = 0$, the original wrap equation (3.4.27) simply becomes

$$\frac{dT_j}{d\theta} = \rho_b A_b\left(r_{j,\text{cm}}\dot{v}_b + \dot{r}_j v_b\right), \tag{3.4.37}$$

and direct integration gives

$$T_j(\theta) = T_{j,\text{in}} + \rho_b A_b\left(r_{j,\text{cm}}\dot{v}_b + \dot{r}_j v_b\right)\theta. \tag{3.4.38}$$

This is also the continuous limit of Equation (3.4.36) as $\lambda_j \to 0$.

The exit of the wrap occurs at $\theta = \phi_j$. Evaluating the tension field there converts the local solution into the boundary relation needed by the complete belt,

Pulley-Wrap Entry-to-Exit Tension Relation **(3.4.39)**

$$
\begin{aligned}
T_{j,\text{out}} = T_{j,\text{in}} \exp\left(-\frac{\lambda_j \phi_j}{\sin\beta}\right) \\
+ \rho_b A_b \left[v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin\beta}{\lambda_j} \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b\right)\right] \\
\times \left[1 - \exp\left(-\frac{\lambda_j \phi_j}{\sin\beta}\right)\right], \qquad \lambda_j \neq 0.
\end{aligned}
$$

The additional terms in this expression come from the transient belt motion retained in the local balances. Their relation to the familiar belt result is seen by holding the pulley geometry fixed and making the belt transport steady, so that $\dot{r}_j = \ddot{r}_j = \dot{v}_b = 0$. The generic wrap equation then reduces to

$$
\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin\beta} \left(T_j - \rho_b A_b v_b^2\right) = 0. \tag{3.4.40}
$$

Thus the inertia-corrected, or tractive, tension $T_j - \rho_b A_b v_b^2$ follows the exponential law

$$
T_j(\theta) - \rho_b A_b v_b^2 = \left(T_{j,\text{in}} - \rho_b A_b v_b^2\right) \exp\left(-\frac{\lambda_j}{\sin\beta}\theta\right). \tag{3.4.41}
$$

At the wrap exit,

$$
\frac{T_{j,\text{out}} - \rho_b A_b v_b^2}{T_{j,\text{in}} - \rho_b A_b v_b^2} = \exp\left(-\frac{\lambda_j \phi_j}{\sin\beta}\right). \tag{3.4.42}
$$

If belt inertia is neglected as well, the remaining relation is

$$
\boxed{\frac{T_{j,\text{out}}}{T_{j,\text{in}}} = \exp\left(-\frac{\lambda_j \phi_j}{\sin\beta}\right).} \tag{3.4.43}
$$

This is the familiar V-belt capstan structure, with the direction of tension change carried by the sign of λ_j. For a wrap along which tension rises in the direction of belt travel, $\lambda_j < 0$ under the present pulley-on-belt sign convention, so the exponential ratio is greater than one. At a friction limit, the magnitude of λ_j becomes the corresponding Coulomb coefficient. The classical exponential law is therefore recovered as a limiting case of the transient wrap balance rather than replaced by it (Gerbert, 1996).

3.4.4 Completing the Pulley Wrap

The tension field in Section 3.4.3 was obtained by eliminating the distributed normal contact from the two local equations of motion. That elimination let the tension be followed from the entrance of the wrap to its exit, but the removed quantity is also the contact through which the belt and pulley press against one another. With $T_j(\theta)$ now known, the radial balance can be read in the opposite direction to recover that contact and complete the local description of the wrap.

From Equation (3.4.25), the normal load accumulated per unit wrap angle is

$$\frac{dN_j}{d\theta} = \frac{1}{\sin\beta}\left[T_j(\theta) - \rho_b A_b\left(v_b^2 - r_{j,\mathrm{cm}}\ddot{r}_j\right)\right]. \tag{3.4.44}$$

Here dN_j is the same physical quantity defined on the infinitesimal belt free body: the sum of the magnitudes of the normal loads applied by the two sheave faces. Equation (3.4.44) therefore recovers how that contact is distributed around the wrap. The total normal resultant over both faces follows by accumulation from the entrance to the exit,

$$N_j = \frac{1}{\sin\beta}\int_0^{\phi_j}\left[T_j(\theta) - \rho_b A_b\left(v_b^2 - r_{j,\mathrm{cm}}\ddot{r}_j\right)\right] d\theta. \tag{3.4.45}$$

The recovered fields also show whether the prescribed wrap is physically admissible. Under Assumption 8 (Prescribed Planar Belt Path), the belt carries tension but not compression, and the belt and sheave faces can press against one another but cannot pull one another together. The complete wrap therefore requires

$$T_j(\theta) \geq 0, \qquad \frac{dN_j}{d\theta} \geq 0, \qquad 0 \leq \theta \leq \phi_j. \tag{3.4.46}$$

A negative tension would invalidate the assumed taut belt path, while a negative normal-load gradient would indicate local lift-off rather than a physical negative belt load.

Because the complete tension field is known from Equation (3.4.36), Equation (3.4.45) already determines N_j. The same resultant can be written more compactly by returning to the tangential equation of motion in Equation (3.4.24). Integrating that balance from $\theta = 0$ to $\theta = \phi_j$ gives

$$\begin{aligned} \int_0^{\phi_j}\frac{dT_j}{d\theta}\,d\theta + \lambda_j\int_0^{\phi_j}\frac{dN_j}{d\theta}\,d\theta &= \rho_b A_b\left(r_{j,\mathrm{cm}}\dot{v}_b + \dot{r}_j v_b\right)\int_0^{\phi_j} d\theta, \\ T_{j,\mathrm{out}} - T_{j,\mathrm{in}} + \lambda_j N_j &= \rho_b A_b \phi_j\left(r_{j,\mathrm{cm}}\dot{v}_b + \dot{r}_j v_b\right). \end{aligned} \tag{3.4.47}$$

The first term is the change in tension through the wrap, the second is the complete tangential contact action, and the right-hand side is the tangential momentum change of the wrapped belt. For nonzero traction utilization, the normal resultant is therefore

Pulley-Wrap Normal Resultant **(3.4.48)**

$$N_j = \frac{T_{j,\mathrm{in}} - T_{j,\mathrm{out}} + \rho_b A_b \phi_j\left(r_{j,\mathrm{cm}}\dot{v}_b + \dot{r}_j v_b\right)}{\lambda_j}, \qquad \lambda_j \neq 0.$$

This compact relation and the radial integral in Equation (3.4.45) describe the same resultant. The second form is especially useful once the entry-to-exit tension relation is known; the first retains the local contact distribution as well.

Zero traction again requires no singular physics. When $\lambda_j = 0$, the tangential balance no longer contains N_j, so it cannot determine the normal loading. The radial balance still does. Substituting the zero-traction tension field from Equation (3.4.38) into Equation (3.4.45) gives

$$\boxed{\begin{aligned} N_j = \frac{\phi_j}{\sin\beta}\left[T_{j,\text{in}} - \rho_b A_b\left(v_b^2 - r_{j,\text{cm}}\ddot{r}_j\right)\right] \\ + \frac{\rho_b A_b \phi_j^2}{2\sin\beta}\left(r_{j,\text{cm}}\dot{v}_b + \dot{r}_j v_b\right). \end{aligned}} \tag{3.4.49}$$

The isolated wrap therefore provides a relation between its tension level and its total normal loading; neither absolute level is fixed by the wrap alone. What has been recovered is the contact resultant that was removed while the tension equation was being solved.

Axial reaction on the movable sheave The connection back to the pulley follows directly from the same normal resultant. Under Assumption 12 (Symmetric Sheave-Face Sharing), each sheave face carries one half of N_j. The face normal is inclined by the sheave half-angle β from the shaft axis, so the axial component exerted by the belt on the movable face is

$$\boxed{F_{j,\text{belt}} = \frac{N_j}{2}\cos\beta, \qquad j \in \{p, s\}.} \tag{3.4.50}$$

For the primary and secondary, this is exactly the opening reaction introduced in Equations (3.2.13) and (3.2.32). The belt force that remained unknown in those axial balances is therefore the axial projection of the wrap contact recovered above.

Torque exerted by the belt The same contact has a tangential component. From the signed traction definition in Equation (3.4.22), the pulley exerts $dF_{t,j} = \lambda_j dN_j$ on the belt. Since λ_j is common to the reduced wrap, the total pulley-on-belt tangential force is

$$F_{t,j} = \lambda_j N_j. \tag{3.4.51}$$

The moment arm used to convert this tangential contact force into pulley torque is the radius of the tension-carrying cord line. This effective radius $r_{j,\text{eff}}$ was defined in Equation (3.1.9). It is distinct from $r_{j,\text{cm}}$, which follows the belt mass and therefore appears in the inertial terms above. The belt exerts the equal-and-opposite tangential force on the pulley, so the signed belt torque used in the rotational balances is

$$\boxed{\tau_j = -r_{j,\text{eff}}\lambda_j N_j, \qquad j \in \{p, s\}.} \tag{3.4.52}$$

No separate primary and secondary sign rule is needed; the traction direction is already carried by λ_j. A nonzero normal resultant can also exist with $\lambda_j = 0$, in which case the pulley remains clamped while the modeled tangential contact transmits no torque.

The belt reaction in the axial equations and the belt torque in the rotational equations are therefore not separate contact descriptions. They are the axial and tangential consequences of the same distributed wrap loading. Conditional on the tension entering the wrap and the instantaneous belt motion, the isolated pulley wrap is now characterized by its tension field, exit tension, normal resultant, and corresponding axial and rotational actions.

3.4.5 Reassembling the Closed Belt

The two straight spans and the two pulley wraps have now been described individually. The span balance in Equation (3.4.4) relates the tensions at the ends of a free span, while the wrap relations in Sections 3.4.3 and 3.4.4 describe how the tension and contact evolve from one tangent point to the next. The actual belt has no free ends. The tension leaving each region is the tension entering the next, so the four local descriptions must now fit together around the single closed loop introduced in Figure 3.11.

Returning to that loop imposes two requirements that no isolated region can supply by itself. First, the tangential actions accumulated around the entire belt must account for the acceleration of the belt as a whole. Second, the tension changes predicted through the two wraps and two spans must be mutually compatible when the path closes on itself.

Following the positive belt-travel direction around that loop fixes the wrap boundary values introduced earlier. The belt enters the primary from the upper span and leaves into the lower span, so

$$T_{p,\mathrm{in}} = T_{p,u}, \qquad T_{p,\mathrm{out}} = T_{p,l}. \tag{3.4.53}$$

The lower span then carries the belt to the secondary, which returns it to the upper span. Hence,

$$T_{s,\mathrm{in}} = T_{s,l}, \qquad T_{s,\mathrm{out}} = T_{s,u}. \tag{3.4.54}$$

With these identifications, every part of the belt can be written in the same sequence in which the belt is traversed. The complete-belt geometry in Section 3.1.2 already showed that the two external tangent spans have the same instantaneous length. Thus $l_u = l_l = l$, using the same span length l introduced there. The two span balances from Equation (3.4.5), together with the integrated wrap balance from Equation (3.4.47), are therefore

$$\begin{aligned}
T_{p,u} - T_{s,u} &= \rho_b A_b l \dot{v}_b, && (3.4.55)\\
T_{p,l} - T_{p,u} + \lambda_p N_p &= \rho_b A_b \phi_p \left(r_{p,\mathrm{cm}} \dot{v}_b + \dot{r}_p v_b \right), && (3.4.56)\\
T_{s,l} - T_{p,l} &= \rho_b A_b l \dot{v}_b, && (3.4.57)\\
T_{s,u} - T_{s,l} + \lambda_s N_s &= \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}} \dot{v}_b + \dot{r}_s v_b \right). && (3.4.58)
\end{aligned}$$

These four relations contain the same four boundary tensions that were introduced when the belt was first divided into spans and wraps. When the four segment balances are added, each boundary tension appears once as the force on one region and once with the opposite sign on its neighbour. Adding Equations (3.4.55) to (3.4.58) indeed cancels every boundary tension and gives

$$\begin{aligned}
\lambda_p N_p + \lambda_s N_s = {} & \rho_b A_b \left(2l + r_{p,\mathrm{cm}} \phi_p + r_{s,\mathrm{cm}} \phi_s \right) \dot{v}_b \\
& + \rho_b A_b v_b \left(\phi_p \dot{r}_p + \phi_s \dot{r}_s \right).
\end{aligned} \tag{3.4.59}$$

The two terms on the right have different origins. The first is the direct transport acceleration of all belt mass contained in the two spans and two wraps. The second comes from the local tangential acceleration created as the wrapped belt moves radially while travelling around the pulleys. That second term was essential in each local wrap balance; whether it survives for the complete belt is decided by the geometry already established in Section 3.1.4.

Within the engaged shift branch, Equations (3.1.36) and (3.1.57) gives

$$\dot{r}_s = -\frac{\phi_p}{\phi_s}\dot{r}_p, \tag{3.4.60}$$

and therefore

$$\phi_p\dot{r}_p + \phi_s\dot{r}_s = 0. \tag{3.4.61}$$

The same relation is trivially satisfied in the primary deadzone, where both belt radii are fixed. Thus, shifting redistributes the local tangential inertia between the two wraps, but contributes no additional net circumferential force to the complete fixed-length belt.

The coefficient multiplying $\dot{v}_b$ in Equation (3.4.59) is likewise the total belt mass represented by the transport coordinate. The outer-surface belt length used in Section 3.1 is

$$L_b = 2l + r_p\phi_p + r_s\phi_s. \tag{3.4.62}$$

The mass centroid lies the fixed distance y_{cm} inward from that surface, so Equation (3.1.14) and $\phi_p + \phi_s = 2\pi$ give the corresponding centroid-path length

$$\begin{aligned} L_{b,\text{cm}} &= 2l + r_{p,\text{cm}}\phi_p + r_{s,\text{cm}}\phi_s \\ &= L_b - y_{\text{cm}}\left(\phi_p + \phi_s\right) = L_b - 2\pi y_{\text{cm}}. \end{aligned} \tag{3.4.63}$$

Both L_b and y_{cm} are constant under the adopted belt geometry, so this centroid-path length is constant as well. Defining the total belt mass represented by the transport coordinate as

$$m_b = \rho_b A_b L_{b,\text{cm}}, \tag{3.4.64}$$

and applying Equation (3.4.61) reduces Equation (3.4.59) to the complete-belt equation of motion,

Complete-Belt Momentum Balance **(3.4.65)**

$$m_b\dot{v}_b = \lambda_p N_p + \lambda_s N_s.$$

This equation has a direct force interpretation. The quantities $\lambda_p N_p$ and $\lambda_s N_s$ are the signed tangential forces exerted by the two pulleys on the belt. Their sum accelerates the total belt mass along the retained transport coordinate. Using the belt-on-pulley torque relation in Equation (3.4.52) gives the equivalent torque form

$$\boxed{m_b \dot{v}_b = -\frac{\tau_p}{r_{p,\mathrm{eff}}} - \frac{\tau_s}{r_{s,\mathrm{eff}}}.} \tag{3.4.66}$$

The complete-belt momentum equation answers what net tangential force changes v_b, but the addition that produced it deliberately removed the internal tension pattern. Each boundary tension appeared once on either side of an internal cut and cancelled. The same four segment equations therefore contain more information than their sum alone: the two straight spans must also be able to connect the boundary tensions predicted by the two pulley wraps.

The two span balances isolate that remaining information cleanly. Because the upper and lower spans have the same instantaneous length and carry the same belt transport acceleration, they require the same tension change,

$$T_{p,u} - T_{s,u} = T_{s,l} - T_{p,l}. \tag{3.4.67}$$

Rearranging gives

$$T_{p,u} + T_{p,l} = T_{s,u} + T_{s,l}. \tag{3.4.68}$$

This is already a different statement from the complete-belt momentum balance. The momentum equation retains only the net tangential action on the belt; the relation above retains information about how the internal tension levels on the two sides of the loop must match. It does not require uniform belt tension. Instead, the two wrap solutions and the two accelerating spans must meet at the same four tangent points without assigning contradictory boundary tensions.

Those four tensions were introduced only to divide the continuous belt into manageable regions, so they should not remain in the final whole-loop relation. For a generic pulley j, the wrap relations already provide the information needed to remove them. From Equation (3.4.47),

$$T_{j,\mathrm{out}} - T_{j,\mathrm{in}} = \rho_b A_b \phi_j \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b\right) - \lambda_j N_j. \tag{3.4.69}$$

The entry-to-exit solution in Equation (3.4.39) supplies the second relation between the same two boundary tensions. Solving those two relations for the entering tension gives, for $\lambda_j \neq 0$,

$$\begin{aligned} T_{j,\mathrm{in}} = \rho_b A_b &\left[v_b^2 - r_{j,\mathrm{cm}} \ddot{r}_j + \frac{\sin\beta}{\lambda_j}\left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b\right)\right] \\ &+ \frac{\lambda_j N_j - \rho_b A_b \phi_j \left(r_{j,\mathrm{cm}} \dot{v}_b + \dot{r}_j v_b\right)}{1 - \exp\left(-\dfrac{\lambda_j \phi_j}{\sin\beta}\right)}. \end{aligned} \tag{3.4.70}$$

Adding Equation (3.4.69) to this result gives the corresponding exit tension. Their sum, which is the combination required by Equation (3.4.68), is therefore

$$
\begin{aligned}
T_{j,\text{in}} + T_{j,\text{out}} = 2\rho_b A_b &\left[v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin\beta}{\lambda_j} \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b \right) \right] \\
&+ \frac{1 + \exp\left(-\dfrac{\lambda_j \phi_j}{\sin\beta} \right)}{1 - \exp\left(-\dfrac{\lambda_j \phi_j}{\sin\beta} \right)} \\
&\quad \times \left[\lambda_j N_j - \rho_b A_b \phi_j \left(r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b \right) \right].
\end{aligned} \tag{3.4.71}
$$

Equation (3.4.71) removes the artificial boundary tensions from the compatibility condition. Applying it once to the primary wrap and once to the secondary wrap gives the second complete-belt relation,

Closed-Loop Tension Compatibility **(3.4.72)**

$$
\begin{aligned}
&2\rho_b A_b \left[v_b^2 - r_{p,\text{cm}} \ddot{r}_p + \frac{\sin\beta}{\lambda_p} \left(r_{p,\text{cm}} \dot{v}_b + \dot{r}_p v_b \right) \right] \\
&\quad + \frac{1 + \exp\left(-\dfrac{\lambda_p \phi_p}{\sin\beta} \right)}{1 - \exp\left(-\dfrac{\lambda_p \phi_p}{\sin\beta} \right)} \left[\lambda_p N_p - \rho_b A_b \phi_p \left(r_{p,\text{cm}} \dot{v}_b + \dot{r}_p v_b \right) \right] \\
&= 2\rho_b A_b \left[v_b^2 - r_{s,\text{cm}} \ddot{r}_s + \frac{\sin\beta}{\lambda_s} \left(r_{s,\text{cm}} \dot{v}_b + \dot{r}_s v_b \right) \right] \\
&\quad + \frac{1 + \exp\left(-\dfrac{\lambda_s \phi_s}{\sin\beta} \right)}{1 - \exp\left(-\dfrac{\lambda_s \phi_s}{\sin\beta} \right)} \left[\lambda_s N_s - \rho_b A_b \phi_s \left(r_{s,\text{cm}} \dot{v}_b + \dot{r}_s v_b \right) \right], \qquad \lambda_p \lambda_s \neq 0.
\end{aligned}
$$

As with the wrap solution itself, the apparent singularity at zero traction is only a consequence of writing the exponential solution in a form divided by λ_j. The original local balances remain regular. Taking the continuous limit of Equation (3.4.71) gives, for any wrap with $\lambda_j = 0$,

$$
\boxed{T_{j,\text{in}} + T_{j,\text{out}} = 2\rho_b A_b \left(v_b^2 - r_{j,\text{cm}} \ddot{r}_j \right) + \frac{2N_j \sin\beta}{\phi_j}.} \tag{3.4.73}
$$

That limiting expression replaces the corresponding primary or secondary side of Equation (3.4.72) whenever its traction utilization is zero.

The closed belt has therefore supplied two independent whole-loop relations in terms of the retained mechanical quantities rather than the four temporary boundary tensions. The transport balance in Equation (3.4.65) determines how the net tangential contact action accelerates the belt as a whole. The tension compatibility in Equation (3.4.72) preserves the independent internal information lost when those segment equations were added, requiring the two wrap tension solutions to be connectable by the same pair of straight spans. Together they complete the closed-belt mechanics that motivated the return to the full loop.

3.4.6 What the Belt Mechanics Establish

The belt now has a complete reduced mechanical description of its own. The straight spans and pulley wraps were separated only because tension can change for different reasons in each region. In a free span, the endpoint tension difference accelerates the belt mass carried between the moving tangent points. The motion of those tangent points does not introduce another circumferential inertia term: once the associated momentum flux is included, the changing span mass and moving-boundary contributions cancel. On a pulley wrap, the belt is instead redirected while exchanging distributed normal and tangential forces with the sheaves, so the endpoint tensions could only be related after the local belt-element dynamics were resolved.

Those local wrap balances show how much of the familiar belt relation survives when the adopted transient kinematics are retained. The radial balance links tension to the normal loading required to redirect and radially accelerate the belt, while the tangential balance links tension change and traction to the circumferential acceleration of the belt material. Combining the two produces one first-order tension equation containing the direct transport acceleration $\dot{v}_b$, the coupling $\dot{r}_j v_b$ created as the wrap radius changes, and the radial acceleration contribution through $v_b^2 - r_{j,\mathrm{cm}}\ddot{r}_j$. When those transient terms vanish, the solution reduces to the inertia-corrected tractive-tension form and then to the classical V-belt capstan relation. Neither the exponential tension law nor the use of belt inertia in that steady relation is new; both belong to the established belt-mechanics lineage (Gerbert, 1996; Messick, 2018). Reduced engineering models may use this familiar structure directly. For example, Skinner (2020) combines an exponential wrap-tension relation with a centrifugal belt-force correction in a quasi-static Baja-CVT force-balance model. The distinction here is therefore not the capstan form itself, but that the transient accelerations implied by the retained belt kinematics remain in the local equations before the wrap is reduced analytically.

Solving the tension field also makes the pulley contact mechanically complete. The normal load was temporarily eliminated to obtain the tension equation; once $T_j(\theta)$ is known, the radial balance recovers that distributed contact and its resultant N_j. Only fields that keep both the belt tension and the distributed normal loading non-negative are compatible with the assumed taut, continuously wrapped path, as stated in Equation (3.4.46). The axial belt reaction and transmitted pulley torque then follow as two components of the same contact resultant rather than as independent constitutive laws. This idea also has strong precedent. Detailed rubber-belt models such as Kong and Parker (2005, 2006) resolve local distributed belt and contact mechanics at substantially greater spatial fidelity, while Messick (2018) uses that type of solution to recover the distributed pulley loading and integrated axial force. Duan et al. (2016) provide an adjacent chain-CVT example in which distributed normal and friction forces are integrated into the axial and rotational dynamics. The present result is therefore not that belt contact can produce both thrust and torque. Its role is to obtain those two pulley actions from the same compact transient wrap description that also governs belt transport.

The final closure appears only after the four belt regions are returned to one continuous loop. Whole-belt compatibility is likewise not a new physical requirement: the steady belt–pulley formulation of Kong and Parker (2005), and the rubber-belt extension developed and validated by Messick (2018), solve the pulley contacts and free spans together as one boundary-value problem. The reduction obtained here is different in form and purpose. The four temporary endpoint tensions disappear when the local regions are reassembled, leaving two independent whole-belt relations. Their sum gives the complete-belt momentum balance, $m_b\dot{v}_b = \lambda_p N_p + \lambda_s N_s$, which determines the net force that changes the retained transport state. A second relation survives because that sum discards the internal tension levels: the closed-loop tension compatibility requires the tension changes through both pulley wraps and both accelerating spans to coexist on one continuous belt. The first relation therefore governs the net circumferential motion of the belt, while the second preserves the independent internal information lost when the four segment balances are simply added.

This reassembly also clarifies the role of the shifting-radius inertia. Terms proportional to $\dot{r}_j v_b$ are required locally because a belt element travelling around a moving wrap has a tangential acceleration even when v_b itself is instantaneously constant. They influence the local tension and normal-contact fields, but their contributions from the two pulley wraps cancel from the complete-belt momentum equation once the fixed-length geometry is enforced. Their disappearance from the global balance is therefore a result of the closed-belt geometry, not an assumption made before the local mechanics are formed. A term can consequently be important to the load distribution within each wrap while producing no additional net circumferential inertia for the complete belt.

The model resolution that makes this compact closure possible also defines what it cannot predict. Gerbert (1996) shows that rubber-belt slip can depend on longitudinal creep, radial compliance, shear deformation, and seating and unseating effects. The distributed models of Kong and Parker (2005, 2006) and Messick (2018) additionally resolve belt deformation, local path and pressure variation, and spatial sliding behaviour at a level intentionally absent here. Transient shift models also retain effects beyond the present one-speed, prescribed-path description: Sorge (2008) resolves radial belt migration and sliding during ratio change, while Julió and Plante (2011) and Ballew (2015) obtain transient belt motion from spatially discretized elements with local inertia and compliance. These approaches can therefore resolve local stress, creep, seating, loss, material-point migration, and contact-state variation within a wrap that the present reduction cannot. Incorporating those effects here would require relaxing the fixed-cross-section, fixed-length, single-transport-coordinate, prescribed-path, and wrap-level traction assumptions, and would replace part of the present analytical closure with additional distributed or spatial states.

The comparison therefore places the present belt model between the usual reduced capstan-style closures and the higher-fidelity distributed or discretized belt formulations. Transient belt mechanics, radial migration, whole-loop compatibility, axial-thrust prediction, and distributed contact are all established ideas individually. Among the reviewed reduced-order rubber V-belt CVT formulations, however, no equivalent combination was identified in which the radial and tangential accelerations implied by a shifting prescribed belt path are carried through an analytical wrap-tension solution, the same contact field supplies both pulley normal reactions and transmitted torques, and the two pulley wraps and two moving straight spans reduce to complete-belt transport and tension-compatibility relations without spatial discretization. The contribution is therefore the particular dynamic closure and level of reduction, not a new capstan law or a higher-fidelity local contact theory.

The physical picture established by this section is consequently compact. Belt tension is no longer an externally assigned tight–slack pair, and the belt is not merely a kinematic constraint between pulley radii. It carries its own transport momentum; its tension evolves through each pulley contact; that same contact supplies the axial reactions required by the sheave balances and the torques required by the shaft balances; and the four local belt regions must satisfy both whole-loop momentum and tension compatibility. Together with the geometry and pulley balances already derived, the remaining accelerations, normal loads, torques, and traction utilizations belong to one simultaneous mechanical problem rather than to separate subsystem closures.

3.5 Dynamic Closure of the Freely Shifting Engaged CVT

The preceding sections have now supplied the mechanical description of each part of the engaged CVT. The axial balances describe motion along the pulley shafts, the rotational balances describe motion about them, and the belt mechanics supply both the contact resultants acting on the pulleys and the two relations that govern the closed belt itself. Each relation was derived on the free body where its origin was clearest. The remaining question is whether, when those relations are brought together, they are sufficient to determine the instantaneous motion of the complete engaged system.

No additional free body remains to be introduced. The issue is now one of closure: which quantities remain unknown after the equations already derived are assembled, and what physical relations are still missing?

3.5.1 Assembling the Engaged Equations

The clearest way to identify what remains is to begin with the equations already available. The primary and secondary pulley assemblies contribute the two rotational balances in Equations (3.3.5) and (3.3.13) and the two axial balances in Equations (3.2.38) and (3.2.39). The belt contributes the complete transport balance in Equation (3.4.65) and the independent closed-loop tension compatibility in Equation (3.4.72). These are six simultaneous mechanical equations: two rotational, two axial, and two belonging to the belt.

The rotational equations were originally written using the belt torques τ_p and τ_s, and the secondary axial balance also contains τ_s. Those torques are no longer independent quantities. The wrap mechanics established

$$\boxed{\tau_j = -r_{j,\text{eff}}\lambda_j N_j, \qquad j \in \{p, s\},} \tag{3.5.1}$$

so this result can be inserted wherever a belt torque appears. The axial belt reactions are likewise already expressed through the normal resultants N_p and N_s. After these substitutions, no separate belt torque or axial-reaction unknown remains.

The six equations therefore contain the eight instantaneous unknowns

$$\boxed{\mathbf{u}_{\text{eng}} = \begin{bmatrix} \dot{\omega}_p & \dot{\omega}_s & \dot{v}_b & \ddot{s} & N_p & N_s & \lambda_p & \lambda_s \end{bmatrix}^{\mathsf{T}}.} \tag{3.5.2}$$

There are thus two fewer equations than unknowns. The missing relations cannot come from another equation of motion. Rotation about both shafts has already been balanced, the shared axial motion has already been balanced on both pulley assemblies, and the retained belt motion has already supplied its two independent whole-loop conditions. The normal resultants N_p and N_s are ordinary contact forces appearing in those balances and are determined as part of the simultaneous mechanical solution.

The remaining quantities, λ_p and λ_s, entered differently. They were introduced in the belt mechanics through $dF_{t,j} = \lambda_j\, dN_j$: each λ_j states how much signed tangential action accompanies the normal contact on one pulley wrap. That definition allowed the belt equations to be derived without yet specifying what instantaneous contact condition fixes the value of λ_j. The two missing relations must therefore come from the tangential belt–pulley contact itself.

The most restrictive tangential contact state is sticking. In that state, the reduced contact law must keep the belt transport speed matched to the representative pulley surface speed. It therefore provides the natural place to determine what the complete coupled problem requires.

3.5.2 Sticking Contact and the Full Coupled Problem

Both primary sheaves rotate at the primary shaft speed ω_p. On the secondary, Assumption 12 (Symmetric Sheave-Face Sharing) specifies the shaft speed ω_s as the representative speed of the aggregated wrap contact. Sticking in this reduced description means that the belt transport speed matches the corresponding shaft-referenced surface speed at the effective radius. For either pulley, the sticking condition is therefore

$$\boxed{v_b = r_{j,\mathrm{eff}}\omega_j.} \tag{3.5.3}$$

During an established shaft-referenced sticking interval this equality is already satisfied by the current state. The instantaneous accelerations must preserve it. Differentiating Equation (3.5.3) gives

$$\boxed{\dot{v}_b - r_{j,\mathrm{eff}}\dot{\omega}_j - \dot{r}_{j,\mathrm{eff}}\omega_j = 0.} \tag{3.5.4}$$

The first two terms compare the circumferential accelerations of the belt and pulley at the current radius. The final term is required because that radius itself moves during a shift. Even if ω_j were instantaneously constant, a changing radius changes the tangential speed of the pulley surface that the belt must follow.

If both pulley contacts stick, applying Equation (3.5.4) at the primary and secondary supplies the two relations missing from the six mechanical equations above. The freely shifting engaged CVT now has eight equations for the eight quantities in Equation (3.5.2). The next question is whether those eight equations can simply be solved together.

Their algebra gives the answer. The six quantities

$$\dot{\omega}_p, \qquad \dot{\omega}_s, \qquad \dot{v}_b, \qquad \ddot{s}, \qquad N_p, \qquad N_s$$

each enter the assembled equations linearly. The nonlinear dependence is confined to λ_p and λ_s. Besides multiplying the normal resultants in the tangential contact forces and pulley torques, the traction utilizations enter the closed-loop tension relation through the analytical wrap solution, including factors of the form

$$\exp\left(-\frac{\lambda_j \phi_j}{\sin\beta}\right) \tag{3.5.5}$$

and the corresponding reciprocal combinations shown in Equation (3.4.72). The explicit zero-traction limit established in Equation (3.4.73) keeps that relation finite at $\lambda_j = 0$, but does not remove its nonlinear dependence on λ_j.

The complete sticking problem is therefore not an 8×8 linear system. Six unknowns belong to a linear mechanical block, while the two static traction utilizations enter that block nonlinearly.

Because the two λ_j values occur inside coupled exponential and reciprocal terms, there is no useful algebraic isolation that reduces the sticking equations to a closed-form pair for λ_p and λ_s. Their values must instead be found from a root condition.

The structure of that root follows directly from the equations. A trial pair (λ_p, λ_s) turns every nonlinear coefficient in the six mechanical equations into a known number. The six remaining unknowns can then be collected as

$$\boxed{\mathbf{z} = \begin{bmatrix} \dot{\omega}_p & \dot{\omega}_s & \dot{v}_b & \ddot{s} & N_p & N_s \end{bmatrix}^{\mathsf{T}}}, \tag{3.5.6}$$

and the six previously derived equations take the linear form

$$\boxed{\mathbf{A}(\lambda_p, \lambda_s)\mathbf{z} = \mathbf{b}(\lambda_p, \lambda_s).} \tag{3.5.7}$$

No new physical equation has been introduced in Equation (3.5.7). Its six rows are simply the two rotational balances, two axial balances, belt transport balance, and belt closed-loop compatibility already identified above. To make that statement explicit, the equations can be assembled directly in the order just listed. The radius accelerations appearing in the closed-loop compatibility are the primary and secondary results already established in Equations (3.1.65) and (3.1.82). Because the effective and mass-centroid radii remain at fixed depths within the belt, their accelerations follow the same results by Equation (3.1.83).

With the columns ordered according to Equation (3.5.6), the coefficient matrix is, for $\lambda_p \lambda_s \neq 0$,

$$\mathbf{A}(\lambda_p, \lambda_s) = \begin{bmatrix} I_{\mathrm{B},p} + I_{p,\mathrm{const}} + J_{f,p}(s) & 0 & 0 & 0 & r_{p,\mathrm{eff}}\lambda_p & 0 \\ 0 & I_{\mathrm{B},s} + I_{s,F} + I_{s,M} & 0 & I_{s,M}\dfrac{d\theta_s}{dx_s}\dfrac{dx_s}{ds} & 0 & r_{s,\mathrm{eff}}\lambda_s \\ 0 & 0 & 0 & m_p + I_f\left(\dfrac{dq_f}{ds}\right)^2 & \dfrac{1}{2}\cos\beta & 0 \\ 0 & I_{s,M}\dfrac{d\theta_s}{dx_s} & 0 & \left[m_s + I_{s,M}\left(\dfrac{d\theta_s}{dx_s}\right)^2\right]\dfrac{dx_s}{ds} & 0 & \dfrac{1}{2}\left[\cos\beta + r_{s,\mathrm{eff}}\lambda_s\dfrac{d\theta_s}{dx_s}\right] \\ 0 & 0 & m_b & 0 & -\lambda_p & -\lambda_s \\ 0 & 0 & \rho_b A_b\left\{ r_{p,\mathrm{cm}}\left[\dfrac{2\sin\beta}{\lambda_p} - \phi_p\dfrac{1+\exp\left(-\dfrac{\lambda_p\phi_p}{\sin\beta}\right)}{1-\exp\left(-\dfrac{\lambda_p\phi_p}{\sin\beta}\right)}\right] - r_{s,\mathrm{cm}}\left[\dfrac{2\sin\beta}{\lambda_s} - \phi_s\dfrac{1+\exp\left(-\dfrac{\lambda_s\phi_s}{\sin\beta}\right)}{1-\exp\left(-\dfrac{\lambda_s\phi_s}{\sin\beta}\right)}\right]\right\} & 2\rho_b A_b\left(r_{s,\mathrm{cm}}\dfrac{dr_s}{ds} - r_{p,\mathrm{cm}}\dfrac{dr_p}{ds}\right) & \lambda_p\dfrac{1+\exp\left(-\dfrac{\lambda_p\phi_p}{\sin\beta}\right)}{1-\exp\left(-\dfrac{\lambda_p\phi_p}{\sin\beta}\right)} & -\lambda_s\dfrac{1+\exp\left(-\dfrac{\lambda_s\phi_s}{\sin\beta}\right)}{1-\exp\left(-\dfrac{\lambda_s\phi_s}{\sin\beta}\right)} \end{bmatrix}. \tag{3.5.8}$$

The corresponding known-side vector is

$$\mathbf{b}(\lambda_p, \lambda_s) = \begin{bmatrix} \tau_{\mathrm{B},p} - \dfrac{dJ_{f,p}}{ds}\dot{s}\,\omega_p \\ \tau_{\mathrm{B},s} - I_{s,M}\left[\dfrac{d\theta_s}{dx_s}\dfrac{d^2x_s}{ds^2} + \dfrac{d^2\theta_s}{dx_s^2}\left(\dfrac{dx_s}{ds}\right)^2\right]\dot{s}^2 \\ \dfrac{1}{2}\omega_p^2\dfrac{dJ_{f,p}}{ds} - I_f\dfrac{dq_f}{ds}\dfrac{d^2q_f}{ds^2}\dot{s}^2 - k_p\left(x_{p,\mathrm{pre}} + s\right) \\ \dfrac{d\theta_s}{dx_s}k_{s,\theta}\left(\theta_{s,\mathrm{pre}} - \theta_s\right) + k_{s,x}\left(x_{s,\mathrm{pre}} - x_s\right) - \left\{\left[m_s + I_{s,M}\left(\dfrac{d\theta_s}{dx_s}\right)^2\right]\dfrac{d^2x_s}{ds^2} + I_{s,M}\dfrac{d\theta_s}{dx_s}\dfrac{d^2\theta_s}{dx_s^2}\left(\dfrac{dx_s}{ds}\right)^2\right\}\dot{s}^2 \\ 0 \\ \left\{2\rho_b A_b\left[v_b^2 - r_{s,\mathrm{cm}}\dfrac{d^2r_s}{ds^2}\dot{s}^2 + \dfrac{\sin\beta}{\lambda_s}\dot{r}_s v_b\right] - \rho_b A_b\phi_s\dfrac{1+\exp\left(-\dfrac{\lambda_s\phi_s}{\sin\beta}\right)}{1-\exp\left(-\dfrac{\lambda_s\phi_s}{\sin\beta}\right)}\dot{r}_s v_b\right\} - \left\{2\rho_b A_b\left[v_b^2 - r_{p,\mathrm{cm}}\dfrac{d^2r_p}{ds^2}\dot{s}^2 + \dfrac{\sin\beta}{\lambda_p}\dot{r}_p v_b\right] - \rho_b A_b\phi_p\dfrac{1+\exp\left(-\dfrac{\lambda_p\phi_p}{\sin\beta}\right)}{1-\exp\left(-\dfrac{\lambda_p\phi_p}{\sin\beta}\right)}\dot{r}_p v_b\right\} \end{bmatrix}. \tag{3.5.9}$$

The first four rows are the pulley balances with Equation (3.5.1) inserted, the fifth is the complete-belt transport balance, and the sixth is the primary wrap boundary sum minus the secondary wrap boundary sum. The latter is the only row whose coefficients retain the exponential wrap dependence on both traction utilizations.

The displayed sixth row uses the nonzero-λ_j form of the wrap solution. If either trial utilization is exactly zero, its contribution is replaced by the continuous limit already derived in Equation (3.4.73). Before taking the primary-minus-secondary difference, that wrap contributes

$$\boxed{\begin{aligned} T_{j,\mathrm{in}} + T_{j,\mathrm{out}} = 2\rho_b A_b \left[v_b^2 - r_{j,\mathrm{cm}} \frac{d^2 r_j}{ds^2} \dot{s}^2 \right] \\ - 2\rho_b A_b r_{j,\mathrm{cm}} \frac{dr_j}{ds} \ddot{s} + \frac{2\sin\beta}{\phi_j} N_j, \qquad \lambda_j = 0, \end{aligned}} \tag{3.5.10}$$

so the same six unknowns remain linear and the matrix stays finite through zero traction. In a numerical evaluation, the divided nonzero-traction form should be replaced by this limit, or by its local series expansion, before its cancelling terms approach floating-point resolution.

Provided the matrix is nonsingular for the trial pair, solving this 6×6 system determines the complete mechanical response associated with that pair.

That response can now be tested against the condition that motivated the static traction in the first place. For pulley j, define the sticking residual

$$R_j(\lambda_p, \lambda_s) = \dot{v}_b - r_{j,\mathrm{eff}} \dot{\omega}_j - \dot{r}_{j,\mathrm{eff}} \omega_j, \tag{3.5.11}$$

where the accelerations on the right are those returned by Equation (3.5.7) for the trial pair. A single trial in the stick–stick root therefore has the sequence

$$\boxed{(\lambda_p^{(k)}, \lambda_s^{(k)}) \;\longrightarrow\; \mathbf{A}(\lambda_p^{(k)}, \lambda_s^{(k)})\mathbf{z}^{(k)} = \mathbf{b}(\lambda_p^{(k)}, \lambda_s^{(k)}) \;\longrightarrow\; (R_p^{(k)}, R_s^{(k)}).} \tag{3.5.12}$$

The trial traction pair first produces one complete mechanical solution. The resulting two residuals then state how much that solution would separate the belt and pulley accelerations at the two contacts. The root search changes the trial values until

$$\boxed{R_p(\lambda_p, \lambda_s) = 0, \qquad R_s(\lambda_p, \lambda_s) = 0.} \tag{3.5.13}$$

When the root exists, the resulting λ_p and λ_s are not prescribed static friction forces. They are the contact reactions required to make the six mechanical balances compatible with adhesion at both pulleys.

A zero residual establishes kinematic compatibility, but static contact can supply only a finite tangential reaction. The required value at each sticking interface must also satisfy

$$\boxed{\left| \lambda_{j,\mathrm{st}}^{\mathrm{req}} \right| \le \mu_s, \qquad j \in \{p, s\}.} \tag{3.5.14}$$

The distinction is important. The root determines what traction would be required to maintain zero relative motion; the inequality determines whether the contact can actually provide it. A converged root outside the static interval is therefore not discarded as a failed numerical solve. It states that the assumed sticking condition is mechanically incompatible with the available static traction.

This identifies what must change when adhesion cannot be supported: the belt and pulley must be allowed to move relative to one another.

3.5.3 Sliding Contact and the Common Mechanical Solve

Once relative sliding exists, the no-slip acceleration condition no longer determines the tangential contact reaction. The relative motion itself fixes the direction of kinetic friction, and the Coulomb law fixes its magnitude. The primary surface speed follows directly from the common motion of its two sheaves. For the secondary, the shaft-referenced surface speed is the representative speed specified in Assumption 12 (Symmetric Sheave-Face Sharing). Accordingly, the relative tangential speed at either pulley j is

$$v_{\mathrm{rel},j} = v_b - r_{j,\mathrm{eff}}\omega_j, \tag{3.5.15}$$

and a sliding interface has

$$\boxed{\begin{aligned} &\sigma_j = -\operatorname{sgn}(v_{\mathrm{rel},j}), \\ &\lambda_j = \sigma_j \mu_k, \qquad v_{\mathrm{rel},j} \neq 0. \end{aligned}} \tag{3.5.16}$$

Here σ_j is the sign of the pulley-on-belt tangential action. The minus sign makes that action oppose the established relative motion. Unlike the static case, λ_j is therefore known directly from a sliding contact state rather than determined as an unknown reaction. At $v_{\mathrm{rel},j} = 0$, no established sliding direction exists, so the appropriate successor branch must instead be selected from the relative acceleration returned by the candidate coupled solutions.

Suppose both contacts slide. Then both traction utilizations are known, so the mechanical solution is obtained by inserting those two values into Equation (3.5.7):

$$\boxed{\mathbf{A}(\sigma_p\mu_k, \sigma_s\mu_k)\mathbf{z} = \mathbf{b}(\sigma_p\mu_k, \sigma_s\mu_k).} \tag{3.5.17}$$

This is the same calculation that appeared inside every trial of the stick–stick root. In sticking, a trial (λ_p, λ_s) was inserted, the 6×6 mechanical system was solved, and the result was then checked against the two no-slip residuals. In sliding, Coulomb friction supplies that traction pair immediately. The same 6×6 system is solved once, and no no-slip residual is imposed because relative motion is permitted.

The two pulley contacts need not share the same contact state. If the primary slides while the secondary sticks, the primary value $\lambda_p = \sigma_p\mu_k$ is known, while λ_s must still take the static value that preserves the secondary no-slip condition. A trial $\lambda_s^{(k)}$ therefore produces the same sequence as before,

$$\boxed{\lambda_s^{(k)} \;\longrightarrow\; \mathbf{A}(\sigma_p\mu_k, \lambda_s^{(k)})\mathbf{z}^{(k)} = \mathbf{b}(\sigma_p\mu_k, \lambda_s^{(k)}) \;\longrightarrow\; R_s^{(k)},} \tag{3.5.18}$$

until $R_s = 0$. Only one static utilization remains unknown, so this is a one-dimensional root rather than the two-dimensional root required when both contacts stick. If the primary sticks while the secondary slides, the same argument gives a one-dimensional root in λ_p with $\lambda_s = \sigma_s \mu_k$.

The four possible engaged contact combinations therefore differ only in how many traction utilizations must be found from static compatibility:

contact state	unknown static λs	outer solve
slip–slip	0	none
stick–slip	1	one-dimensional root
slip–stick	1	one-dimensional root
stick–stick	2	two-dimensional root

(3.5.19)

Every entry in the final column uses the same 6×6 mechanical solve at each evaluation; the table counts only the additional search required to obtain static traction utilizations. The branching therefore does not create four different mechanical models. It changes how λ_p and λ_s are obtained. A sliding contact supplies its value directly from kinetic friction. A sticking contact leaves its value unknown and supplies a no-slip condition instead, so that value must be found by repeatedly applying the same mechanical solve until the condition is satisfied.

3.5.4 Stick–Slip Admissibility and Contact-State Changes

The equations above determine an instantaneous solution for an assumed contact combination. The admissibility considered here is specifically whether each wrap can remain in its assumed tangential stick or slip state. That state can persist only while the solution remains consistent with the contact condition used to construct it.

For a sticking interface, the relevant quantity is the static utilization required by the no-slip root. While Equation (3.5.14) is satisfied, the contact can supply the tangential reaction needed to preserve adhesion. If continued sticking would require

$$\left| \lambda_{j,\mathrm{st}}^{\mathrm{req}} \right| > \mu_s, \tag{3.5.20}$$

then no admissible static reaction can satisfy the no-slip condition. The interface must release into sliding contact, where its traction is instead set by Equation (3.5.16) with a direction consistent with the emerging relative motion. At the release instant, $v_{\mathrm{rel},j} = 0$, so this direction cannot be read from the current speed. For a candidate kinetic sign σ_j, the coupled sliding closure must instead predict a relative acceleration into that branch. Because $R_j = \dot{v}_{\mathrm{rel},j}$ at zero relative speed, the branch-consistency condition is

$$\boxed{\sigma_j R_j < 0.} \tag{3.5.21}$$

That release changes the coupled solution, not merely one entry in it. The new kinetic λ_j changes the belt tangential force and pulley torque, which changes the accelerations, normal resultants, and closed-loop tension balance. If the other pulley remains a candidate for sticking, its required static utilization must therefore be solved again under the new contact combination. Loss of adhesion at one interface does not by itself require loss of adhesion at the other.

A sliding interface reaches the opposite question when its relative speed in Equation (3.5.15) returns to zero. At that instant the belt and pulley no longer have an established velocity difference, so adhesion is again kinematically possible. The corresponding sticking closure is then formed at that same state. If its root gives a required static utilization within Equation (3.5.14), the contact can reattach. If not, zero relative speed does not create a sustainable sticking state and the candidate kinetic branches are evaluated using Equation (3.5.21); sliding continues on the branch whose coupled solution produces a consistent subsequent relative motion.

The two transition tests therefore ask complementary questions. A sticking contact asks whether the required static reaction still lies within the available traction capacity. A sliding contact asks, when its relative motion vanishes, whether an admissible static reaction exists that can keep it there. Each candidate contact state is evaluated by the same coupled mechanical closure developed above.

3.5.5 What the Engaged Closure Establishes

For an assumed tangential contact combination, the freely shifting engaged CVT has a complete instantaneous closure whenever the common mechanical matrix is nonsingular and any required static-traction root exists. The two pulley assemblies and the belt contribute six mechanical equations, while the two tangential contacts supply the remaining information needed to close the system.

Sticking reveals the structure of that closure most clearly. The two no-slip acceleration conditions complete the eight-equation problem, but the eight unknowns do not enter in the same way. The pulley accelerations, belt acceleration, shift acceleration, and two normal resultants form a linear six-variable mechanical block. The two traction utilizations enter that block nonlinearly through the contact resultants and analytical wrap solution. Their required static values are therefore found from a low-dimensional root: each trial traction pair is inserted into the same 6×6 mechanical system, and the resulting accelerations are tested against the no-slip conditions.

Sliding changes only how the traction utilization is supplied. Kinetic friction fixes λ_j directly from the established relative-motion direction, so a sliding contact removes one static unknown and one no-slip root condition. The mechanical calculation itself is unchanged. Two sliding contacts require one 6×6 solve, one sticking contact requires a one-dimensional root whose every trial performs that same solve, and two sticking contacts require a two-dimensional root built from the same calculation.

The static friction limit then decides whether a mathematically compatible sticking solution is physically supportable. The root determines the traction required to preserve adhesion; $|\lambda_j^{\mathrm{req}}| \le \mu_s$ determines whether the interface can supply it. Conversely, a sliding interface can return to sticking only when its relative speed reaches zero and the corresponding static requirement is admissible. Because either change in contact state alters the complete coupled solution, the remaining contact must be reevaluated rather than carrying forward its previous traction requirement.

For any stick–slip-admissible freely shifting contact combination with a nonsingular mechanical solve and an applicable static root, the current state and applied shaft boundaries therefore determine $\dot{\omega}_p$, $\dot{\omega}_s$, $\dot{v}_b$, $\ddot{s}$, N_p, N_s, λ_p, and λ_s. The corresponding belt torques and axial reactions then follow from the contact resultants already derived. The geometry, pulley balances, belt mechanics, and tangential contact conditions have thus become one simultaneous description of the freely shifting engaged CVT.

3.6 Shift Boundaries and Regime Transitions

The engaged closure of the preceding section determines the instantaneous CVT motion while both pulley contacts exist and the shift coordinate is free. That solution can be advanced only until the mechanism reaches a physical boundary or an active unilateral constraint can no longer supply an admissible reaction. The remaining task is therefore to define how the same pulley, belt, and contact equations reduce at the shift boundaries and how the state is transferred when the active mechanical arrangement changes.

3.6.1 Shift Domain and Structural Regimes

The geometry developed in Section 3.1 already establishes the three positions needed here. The fully open primary is the coordinate origin, first primary belt contact occurs at $x_{p,\mathrm{dz}}$, and the maximum permitted primary travel is $x_{p,\max}$. Because the shared shift coordinate satisfies $s \equiv x_p$ by Equation (3.1.31), write

$$s_{\mathrm{dz}} \equiv x_{p,\mathrm{dz}}, \qquad s_{\max} \equiv x_{p,\max}. \tag{3.6.1}$$

The admissible shift domain is therefore

$$\boxed{0 \leq s \leq s_{\max}, \qquad s = s_{\mathrm{dz}} \text{ is the primary engagement boundary.}} \tag{3.6.2}$$

The upper limit $s_{\max}$ is fixed by the permitted travel of the selected hardware. It is not inferred again from the belt width in this section. The belt dimensions have already entered the pulley-radius and belt-loop geometry; the travel limit states where that geometry is physically terminated.

For $0 \leq s < s_{\mathrm{dz}}$, the primary movable sheave has not reached the belt and the deadzone geometry applies. For $s_{\mathrm{dz}} < s \leq s_{\max}$, both pulley contacts exist and the engaged geometry applies. At $s = s_{\mathrm{dz}}$, the positions calculated from the two descriptions agree, but their allowed motions do not. Immediately before engagement, the belt radii and secondary position are fixed while the primary crosses the remaining clearance. Immediately after engagement, further primary motion changes both pulley radii and opens the secondary. In the one-sided notation introduced by these two mechanical arrangements,

$$r_j^-(s_{\mathrm{dz}}) = r_j^+(s_{\mathrm{dz}}), \qquad \left.\frac{dr_j}{ds}\right|_- \neq \left.\frac{dr_j}{ds}\right|_+ \quad \text{in general,} \tag{3.6.3}$$

where superscripts $-$ and $+$ denote the deadzone- and engaged-side mappings, respectively. The coordinate is therefore position-continuous at engagement, but the tangent relating $\dot{s}$ to the physical component velocities changes. This distinction is what makes engagement a mechanical transition rather than an ordinary point inside one set of differential equations.

Position alone also does not determine whether a boundary is active. A state at $s = s_{\mathrm{dz}}$, for example, may be leaving the boundary in free engaged shift, held there by the fully closed secondary stop, or opening into the deadzone after primary separation. The mechanical regime introduced in Section 2.5 records that additional information. Over the complete shift domain, five structural arrangements are retained: free and lower-stop motion in the deadzone, together with free, low-ratio-seat, and upper-stop motion while engaged. Each engaged arrangement also carries one of the tangential contact branches derived in Section 3.5.

The engaged arrangements follow most directly from the closure already in hand.

3.6.2 Continuous Engaged Regimes

The six mechanical rows assembled in Equation (3.5.7) contain two axial balances. A free shift retains both. When a sheave is held at a hardware stop, the axial balance belonging to that stopped member is replaced by the kinematic row $\ddot{s} = 0$. After the constrained mechanical solution is known, the omitted axial balance is evaluated as a residual; that residual is the reaction the stop must supply.

It is useful to state the reaction signs directly from the two movable-sheave free bodies. Define the non-stop closing-force residual of each local balance by

$$
\begin{aligned}
F_{p,\text{free}} &\equiv F_{p,\text{fly}} - F_{p,\text{spring}} - F_{p,\text{belt}}, \\
F_{s,\text{free}} &\equiv F_{s,\text{helix}} + F_{s,\text{spring}} - F_{s,\text{belt}}.
\end{aligned}
\tag{3.6.4}
$$

These are the right-hand sides of Equations (3.2.1) and (3.2.18) before a stop reaction is added. At a constrained state they are evaluated using the accelerations, normal resultants, and belt torques returned by that constrained closure.

Free Engaged Shift

When neither axial stop is active, no boundary reaction acts. The complete instantaneous problem is exactly the freely shifting engaged closure of Section 3.5. Its mechanical unknowns remain

$$
\dot{\omega}_p, \qquad \dot{\omega}_s, \qquad \dot{v}_b, \qquad \ddot{s}, \qquad N_p, \qquad N_s,
$$

while λ_p and λ_s are supplied by the applicable stick–slip construction. Away from the boundaries this regime occupies

$$
s_{\text{dz}} < s < s_{\text{max}}. \tag{3.6.5}
$$

The same free closure may also be evaluated at either endpoint when the corresponding stop is inactive and the motion points back into the engaged domain. The strict interval in Equation (3.6.5) therefore locates its ordinary interior; it is not a rule that every endpoint state must be constrained.

Engaged Low-Ratio Seat

At minimum engaged ratio, the secondary movable sheave is fully closed against its hardware stop. That secondary stop, rather than an unresolved reaction in the primary balance, prevents motion toward a ratio below the engagement geometry. During a sustained seated interval,

$$
\boxed{s = s_{\text{dz}}, \qquad \dot{s} = 0, \qquad \ddot{s} = 0.} \tag{3.6.6}
$$

Both belt–pulley contacts remain present. The rotational balances, primary axial balance, complete-belt transport relation, closed-loop tension compatibility, and the two tangential contact relations therefore remain the same equations used in free engaged shift. Only the secondary axial row is replaced by $\ddot{s} = 0$. Since $\dot{s} = \ddot{s} = 0$, the radius rates and accelerations established in Sections 3.1.4 and 3.1.5 also vanish at the seat.

The omitted secondary equation recovers the force exerted by the fully closed secondary stop. A positive reaction acts in the local opening direction and balances a non-stop tendency to close the secondary farther. With the sign in Equation (3.6.4), its magnitude is

$$\boxed{F_{\text{stop,LR}} = F_{s,\text{free}}\big|_{s=s_{\text{dz}},\,\dot{s}=\ddot{s}=0}\,.} \tag{3.6.7}$$

The stop is unilateral: it can push the secondary open, but it cannot pull the secondary closed. The seated constraint is therefore admissible only while

$$\boxed{F_{\text{stop,LR}} \geq 0.} \tag{3.6.8}$$

If the constrained solution gives $F_{\text{stop,LR}} < 0$, the secondary would have to be pulled against its stop. The constraint releases and the free engaged closure supplies the subsequent upshift acceleration.

The stop reaction and the belt-face normal resultant are different physical loads. The belt still opens the movable secondary through $F_{s,\text{belt}} = N_s \cos\beta/2$, while $F_{\text{stop,LR}}$ acts between the secondary hardware and its travel stop. The latter is not added to N_s and does not increase the secondary traction capacity.

Engaged Upper Primary Stop

At maximum shift, the primary movable assembly reaches its closing travel limit. A continuous upper-stop interval satisfies

$$\boxed{s = s_{\max}, \qquad \dot{s} = 0, \qquad \ddot{s} = 0.} \tag{3.6.9}$$

The engaged contact and belt equations again remain active. Here the primary axial row is replaced by $\ddot{s} = 0$, while the secondary axial balance remains part of the simultaneous mechanical solve. The omitted primary row recovers the opening-direction force supplied by the upper stop:

$$\boxed{F_{\text{stop,max}} = F_{p,\text{free}}\big|_{s=s_{\max},\,\dot{s}=\ddot{s}=0}\,.} \tag{3.6.10}$$

A positive value means the unconstrained primary would continue closing and the stop must push it open. The upper-stop regime is therefore admissible while

$$\boxed{F_{\text{stop,max}} \geq 0.} \tag{3.6.11}$$

When this reaction reaches zero and the free tendency is toward decreasing s, the stop releases without an impulse. The free engaged equations are then reevaluated at the same position and velocity state.

3.6.3 Continuous Deadzone Regimes

When the primary sheaves are separated from the belt, the primary contact resultant and belt torque are absent:

$$\boxed{N_p = 0, \qquad \tau_p = 0.} \tag{3.6.12}$$

No primary traction utilization exists in this topology. The two-wrap tension closure also no longer describes the slack primary side. Under Assumption 9 (Reduced Belt Motion While the Primary Is Disengaged), the entire belt is instead carried by the fully closed secondary at its effective radius. Define

$$r_{s,\text{eff},\text{dz}} \equiv r_{s,\text{eff}}(s_{\text{dz}}). \tag{3.6.13}$$

The deadzone lock is then

$$\boxed{v_b = r_{s,\text{eff},\text{dz}}\omega_s.} \tag{3.6.14}$$

The secondary radius is fixed throughout the deadzone, so differentiating the constraint gives

$$\boxed{\dot{v}_b = r_{s,\text{eff},\text{dz}}\dot{\omega}_s.} \tag{3.6.15}$$

The belt transport inertia can consequently be reflected to the secondary shaft. The secondary and belt obey

$$\boxed{\tau_{\text{B},s} = \left[I_{\text{B},s} + I_{s,F} + I_{s,M} + m_b r_{s,\text{eff},\text{dz}}^2\right]\dot{\omega}_s.} \tag{3.6.16}$$

The effective radius appears in both Equations (3.6.14) and (3.6.16) because it is the tangential no-slip radius relating belt transport to shaft rotation. The belt mass itself remains the complete belt mass m_b; only its transport speed has been referred to the shaft.

Free Primary Motion Through the Deadzone

For

$$0 < s < s_{\text{dz}}, \tag{3.6.17}$$

the primary movable assembly is free. Its previously derived rotational and axial balances are evaluated with the belt quantities in Equation (3.6.12) removed. Those two balances determine $\dot{\omega}_p$ and $\ddot{s}$ from the same primary mechanism used while engaged; entering the deadzone does not freeze the flyweights, ramp, spring, or retained primary inertia. The secondary relation in Equation (3.6.16) determines $\dot{\omega}_s$, and Equation (3.6.15) then determines $\dot{v}_b$.

Thus the free deadzone returns

$$\boxed{\dot{\omega}_p, \qquad \dot{\omega}_s, \qquad \dot{v}_b, \qquad \ddot{s},} \tag{3.6.18}$$

without solving for N_p, N_s, λ_p, λ_s, or a closed-loop tension field. The deadzone belt–secondary lock replaces those contact-level quantities rather than introducing another contact branch.

As in the engaged case, the strict interval in Equation (3.6.17) identifies the ordinary interior. The free deadzone closure may also be evaluated at $s = 0$ while the lower stop is releasing, or at $s = s_{\mathrm{dz}}$ while the primary is opening away from the engagement boundary.

Fully Open Primary Stop

At the fully open primary stop,

$$\boxed{s = 0, \qquad \dot{s} = 0, \qquad \ddot{s} = 0.} \tag{3.6.19}$$

The primary remains disengaged and the secondary–belt lock remains active. The primary axial row is replaced by $\ddot{s} = 0$; its rotational balance is evaluated at fixed shift, and the omitted axial row recovers the force exerted by the lower stop. This stop pushes in the positive, closing direction, so

$$\boxed{F_{\mathrm{stop},0} = -\left. F_{p,\mathrm{free}} \right|_{s=0,\, \dot{s}=\ddot{s}=0,\, N_p=0} .} \tag{3.6.20}$$

The minus sign appears because the lower stop supports an unconstrained opening tendency, opposite the positive primary closing direction. Its unilateral admissibility condition is

$$\boxed{F_{\mathrm{stop},0} \geq 0.} \tag{3.6.21}$$

If this reaction becomes negative, holding the primary open would require the stop to pull. The constraint releases and the free deadzone closure is evaluated at the same state.

The equations above define continuous evolution within each structural arrangement. Reaching a boundary with finite speed requires one additional piece: the velocities must be transferred into the tangent space allowed by the new arrangement.

3.6.4 Momentum-Consistent Boundary Transitions

Collect the four retained generalized velocities as

$$\boldsymbol{\nu} = \begin{bmatrix} \omega_p & \omega_s & v_b & \dot{s} \end{bmatrix}^{\mathsf{T}}. \tag{3.6.22}$$

At a fixed event configuration, these four quantities determine the physical velocities of every retained inertial component. Let

$$\mathbf{y}_q = \mathbf{J}_q \boldsymbol{\nu}, \tag{3.6.23}$$

collect those component velocities in mechanical regime q. The rows of $\mathbf{J}_q$ contain the kinematic maps already derived for shaft rotation, primary translation and flyweight motion, secondary translation and helix-relative rotation, and belt transport. The diagonal matrix $\mathbf{W}$ contains the corresponding masses and inertias, including the equivalent inertias supplied through the shaft interfaces. The kinetic energy represented by a regime is therefore

$$\mathcal{T}_q = \frac{1}{2}\boldsymbol{\nu}^\mathsf{T}\mathbf{J}_q^\mathsf{T}\mathbf{W}\mathbf{J}_q\boldsymbol{\nu}. \tag{3.6.24}$$

The physical components do not change identity when a boundary is crossed, but the matrix mapping the reduced velocities into their motion can change. At engagement, for example, the deadzone-side mapping holds the secondary axial and helix motion fixed, whereas the engaged-side mapping makes those motions depend on $\dot{s}$. Copying the four entries of $\boldsymbol{\nu}$ across that change would not, in general, preserve the momentum of the physical components and can create kinetic energy.

Let $\mathbf{C}_+\boldsymbol{\nu}^+ = \mathbf{0}$ collect any velocity constraints active immediately after an event. Projecting the incoming physical momentum onto every admissible post-event virtual velocity gives

$$\mathbf{J}_+^\mathsf{T}\mathbf{W}\left(\mathbf{J}_+\boldsymbol{\nu}^+ - \mathbf{J}_-\boldsymbol{\nu}^-\right) + \mathbf{C}_+^\mathsf{T}\boldsymbol{\Gamma} = \mathbf{0}, \tag{3.6.25}$$

where $\boldsymbol{\Gamma}$ contains the impulses enforcing the newly active constraints. Together with $\mathbf{C}_+\boldsymbol{\nu}^+ = \mathbf{0}$, this gives the mass-metric projection

$$\boxed{\begin{bmatrix} \mathbf{J}_+^\mathsf{T}\mathbf{W}\mathbf{J}_+ & \mathbf{C}_+^\mathsf{T} \\ \mathbf{C}_+ & \mathbf{0} \end{bmatrix}\begin{bmatrix} \boldsymbol{\nu}^+ \\ \boldsymbol{\Gamma} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_+^\mathsf{T}\mathbf{W}\mathbf{J}_-\boldsymbol{\nu}^- \\ \mathbf{0} \end{bmatrix}.} \tag{3.6.26}$$

The configuration remains continuous through this instantaneous event, so $s^+ = s^-$. The components of $\boldsymbol{\nu}$, however, are solved together rather than copied or reset independently. Under the perfectly inelastic event idealization of Assumption 13 (Instantaneous, Perfectly Inelastic Constraint Activation), the projection satisfies

$$\boxed{\mathcal{T}^+ \leq \mathcal{T}^-.} \tag{3.6.27}$$

Applied torques and other bounded forces contribute no finite impulse over the unresolved event duration. Their associated equivalent inertias remain in $\mathbf{W}$, but their continuous torques return only when evolution resumes.

For the ordering in Equation (3.6.22), the elementary velocity constraints used below have rows

$$\begin{aligned} \mathbf{c}_{\text{stop}} &= \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}, \\ \mathbf{c}_{p,\text{stick}} &= \begin{bmatrix} -r_{p,\text{eff}} & 0 & 1 & 0 \end{bmatrix}, \\ \mathbf{c}_{s,\text{stick}} &= \begin{bmatrix} 0 & -r_{s,\text{eff}} & 1 & 0 \end{bmatrix}. \end{aligned} \tag{3.6.28}$$

The stop row imposes $\dot{s}^+ = 0$. The other two rows impose the shaft-referenced no-slip conditions of Equation (3.5.3). In accordance with Assumption 14 (Tangential Contact Through Boundary Events), a contact already sticking before the event retains its corresponding row during the projection. A sliding contact does not. A newly created primary contact is likewise not assigned an impulsive stick row merely because normal contact has formed.

3.6.5 Transitions at the Engagement Boundary

The two directions through $s = s_{\mathrm{dz}}$ describe different physical events. Closing motion from the deadzone creates primary belt contact and activates the engaged geometry. Opening motion from the engaged side brings the secondary movable sheave back to its fully closed stop and may allow the primary to separate.

Primary Engagement

When a deadzone trajectory reaches $s = s_{\mathrm{dz}}$ with $\dot{s}^- > 0$, the pre-event map in Equation (3.6.26) is the deadzone-side $\mathbf{J}_-$, and the post-event map is the engaged-side $\mathbf{J}_+$. The deadzone secondary–belt lock already exists and is retained through the capture, so $\mathbf{C}_+$ includes the same secondary no-slip row $\mathbf{c}_{s,\mathrm{stick}}$. It does not include $\mathbf{c}_{p,\mathrm{stick}}$, because the primary contact is new.

The event therefore preserves the engagement position while redistributing the incoming momentum among primary rotation, secondary rotation, belt transport, secondary helix motion, and the remaining shift motion. In particular,

$$\boxed{s^+ = s^- = s_{\mathrm{dz}}, \qquad \boldsymbol{\nu}^+ = \Delta_{\mathrm{dz}\to\mathrm{eng}}\left(\boldsymbol{\nu}^-\right),} \tag{3.6.29}$$

where $\Delta_{\mathrm{dz}\to\mathrm{eng}}$ is the projection in Equation (3.6.26) with the constraints just described. The post-event shift speed may remain positive, but it is not assumed equal to its incoming value.

The engaged closure is then solved at the projected state. The new primary relative speed from Equation (3.5.15) determines whether its continuous contact begins in slip or is a candidate for stick. A new normal contact therefore activates the primary belt force and torque without artificially gluing the two tangential velocities together.

Backshift to Minimum Ratio

When free engaged motion reaches $s = s_{\mathrm{dz}}$ with $\dot{s}^- < 0$, the secondary movable sheave reaches its fully closed metal stop. This is not the reverse of first engagement. The impact arrests the secondary axial and helix-relative motion, and the corresponding momentum can be transferred into the secondary shaft and belt. The primary movable assembly is not itself attached to that stop and must not have its velocity independently set to zero.

Under the present rigid topology, a finite-speed return continues through primary separation while retaining the primary motion compatible with the deadzone topology. That transition uses Equation (3.6.26) with the engaged map before the event, the deadzone map afterward, and the secondary–belt lock imposed on the post-event velocity. If the motion instead reaches the zero-speed limiting state while primary contact and the secondary stop reaction remain admissible, the sustained low-ratio seat of Section 3.6.2 becomes the continuous successor.

The two possible departures from an established seat are distinct. If $F_{\mathrm{stop,LR}}$ reaches zero and the free engaged tendency is toward increasing s, the secondary stop releases into free engaged upshift. If the primary contact instead separates, the system enters the deadzone and the secondary–belt lock is activated by the same momentum-consistent projection.

Primary separation is not authorized by one signed actuator force or by $N_p = 0$ alone. At the seated boundary it requires both

$$\boxed{N_p = 0, \qquad \ddot{s}_{\mathrm{cf}} < 0,} \tag{3.6.30}$$

where $\ddot{s}_{\mathrm{cf}}$ is the acceleration obtained from the contact-free primary deadzone balance at the same state. The first condition removes the compressive belt support; the second verifies that, once that support is removed, the primary actually accelerates open. Requiring both prevents a momentary zero normal resultant from causing disengagement when the free mechanism would immediately close the contact again.

Remark (Rigid low-ratio capture)

A perfectly rigid, perfectly inelastic description can generate a shrinking sequence of secondary-stop impacts and primary make–break events as the remaining shift kinetic energy approaches zero. This sequence introduces no additional continuous mechanical regime. A numerical implementation may complete the limiting capture once the remaining removable kinetic energy is indistinguishable from zero at its working precision; that threshold is a numerical completion criterion, not a new physical parameter.

3.6.6 Metal-Stop Arrival and Release

The lower primary stop is reached from free deadzone motion at $s = 0$ with $\dot{s}^- < 0$. The pre- and post-event topology is deadzone, while $\mathbf{C}_+$ contains both the stop row and the existing secondary–belt lock. The projection sets the compatible post-event shift speed to zero without deleting momentum stored in the other retained components. The lower-stop closure is then solved and $F_{\mathrm{stop},0}$ is recovered. If that reaction is negative, the impact still occurred, but the stop cannot support the following continuous motion and releases immediately into free deadzone motion.

The upper primary stop is reached from engaged motion at $s = s_{\max}$ with $\dot{s}^- > 0$. Here the topology remains engaged, the stop row is added, and any primary or secondary contact already sticking retains its no-slip row through the impact. The complete velocity projection is required because removing $\dot{s}$ also removes the secondary movable sheave's helix-relative velocity. Its angular momentum is redistributed through the same impulse balance rather than being discarded by a scalar $\dot{s} \leftarrow 0$ reset.

After the upper-stop projection, the contact branch is reevaluated using the fixed-shift closure and $F_{\mathrm{stop},\max}$ is recovered. A negative reaction causes immediate release into free engaged motion. During an established stop interval, a reaction reaching zero produces a nonimpulsive release: the position and velocities pass through unchanged, and only the continuous acceleration law changes.

The physical event surfaces are the exact boundaries

$$s = 0, \qquad s = s_{\mathrm{dz}}, \qquad s = s_{\max}. \tag{3.6.31}$$

Their crossing direction distinguishes arrival from departure and engagement from backshift. Numerical event localization may identify a value differing from one of these surfaces by roundoff, but that does not move the physical boundary or create a tolerance-width regime.

3.6.7 Successor Admissibility and Complete Regime Map

An event does not end when a candidate successor label has been chosen. The mechanical closure associated with that successor must be solved at the event state, because changing a shift constraint or tangential contact branch changes the accelerations, normal resultants, required static tractions, and recovered stop reactions together.

The successor can begin a continuous segment only when all conditions that define that regime are satisfied:

- every active metal-stop or low-ratio-seat reaction is nonnegative;
- both normal resultants are nonnegative in an engaged regime, with primary separation at the low-ratio boundary governed by Equation (3.6.30);
- every sticking or sliding contact satisfies the branch conditions of Section 3.5.4; and
- every imposed velocity constraint is satisfied by the post-event state.

If a candidate contact change makes an already-active stop reaction negative, the stop releases at the same event time. Likewise, if a new fixed-shift closure changes the admissible contact branch, that branch is selected before continuous integration resumes. The model therefore never begins a segment under a constraint that would have to pull or under a tangential contact state that its own coupled solution cannot support.

The nonnegative belt tension and distributed normal loading required by Assumption 8 (Prescribed Planar Belt Path) remain validity conditions on every engaged solution. They do not create additional structural regimes in the present model: violation means that the prescribed engaged belt path itself is no longer an admissible description.

Table 3.1 collects the five structural arrangements and the quantities returned by their continuous closures. The three engaged rows are each paired with one of the four contact combinations in Equation (3.5.19); those contact combinations are subregimes of the same structural mechanics rather than additional pulley configurations.

Structural regime	**Active mechanical condition**	**Instantaneous CVT results**	**Continuation condition**
Deadzone lower stop	Primary disengaged; secondary–belt lock; $s = 0$, $\dot{s} = \ddot{s} = 0$	$\dot{\omega}_p$, $\dot{\omega}_s$, $\dot{v}_b$, and $F_{\text{stop},0}$	$F_{\text{stop},0} \geq 0$
Deadzone free	Primary disengaged; secondary–belt lock; shift unconstrained	$\dot{\omega}_p$, $\dot{\omega}_s$, $\dot{v}_b$, and $\ddot{s}$	State remains inside the deadzone until a directed boundary event
Engaged low-ratio seat	Both pulley contacts; fully closed secondary stop; $s = s_{\text{dz}}$, $\dot{s} = \ddot{s} = 0$	$\dot{\omega}_p$, $\dot{\omega}_s$, $\dot{v}_b$, N_p, N_s, λ_p, λ_s, and $F_{\text{stop,LR}}$	Admissible contact branch, $F_{\text{stop,LR}} \geq 0$, and no primary separation
Engaged free shift	Both pulley contacts; shift unconstrained	The eight quantities in Equation (3.5.2)	Admissible contact branch and no directed shift-boundary arrival
Engaged upper stop	Both pulley contacts; primary upper stop; $s = s_{\max}$, $\dot{s} = \ddot{s} = 0$	$\dot{\omega}_p$, $\dot{\omega}_s$, $\dot{v}_b$, N_p, N_s, λ_p, λ_s, and $F_{\text{stop,max}}$	Admissible contact branch and $F_{\text{stop,max}} \geq 0$

Table 3.1: Instantaneous CVT results and continuation conditions for the five structural operating regimes. Quantities fixed by an active constraint, such as $\ddot{s} = 0$, are shown in the mechanical-condition column rather than as free results.

The transition graph is consequently branched rather than a reversible linear sequence. Deadzone engagement can capture directly into free engaged motion. Return to the low-ratio boundary can produce secondary-stop impact followed by primary separation, while an established low-ratio seat can release either to free engaged upshift or to the deadzone for different physical reasons. The upper and lower metal stops likewise retain the system only when their post-event reactions remain compressive.

If $s_{\text{dz}} = 0$, the deadzone has zero width. The two deadzone rows in Table 3.1 then disappear, and the shared lower boundary is the engaged low-ratio seat. No separate neutral successor is introduced.

The complete CVT evolution is now determined by one consistent rule. The current state and regime select a continuous closure; that closure advances the state until one of its defining conditions reaches a boundary; any newly active kinematic constraint then transfers the retained momentum through Equation (3.6.26); and the successor closure is accepted only after its own contact and unilateral-reaction conditions have been satisfied.

Chapter 4

Results and Discussion

4.1 Simulation Results

This section compares four launch simulations: a nominal baseline case, a lighter-flyweight case, a steeper-helix case with reduced secondary torque reactivity, and a combined case with both a shallower initial primary ramp and a steeper helix. The purpose of these results is not to match experimental vehicle data directly, but to show that the model produces mechanically interpretable launch behaviour and responds to tuning changes in physically sensible ways.

The results are organized around the launch shift curves of these four cases. These curves first provide an overall picture of the launch, including engagement, low-ratio operation, active upshift, and high-ratio behaviour. The ratio terminology follows ??: low ratio means larger R (greater speed reduction), and high ratio means smaller R. following subsections then explain those trends by examining the axial force balance across the CVT and the resulting torque-transfer and slip behaviour. In particular, the combined case is included to show a clearly slip-dominated launch, in which the model predicts loss of the low-ratio hold and sustained slip through the early portion of the launch.

Taken together, the four cases are intended to show three things: first, that the baseline tuning produces a coherent and interpretable launch trajectory; second, that changes to primary and secondary tuning shift the launch behaviour in the expected direction; and third, that when the tuning is made sufficiently unfavorable, the model captures the transition to prolonged

slipping rather than continuing to predict an unrealistic no-slip response. The implementation used to generate these simulations, along with repository, release, and data-access details, is documented in Section 4.2.

4.1.1 Simulation Cases and Phase Convention

The results section compares four full-throttle launch simulations from rest. One case is treated as the nominal baseline, while the remaining three are targeted tuning variations chosen to isolate specific changes in CVT behaviour. All cases use the same vehicle, gearbox, tire, and engine model parameters; they differ only in the clutch tuning parameters noted below.

Table 4.1: Launch cases considered in the results section. All parameters not listed remain equal to the nominal baseline.

Case	Description	Purpose in results section
D	Default tuning	Baseline reference case
F	Lighter flyweights	Shows the effect of reduced primary centrifugal clamping
H	Steeper helix	Shows the effect of reduced secondary torque reactivity
C	Shallower initial primary ramp + steeper helix	Produces a strongly slip-dominated launch

To keep the discussion consistent across the different plots, the launch is described in terms of a small number of recurring operating phases. These phases are not introduced as separate events for each case, but as common regions of behaviour used to interpret the launch trajectories and their underlying force and torque balances.

Throughout the following subsections, these phases will be indicated directly on the plots using low-opacity background shading. The same color convention is used wherever possible so that the reader can track the evolution of each case across the shift, clamping-force, and torque-transfer results.

The phase colors are

- **engagement** — light blue,
- **low-ratio operation** — light green,
- **active upshift** — light yellow, and
- **high-ratio operation** — light red.

Not every case is expected to exhibit all four phases equally clearly. The phase shading is therefore used as a visual guide to the dominant operating regime, rather than as a claim that every boundary is perfectly sharp.

4.1.2 Launch Shift Curves

The overall launch behaviour of the four tuning cases is first compared in Figure 4.1. Each trajectory is plotted in engine-speed–vehicle-speed coordinates, so that engagement, ratio change, and the final high-ratio condition can all be viewed on the same axes. All four simulations begin from the same launch initial conditions,

$$\omega_p(0) = 1800\,\text{rpm}, \qquad \omega_s(0) = 0,$$

where ω_p is the primary angular speed and ω_s is the driven angular speed corresponding to zero vehicle speed. From this state, the system is released into a full-throttle forward simulation using the complete coupled model developed in the preceding sections.

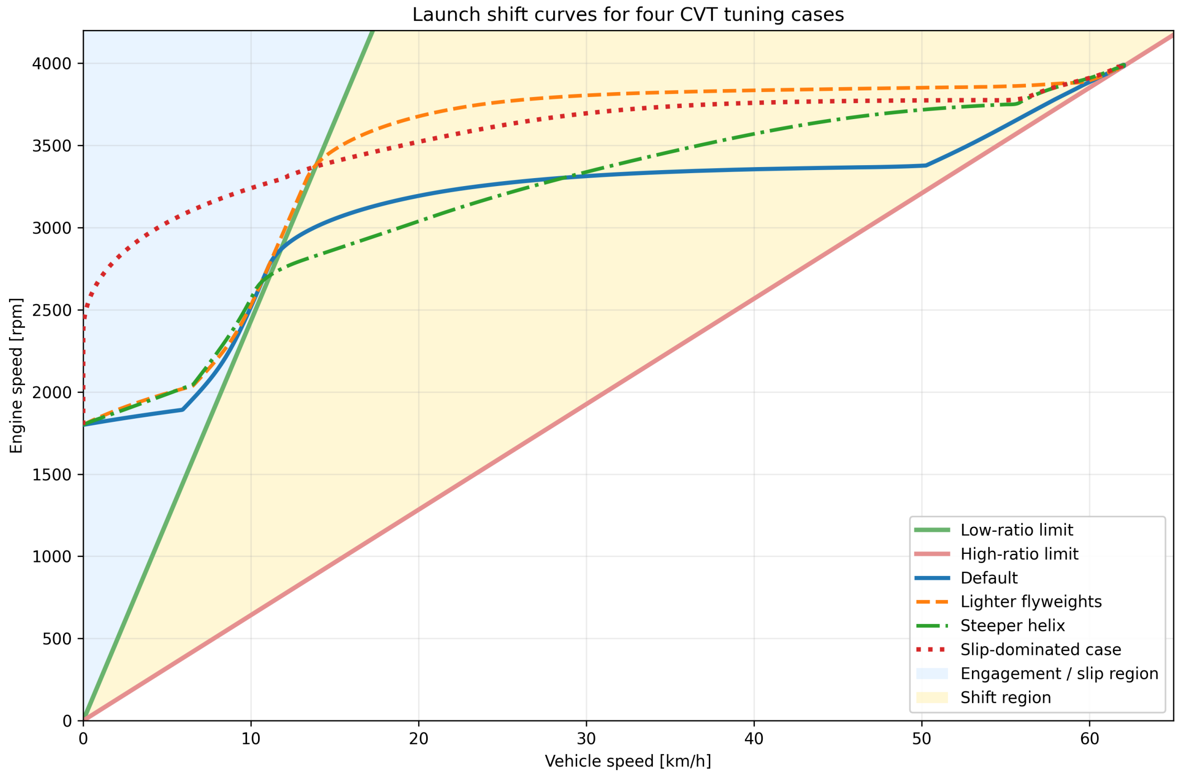

Figure 4.1: Engine-speed–vehicle-speed launch trajectories for the four tuning cases. The green line is the low-ratio limit, corresponding to a fully unshifted CVT. The red line is the high-ratio limit, corresponding to a fully shifted CVT. The blue shaded region to the left of the low-ratio line corresponds to launch engagement, where slip must be present. The yellow region between the two limit lines is the no-slip shift region.

The two reference lines are the kinematic no-slip limits of the CVT. The **low-ratio line** represents the fully unshifted transmission, where the CVT is in its torque-multiplying launch condition. The **high-ratio line** represents the fully shifted transmission, where the CVT has reached its final speed-increasing condition. Between these two limits lies the admissible shift region. A trajectory cannot move to the right of the high-ratio line, and any portion of a trajectory lying to the left of the low-ratio line must involve slip, since even the fully unshifted CVT cannot support such a low vehicle speed for the corresponding engine speed.

Read from left to right, the figure therefore shows the full launch sequence. The launch begins in the blue engagement region, where the vehicle is still too slow for no-slip operation. After crossing the low-ratio boundary, the trajectory enters the shift region. In a well-behaved launch, the CVT then tends to move approximately horizontally across this region, increasing vehicle speed while holding engine speed near a preferred operating value. This nearly horizontal portion will be referred to here as the *flat-shift region*. As the shift completes, the trajectory approaches the high-ratio line, which forms the final no-slip boundary of the launch.

With that interpretation in place, the differences between the four tuning cases are already visible. The **default** case shows the clearest baseline behaviour: a short engagement region, a brief low-ratio hold, and then a relatively flat shift through the middle of the plot before approaching the high-ratio boundary. This is the most balanced of the four trajectories, since it neither leaves low ratio too early nor rises excessively in engine speed during the main shift.

The **lighter-flyweight** case retains low ratio for longer before crossing into the shift region. Its flat-shift portion is also noticeably higher in engine speed than the default case, so the launch spends more of the main shift at elevated primary speed. The overall shape is still recognizable as a conventional launch trajectory, but the shift is delayed and carried out at a higher engine-speed level.

The **steeper-helix** case behaves differently. It leaves the low-ratio boundary much earlier, so the launch shows little sustained low-ratio hold. Its shift path is also less flat: instead of moving nearly horizontally across the shift region, it climbs upward more strongly as vehicle speed increases. This indicates that the launch does not settle into the same nearly constant engine-speed shift seen in the default case.

The **combined slip-dominated** case is the most extreme. It remains in the engagement region for much longer, showing no clear low-ratio hold before entering the shift region. Once it does cross into the no-slip range, its shift still occurs at a comparatively high engine speed, and the early launch is much more dominated by slip than by clean low-ratio operation.

At this stage, Figure 4.1 is intended to establish the overall shape of the four launches and the main qualitative differences between them. The key observations are the amount of low-ratio retention, the height and flatness of the main shift segment, and the extent of the initial slip-dominated portion of the launch. The following subsections explain these differences in terms of the underlying axial force balance and torque-transfer behaviour of the CVT.

4.1.3 Axial Force Balance Through the Launch

The shift-curve trajectories of Figure 4.1 indicate when each case remains near low ratio, when it begins to upshift, and how quickly it reaches high ratio. To explain those differences mechanically, Figure 4.2 compares the total axial force acting on the primary and secondary pulleys for each tuning case, with the same launch phases shaded in the background.

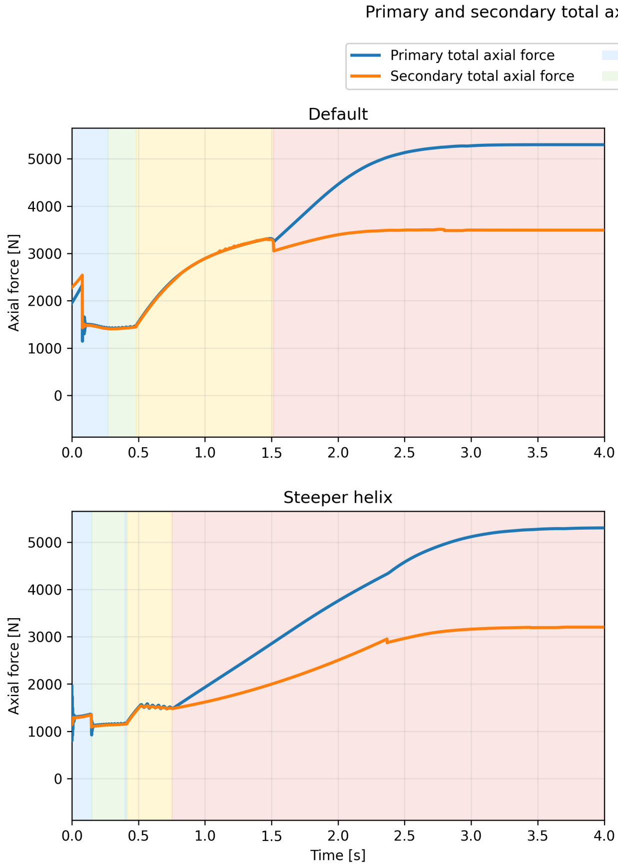

Figure 4.2: Primary and secondary total axial forces during launch for the four tuning cases. Blue curves show total primary axial force, orange curves show total secondary axial force. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as force-balance diagrams for the shift. While the secondary axial force exceeds the primary axial force, the transmission is held toward low ratio. Once the primary force overtakes the secondary, the pulley pair is driven into upshift. The timing of that crossover therefore determines how long the launch remains near low ratio and how early the transmission begins moving through its ratio range.

The **default** case shows the most balanced behaviour. During the initial clutching phase, the secondary force begins noticeably above the primary force, so the transmission is held near its low-ratio condition rather than shifting immediately. This produces the short but clear low-ratio hold seen previously in Figure 4.1. As the launch progresses, the primary axial force rises and eventually overtakes the secondary, initiating the main upshift. Importantly, the secondary force does not remain fixed while this happens. It also rises throughout the shift, so the system continues to resist ratio change even after upshift begins. This is mechanically important: if the secondary force stayed nearly constant while the primary force rose rapidly, the transmission would sweep through the shift range much more abruptly. Instead, both sides grow together, giving the default case its smoother and more controlled shift.

The **lighter-flyweight** case makes this point especially clear. Relative to the default case, the primary force is reduced in the early launch, while the secondary remains comparatively strong.

The initial force gap is therefore larger, meaning the low-ratio condition is held more firmly and for longer. This matches the longer low-ratio segment seen in the shift curves. Later in the launch, the primary still rises enough to drive upshift, but it does so later and at a higher engine-speed level than in the default case. In other words, this case does not fundamentally change the character of the shift; it mainly delays it by weakening the primary closing tendency during engagement and early launch.

The **steeper-helix** case behaves differently. Here, the initial difference between secondary and primary force is very small, and the primary overtakes the secondary much earlier. This explains why the corresponding shift curve showed little sustained low-ratio hold and entered the shifting region so quickly. We also see a differently shaped force evolution throughout the shift itself. In the default and lighter-flyweight cases, both forces continue rising together in a gradual way through the main shift region. In the steeper-helix case, both forces rise sharply early on and then level off much sooner. The two forces still remain relatively close to one another, but the overall shape they trace is different. This helps explain why the earlier shift curve did not display the same flat-shift behaviour as the default case. Rather than maintaining a long, nearly constant-speed shift, the transmission shows a more rising shift trajectory, consistent with this less gradual force evolution.

The **slip-dominated** case is the most extreme. Its early launch is characterized by a prolonged clutching phase and no meaningful low-ratio hold. This is consistent with the earlier shift-curve figure, where the trajectory spent much longer in the slip-dominated region before entering the no-slip shift region. Importantly, the beginning of geometric shift does not imply that the system has already stopped slipping. It only indicates that the shift coordinate has begun to move away from its initial value. The more important feature in this case is the low magnitude of both axial forces in the early launch. Compared with the other cases, neither pulley develops a strong axial loading during the initial portion of the event. This weak overall clamping is a key reason why this case behaves poorly, and it will be seen more directly in the following torque-transfer plots where the associated slip remains elevated for much longer.

One additional feature is visible near the beginning of each case. The early spike in secondary axial force is the torque-reactive contribution of the secondary. It appears most strongly during the initial launch when transmitted torque rises rapidly, and it provides an immediate resisting effect against premature upshift. This feature is especially useful because it shows that the secondary is responding dynamically to transmitted torque rather than acting as a static spring-only restraint. The influence of this torque reactivity will be seen more directly in the following torque-transfer results, but its signature is already visible here in the early force histories.

Taken together, Figure 4.2 provides the mechanical explanation for the differences observed in Figure 4.1. A larger initial gap between secondary and primary force corresponds to stronger low-ratio retention, as seen in the lighter-flyweight case. A very small gap corresponds to premature upshift, as seen in the steeper-helix case. The default case lies between these extremes and therefore produces the most balanced launch. Perhaps most importantly, the figure shows that the shift is not driven by the primary alone. The secondary rises with it throughout the launch, and the actual shift behaviour emerges from the evolving balance between the two rather than from either force in isolation. At the same time, this figure only explains the shift tendency and overall clamping levels; it has not yet quantified the resulting slip behaviour, which is addressed next.

4.1.4 Torque Bounds, No-Slip Demand, and Coupling Torque

The force-balance results of Figure 4.2 explain when each case tends to remain at low ratio and when it begins to shift. However, force balance alone does not determine whether the belt is actually able to transmit the torque required for sticking. To examine that question, Figure 4.3 compares four quantities during the launch: the primary and secondary traction bounds, the no-slip coupling torque τ_{ns}, and the actual transmitted coupling torque τ_c.

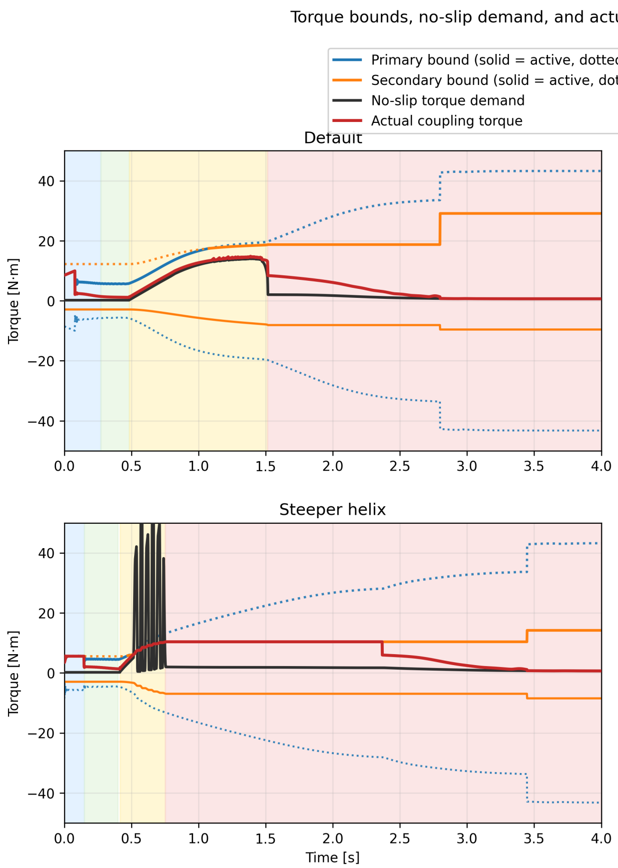

Figure 4.3: Torque bounds, no-slip demand, and actual coupling torque for the four tuning cases. Blue and orange curves show the primary and secondary bounds, with the active bound drawn solid and the inactive bound dotted. The black curve shows the no-slip torque demand τ_{ns}, and the red curve shows the actual transmitted coupling torque τ_c. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as traction-capacity diagrams. At any instant, the belt can only remain in stick if the required no-slip torque lies inside the admissible interval defined by the active upper and lower bounds. When τ_{ns} lies inside that interval, the actual coupling torque can follow the no-slip demand closely. When τ_{ns} exceeds the available bounds, sticking is no longer admissible and the transmitted torque must saturate at the corresponding active limit.

A first broad trend is immediately visible. The cases that settle out of slip early in the launch are precisely the cases with larger admissible positive torque capacity during the initial clutching and low-ratio phases. In the **default** and **lighter-flyweight** cases, the upper torque bound rises to a relatively large value early in the launch. This gives the system enough admissible coupling torque to reduce the belt-speed mismatch and recover toward sticking. By contrast, the **slip-dominated** case begins with essentially no admissible positive torque and only develops usable capacity more gradually. That lower early torque capacity directly explains why the slip-dominated case spends much longer in the clutching region.

This same distinction appears in the evolution of τ_c. In the cases that recover stick early, the actual coupling torque relaxes toward and then tracks the no-slip demand through the end of clutching and into low-ratio operation. Once low-ratio launch is established, the red and black curves remain close, indicating that the required torque stays admissible and sustained slip is not reintroduced immediately.

A second important pattern appears once the transmission begins to shift. In all four cases, the no-slip torque rises during the shift phase. This is consistent with the form of τ_{ns}, in which the ratio-rate term becomes active once the CVT begins moving away from its initial ratio. In the **default** and **lighter-flyweight** cases, this rise remains below the admissible upper torque bound, so the actual coupling torque continues to follow the no-slip demand and large renewed slip does not occur. In the **steeper-helix** and **slip-dominated** cases, however, τ_{ns} rises above the admissible upper bound during the shift. At that point, sticking is no longer feasible, the actual coupling torque saturates at the active maximum, and renewed slip appears until the state returns closer to the sticking condition.

The **steeper-helix** and **slip-dominated** cases also show visibly oscillatory behaviour in the no-slip torque during the shift onset. This is consistent with the oscillations already seen in the preceding axial-force plots. The most direct source is the ratio-rate contribution to τ_{ns}: if the shift dynamics are not smooth, then the implied no-slip torque need not evolve smoothly either. In these two poorer tuning cases, the shift motion appears more weakly damped, so the no-slip torque exhibits stronger excursions during the transition into active shifting. A plausible reason is that the present model neglects parasitic losses and other non-critical frictional effects, which in a real system would provide additional damping and suppress some of this oscillatory behaviour. This is therefore a useful limitation to note: the model captures the overall stick–slip transitions and the active limiting mechanism, but the transient oscillations in the most aggressive cases are likely amplified by the intentionally idealized loss assumptions.

Another common feature is the gradual relaxation of the actual coupling torque toward the no-slip demand rather than an instantaneous snap to it. This comes from the regularized transition law used in the simulation, discussed in Appendix B.3, which replaces a discontinuous switch with a smooth tanh-based approximation. Its purpose is numerical: it reduces chattering and improves stability near the stick–slip transition. The effect is visible here as a slight lag in the recovery of perfect sticking, but it does not change the main qualitative behaviour of the four cases. The same limiting patterns are still present, and the cases that lose admissibility remain clearly distinguishable from those that do not.

Finally, each case shows a noticeable jump in the admissible torque bound once full sticking is re-established near the end of the launch. This reflects the change in limiting behaviour used by the model after slip is removed, together with the corresponding increase in effective friction behaviour assumed in the sticking regime. The important point for the present discussion is that this jump occurs only after the system has already relaxed back toward the no-slip state; it is therefore a consequence of stick recovery rather than its cause.

Taken together, Figure 4.3 closes the interpretation begun in the previous two subsections. The shift curves showed the overall launch behaviour, and the axial-force plots explained the tendency to hold or leave low ratio. The present figure then shows whether the required coupling torque is actually admissible throughout those phases. The default and lighter-flyweight cases maintain sufficient positive torque capacity to recover stick early and remain near the no-slip demand through the main shift. The steeper-helix and slip-dominated cases do not: in both, the no-slip torque exceeds the available upper bound during the shift, forcing the actual coupling torque to saturate and reintroducing slip. In this way, the torque plots both confirm the solid patterns seen earlier and reveal one of the model's clearest shortcomings: the most weakly damped cases display exaggerated oscillatory transients due to the idealized loss assumptions.

4.1.5 Launch Performance Comparison

The preceding subsections established the mechanical differences between the four tuning cases. The shift-curve plots showed the overall launch trajectory, the axial-force plots explained the tendency to hold or leave low ratio, and the torque-bound plots showed whether the required coupling torque remained admissible throughout the launch. The final step is to compare how those differences affect overall launch performance.

Two simple metrics are used here: the time required to travel 50 m, and the time required to reach 60 km/h. These are useful because they summarize the launch in two complementary ways. The 50 m time reflects the integrated quality of the launch as a whole, while the 60 km/h time emphasizes how quickly the drivetrain can build vehicle speed through the main acceleration phase.

Table 4.2: Launch performance metrics for the four tuning cases.

Case	**Time to** 50 m [s]	**Time to** 60 km/h [s]
Default	3.93	2.27
Lighter flyweights	4.26	2.78
Steeper helix	4.19	2.82
Slip-dominated case	4.41	2.98

The default case performs best on both metrics, which is consistent with the earlier mechanism-level results. It develops enough admissible coupling torque to reduce slip early, retains low ratio long enough to launch effectively, and then transitions into a controlled shift without reintroducing large sustained slip. This is the most balanced use of the CVT: the transmission first allows a strong initial torque transfer into the driveline, and then continues accelerating the vehicle while keeping the shift progression smooth and admissible.

This point is worth emphasizing. The coupling torque plays two roles at once. It is the torque that loads the engine through the transmission, limiting how freely the primary (and therefore the engine) can accelerate, and it is also the torque delivered through the CVT that accelerates the vehicle from rest. A strong early coupling torque is therefore beneficial when it is admissible, because it quickly pulls the belt-speed mismatch down and simultaneously applies useful drive torque to the vehicle. The default case benefits from exactly this behaviour: it uses the initial slip region to establish a relatively large transmitted torque early in the launch, which helps "kick-start" vehicle acceleration before the main shift begins.

The lighter-flyweight case performs worse on both metrics, even though its shift remains comparatively well behaved. This matches the earlier interpretation that its main drawback is delayed primary loading during the early launch. By holding low ratio longer but developing less aggressive early coupling, it sacrifices initial acceleration and does not recover that loss later.

The steeper-helix case shows a different trade-off. It leaves low ratio too early and re-enters slip during the shift, which degrades the quality of the launch. However, because it still develops appreciable coupling torque early in the event, its 50 m time remains slightly better than the lighter-flyweight case. Its slower time to 60 km/h, on the other hand, is consistent with the less stable shift behaviour and the renewed slip seen in the torque-bound plots.

The slip-dominated case performs worst overall. This is the expected result from the earlier figures: its early clamping forces are weakest, its admissible torque rises too slowly, and its launch remains slip-dominated for too long. As a result, it spends too much of the start of the event establishing torque transfer rather than converting that torque cleanly into forward acceleration.

Overall, these performance metrics support the mechanism-level interpretation developed throughout the section. The best launch is not produced by simply holding low ratio as long as possible, nor by shifting as early as possible. Instead, good performance requires a balance: sufficient early admissible coupling torque to accelerate the vehicle strongly from rest, enough low-ratio retention to avoid premature upshift, and a smooth enough shift that the required no-slip torque does not repeatedly exceed the available traction limits.

4.2 Implementation and Reproducibility

The source code for the CVT simulator used in this work is available at `https://github.com/gr812b/CVT-Simulator`. This implementation includes the mathematical model developed in the main derivation sections along with the practical numerical regularization strategy described in Section B.3.

To support reproducibility, the exact version used to generate the figures and tables in this manuscript will be archived as a tagged release:

- repository release (to be finalized): `https://github.com/gr812b/CVT-Simulator/releases/tag/vX.Y.Z`,
- archival DOI landing page (placeholder): `https://doi.org/XX.XXXX/zenodo.XXXXXXX`.

Unless otherwise stated, simulation cases in Section 4.1 correspond to that archived release. The repository README and release notes describe the execution workflow, configuration files, and plotting scripts used to produce the reported results.

An interactive web deployment of the simulator is also available for demonstration purposes: `https://cvt.kaiarseneau.dev`. This live endpoint is provided for accessibility and does not replace the archived release as the canonical reproducibility artifact.

4.3 Conclusion

This work developed a unified first-principles dynamic model for a mechanically actuated dry rubber V-belt CVT and its coupling to the vehicle drivetrain. Starting from pulley geometry, belt-length closure, actuator mechanics, and rotational dynamics, the derivation produced a closed nonlinear state-space formulation in which ratio evolution, torque transmission, slip admissibility, and vehicle response are solved consistently in time.

The completed formulation brings together several mechanisms that strongly govern mechanically actuated CVT behaviour: primary flyweight actuation, secondary torque-reactive helix clamping, belt geometric closure, axial shift dynamics, traction-limited torque transfer, and longitudinal drivetrain coupling. The main outcome is a single dynamic framework in which these effects interact directly, while remaining mechanically interpretable in terms of the underlying pulley forces, ratio change, and torque limits.

The simulation results showed coherent launch behaviour across the tuning cases considered. The shift-curve results captured distinct differences in engagement, low-ratio retention, main-shift shape, and approach to high ratio. The axial-force results showed that these differences could be traced back to the evolving balance between primary and secondary clamping. The torque-bound results then showed when the no-slip coupling demand remained admissible and when renewed slip was forced by the available traction limits. Together, these results formed a consistent mechanism chain from tuning changes, to force balance, to torque admissibility, to overall launch behaviour.

An important outcome of the study is that the model preserved physical intuition across multiple levels. Changes in tuning did not simply produce different output curves; they produced changes that remained understandable in mechanical terms. Stronger or weaker low-ratio retention appeared through the initial clamping-force gap, earlier or later upshift appeared through the timing of force crossover, and reintroduced slip appeared when the no-slip torque rose beyond the admissible coupling bounds. This consistency gives confidence that the formulation is capturing the dominant interactions it was intended to represent.

The results also highlighted useful limitations of the present model. In the more aggressive cases, the no-slip torque demand exhibited visibly underdamped oscillatory transients during shift onset. These were consistent with irregular shift-rate evolution in the model and suggest that the idealized loss assumptions, particularly the neglect of parasitic losses and other non-critical frictional effects, leave the system with less damping than would be expected in real hardware. The formulation also remains limited to one-dimensional vehicle motion and does not yet include thermal effects, wear progression, or more detailed tribological contact evolution.

Overall, the present work provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behaviour during transient launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behaviour remains traceable back to clear mechanical causes. With further parameter identification, experimental validation, and extension to include more realistic damping and loss mechanisms, the framework developed here can serve as a useful basis for both engineering interpretation and future design-oriented CVT simulation.

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Appendix A

Belt Length Small-Angle Approximation

This appendix quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models. Using parameter values representative of the physical system considered in this work, we demonstrate that the approximation introduces substantial error in the computed center-to-center distance, then further in the ratio, which is critical for accurate dynamic modeling.

A.1 Standard Approximation

Many simplified treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this approximation, the belt-length constraint **??** reduces to

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{A.1.1}$$

A.2 Derivation of the Approximate Center Distance

To determine the center-to-center distance at the initial configuration, we substitute $r_p = r_{p,\text{outer,min}}$ and $r_s = r_{s,\text{outer,max}}$ into Equation (A.1.1) and solve for C.

Step 1: Isolate the square root. Rearranging Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) = 2\sqrt{C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}. \tag{A.2.1}$$

Step 2: Square both sides.

$$\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2 = 4\left[C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2\right]. \tag{A.2.2}$$

Step 3: Solve for C^2.

$$C^2 = \frac{\big[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\big]^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2. \tag{A.2.3}$$

Step 4: Take the square root. Expanding the squared term in the numerator:

$$\big[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\big]^2 = L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2,$$

the approximate center distance becomes

$$\boxed{C_{\text{approx}} = \sqrt{\frac{L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}.} \tag{A.2.4}$$

This closed-form expression avoids the need for numerical root finding but neglects the $\sin^{-1}$ term in the exact constraint.

A.3 Error Analysis

We now evaluate the approximation error using parameter values from the physical CVT system considered in this work.

System Parameters

Parameter	Value	SI Value
Belt length L_b	37.538 in	0.9535 m
Belt height h_b	0.613 in	0.015 57 m
Groove half-angle β	11.5°	0.2007 rad
Inner primary radius	0.8125 in	0.020 637 5 m
Primary outer radius $r_{p,\text{outer,min}}$	1.4255 in	0.036 21 m
Secondary outer radius $r_{s,\text{outer,max}}$	4.0 in	0.1016 m

The primary outer radius is computed as the inner radius plus belt height: $r_{p,\text{outer,min}} = 0.8125 + 0.613 = 1.4255$ in.

Computed Center Distances Solving the exact belt-length constraint **??** numerically yields

$$C_{\text{exact}} = 0.251\,74\,\text{m} \approx 9.911\,\text{in}.$$

Evaluating the approximate formula Equation (A.2.4) yields

$$C_{\text{approx}} = 0.268\,37\,\text{m} \approx 10.566\,\text{in}.$$

Error Metrics The absolute and relative errors are:

$$\Delta C = C_{\text{approx}} - C_{\text{exact}} = 0.016\,63\,\text{m} = 0.655\,\text{in}, \tag{A.3.1}$$

$$\varepsilon_{\text{rel}} = \frac{\Delta C}{C_{\text{exact}}} = \frac{0.01663}{0.25174} = 6.61\%. \tag{A.3.2}$$

Assessment The small-angle approximation introduces a relative error of nearly 7% in the center-to-center distance for this system. This error arises because the ratio

$$\frac{r_{s,\text{outer,max}} - r_{p,\text{outer,min}}}{C} = \frac{0.1016 - 0.03621}{0.25174} = 0.260$$

is not small. The assumption $\lambda \approx 0$ is therefore not justified for this CVT geometry.

Since the center distance C propagates into all subsequent geometric calculations—wrap angles, tangent span lengths, and derived quantities such as the transmission ratio—a 7% error in C can compound into larger errors in predicted system behavior.

A.4 Propagation to Secondary Radius and Transmission Ratio

To demonstrate how the center distance error propagates into operationally relevant quantities, we now examine the secondary radius and CVT ratio at two shift positions: zero shift and maximum shift.

A.4.1 Derivation of the Approximate Secondary Radius

Given a known primary outer radius $r_{p,\text{outer}}$ and center distance C, the secondary outer radius can be obtained by solving the belt-length constraint. Under the small-angle approximation Equation (A.1.1), this yields a closed-form expression.

Step 1: Isolate the square root. From Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer}} + r_{s,\text{outer}}) = 2\sqrt{C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2}. \tag{A.4.1}$$

Step 2: Square both sides.

$$\left[L_b - \pi r_{p,\text{outer}} - \pi r_{s,\text{outer}}\right]^2 = 4\left[C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2\right]. \tag{A.4.2}$$

Step 3: Expand and collect terms. Expanding the left side:

$$(L_b - \pi r_{p,\text{outer}})^2 - 2\pi(L_b - \pi r_{p,\text{outer}})r_{s,\text{outer}} + \pi^2 r_{s,\text{outer}}^2.$$

Expanding the right side:

$$4C^2 - 4r_{s,\text{outer}}^2 + 8r_{p,\text{outer}}r_{s,\text{outer}} - 4r_{p,\text{outer}}^2.$$

Collecting powers of $r_{s,\text{outer}}$ yields a quadratic equation:

$$(\pi^2+4)r_{s,\text{outer}}^2 - 2\left[\pi L_b - \pi^2 r_{p,\text{outer}} + 4r_{p,\text{outer}}\right]r_{s,\text{outer}} + \left[(L_b - \pi r_{p,\text{outer}})^2 + 4r_{p,\text{outer}}^2 - 4C^2\right] = 0. \tag{A.4.3}$$

Step 4: Apply the quadratic formula. The physically meaningful root (smaller secondary radius for overdrive configuration) is

$$\boxed{r_{s,\text{approx}} = \frac{\pi L_b + (4-\pi^2)r_{p,\text{outer}} - 2\sqrt{(\pi^2+4)C^2 - L_b^2 + 4\pi L_b r_{p,\text{outer}} - 4\pi^2 r_{p,\text{outer}}^2}}{\pi^2+4}.} \tag{A.4.4}$$

This closed-form expression enables direct computation of the approximate secondary radius without numerical root finding.

A.4.2 Zero Shift Configuration

At zero shift, the CVT is in its lowest ratio (maximum reduction) configuration. This is also the geometry used to compute the approximate center distance C_{approx}, so much of that initial center-distance error cancels when the same configuration is used again to recover the secondary radius.

Primary Radius at Zero Shift At $s = 0$, the primary is within the deadzone, so from **??**:

$$r_{p,\text{outer}}(0) = r_{p,\text{outer,min}} = 0.036\,21\,\text{m}.$$

Secondary Radius Comparison Solving the exact belt-length constraint with C_{exact} yields

$$r_{s,\text{exact}} = 0.101\,60\,\text{m}.$$

Evaluating the approximate formula Equation (A.4.4) with $C_{\text{approx}} = 0.268\,37\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.101\,60\,\text{m}.$$

Transmission Ratio Error Computing effective radii with $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	Exact	Approximate
$r_{p,\text{eff}}$	0.0284 m	0.0284 m
$r_{s,\text{eff}}$	0.0938 m	0.0938 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	3.300	3.300

The ratio error at zero shift is

$$\varepsilon_R = \frac{|3.300 - 3.300|}{3.300} \approx 0.000\%.$$

Thus, although C_{approx} is already 6.61% too large, the induced error has little visible effect at minimum shift because the radius calculation is being evaluated at the same operating point used to construct that approximate center distance in the first place.

Figure A.1: Scaled diagram of the CVT at zero shift ($s = 0$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The two solutions nearly overlap at this operating point, indicating negligible ratio error in this case.

A.4.3 Maximum Shift Configuration

At maximum shift, the primary and secondary pulleys are closest in size, representing a near-crossover configuration.

Primary Radius at Maximum Shift Using ?? with $s = s_{\max} = 0.75\,\text{in} = 0.019\,05\,\text{m}$ and deadzone $s_{\text{dz}} = 0.002\,489\,2\,\text{m}$:

$$\begin{aligned} r_{p,\text{outer}}(s_{\max}) &= r_{p,\text{outer,min}} + \frac{s_{\max} - s_{\text{dz}}}{2\tan\beta} \\ &= 0.03621 + \frac{0.01905 - 0.0024892}{2 \times \tan(11.5°)} \\ &= 0.03621 + \frac{0.01656}{0.4070} \\ &= 0.0769\,\text{m}. \end{aligned} \tag{A.4.5}$$

Secondary Radius Comparison Solving the exact belt-length constraint ?? numerically with $C_{\text{exact}} = 0.251\,74\,\text{m}$ yields

$$r_{s,\text{exact}} = 0.066\,73\,\text{m}.$$

Evaluating the approximate formula Equation (A.4.4) with $C_{\text{approx}} = 0.268\,37\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.056\,26\,\text{m}.$$

Transmission Ratio Error The CVT ratio is defined using effective radii, which account for belt seating depth (??):

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}.$$

With $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	**Exact**	**Approximate**
$r_{p,\text{eff}}$	0.0691 m	0.0691 m
$r_{s,\text{eff}}$	0.0589 m	0.0485 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	0.853	0.701

Both models predict overdrive ($R < 1$), but the approximation substantially underestimates the ratio. The ratio error is

$$\varepsilon_R = \frac{|0.701 - 0.853|}{0.853} = 17.8\%.$$

Unlike the minimum-shift case, this operating point does not benefit from the same cancellation, so the center-distance error propagates into a much larger error in secondary radius and transmission ratio.

Figure A.2: Scaled diagram of the CVT at maximum shift ($s = s_{\max}$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The approximation underestimates the secondary radius, leading to a lower predicted ratio than the exact model.

A.5 Summary of Approximation Errors

Configuration	R_{exact}	R_{approx}	**Ratio Error**
Zero shift ($s = 0$)	3.300	3.300	0.000%
Maximum shift ($s = s_{\text{max}}$)	0.853	0.701	17.8%

The small-angle approximation error is strongly operating-point dependent. At zero shift, most of the center-distance error cancels because the approximate center distance was itself constructed from that same geometry. At maximum shift, that cancellation is lost and the ratio error becomes substantial.

These errors arise from the initial 6.61% error in center distance, which propagates nonlinearly through the belt-length geometry in a state-dependent way. Since the transmission ratio directly enters the rotational equations of motion (**??**), this approximation can still produce materially inaccurate dynamics near maximum shift.

This analysis justifies the use of exact numerical solutions for all geometric computations in this work.

Appendix B

Numerical Methods Summary

This section documents the numerical algorithms used for belt-geometry root finding and dynamic simulation, including the specific tolerances, bracketing strategies, and solver configurations employed in the implementation.

B.1 Root Finding for Belt-Length Constraint

Given a known primary outer radius r_1 and fixed center-to-center distance C, the secondary outer radius r_2 is the unique root of the belt-length constraint. This scalar root problem is obtained by rearranging the exact belt length closure from **??** into the residual form:

$$g(r_2;\, r_1) = \pi(r_1 + r_2) + 2(r_2 - r_1)\arcsin\left(\frac{r_2 - r_1}{C}\right) + 2\sqrt{C^2 - (r_2 - r_1)^2} - L_b = 0, \tag{B.1.1}$$

where L_b is the fixed belt length. A separate invocation solves for the center-to-center distance C itself at initialization, using the same equation evaluated at the boundary radii $(r_{1,\min},\, r_{2,\max})$ with C as the unknown; this is exactly the initialization step implied by **??** before the ratio law **??** can be evaluated over the shift range.

Bracketing strategy. The arcsin term in Equation (B.1.1) is real only when $|r_2 - r_1| \leq C$, so the natural domain for r_2 is the open interval $(r_1 - C,\, r_1 + C)$. Within this interval a sign change is guaranteed: at the lower boundary $r_2 \to r_1 - C$, the arcsin saturates to $-\pi/2$ and the square root vanishes, giving $g \to 2\pi r_1 - 2\pi C - L_b < 0$ for all physically valid parameters. At the upper boundary $r_2 \to r_1 + C$, by symmetry $g \to 2\pi r_1 + 2\pi C - L_b > 0$, satisfied whenever the belt length is less than the circumference subtended by the wider pulley pair — a requirement any real CVT must satisfy. A small safety margin $\varepsilon = 10^{-9}$ m is applied at both ends to keep arcsin away from its domain boundary where the function becomes numerically steep.

For center-to-center distance solving, the bracket is instead the interval $(\Delta r + \varepsilon,\ L_b/2)$, where $\Delta r = |r_{2,\text{eff}} - r_{1,\text{eff}}|$ is the effective-radius difference at the boundary configuration. The lower bound ensures arcsin is defined; the upper bound is a conservative geometric limit (a straight two-span belt cannot exceed L_b across a gap of $L_b/2$). Both brackets are validated by an explicit sign-change check before invoking the solver, so any failure to bracket immediately identifies an infeasible geometry rather than producing a silent wrong answer.

Brent's method. Both roots are found using `scipy.optimize.brentq`, an implementation of Brent's algorithm Brent (1973). The method combines three strategies: bisection (guaranteed bracket-halving), secant interpolation (superlinear convergence near smooth roots), and inverse quadratic interpolation (order-≈ 1.84 convergence when three previous iterates are well-conditioned). Whenever the faster interpolation steps would place the next iterate outside the current bracket, the algorithm falls back to bisection, preserving the worst-case $O(\log_2(1/\varepsilon))$ guarantee of pure bisection. Because the sign-change brackets described above are guaranteed for all valid CVT geometries, `brentq` is certain to converge. Both root solves use an absolute tolerance of 10^{-9} m, well below any physically meaningful length precision.

B.2 ODE Integration for Dynamic Simulation

The full CVT state vector $\mathbf{x} = (\omega_p, \omega_s, s, \dot{s}, v_b)$ is integrated using `scipy.integrate.solve_ivp`. The governing system is written as a first-order ODE system in **??**, with rotational sub-equations **??** and **??**, axial dynamics **??**, belt-speed dynamics for v_b, and algebraic geometry closures **??** and **??**. Although **??** is second-order in s, introducing $(s, \dot{s})$ as separate states converts the complete model to first-order form for time integration. The time integrator is used with the `RK45` method — the Dormand–Prince explicit embedded Runge–Kutta pair Dormand and Prince (1980). The method propagates the state with a 5th-order formula and simultaneously computes a 4th-order companion estimate for local error control, using the same six function evaluations. The tableau is FSAL (first-same-as-last): the final stage of each accepted step is reused as the first stage of the next, so only five new evaluations are needed per step in steady operation.

Adaptive step size control. At each candidate step the per-component local error estimate $\hat{e}_i$ is compared to a mixed tolerance threshold. A step is accepted when

$$\max_i \frac{|\hat{e}_i|}{\mathrm{atol} + \mathrm{rtol} \cdot |y_i|} \leq 1, \tag{B.2.1}$$

and the next step size is scaled proportionally to $(\mathrm{tol}/\hat{e})^{1/5}$. The tolerances used in this work are $\mathrm{atol} = 10^{-6}$ and $\mathrm{rtol} = 10^{-4}$ (Table B.1). The absolute floor prevents spurious step rejection when a state variable passes through zero (e.g. $\dot{s} \approx 0$ at shift equilibrium or $\omega_p \approx 0$ at engine start), where a purely relative criterion would demand unreachable accuracy.

Parameter	**Value**	**Interpretation**
`atol`	10^{-6}	absolute floor: 1 µm on s, 1 µrad/s on ω
`rtol`	10^{-4}	relative tolerance: 0.01% of current state magnitude
`t_eval`	10 000 pts	uniform output grid over 30 s

Table B.1: ODE solver tolerance parameters.

Event detection. Phase transitions (shift engagement, boundary contact, back-shift onset) are located using `solve_ivp`'s event mechanism. These events correspond to qualitative changes in the dynamics around the bounded shift coordinate used in **??** and the full state evolution in **??**. After each accepted step the solver evaluates the event functions on a degree-4 dense-output interpolant and applies a root-finding pass to detect any sign change within the step. When a crossing is found it is bracketed at sub-tolerance precision before being reported, so transitions are located with the same accuracy as the integration tolerance rather than being snapped to the coarse output grid.

Phase splitting at hard constraints. The shift bounds $s \in [0, s_{\max}]$ with $\dot{s} = 0$ at both endpoints are handled by partitioning the trajectory into separate `solve_ivp` calls rather than imposing inequality constraints inside the ODE. Each phase — normal shifting, clamped-at-maximum, and back-shifting — presents a smooth right-hand side to the integrator, allowing RK45 to operate at full 5th-order accuracy throughout. The terminal event of each phase supplies the initial condition of the next. This is numerically cleaner than directly clamping **??** inside a single solve because it preserves a smooth right-hand side within each phase. Up to ten alternating full-shift / back-shift cycles are supported before the simulation terminates, preventing unbounded phase loops for unusual parameter regimes.

B.3 Slip-Law Regularization and Blend Parameter Selection

The traction law in **??** is physically clear but discontinuous at $v_{\mathrm{rel}} = 0$, which can induce step-size chatter near stick–slip transitions. A practical regularized formulation uses three ideas: a smooth saturation law, a clamped no-slip torque request, and a blending rule between those two limits.

First define a smooth saturation torque

$$\tau_{\mathrm{sat}} = \tau_{\max} \tanh\left(\frac{v_{\mathrm{rel}}}{v_{\mathrm{smooth}}}\right), \tag{B.3.1}$$

where $v_{\mathrm{smooth}} > 0$ sets the transition width in slip-speed units. For $|v_{\mathrm{rel}}| \gg v_{\mathrm{smooth}}$, this approaches the usual friction-limited value $\pm\tau_{\max}$, while near zero it removes the sharp sign discontinuity.

Next, clamp the no-slip torque request:

$$\tau_{\mathrm{req}} = \mathrm{clip}(\tau_{\mathrm{ns}},\, -\tau_{\max},\, \tau_{\max})\,. \tag{B.3.2}$$

This is the largest admissible torque consistent with the no-slip request, without allowing the requested value to exceed traction capacity.

Then define the blend factor

$$\alpha = \mathrm{clip}\left(\frac{|v_{\mathrm{rel}}|}{v_{\mathrm{blend}}},\, 0,\, 1\right), \tag{B.3.3}$$

with $v_{\mathrm{blend}} > 0$ defining the width of the blending region, and use the transmitted torque

$$\tau_c = (1 - \alpha)\, \tau_{\mathrm{req}} + \alpha\, \tau_{\mathrm{sat}}. \tag{B.3.4}$$

This has a useful interpretation. For very small mismatch, $\alpha \approx 0$, so the model stays close to the clamped no-slip demand. For larger mismatch, $\alpha \to 1$, so the law transitions to the smooth Coulomb limit. In other words, the regularized model behaves like a requested no-slip torque near stick and like a friction-limited sliding law once the mismatch becomes appreciable.

How to choose v_{smooth}. Choose the smoothing scale small enough that $\tanh(v_{\mathrm{rel}}/v_{\mathrm{smooth}})$ is close to a sign law through most of the slip regime, but not so small that the solver again sees an effectively discontinuous transition. A practical workflow is:

1. Start with v_{smooth} around 0.5–2% of a representative relative-speed magnitude seen during actual slip.
2. Decrease it until predicted macro outputs (ratio trajectory, launch time, peak speeds) stop changing materially.
3. Increase it slightly if event localization or adaptive step-size behavior becomes noisy near $v_{\text{rel}} \approx 0$.

The blend scale v_{blend} should generally be chosen greater than or equal to v_{smooth}, so that the handoff from clamped demand to the smooth Coulomb limit is not sharper than the smoothing law itself.

Appendix C

Rate and Acceleration of the Geometric CVT Ratio

The main derivation uses the individual pulley-radius rates and accelerations from Section 3.1.4. Those same results can be combined to show how quickly the available geometric CVT ratio changes as the sheaves move.

Recall the geometric CVT ratio from **??**:

$$R_g = \frac{r_s}{r_p}, \tag{C.0.1}$$

The individual primary and secondary radius rates and accelerations have already been established in Section 3.1.4. This appendix only combines those results to show how the available geometric ratio itself changes as the sheaves move.

C.0.1 Rate of Change of the Geometric CVT Ratio

Both the numerator and denominator in Equation (C.0.1) can change with time. Using the quotient rule,

$$\begin{aligned}
\dot{R}_g &= \frac{d}{dt}\left(\frac{r_s}{r_p}\right) \\
&= \frac{\dot{r}_s r_p - r_s \dot{r}_p}{r_p^2} \\
&= \frac{\dot{r}_s}{r_p} - \frac{r_s}{r_p^2}\dot{r}_p.
\end{aligned} \tag{C.0.2}$$

The secondary-radius result in Equation (3.1.57) supplies the coupled secondary motion. Within the active-shift branch,

$$\dot{r}_s = -\frac{\phi_p}{\phi_s}\dot{r}_p. \tag{C.0.3}$$

Substituting this into Equation (C.0.2) gives

$$\begin{aligned}\dot{R}_g &= \frac{1}{r_p}\left(-\frac{\phi_p}{\phi_s}\dot{r}_p\right) - \frac{r_s}{r_p^2}\dot{r}_p \\ &= -\frac{\dot{r}_p}{r_p}\left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p}\right) \\ &= -\frac{\dot{r}_p}{r_p}\left(\frac{\phi_p}{\phi_s} + R_g\right).\end{aligned} \tag{C.0.4}$$

Finally, Equation (3.1.36) gives $\dot{r}_p = \dot{s}/(2\tan\beta)$ in the active-shift branch. Therefore,

Rate of Change of Geometric CVT Ratio **(C.0.5)**

$$\dot{R}_g = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ -\dfrac{\dot{s}}{2\tan\beta\, r_p}\left(\dfrac{\phi_p}{\phi_s} + R_g\right), & s_{\mathrm{dz}} < s \le s_{\max}. \end{cases}$$

For an upshift, $\dot{s} > 0$, so the available geometric reduction decreases as the belt moves outward on the primary pulley.

C.0.2 Geometry Families in the Radius Plane

The preceding result gives the local rate at which the geometric ratio changes once the pulley geometry is known. To interpret that result in terms of actual CVT design choices, it is useful to first ask which pulley-radius combinations are available in the first place.

Recall once more that the geometric CVT ratio is

$$R_g = \frac{r_s}{r_p}. \tag{C.0.6}$$

Thus, every point in the (r_p, r_s) plane corresponds to one geometric ratio, and every constant-ratio contour is a line of geometrically similar pulley states. This view is helpful because it separates three different design questions that are easy to blur together:

1. What geometric ratio does a given radius pair provide?
2. Which radius pairs are compatible with a selected belt and installed geometry?
3. How does the chosen compatible path influence the way ratio changes as the CVT shifts?

The first question is answered by the constant-R_g contours. The other two come from the belt-length compatibility relation. Once one of the global geometric quantities—belt length L_b or center-to-center distance C—is fixed, the compatibility relation defines a family of allowable radius pairs in the same plane.

Figure C.1 compares these two views directly. The left panel fixes the center distance and varies the belt length, while the right panel fixes the belt length and varies the center distance. In both panels, the solid contours are constant geometric ratio. The dashed families are the geometrically compatible radius pairs selected by the remaining design choice.

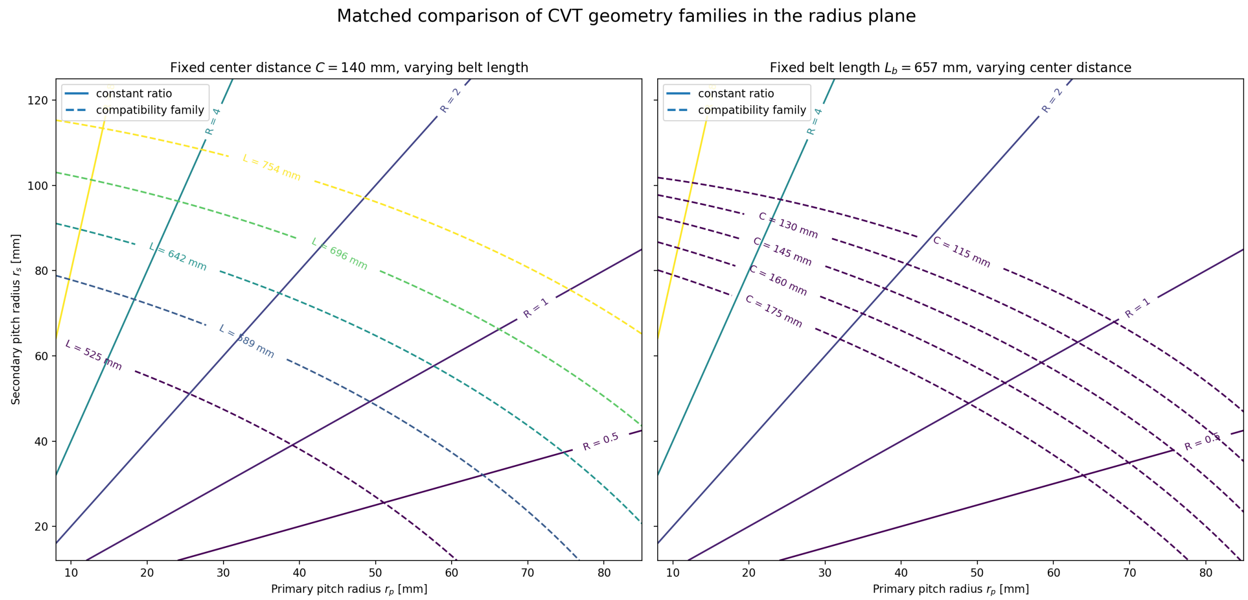


Figure C.1: Matched comparison of geometry families in the pulley-radius plane. In both panels, the solid contours are constant geometric ratio $R_g = r_s/r_p$. In (a), the center-to-center distance is fixed and the dashed families correspond to different belt lengths. In (b), the belt length is fixed and the dashed families correspond to different center-to-center distances.

This figure makes two points especially clear.

First, meeting a target ratio aperture is not only a matter of choosing two endpoint ratios. The selected belt and installed geometry determine which radius pairs are even reachable. A longer or shorter belt shifts the compatible family through the plane, and so does a change in center distance. Since the constant-ratio lines are not parallel to those compatibility families, changing L_b or C changes not only the reachable endpoints, but also the geometric route between them.

Second, the figure makes the roles of belt selection and packaging easier to distinguish. The belt length is largely a component-selection choice, while the center distance is largely a packaging choice. Both appear in the same compatibility relation, but they move the admissible family in different ways. This is why a target ratio range cannot be discussed cleanly without also specifying either the chosen belt or the available center-to-center distance.

In this sense, Figure C.1 is the static geometric view of the design problem. It shows what ratio states are available. The next question is dynamic in a geometric sense: once the CVT moves along one of those compatible paths, how much ratio change does a small increment of axial travel produce?

C.0.3 Exploring the Local Ratio-Change Map

The expression for $\dot{R}_g$ gives the instantaneous rate of change of the geometric ratio. To compare different parts of the geometry without committing to a particular time history, it is useful to divide out the shift speed and look instead at the local ratio change produced by a small increment of axial travel:

$$\frac{dR_g}{ds} = -\frac{1}{2\tan\beta\, r_p}\left(\frac{\phi_p}{\phi_s} + R_g\right). \tag{C.0.7}$$

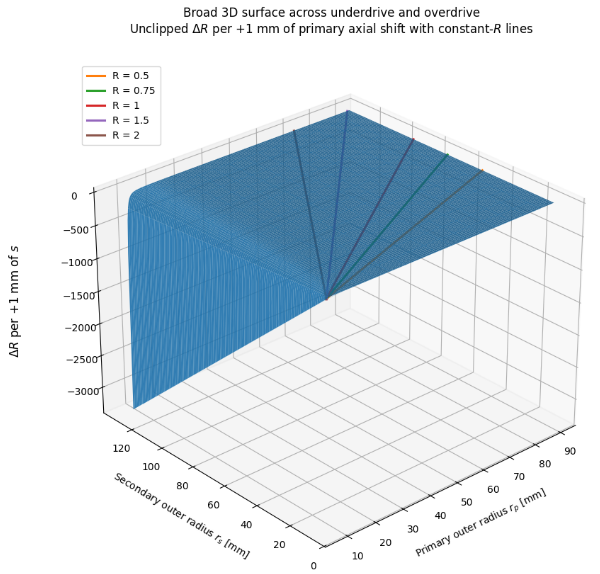
For a small positive increment Δs,

$$\Delta R_g \approx \frac{dR_g}{ds}\Delta s. \tag{C.0.8}$$

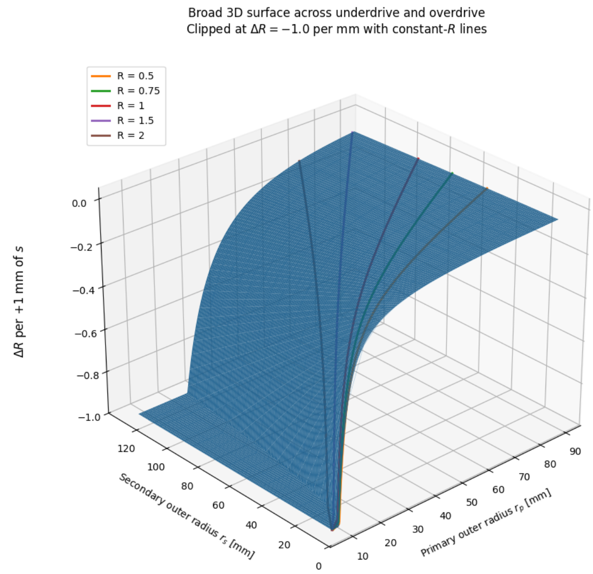
The plots below take $\Delta s = 1\,\mathrm{mm}$, so the displayed quantity may be read as the local change in geometric ratio caused by an additional 1 mm of primary closure. The negative sign simply reflects the usual upshift direction: increasing s reduces the geometric reduction ratio.

A sensitivity field over the radius plane The first view is a broad surface over the radius plane. Here, r_p and r_s are treated as independent coordinates so that every part of the geometry can be inspected. A real CVT does not wander arbitrarily across this surface; its fixed belt length and center distance constrain it to one belt-length-compatible path. Even so, the full surface is valuable because it shows where the geometry itself makes ratio change especially strong or weak.

Figure C.2 shows two versions of this surface. The unclipped view exposes the dominant trend: the local ratio change becomes much larger in magnitude as the primary radius becomes small. This follows directly from the explicit $1/r_p$ factor in Equation (C.0.7), and is reinforced by the fact that the geometric ratio $R_g = r_s/r_p$ is also largest in that same region. The clipped view hides the most extreme values so that the otherwise compressed mid-range curvature can be seen more clearly. The orange curve marks the $R_g = 1$ locus. It is a convenient reference, but not a special boundary in the local ratio-change law itself.



(a) Broad radius domain. The full scale exposes the rapidly increasing magnitude of local ratio change near small r_p.



(b) The same domain with values below -1 per 1 mm clipped, exposing the curvature through the mid-range.

Figure C.2: Local geometric-ratio change for a 1 mm increase in primary axial closure across a broad pulley-radius domain. The orange curve marks the $R_g = 1$ locus.

How belt choice and installed geometry select a path The broad surface is useful, but a particular CVT design only experiences the part of that field lying along its own compatible geometry path. This is where the earlier radius-plane view becomes directly relevant. Once a belt length and center distance are fixed, the CVT is constrained to one compatibility curve in the (r_p, r_s) plane. As the primary sheave closes, the operating point moves along that curve and samples the local ratio-change field point by point.

Figure C.3 overlays three representative belt-length-compatible paths on the local ratio-change surface. Each path corresponds to a different belt-length/reference-geometry choice. The figure shows that two designs with similar overall ratio aperture can still distribute their ratio change very differently over the available shift travel. One path may enter the steep small-r_p region early, concentrating a large amount of ratio change into a short portion of the stroke. Another may spend more of the stroke in the flatter region, distributing the same overall change more gradually.

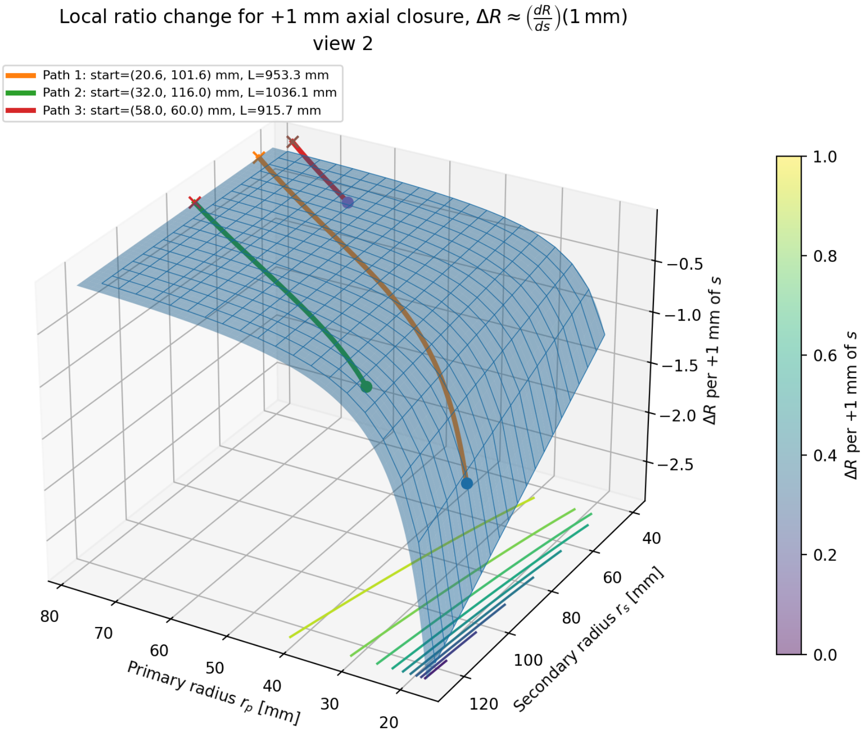
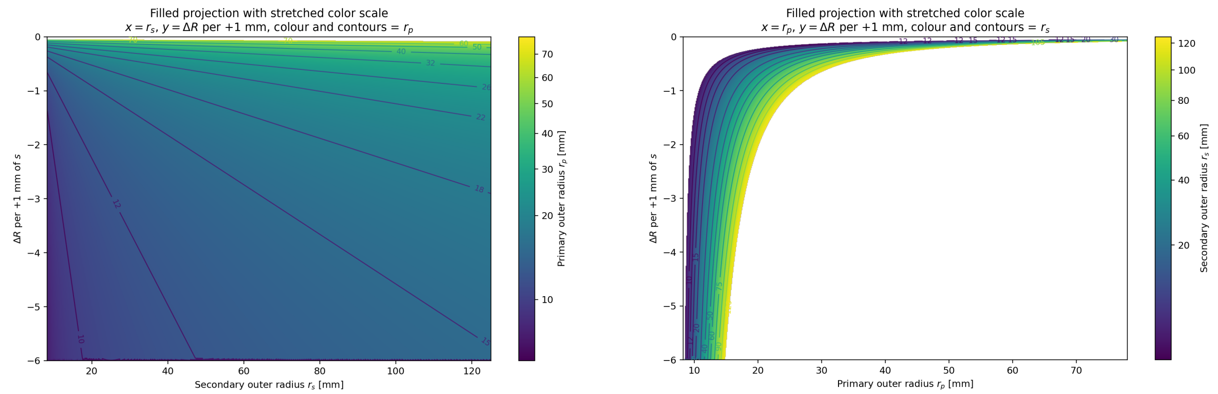


Figure C.3: Representative fixed-belt-length paths through the local ratio-change surface. Each coloured curve is a geometrically compatible pulley-radius path for a different belt-length/reference-geometry choice. The surface value gives the local change in geometric ratio associated with an additional 1 mm of primary closure at that point.

This is the key design value of the map: it does not merely say what overall ratio range is available. It shows *where along the shift stroke* that ratio change is spent.

Separating the roles of the two pulley radii The two projections in Figure C.4 help isolate the individual effects of r_p and r_s. In Figure C.4(a), the horizontal axis is the secondary radius and the contours identify the primary radius. For a fixed primary radius, increasing r_s increases the geometric ratio and makes the local ratio change more negative. The effect is visible, but the contour spacing shows that it remains secondary to the influence of the primary radius.

Figure C.4(b) makes the primary-radius effect more explicit. Every secondary-radius contour bends sharply downward as r_p approaches its lower limit, then flattens at larger radius. This is the visual form of the $1/r_p$ dependence in Equation (C.0.7). Thus, the small-primary-radius region is not only where the geometric reduction is large; it is also where the ratio becomes most sensitive to additional axial motion.



(a) Local ratio change against secondary radius. Colour and contours identify the primary radius.

(b) Local ratio change against primary radius. Colour and contours identify the secondary radius.

Figure C.4: Two projections of the local ratio-change surface for a 1 mm primary closure. The projections separate the contributions of the two radii and show the dominant sensitivity to small primary radius.

Design interpretation Taken together, Figure C.1 and Figure C.2–Figure C.4 give a more complete geometric design picture than endpoint ratios alone.

The radius-plane view answers the question of feasibility. It makes clear how a target ratio aperture depends on the selected belt length and the available center-to-center distance. A longer or shorter belt can move the admissible family enough to change both the reachable ratio range and the portion of the plane through which the CVT shifts. Likewise, a packaging change in center distance can move the same belt into a very different family of compatible radius pairs.

The ratio-change view answers the question of distribution. Once a design selects its belt and installed geometry, the compatible path through the plane determines where ratio change is concentrated. A path that reaches very small r_p can produce large ratio change in little axial travel, which may be attractive when launch or climbing performance demands strong reduction with limited shift stroke. A path that stays farther from that corner distributes ratio change more gently and may be easier to control or to package within a desired sheave travel.

This does *not* mean that the smallest possible r_p is always best. The present plots are geometric only. They show the leverage of the geometry, but not its cost. Experimental work by Chen et al. (1998) showed that, for their rubber-belt CVT, the highest transmission efficiency occurred near a 1:1 speed ratio, reminding us that the most attractive geometric reduction state is not automatically the most efficient operating state. More directly, Childs and Cowburn (1987) identified additional loss penalties at small pulley radii in V-belt drives, with belt-carcass warping and shearing proposed as important contributors. In other words, the same small-r_p region that offers strong geometric leverage is also the region where the belt is bent and seated most severely.

For that reason, these plots are best used as an early design lens rather than a final design verdict. They help screen combinations of belt length, center distance, and pulley radii by showing not only how much ratio range is available, but how that range is distributed over the stroke. Later traction, loss, and dynamic analyses can then be used to judge whether the geometrically attractive part of the design space is physically supportable (Gerbert, 1996; Bertini et al., 2014; Messick, 2018).

C.0.4 Acceleration of the Geometric CVT Ratio

The ratio rate can change for two reasons. The primary sheave may accelerate, and the pulley geometry continues to change as the belt moves. Differentiate Equation (C.0.2) term by term to capture both effects.

For the first term, $\dot{r}_s/r_p$, use the quotient rule:

$$\frac{d}{dt}\left(\frac{\dot{r}_s}{r_p}\right) = \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s\dot{r}_p}{r_p^2}. \tag{C.0.9}$$

For the second term, $r_s\dot{r}_p/r_p^2$, all three factors may change. Using the product rule,

$$\frac{d}{dt}\left(\frac{r_s\dot{r}_p}{r_p^2}\right) = \frac{\dot{r}_s\dot{r}_p}{r_p^2} + \frac{r_s\ddot{r}_p}{r_p^2} - \frac{2r_s\dot{r}_p^2}{r_p^3}. \tag{C.0.10}$$

Subtracting Equation (C.0.10) from Equation (C.0.9) gives

$$\begin{aligned}
\ddot{R}_g &= \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s\dot{r}_p}{r_p^2} - \left[\frac{\dot{r}_s\dot{r}_p}{r_p^2} + \frac{r_s\ddot{r}_p}{r_p^2} - \frac{2r_s\dot{r}_p^2}{r_p^3}\right] \\
&= \frac{\ddot{r}_s}{r_p} - \frac{2\dot{r}_s\dot{r}_p}{r_p^2} - \frac{r_s\ddot{r}_p}{r_p^2} + \frac{2r_s\dot{r}_p^2}{r_p^3}.
\end{aligned} \tag{C.0.11}$$

The main derivation gives the required coupled secondary motion in **????**:

$$\dot{r}_s = -\frac{\phi_p}{\phi_s}\dot{r}_p, \tag{C.0.12}$$

$$\ddot{r}_s = -\frac{\phi_p}{\phi_s}\ddot{r}_p - \frac{8\pi^2}{\phi_s^3\sqrt{C^2-(r_s-r_p)^2}}\dot{r}_p^2. \tag{C.0.13}$$

Substituting these into Equation (C.0.11) gives

$$\begin{aligned}
\ddot{R}_g = \frac{1}{r_p}&\left[-\frac{\phi_p}{\phi_s}\ddot{r}_p - \frac{8\pi^2}{\phi_s^3\sqrt{C^2-(r_s-r_p)^2}}\dot{r}_p^2\right] - \frac{2}{r_p^2}\left[-\frac{\phi_p}{\phi_s}\dot{r}_p\right]\dot{r}_p \\
&- \frac{r_s}{r_p^2}\ddot{r}_p + \frac{2r_s}{r_p^3}\dot{r}_p^2 \\
= &-\frac{1}{r_p}\left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p}\right)\ddot{r}_p + \left[\frac{2}{r_p^2}\left(\frac{\phi_p}{\phi_s} + \frac{r_s}{r_p}\right) - \frac{8\pi^2}{\phi_s^3 r_p\sqrt{C^2-(r_s-r_p)^2}}\right]\dot{r}_p^2 \\
= &-\frac{1}{r_p}\left(\frac{\phi_p}{\phi_s} + R_g\right)\ddot{r}_p + \left[\frac{2}{r_p^2}\left(\frac{\phi_p}{\phi_s} + R_g\right) - \frac{8\pi^2}{\phi_s^3 r_p\sqrt{C^2-(r_s-r_p)^2}}\right]\dot{r}_p^2.
\end{aligned} \tag{C.0.14}$$

Finally, use the primary radius rate and acceleration from Equations (3.1.36) and (3.1.65):

Acceleration of Geometric CVT Ratio **(C.0.15)**

$$\ddot{R}_g = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ -\dfrac{\ddot{s}}{2\tan\beta\, r_p}\left(\dfrac{\phi_p}{\phi_s} + R_g\right) + \dfrac{\dot{s}^2}{2\tan^2\beta\, r_p^2}\left(\dfrac{\phi_p}{\phi_s} + R_g\right) - \dfrac{2\pi^2 \dot{s}^2}{\phi_s^3 r_p \tan^2\beta\sqrt{C^2 - (r_s - r_p)^2}}, & s_{\mathrm{dz}} < s \le s_{\max}. \end{cases}$$

The ratio acceleration has two sources. The term proportional to $\ddot{s}$ is the direct conversion of primary-sheave acceleration into ratio acceleration. Its magnitude is scaled by the local ratio sensitivity dR_g/ds, so the same axial acceleration produces a larger ratio response where the geometric ratio changes rapidly with shift.

The terms proportional to $\dot{s}^2$ arise even when the primary sheave moves at constant axial speed. They reflect the curvature of the geometric relation $R_g(s)$: as the belt moves through the sheaves, the ratio change produced by each additional increment of axial travel need not remain constant. In the usual upshift direction, the large ratio sensitivity associated with small primary radius progressively weakens as the primary radius grows. The ratio therefore often changes rapidly at the beginning of the shift and then eases off, even without a change in axial shift speed.

Appendix D

Construction of Primary Flyweight Geometry Maps

The primary axial and rotational balances developed in the main derivation do not require the detailed flyweight geometry directly. They require only the configuration maps

$$q_f = q_f(x_p), \qquad J_{f,p} = J_{f,p}(x_p), \tag{D.0.1}$$

together with the constant pivot-axis inertia I_f. The first map describes how the flyweight pose changes as the movable primary sheave translates. The second describes how that motion redistributes the flyweight mass about the primary shaft.

These maps form the interface between a particular physical actuator and the general primary equations. Different flyweight mechanisms may therefore be introduced by deriving different geometry maps while leaving the surrounding CVT formulation unchanged.

The first mechanism class treated here is a rigid flyweight pinned to the fixed primary assembly, with a roller near its outer end contacting a ramp rigidly attached to the movable sheave. The ramp may have a general smooth profile and may itself be assembled from several smooth profile segments.

D.1 Pivoted Flyweight with a Roller Follower

Consider the radial–axial plane of one primary flyweight. Let $+\boldsymbol{e}_x$ denote the positive primary closing direction and $+\boldsymbol{e}_r$ the radially outward direction. The flyweight pivots about the fixed point

$$P = (x_P, r_P). \tag{D.1.1}$$

Let C denote the center of the roller bearing and let

$$\ell = |PC| \tag{D.1.2}$$

be the fixed distance from the flyweight pivot to the roller center. The roller has radius R_{roll}.

The flyweight pose is described by the same coordinate q_f used in the main derivation, measured from $+\boldsymbol{e}_x$ to the body-fixed line from P to C. The roller-center position implied by a flyweight angle q_f is therefore

$$\boxed{\boldsymbol{C}(q_f) = \begin{bmatrix} x_P + \ell \cos q_f \\ r_P + \ell \sin q_f \end{bmatrix}.} \tag{D.1.3}$$

The roller center is therefore constrained to a circle of radius ℓ about the pivot. The ramp contact selects the point on this circle that is compatible with the current movable-sheave position.

D.1.1 General Ramp Representation

Rather than assigning the flyweight radius directly as a function of primary shift, the physical ramp itself is described. Introduce a local axial coordinate ξ fixed to the movable ramp carrier and write the ramp as the graph

$$\boxed{\eta = f(\xi), \qquad \xi \in \mathcal{I},} \tag{D.1.4}$$

where η is the radial coordinate in the same local plane.

If the reference axial location of the ramp carrier is $x_{R,0}$, translation of the movable sheave through x_p places the physical ramp at

$$\boxed{\boldsymbol{\Gamma}(\xi; x_p) = \begin{bmatrix} x_{R,0} + x_p + \xi \\ f(\xi) \end{bmatrix}.} \tag{D.1.5}$$

The profile f may be any sufficiently smooth function satisfying the geometric admissibility conditions developed below. Simple profile families remain useful as convenient ways to define the physical ramp. For example,

$$f(\xi) = a + b\xi \tag{D.1.6}$$

gives a straight ramp, while a circular-arc segment may be written

$$f(\xi) = r_c + \sigma_c \sqrt{R_c^2 - (\xi - \xi_c)^2}, \qquad \sigma_c \in \{-1, +1\}, \tag{D.1.7}$$

over an interval that remains on one branch of the circle. A hyperbolic segment may similarly be written

$$f(\xi) = r_c + \sigma_h \sqrt{K\left[(\xi - \xi_c)^2 + \xi_0^2\right]}, \tag{D.1.8}$$

and polynomial, spline, or CAD-derived profiles may be used in exactly the same way.

These shapes are used here only to specify the physical contact surface. No force law is implied directly by the choice of f. In particular, the radial motion of the flyweight mass is not generally equal to $f(x_p)$. For the pivoted mechanism considered here, that motion emerges only after the roller radius, arm length, pivot location, and contact geometry have been applied.

Piecewise profiles are permitted. If

$$f(\xi) = f_i(\xi), \qquad \xi_i \leq \xi \leq \xi_{i+1}, \tag{D.1.9}$$

the pieces must join smoothly enough that the resulting flyweight map satisfies the differentiability requirements of the main model. A geometric corner is not admissible within the active ramp region because the roller normal would change discontinuously there, producing a discontinuity in the flyweight motion ratio. In practice, transitions between straight lines, circular arcs, or other simple primitives may be replaced by smooth polynomial or spline blends.

D.1.2 From the Physical Ramp to the Roller-Center Locus

The roller center does not lie on the physical ramp. At contact it lies one roller radius away in the local normal direction.

The unit tangent of the ramp is

$$\widehat{\boldsymbol{t}}(\xi) = \frac{1}{\sqrt{1+[f'(\xi)]^2}} \begin{bmatrix} 1 \\ f'(\xi) \end{bmatrix}, \tag{D.1.10}$$

and a corresponding unit normal is

$$\widehat{\boldsymbol{n}}(\xi) = \sigma_n \frac{1}{\sqrt{1+[f'(\xi)]^2}} \begin{bmatrix} -f'(\xi) \\ 1 \end{bmatrix}, \qquad \sigma_n \in \{-1, +1\}, \tag{D.1.11}$$

where σ_n selects the side of the ramp on which the physical roller lies.

The locus traced by the center of a roller tangent to the ramp is therefore the offset curve

$$\boxed{\boldsymbol{\Gamma}_R(\xi; x_p) = \boldsymbol{\Gamma}(\xi; x_p) + R_{\text{roll}} \widehat{\boldsymbol{n}}(\xi).} \tag{D.1.12}$$

This removes the roller radius from the remaining contact problem. The physical mechanism may now be viewed as the intersection of two curves: a circle of radius ℓ centered at the fixed pivot P, and the roller-center ramp $\boldsymbol{\Gamma}_R$, which translates axially with the movable sheave.

D.1.3 Solving for the Flyweight Angle

A point on the roller-center locus is compatible with the rigid flyweight when its distance from P is exactly ℓ. Define

$$\boxed{g(\xi, x_p) = \|\boldsymbol{\Gamma}_R(\xi; x_p) - \boldsymbol{P}\|^2 - \ell^2.} \tag{D.1.13}$$

For a prescribed movable-sheave position x_p, the contact coordinate $\xi^\star$ is obtained from

$$\boxed{g(\xi^\star, x_p) = 0.} \tag{D.1.14}$$

The corresponding roller center is

$$\boldsymbol{C}(x_p) = \boldsymbol{\Gamma}_R(\xi^\star(x_p); x_p), \tag{D.1.15}$$

and the flyweight angle follows immediately:

$$\boxed{q_f(x_p) = \operatorname{atan2}\left(C_r(x_p) - r_P,\, C_x(x_p) - x_P\right).} \tag{D.1.16}$$

Thus the physical ramp profile produces the required configuration map through

$$\boxed{f(\xi) \longrightarrow \boldsymbol{\Gamma}_R(\xi; x_p) \longrightarrow \xi^\star(x_p) \longrightarrow q_f(x_p).} \tag{D.1.17}$$

The roller radius enters through the offset curve, while the arm length and pivot location enter through the circle constraint. None of these quantities needs to be absorbed into an empirical ramp factor.

D.1.4 Geometric Admissibility

Equation (D.1.14) is only an algebraic contact condition. A mathematically valid root need not represent an acceptable physical mechanism. The ramp geometry is therefore restricted so that one continuous, non-interfering contact configuration exists throughout the required primary travel

$$x_p \in \mathcal{X}_p. \tag{D.1.18}$$

The following conditions define the admissible geometry used here.

Existence. For every $x_p \in \mathcal{X}_p$, Equation (D.1.14) must possess a contact solution $\xi^\star$ inside the active ramp domain.

Unique physical contact. The roller may touch the ramp at only one geometric point. For the center obtained from Equation (D.1.15),

$$\boxed{\|\boldsymbol{C}(x_p) - \boldsymbol{\Gamma}(\xi; x_p)\| \geq R_{\text{roll}} \qquad \forall\, \xi \in \mathcal{I},} \tag{D.1.19}$$

with equality only at the physical contact point $\xi^\star$. Equality at two parameter values representing the same smooth junction is equivalent to one geometric contact and is not considered a second contact.

Equation (D.1.19) prevents both penetration into another portion of the ramp and separated two-point support of the same roller. The model therefore does not choose between two simultaneously admissible contacts by proximity or by a numerical root-selection rule. A ramp that permits two distinct physical contacts at the same x_p lies outside the present single-contact mechanism class.

Regular roller offset. The offset curve itself must remain regular. Let $\kappa_n(\xi)$ denote the signed ramp curvature measured toward the selected roller normal. The local offset becomes singular if the roller radius equals the corresponding radius of curvature. The admissible profile therefore satisfies

$$\boxed{1 - R_{\text{roll}}\kappa_n(\xi) \neq 0} \tag{D.1.20}$$

throughout the contacted region. In practical profile design, a positive margin from zero is preferable to operation arbitrarily close to this limit.

Nonsingular mechanism branch. The circle and roller-center locus must intersect transversely. From Equation (D.1.13),

$$\boxed{\frac{\partial g}{\partial \xi} \neq 0} \tag{D.1.21}$$

along the operating branch.

If $g_\xi = 0$, the circle is tangent to the roller-center locus. The implicit map $\xi^\star(x_p)$ then loses regularity and the mechanism approaches a fold or dead-center configuration. Such a point can create an unbounded motion ratio or cause the existing contact branch to disappear.

Correct motion direction and smoothness. The resulting map must satisfy the requirements imposed by the primary dynamics,

$$\boxed{q_f \in C^2, \qquad 0 < \frac{dq_f}{dx_p} < \infty.} \tag{D.1.22}$$

Thus primary closure corresponds continuously to outward flyweight rotation, without a reversal or singular motion ratio.

Together, Equations (D.1.19)–(D.1.22) provide a stronger criterion than merely selecting the root nearest the previous time step. Continuation from a known assembled state remains useful numerically, but an admissible ramp is designed so that the physical solution itself is unique throughout the modeled travel.

Ramp endpoints may coincide with physical limits of the actuator. Such endpoints are acceptable provided the contact remains regular when approached from inside the modeled interval; the map is not extended beyond a physical stop.

D.1.5 Derivatives of the Geometry Map

The primary force law requires both dq_f/dx_p and d^2q_f/dx_p^2. These derivatives may be obtained from the same implicit contact equation rather than by finite-differencing a sampled angle map.

Differentiating

$$g(\xi^\star(x_p), x_p) = 0 \tag{D.1.23}$$

gives

$$\boxed{\frac{d\xi^\star}{dx_p} = -\frac{g_{x_p}}{g_\xi}.} \tag{D.1.24}$$

Differentiating again gives

$$\boxed{\frac{d^2\xi^\star}{dx_p^2} = -\frac{g_{x_p x_p} + 2g_{x_p\xi}\dfrac{d\xi^\star}{dx_p} + g_{\xi\xi}\left(\dfrac{d\xi^\star}{dx_p}\right)^2}{g_\xi}.} \tag{D.1.25}$$

The denominator in both expressions shows directly why the nonsingularity condition in Equation (D.1.21) is required.

Let

$$\boldsymbol{v}(x_p) = \boldsymbol{C}(x_p) - \boldsymbol{P} = \begin{bmatrix} v_x \\ v_r \end{bmatrix}, \qquad \|\boldsymbol{v}\| = \ell. \tag{D.1.26}$$

Since

$$q_f = \operatorname{atan2}(v_r, v_x), \tag{D.1.27}$$

its first derivative is

$$\boxed{\frac{dq_f}{dx_p} = \frac{v_x\dfrac{dv_r}{dx_p} - v_r\dfrac{dv_x}{dx_p}}{\ell^2}.} \tag{D.1.28}$$

Because $\|\boldsymbol{v}\| = \ell$ is constant, the second derivative simplifies to

$$\boxed{\frac{d^2 q_f}{dx_p^2} = \frac{v_x \dfrac{d^2 v_r}{dx_p^2} - v_r \dfrac{d^2 v_x}{dx_p^2}}{\ell^2}.} \tag{D.1.29}$$

The required derivatives of $\boldsymbol{v}$ follow from Equation (D.1.15):

$$\frac{d\boldsymbol{C}}{dx_p} = \boldsymbol{e}_x + \frac{\partial \boldsymbol{\Gamma}_R}{\partial \xi} \frac{d\xi^\star}{dx_p}, \tag{D.1.30}$$

and

$$\frac{d^2\boldsymbol{C}}{dx_p^2} = \frac{\partial^2 \boldsymbol{\Gamma}_R}{\partial \xi^2} \left(\frac{d\xi^\star}{dx_p} \right)^2 + \frac{\partial \boldsymbol{\Gamma}_R}{\partial \xi} \frac{d^2 \xi^\star}{dx_p^2}. \tag{D.1.31}$$

Equations (D.1.24)–(D.1.31) therefore provide the complete C^2 flyweight-angle map required by the transient primary balance.

D.2 Shaft-Axis Inertia from the Flyweight Mass Geometry

Once $q_f(x_p)$ is known, constructing $J_{f,p}(x_p)$ is comparatively direct.

Consider one rigid flyweight and introduce body-fixed coordinates (u, v, z) relative to its pivot. Let the u-axis coincide with the reference line defining q_f, let v lie in the axial–radial plane, and let z point circumferentially along the pivot axis.

Rotation through q_f gives the radial coordinate of a mass element dm as

$$r(q_f) = r_P + u \sin q_f + v \cos q_f. \tag{D.2.1}$$

Its squared perpendicular distance from the primary shaft is

$$r_\perp^2 = (r_P + u \sin q_f + v \cos q_f)^2 + z^2. \tag{D.2.2}$$

For n_f identical flyweights arranged symmetrically around the primary, their complete shaft-axis inertia is therefore

$$\boxed{J_{f,p}(q_f) = n_f \int_{\mathcal{B}} \left[(r_P + u \sin q_f + v \cos q_f)^2 + z^2 \right] dm.} \tag{D.2.3}$$

Composition with the geometry map gives the quantity required by the main model:

$$\boxed{J_{f,p}(x_p) = J_{f,p}(q_f(x_p)).} \tag{D.2.4}$$

The same body description gives the constant pivot-axis inertia

$$\boxed{I_f = n_f \int_{\mathcal{B}} \left(u^2 + v^2\right) dm.} \tag{D.2.5}$$

Any freely spinning motion of the roller about its own bearing axis is excluded from Equation (D.2.5). If roller spin losses or roller rotational dynamics are of interest, that motion constitutes an additional internal degree of freedom rather than part of the rigid flyweight coordinate q_f.

D.2.1 Uniform Arm with Concentrated End Mass

For the common case in which most of the moving mass is concentrated along the line from P to the roller center, the general inertia expression simplifies considerably.

Let a uniform arm of mass m_a extend from $u = 0$ to $u = \ell$, and let an additional mass m_e, representing the weights, roller, bolt, and associated outer hardware, be concentrated at $u = \ell$. Taking $v = 0$ for this simplified geometry gives, for one flyweight,

$$\boxed{J_{f,p}^{(1)}(q_f) = m_a \left[r_P^2 + r_P \ell \sin q_f + \frac{\ell^2}{3} \sin^2 q_f \right] + m_e \left(r_P + \ell \sin q_f \right)^2 + J_{\text{const}}.} \tag{D.2.6}$$

Here J_{const} collects configuration-independent contributions. These contributions may be retained in the total shaft inertia but vanish from $dJ_{f,p}/dx_p$.

It is useful to collect the same result in terms of the first and second mass moments measured from the pivot,

$$S_1 = \int u \, dm, \qquad S_2 = \int u^2 \, dm. \tag{D.2.7}$$

For the uniform arm and concentrated end mass,

$$S_1 = \frac{m_a \ell}{2} + m_e \ell, \tag{D.2.8}$$

and

$$S_2 = \frac{m_a \ell^2}{3} + m_e \ell^2. \tag{D.2.9}$$

Writing $M = m_a + m_e$, Equation (D.2.6) becomes

$$\boxed{J_{f,p}^{(1)}(q_f) = M r_P^2 + 2 r_P S_1 \sin q_f + S_2 \sin^2 q_f + J_{\text{const}}.} \tag{D.2.10}$$

For the complete symmetric flyweight set,

$$\boxed{J_{f,p}(x_p) = n_f J_{f,p}^{(1)}(q_f(x_p)) \, .} \tag{D.2.11}$$

Likewise, under the same slender-arm and concentrated-end-mass approximation,

$$\boxed{I_f = n_f \left(\frac{m_a \ell^2}{3} + m_e \ell^2 \right),} \tag{D.2.12}$$

with any additional pivot-axis inertia of hardware rigidly rotating with the flyweight added if required.

Mass located very near the pivot illustrates why this representation is useful. A mass centered at P remains at the constant shaft radius r_P. It therefore contributes to the absolute shaft inertia $J_{f,p}$, but not to its configuration derivative and hence not to the centrifugal closing term. Its contribution to I_f is similarly small if its extent about the pivot is small.

D.3 Resulting Mechanism Interface

The geometry developed above reduces the complete pivoted flyweight and roller-ramp mechanism to the same quantities assumed in the general primary derivation:

$$\boxed{q_f = q_f(x_p), \qquad J_{f,p} = J_{f,p}(x_p), \qquad I_f = \text{constant.}} \tag{D.3.1}$$

The first map follows from rigid-arm compatibility with the roller-center offset of the translating ramp. The second follows by evaluating the flyweight mass distribution at that pose. Their required derivatives follow directly from the same construction.

For the present mechanism class, the complete chain is therefore

$$\boxed{f(\xi) \longrightarrow \mathbf{\Gamma}_R \longrightarrow q_f(x_p) \longrightarrow J_{f,p}(x_p).} \tag{D.3.2}$$

The physical ramp is consequently separated from the dynamical quantities it ultimately produces. Straight, circular, hyperbolic, spline, or piecewise profiles are merely different choices of f; the roller and pivot geometry determine how each such surface becomes a flyweight motion, while the mass geometry determines how that motion becomes a changing shaft-axis inertia.

This separation is also what allows other primary-actuator geometries to be introduced without altering the equations in the main derivation. A different mechanism need only provide its own construction of the same interface in Equation (D.3.1), subject to the same map-level admissibility requirements.

D.4 Alternative Flyweight Geometry

Placeholder.

A second primary flyweight mechanism class will be developed here. Its internal geometry, contact constraints, and admissibility conditions may differ from the pivoted roller-follower mechanism above, but the derivation will terminate at the same actuator interface:

$$\boxed{q_f = q_f(x_p), \qquad J_{f,p} = J_{f,p}(x_p), \qquad I_f = \text{constant.}} \tag{D.4.1}$$

The purpose of this section is therefore not to alter the primary force or rotational equations, but only to supply a different mechanism-specific construction of the quantities in Equation (D.4.1). Any future geometry introduced here should establish:

1. the physical configuration constraint that determines $q_f(x_p)$;
2. the admissibility conditions required for that map to remain unique and regular over the modeled travel;
3. the derivatives dq_f/dx_p and d^2q_f/dx_p^2;
4. the corresponding shaft-axis inertia $J_{f,p}(x_p)$; and
5. the constant pivot- or mechanism-coordinate inertia I_f, where applicable.

Once these quantities are available, the mechanism can be substituted into the same general primary equations without any further change to the CVT model.